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Solution

PIRACY IS NOT GOOD

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SEO Made Easy

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APTECH LIMITED

Contact E-mail: ov-support@onlinevarsity.com

First Edition - 2014

Aptech®
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Dear Learner,

We congratulate you on your decision to pursue an Aptech course.

Aptech Ltd. designs its courses using a sound instructional design model – from conceptualization to execution, incorporating the following key aspects:

- Scanning the user system and needs assessment

Needs assessment is carried out to find the educational and training needs of the learner.

Technology trends are regularly scanned and tracked by core teams at Aptech Ltd. TAG* analyzes these on a monthly basis to understand the emerging technology training needs for the Industry.

An annual Industry Recruitment Profile Survey# is conducted during August - October to understand the technologies that Industries would be adapting in the next 2 to 3 years. An analysis of these trends & recruitment needs is then carried out to understand the skill requirements for different roles & career opportunities.

The skill requirements are then mapped with the learner profile (user system) to derive the Learning objectives for the different roles.

- Needs analysis and design of curriculum

The Learning objectives are then analyzed and translated into learning tasks. Each learning task or activity is analyzed in terms of knowledge, skills and attitudes that are required to perform that task. Teachers and domain experts do this jointly. These are then grouped in clusters to form the subjects to be covered by the curriculum.

In addition, the society, the teachers, and the industry expect certain knowledge and skills that are related to abilities such as learning-to-learn, thinking, adaptability, problem solving, positive attitude etc. These competencies would cover both cognitive and affective domains.

A precedence diagram for the subjects is drawn where the prerequisites for each subject are graphically illustrated. The number of levels in this diagram is determined by the duration of the course in terms of number of semesters etc. Using the precedence diagram and the time duration for each subject, the curriculum is organized.

- Design & development of instructional materials

The content outlines are developed by including additional topics that are required for the completion of the domain and for the logical development of the competencies identified. Evaluation strategy and scheme is developed for the subject. The topics are arranged/ organized in a meaningful sequence.

The detailed instructional material – Training aids, Learner material, reference material, project guidelines, etc.- are then developed. Rigorous quality checks are conducted at every stage.

- Strategies for delivery of instruction

  Careful consideration is given for the integral development of abilities like thinking, problem solving, learning-to-learn etc. by selecting appropriate instructional strategies (training methodology), instructional activities and instructional materials.

  The area of IT is fast changing and nebulous. Hence, considerable flexibility is provided in the instructional process by specially including creative activities with group interaction between the students and the trainer. The positive aspects of Web based learning –acquiring information, organizing information and acting on the basis of insufficient information are some of the aspects, which are incorporated, in the instructional process.

- Assessment of learning

  The learning is assessed through different modes – tests, assignments & projects. The assessment system is designed to evaluate the level of knowledge & skills as defined by the learning objectives.

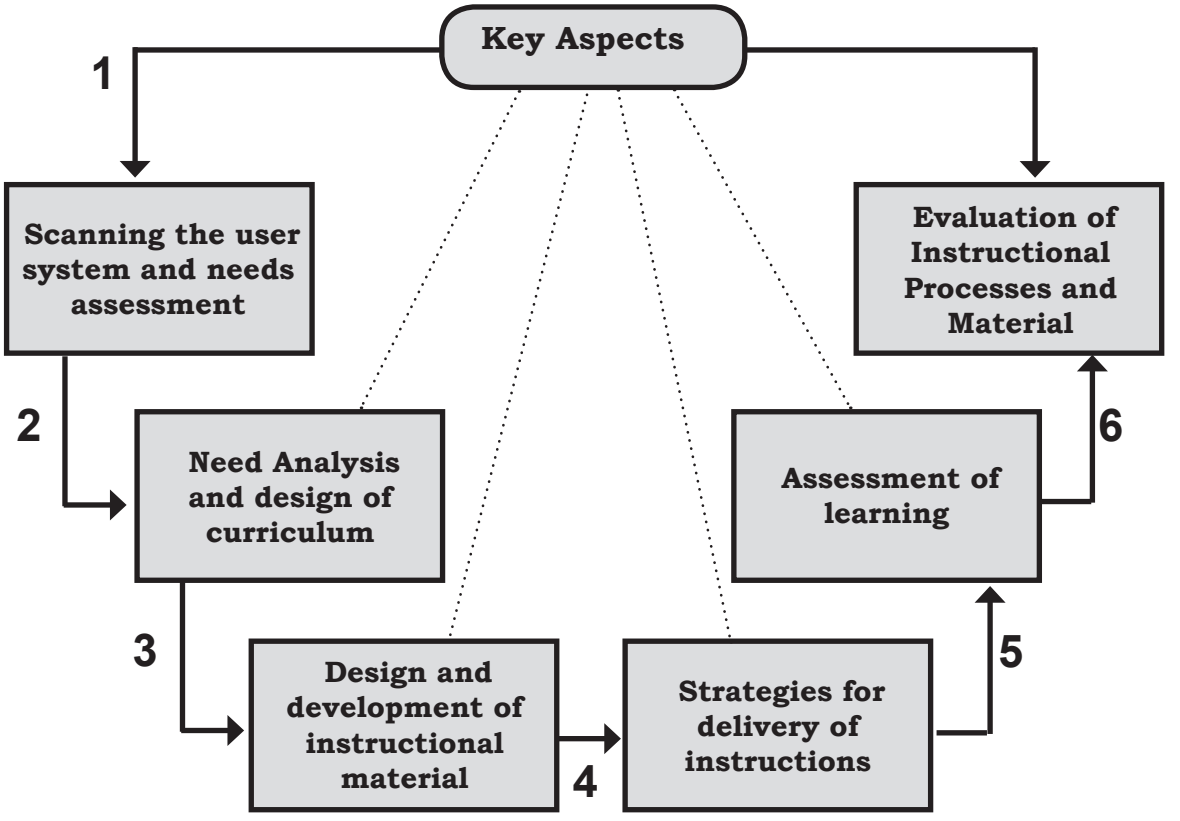
- Evaluation of instructional process and instructional materials

  The instructional process is backed by an elaborate monitoring system to evaluate - on-time delivery, understanding of a subject module, ability of the instructor to impart learning. As an integral part of this process, we request you to kindly send us your feedback in the reply pre-paid form appended at the end of each module.

*TAG – Technology & Academics Group comprises of members from Aptech Ltd., professors from reputed Academic Institutions, Senior Managers from Industry, Technical gurus from Software Majors & representatives from regulatory organizations/forums.

Technology heads of Aptech Ltd. meet on a monthly basis to share and evaluate the technology trends. The group interfaces with the representatives of the TAG thrice a year to review and validate the technology and academic directions and endeavors of Aptech Ltd.

Aptech New Products Design Model



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Preface

Consumers of Web site contents and online shopping customers often make use of search engines for their queries. Today, the number of Web sites across the world exceeds 200 million. On an average, there are over 60 billion search queries per month on Google alone. Given such statistics, it is crucial that a business or even an individual should improve traffic to their Web site through various strategies. Therefore, optimization of Web sites for a search engine becomes extremely important as search engines can drive traffic to the sites. Search Engine Optimization (SEO) is an important Internet marketing strategy that can help to achieve this.

This book begins with an introduction to Google Search and the Google Search Engine. Then, the book explains SEO and various techniques used in SEO. These techniques include on-page optimization, off-page optimization customization, SEO tools, Search Engine Marketing, and mobile SEO.

This book is the result of a concentrated effort of the Design Team, which is continuously striving to bring you the best and the latest in Information Technology. The process of design has been a part of the ISO 9001 certification for Aptech-IT Division, Education Support Services. As part of Aptech's quality drive, this team does intensive research and curriculum enrichment to keep it in line with industry trends.

We will be glad to receive your suggestions.

Design Team

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01 Introduction to Google Search

Welcome to the Session, **Introduction to Google Search.**

This session introduces Google search. The session begins by introducing the Internet and different terminologies related to it. The session also describes the World Wide Web (WWW) and various parts of a browser. It then explains the Google search and different search techniques.

In this Session, you will learn to:

- Define the Internet and different terminologies related to it
- Explain the various parts of a browser
- Describe the terms Web Address, Web pages, and Web site
- Define and describe search engines
- Explain the Google search
- Describe various search techniques and strategies

Session
01

Introduction to Google Search

1.1 Introduction

Google Inc. is a multinational giant in the field of Internet search, communication, social networking, and advertising technologies. One of its most popular products is the Google Search, which for simplicity, is also called Google.

Google Search is used to search information from all over the world. This information can include Web pages, images, videos, news, maps, and so on.

Google is one of the most popular search engines among Internet users, handling over 70% of the word's online requests.

1.2 Introduction to the Internet

The Internet is a network of computers connected globally. It connects universities, government organizations, commercial, and other computers in over 150 countries all over the world. There are a number of networks, thousands of computers, and billions of users on the Internet that expands daily.

With the Internet, apart from accessing the World Wide Web, users can send electronic mail (e-mail), chat with friends globally, shop online, check their bank accounts, work from home, and search for any information worldwide on a number of topics.

Figure 1.1 shows a graphical depiction of the Internet.

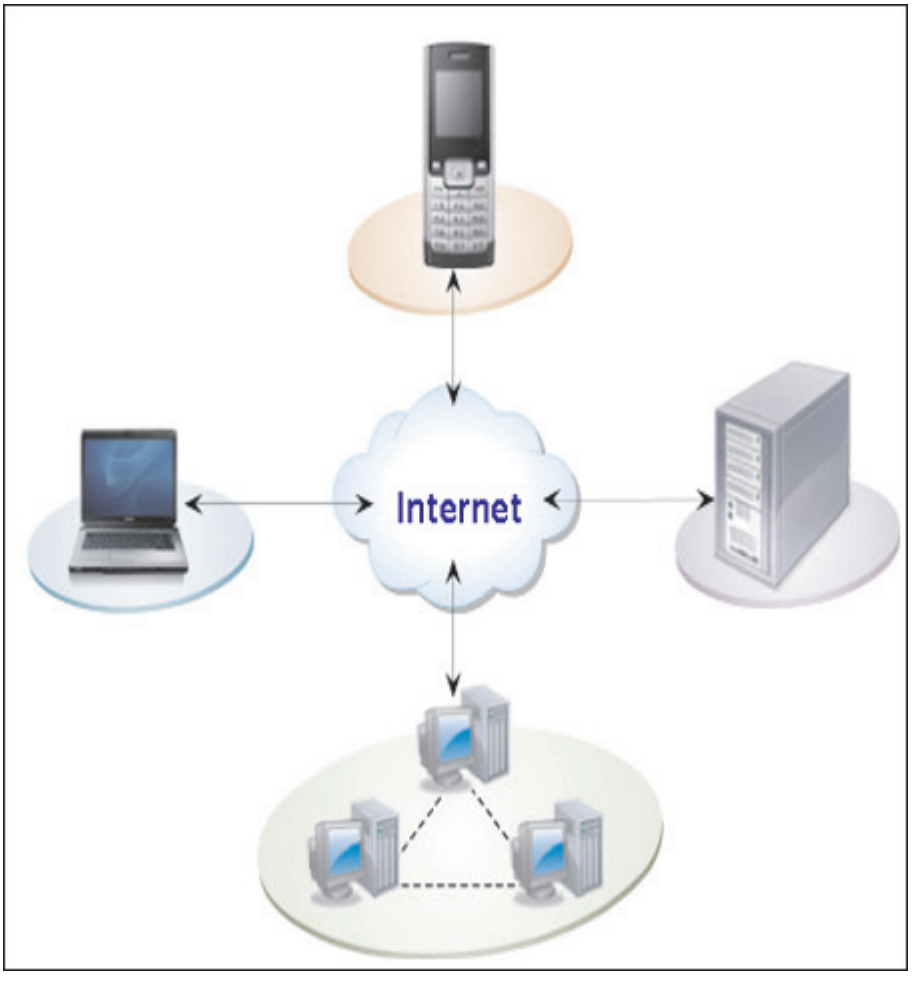

Figure 1.1: Internet

1.3 Terminologies Related to the Internet

The key terminologies related to the Internet are as follows:

- **Internet Service Providers (ISPs)**

  The companies that provide users commercial access to the Internet are known as Internet Service Providers (ISPs). An ISP offers various services such as storing files of customers at remote locations. For example, America Online and Comcast are popular ISPs in the USA.

  Depending on the type of Internet connection (dialup or broadband), ISPs provide certain hardware and software along with a username and password that need to be configured to access the Internet. The ISP charges a monthly fee in return for the service it provides.

  ISPs provide Internet services to both individuals as well as large companies.

- **Internet Protocol (IP) Address**

  An IP address is a method used for identifying the Web site on the computer network.

Session
01

Introduction to Google Search

A Web site is a collection of pages accessed over the Internet. For example, the IP address of the Web site, **www.aptechlearningservices.com** may be 220.226.186.67.

Every computer or mobile device that accesses the Internet is assigned an IP address. The IP address acts like a license number for users to use the services of the Internet. It facilitates tracing the actions of Internet users.

- **Domain Name System (DNS)**
  DNS is referred to as the hierarchal naming system used for computers, resources, and services that are available on the Internet. The DNS translates the host names of a computer to IP address. Domain names or Web site names are assigned to users through DNS. For example, **www.aptechlearningservices.com** is a domain name.

- **Uniform Resource Locator (URL)**
  A URL specifies the location of a resource available on the Internet. A URL contains the path and the address of the Web page. A Web page is a document on a Web site. A URL of a Web page together with the IP address defines its name and location, which can be used to refer the page at a later time and even bookmark it.

- **Proxy Server**
  In a network, client machines may be connected to a proxy server which forwards the client requests to different servers and responds back to the clients. The proxy server acts like a middleman between other servers and the clients.

- **HyperText Transfer Protocol (HTTP)**
  The HTTP protocol is used for transferring information over the Internet. The client makes an HTTP request that uses a Web browser to which an HTTP response is sent from the server.

- **HyperText Transfer Protocol Secured (HTTPS)**
  The HTTPS protocol is similar to the HTTP protocol and is also used for transferring information over the Internet, albeit in a secure manner. S stands for secured. Web pages that use this protocol use a special layer of encryption to hide any personal information and passwords that might be transmitting over the Internet. Typically, browser-based mails and Web sites and pages requiring secure login, such as bank Web sites, use the HTTPS protocol.

- **Cache**
  A cache maintains a list of recently visited Web pages in the Web browser. This can speed up operations for users who visit the same sites frequently.

- **Web Server**
  A Web server accepts requests from the Web clients and provides them with a response.

- **Web Cookie**
  A Web cookie is a small piece of text which is exchanged between a client and a server. This Web cookie is also referred as HTTP cookie. The cookie is then sent by the Web server to the Web client and is returned without making any changes by the client each time it is accessed by the server.

- **Session**
  A session involves sharing of information between the computer and its user. The session is established only for a particular period of time.

- **Hyperlink**
  A hyperlink is a reference in a hypertext or Web document to another document or to another section of the document. A hyperlink is used to redirect the user from one page to another.

- **HTML and XML**
  HTML or HyperText Markup Language is the programming language that is used to create Web pages. The HTML contains commands that the Web browser uses to display information in the desired format. The commands in HTML are called HTML tags. `<body></body>`, `<a href="www.google.com"></a>`, and `<title></title>` are examples of HTML tags.

  XML or eXtensible Markup Language is a language that is used for cataloging and organizing Web page content in a classified manner, similar to a database. Similar to HTML, XML commands can be used display the textual content in Web pages. `<entry>`, `<address>`, and `<city>` are examples of XML commands.

- **World Wide Web (WWW)**
  The WWW is a collection of interconnected documents that can be accessed over the Internet. The WWW contains millions of Web pages that consist of games, images, text, voice, and videos. It is a User Interface (UI) to the Internet. Users can use the WWW without the need to know where the computers are located or how many computers exist.

- **Browser**
  A browser is a software program which facilitates the user to search and view different types of information available on the Internet. This information could be on any topic such as Web sites, videos, images, and audio information. Web browsers are one of the most commonly and widely used software. The main function of a browser is to present the Web resource that the user selects for requesting from the Web server and display that on the browser window. The major browsers used are Chrome, Internet Explorer, Safari, Firefox, and Opera.

1.4 Web Address

A Web address is used to identify address of the file or Web page on the Internet. When the user types the Web address in the Address bar, he or she can access that file on the Internet. Figure 1.2 displays the example of a Web address in a browser window.

An address string contains the access protocol, domain name, and the path to the file or resource on the host computer. Typing the address string in the Address bar allows users to access the require resource over the Internet.

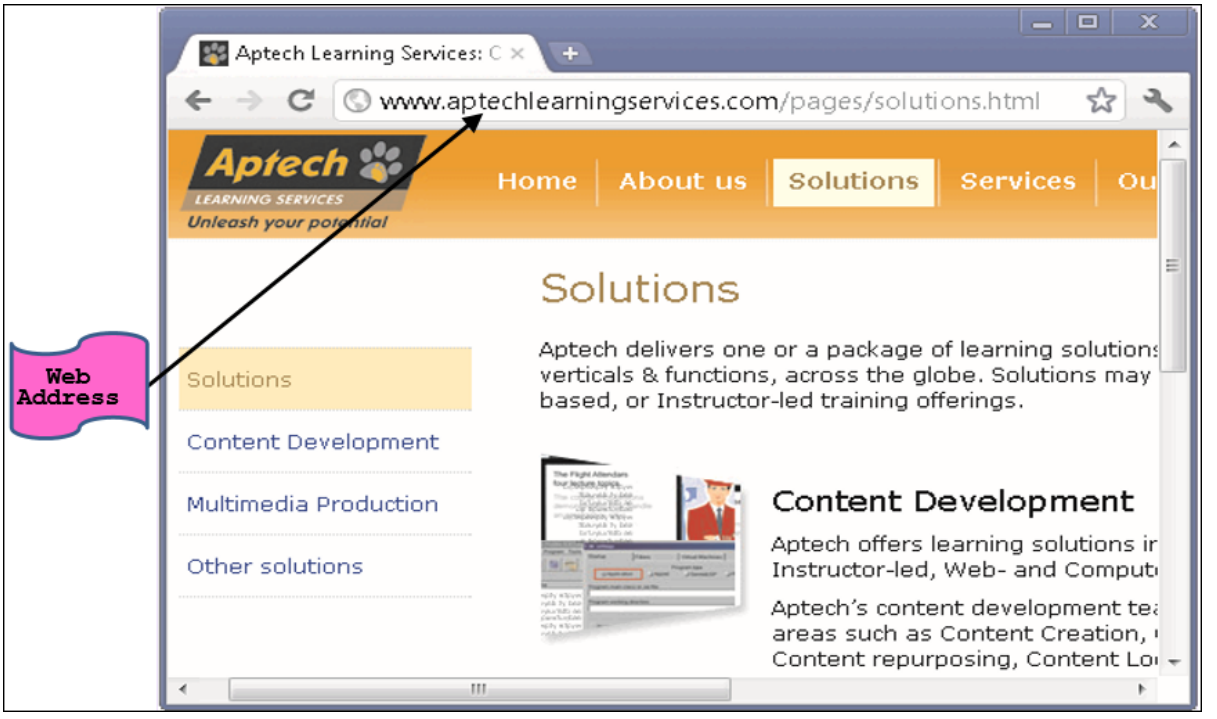

Figure 1.2: Web Address

1.5 Web Pages

A Web page represents a document that resides at remote location. Every Web page is representing different types of information presented to the users in aesthetic manner and readable manner. These Web pages are available on the Internet and are accessible to all Internet users. Every Web page has a Web address. Figure 1.3 displays an example of a Web page.

A Web page is accessed by entering its URL in the Address bar of a browser. Information in a Web page may be presented as a combination of text, images, graphics, and hyperlinks.

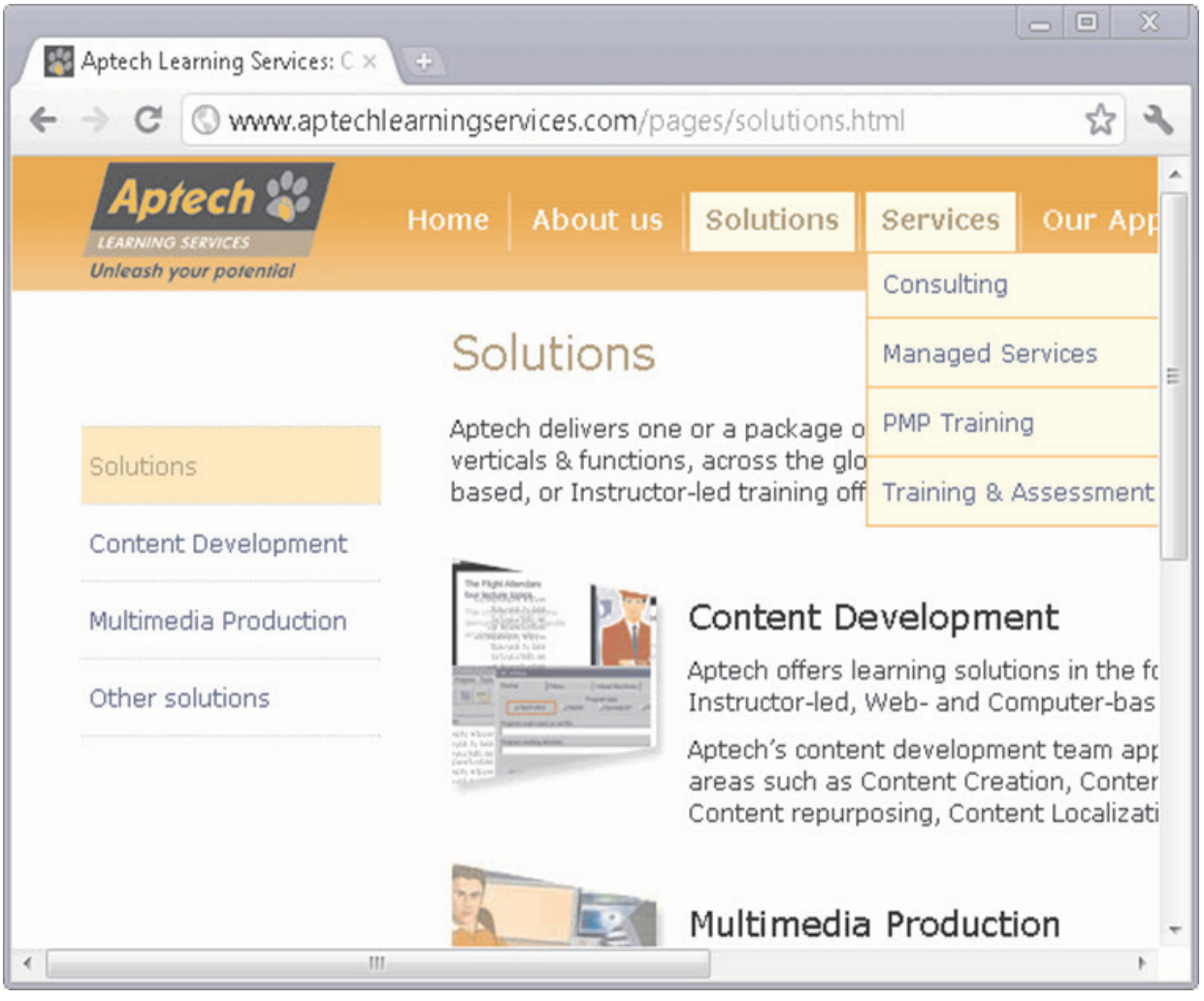

Figure 1.3: Web Page

A Web page has a fantastic feature of allowing embedded references to other pages.

1.6 Web Site

A collection of Web pages that are accessed over the Internet is called a Web site. A Web site can contain one or more pages. When a user types a Web address, the browser navigates to the relevant page of the Web site specified by the Web address. The Web page has various links that helps the user to navigate through number of pages in the Web site. The Web site will introduce the user to what the site is offering, such as some information, or different types of services, and so on.

Putting up a Web site requires two things: a domain name and a Web host. Domain name is the address of the Web site. For example, **www.myWebsite.com** is the domain name of the Web site. A Web host is a computer server on the Internet that stores the information to be displayed on the Web site.

Figure 1.4 displays an example of a Web site.

Figure 1.4: Web Site

1.7 Web Site versus Web Pages

Table 1.1 lists the differences between Web sites and Web pages.

Web Site	Web Pages
Is a collection of Web pages	Is a sub part of a Web site
Belongs to owner or the domain	Belongs to the Web site
Is designed to convey information on a topic or theme	Contains links to other pages in the Web site

Table 1.1: Web Site versus Web Pages

1.8 Google Chrome

Google Chrome, also called Chrome, is a freeware Web browser developed by Google. It is one of the most popular and fastest browsers available. The user can chat, search, shop, read news, and perform many more operations using this browser. Some of the key features of this browser are as follows:

- **Simplicity**
  Chrome has features that are simple and easy to use.

- **Security**
  Chrome is secured and safe on the Web with latest security updates.
- **Speed**
  Chrome is faster, starts quickly on the user's desktop, and allows running of complex Web applications.
- **Single box for Search, Address, and History**
  Unlike other Web browsers, Chrome does not have separate boxes for specifying the address, searching, and view history. A single bar placed below the Toolbar serves all three purposes. So, typing text like 'techrepublic' in the box will suggest the Google search engine results for the words, the Web address **techrepublic.com**, as well as the recently visited pages.

  Figure 1.5 shows an example of a single box for search, address, and history.

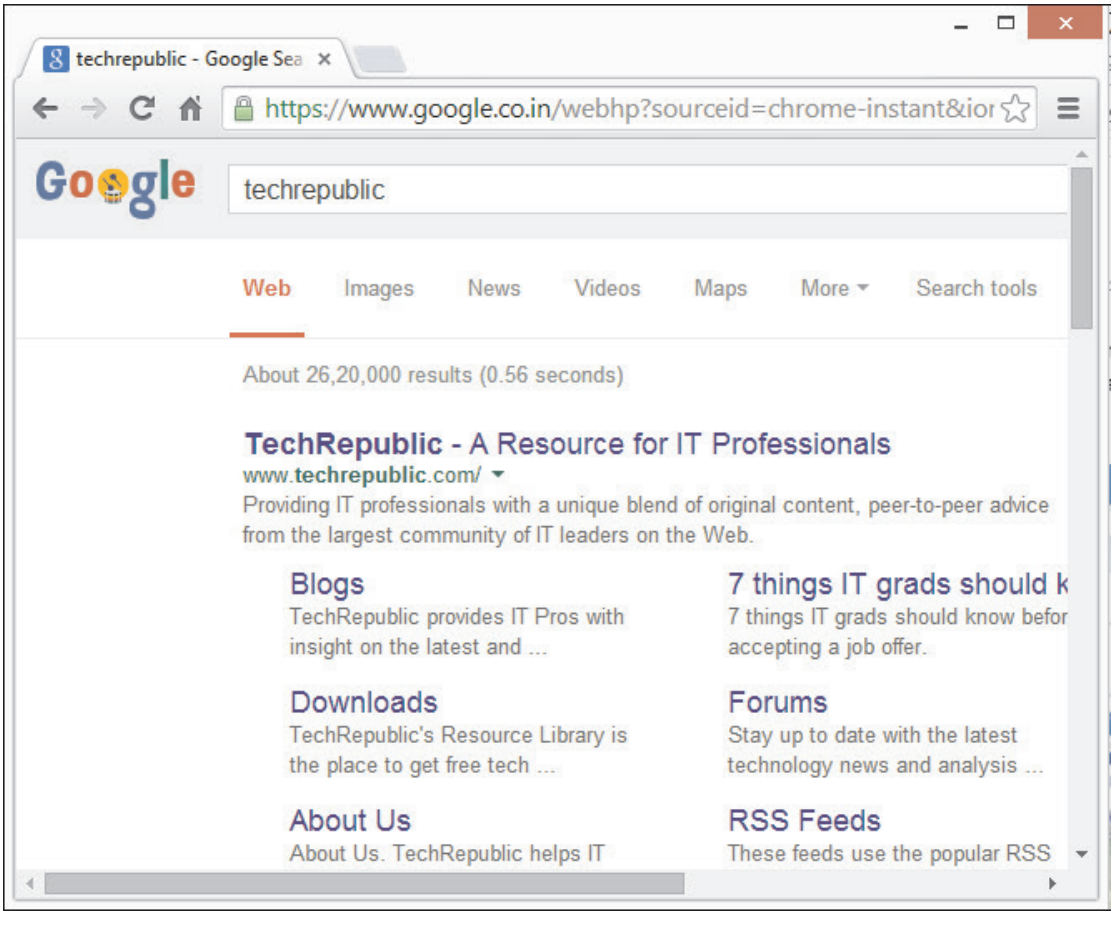

Figure 1.5: One Box for Everything Feature

- **Incognito Browsing**
  Chrome provides an **Incognito browsing** mode where the browser does not store any history about the Web sites visited, the form fields, or cookies. This mode is very useful when users are working on shared computers in a public environment.

- **Pin Frequently Used Tabs**

Chrome allows users to pin tabs containing their favorite or most frequently visited Web pages. This helps keep the Tab bar uncluttered and also provide quick access to the Web pages. In figure 1.6, eight tabs have been pinned to the browser.

To pin a tab, right-click the tab and select the Pin tab option from the menu.

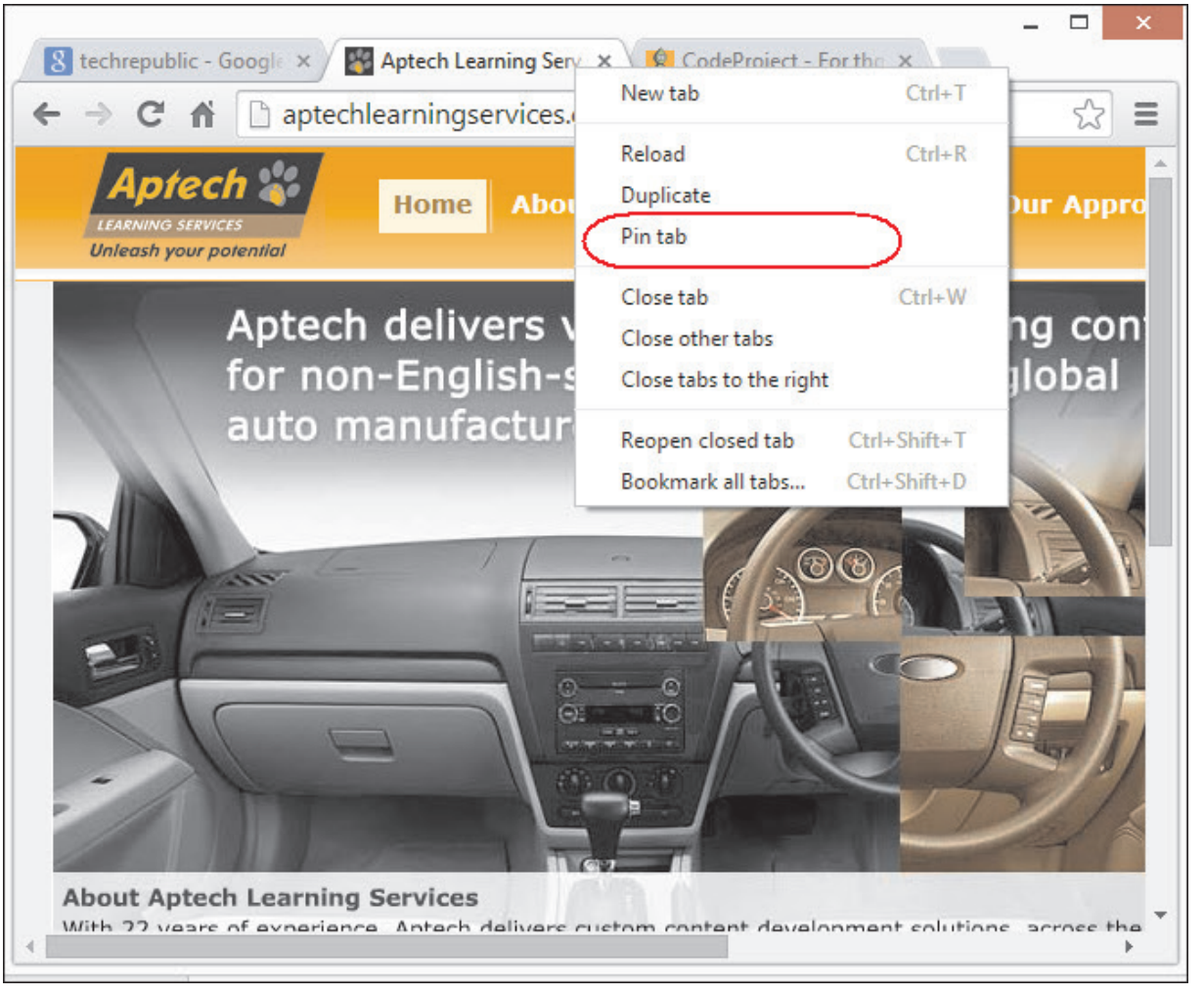

Figure: 1.6: Pin Tabs

- **Keyboard Shortcuts**

Chrome provides several keyboard shortcuts to help users work with the browser faster and improve productivity. Following are some of the most useful shortcuts:

- **Ctrl + 1-8:** Pressing **Ctrl** and any number 1 through 8 will move to the corresponding tab in the Tab bar
- **Ctrl + 1-8:** Pressing Ctrl and any number 1 through 8 will move to the corresponding tab in the Tab bar

- **Ctrl + 9:** Switch to the last tab
- **Ctrl + H:** Show history
- **Ctrl + J:** Open Downloads window
- **Ctrl + K:** Move the cursor to the Omnibox (Address bar)
- **Ctrl + T:** Open a new tab

- **The Chrome Task Manager**
  Each tab in the Chrome browser runs in its own sandbox. Therefore, it is not possible for users to determine the overall system resources being used by each tab from the Windows Task Manager. Windows Task Manager will simply show an instance of Google Chrome running for each open tab. The Chrome Task Manager is a useful tool that the users can use to check the memory and CPU resources being used by each tab and extension running in the browser. To open the Chrome Task Manager, click the Menu ( ) icon in the top-right corner of the browser, point to Tools, and then click Task Manager in the menu.

1.8.1 Downloading and Installing Google Chrome

If Google Chrome is not installed on local machine, download and install Google Chrome using the following steps:

1. Open a new window in the currently installed browser.
2. Type **http://www.google.com/chrome** in the Address bar.
3. Press **ENTER**. The Google Chrome download page is displayed.
4. Click **Download Google Chrome**. The Chrome License Agreement page is displayed.
5. Click **Accept** and **Install**. A dialog box will be displayed with a progress bar indicating that Google Chrome is being installed. After the installation is complete, Google Chrome prompts users to select a search engine.
6. Click **Choose** under Google search. The Google Chrome First Run window is displayed. Google Chrome is now successfully installed.

1.8.2 Opening a Web Site with Google Chrome

Perform the following steps to open a Web site in Google Chrome:

1. Click **Start** button.
2. Point to **Programs** on the **Start** menu if Classic view is selected for the **Start** menu. Otherwise, point to **All Programs**.

3. Select the **Google Chrome** group and click **Google Chrome**. This loads the Google Chrome browser.

4. Type **www.aptechlearningservices.com** in the Address bar. This is the Web address of the Web site.

5. Click **About us** on the home page. The page is displayed as seen in figure 1.7.

Figure 1.7: Opening a Web Site

1.9 Parts of a Browser

Following are the various parts that make up a Web browser:

- **Status Bar**
  The status bar is displayed at the bottom of the browser window. This bar describes the status of what the user is doing at each and every moment. The status bar displays the load speed and the address of the URL where the mouse is hovered.

- **Address Bar**
  The Address bar is available at the top of the browser window and its aim is to display the URL or the Web site address.

- **Title Bar**
  The aim of the title bar is to display the title of the Web page that the user is currently working with. The title bar is at the topmost part of the browser and is blue in color for major browsers.

- **Toolbar Icons**
  The toolbar is located under the title bar of the browser window.

The toolbar has the Back button, Home button, the Refresh button, and so forth.

- **Display Window**
  The display window is the place where the current Web site accessed by the user is seen.

- **Scroll Bars**
  The scroll bar is displayed at the right side of the display window of the browser. This bar allows the user to move the Web page up and down. Using this, the user can view the whole information available on display window of the Web page.

Though most browsers have common items, the items displayed on a browser home page may differ from browser to browser.

Figure 1.8 displays the various parts of Google Chrome.

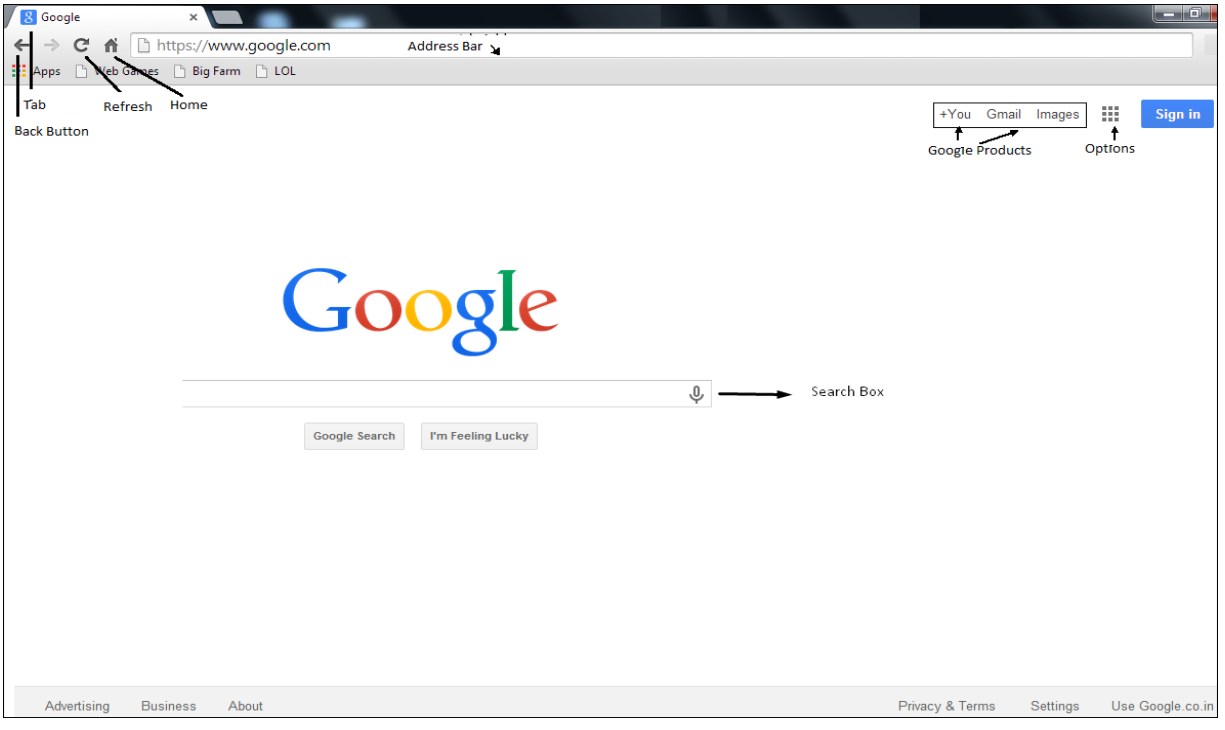

Figure 1.8: Parts of the Google Chrome Browser

1.10 Introduction to Search Engine

A Web search engine searches and retrieves information on the WWW and other Web-based servers.

This information is presented in a list format and is called Search Engine Results Pages (SERPs). The results can list Web pages, images, news, videos, and other types of information. A search engine can also display different types of files.

Search engines also collect information from databases and directories. Web search engines collect and maintain real-time data using a specific algorithm. Some of the popular search engines are Ask, Bing, Google, Yahoo!, MSN, and so on.

1.11 Introduction to Google Search

Currently, the most popular search engine on the Web is the Google search that is owned by Google Inc.

Google was founded by Larry Page and Sergey Brin in 1997. Google has consistently maintained its ranking as the number one search engine for the past several years. It has also built a reputation for providing the most relevant and fast search results.

A results page is displayed when a user searches for specific information.

Following steps are used to perform a search with Google Chrome:

1. Open **Google Chrome** window or tab.
2. Type **Google Products** in the search text box.
3. Click **Search** on the home page.

The different parts of the Google results page are shown in figure 1.9.

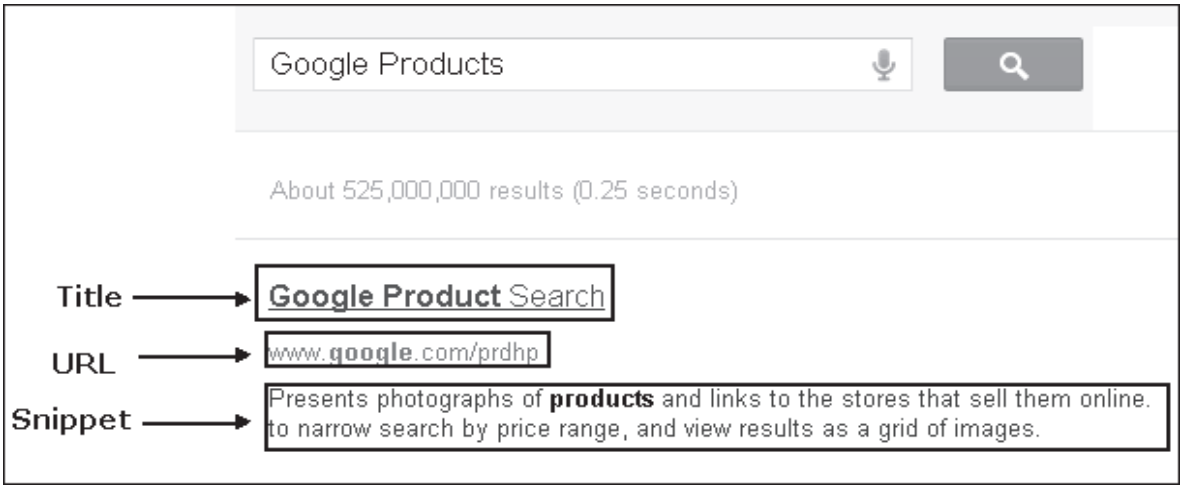

Figure 1.9: Parts of a Google Results Page

Google is mainly used to search information on the Web. Google search also supports searching for other types of data such as images and videos. Google has recently introduced its voice search service.

1.11.1 Features of Google Search

There are various features in the Google search that are unique.

The key features of Google search are as follows:

- **I'm Feeling Lucky** - The Google home page provides the **I'm Feeling Lucky** button which takes the user directly to the first search result page that would have appeared in normal search. For example, if the user wants to search for a Web site such as **www.wikipedia.org** and the user knows that the first result that would be displayed on the results page will be the same, then the user can click the **I'm Feeling Lucky** button to directly go to **www.wikipedia.org**, thus, bypassing the results page.

Perform the following steps:

- Open a new Google Chrome window or tab. By default, the **I'm Feeling Lucky** button is displayed as shown in figure 1.10.

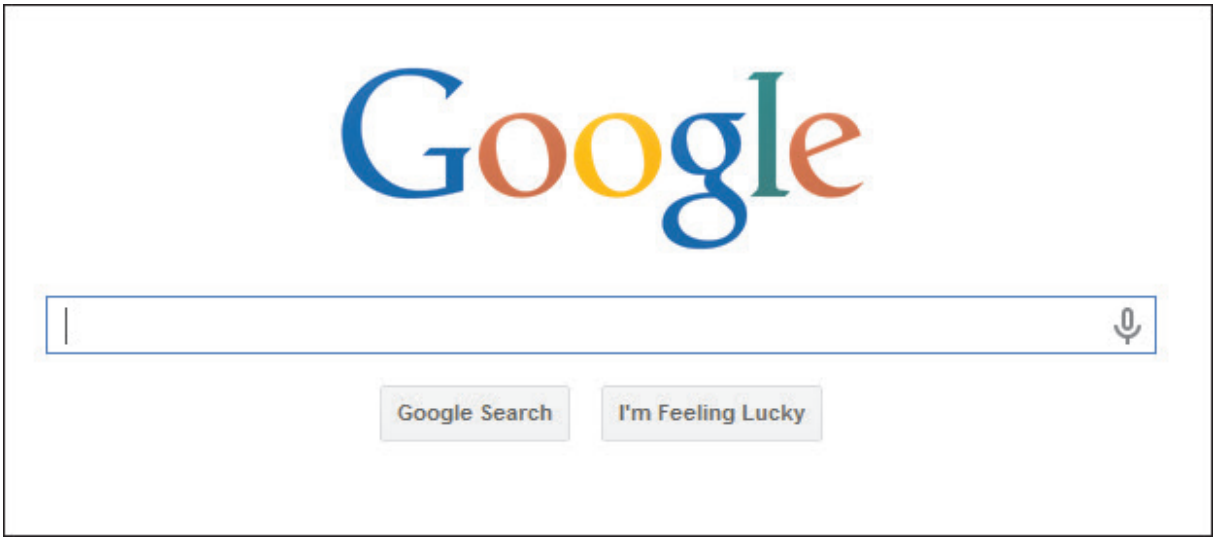

Figure 1.10: I'm Feeling Lucky Button

- Type a phrase to search for.
- Click **I'm Feeling Lucky** on the home page. This takes the user directly to the first search result page that would have appeared in normal search.

  Google Instant is a special enhancement to the search engine which shows results as soon as typing begins even before the **ENTER** key is pressed.

  If a user has enabled Google Instant, the **I'm Feeling Lucky** button will disappear when the user starts typing. In that case, the user can hover the cursor over the search list, where a link 'I'm Feeling Lucky' will appear at the right of the search list.

- **Advanced Search** - When keywords are typed into Google search, the keyword syntax is broken into simple text that helps in retrieving better and faster search results. Boolean operators such as quotation marks ("), plus (+), and so on can also be used in the search query. For example, to search for a recipe that should include potatoes and onions, the keywords can be typed as 'recipe + potatoes + onions'.

- **Query Expansion** - Google search uses the query expansion technique to improve the quality of the search results that are retrieved.

  Some of the techniques used in query expansion are as follows:

  - **Stemming** - This technique searches for the root of the keyword and searches for terms that can be related or similar. For example, the words 'fishing', 'fisherman', and 'fishy' will lead back to the root word 'fish'.
  - **Reordering** - The terms that are searched can also be reordered to provide relevant search results.
  - **Demographic Data** - Data related to public affairs of different countries such as population data, Gross Domestic Product (GDP), per capita income, and so on can be found on Google.
  - **Sports Updates** - Google can be used to know the scores and schedules of sporting events happening around the world.
  - **Area Code** - An area or zip code can also be found by typing a location as the keyword in Google.
  - **Unit Conversion** - Google can also be used for unit conversion. For example, Google can be used to convert kilometers to miles.
  - **Flight Schedules** - Google allows the user to view the flight schedules for a particular destination. The user can select the location and view the specified route as per his/her requirements.
  - **Spell Check** - Google search also provides a spelling check in case a keyword is typed incorrectly. It also provides a suggestion for the correct word as shown in figure 1.11.

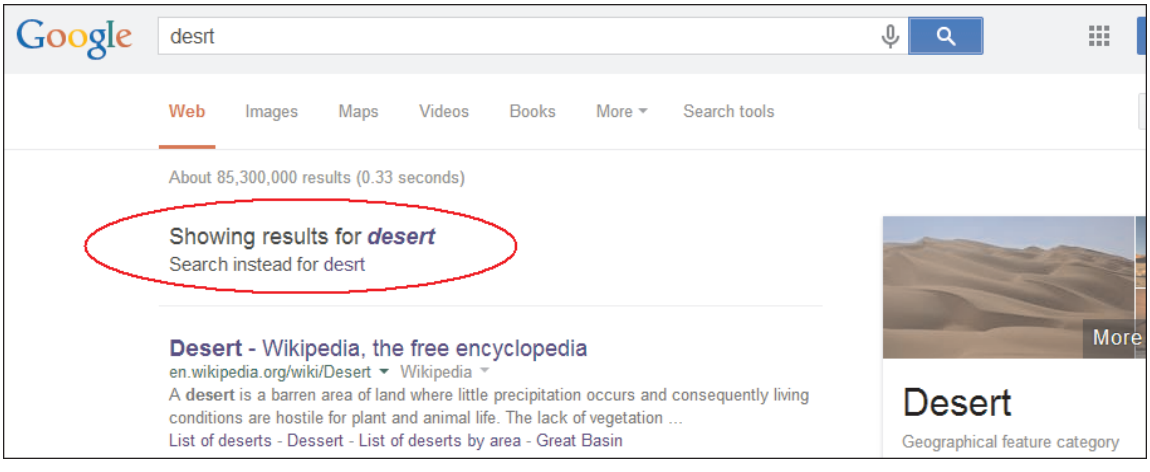

Figure 1.11: Automatic Spell Checking in Google

1.12 Search Techniques and Strategies

A user can use various techniques and strategies to search for relevant content on Google.

1.12.1 Basic Search Techniques

Some of these techniques are as follows:

- **Use Keywords** - Keywords are characteristics or attributes through which Web sites can be identified. Users can use keywords to find certain information. Search engines use keywords on the basis of which they retrieve the required information in the search results. Some considerations for selecting keywords are as follows:
  - Arrange the keywords on priority basis
  - Use the words that appear frequently on the Web page
  - Use few selective terms to describe the site
  - Use descriptive words
  - Avoid long phrases and sentences
- **Exclude Words** - The minus (-) operator can be used to exclude certain words from the search criteria. Perform the following steps to exclude words:

1. Open a new **Google Chrome** window or tab.
2. Type **treadmill -gym** in the search text box.
3. Click **Search**.

Figure 1.12 shows how to exclude certain words from the search criteria.

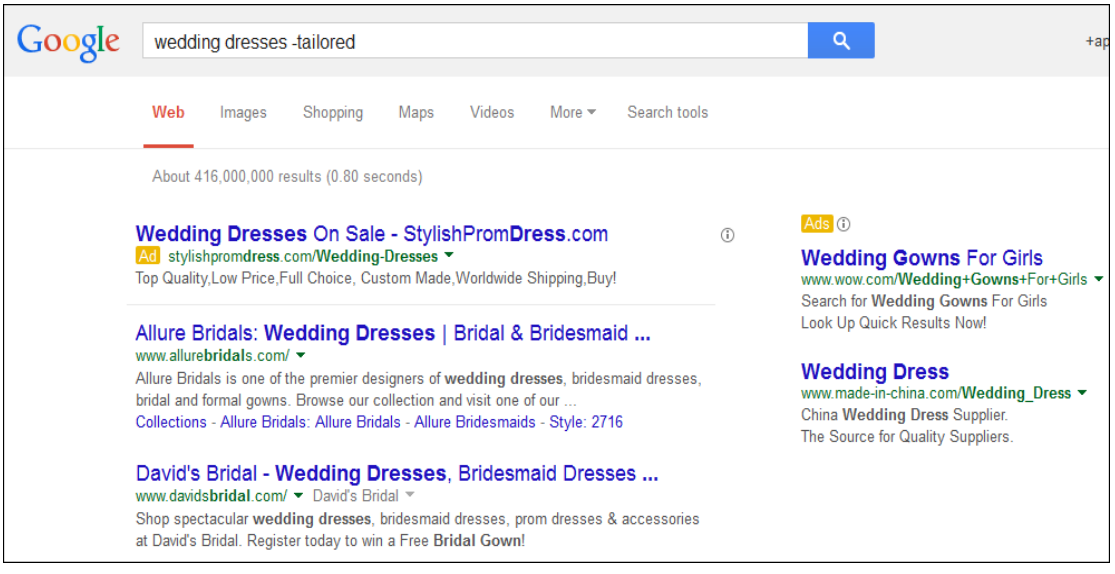

Figure 1.12: Excluding Words

This will return search results containing wedding dresses but excluding the word 'tailored'.

- **Search for Exact Phrases** - Google provides the facility to search for the exact phrase or keyword.

This restricts showing pages that can be related but may not have the word in the same order. This can be done by using double quotes (""). For example, if a person searches for Isaac Thomas, the results will include all persons with the name. It may also include names such as Issac John Thomas and Isaac Roger Thomas. However, a keyword search of "Isaac Thomas" with double quotes would show information specific to only these two words. This can save time when a user knows exactly what to look for.

1.12.2 Quick Search Techniques

Google identifies some useful details that can be found only if the user knows the right approach to query. Some handy and quick techniques for a faster search are as follows:

- **Currency Conversion Search**

  The status bar is displayed at the bottom of the browser window. This bar describes the status of what the user is doing at each and every moment. The status bar displays the load speed and the address of the URL where the mouse is hovered.

  Perform the following steps to view sites that help convert currency:

  1. Open a new Google Chrome window or tab.
  2. Type 30 USD to Euro in the search text box.
  3. Click **Search**.

Figure 1.13 shows the currency conversion results.

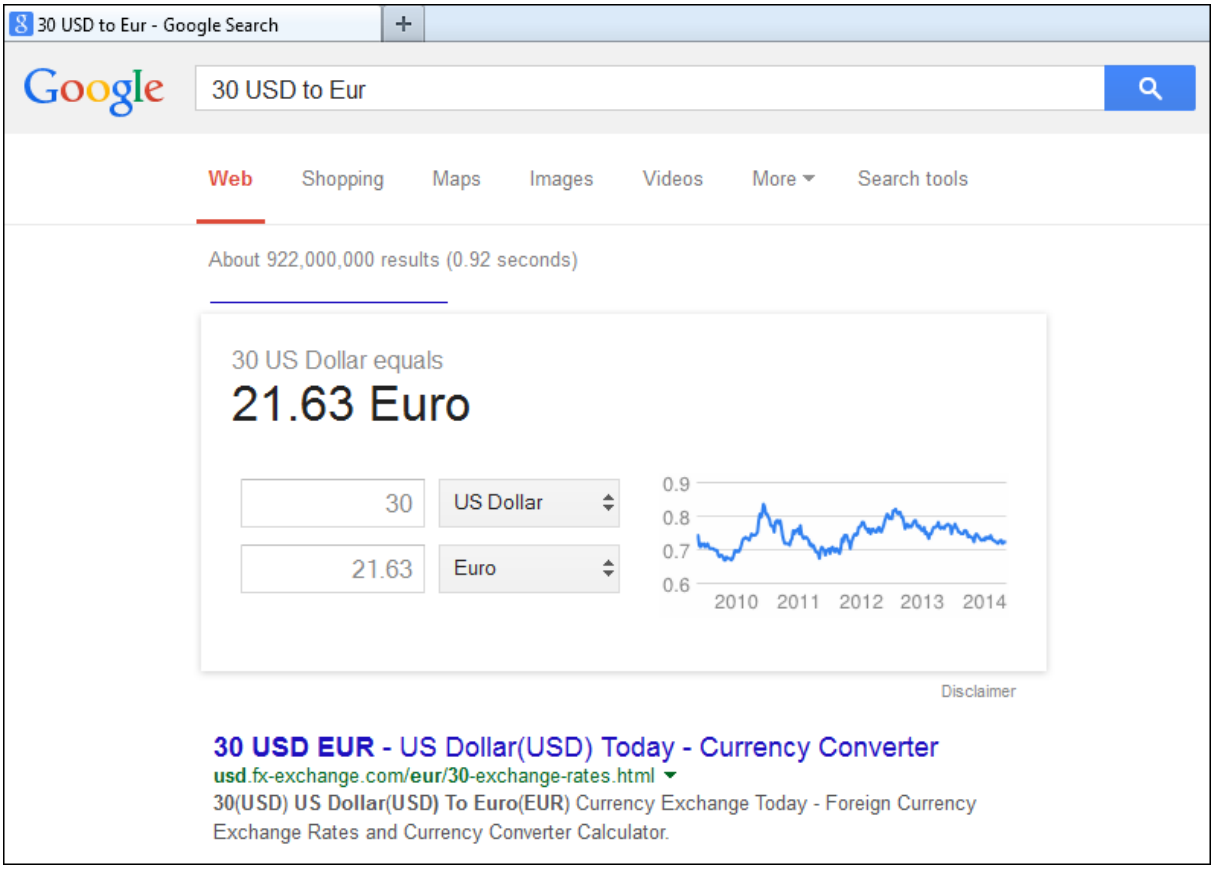

Figure 1.13: Currency Conversion Results

- **Climate Search**

  The user can even view the four days weather forecast worldwide for many cities using Google search.

  Perform the following steps to view the weather forecast:

  1. Open a new **Google Chrome** window or tab.
  2. Type **weather russia** in the search text box.
  3. Click **Search**.

Figure 1.14 shows the weather forecast for Moscow, Russia.

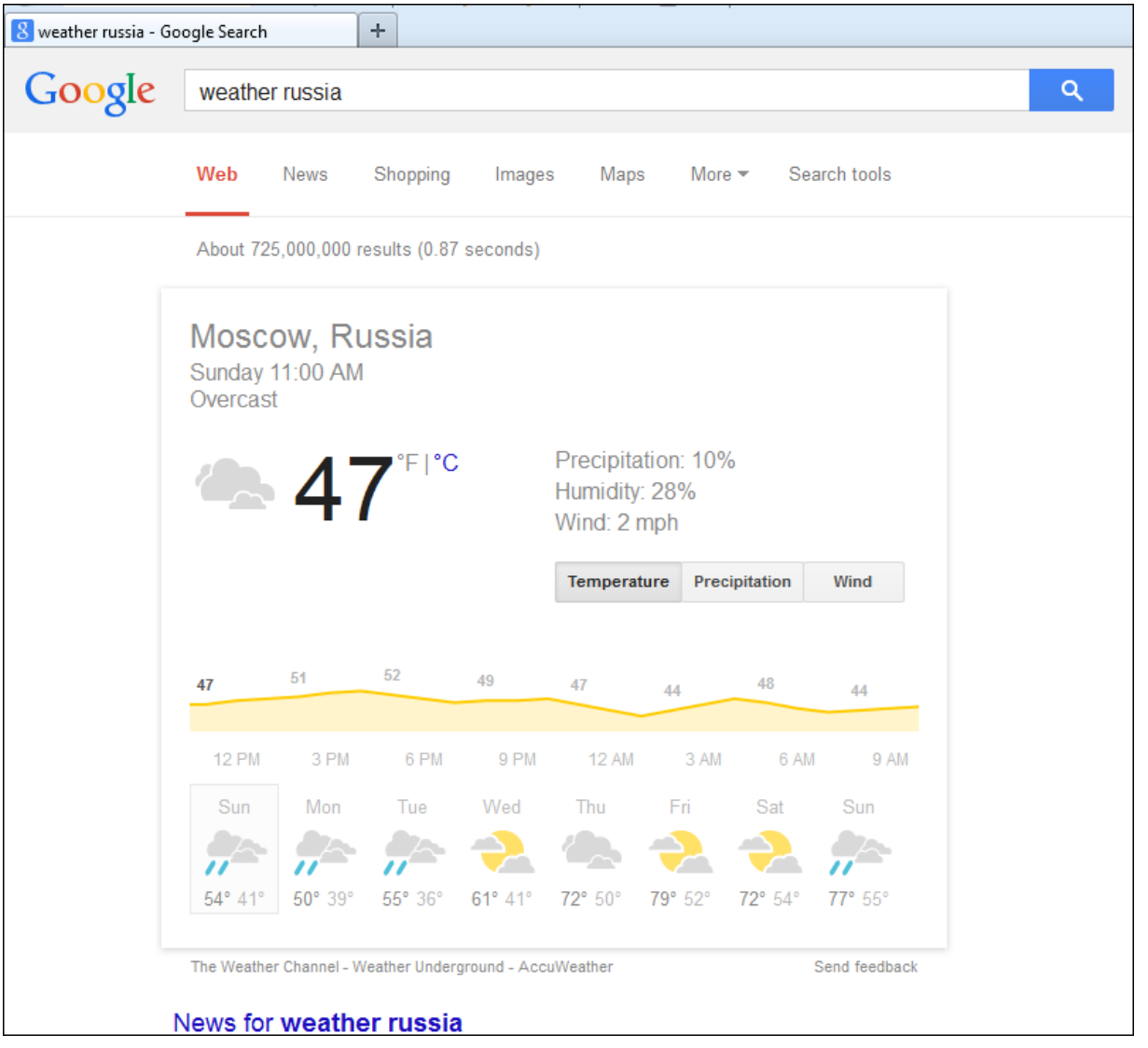

Figure 1.14: Weather Forecast for Moscow, Russia

- **Current Local Time Search**

The user can view the current local time for any city across the world.

Perform the following steps to view the local time in Melbourne:

1. Open a new **Google Chrome** window or tab.
2. Type **time melbourne** in the search text box.
3. Click **Search**.

Figure 1.15 shows the local time for Melbourne.

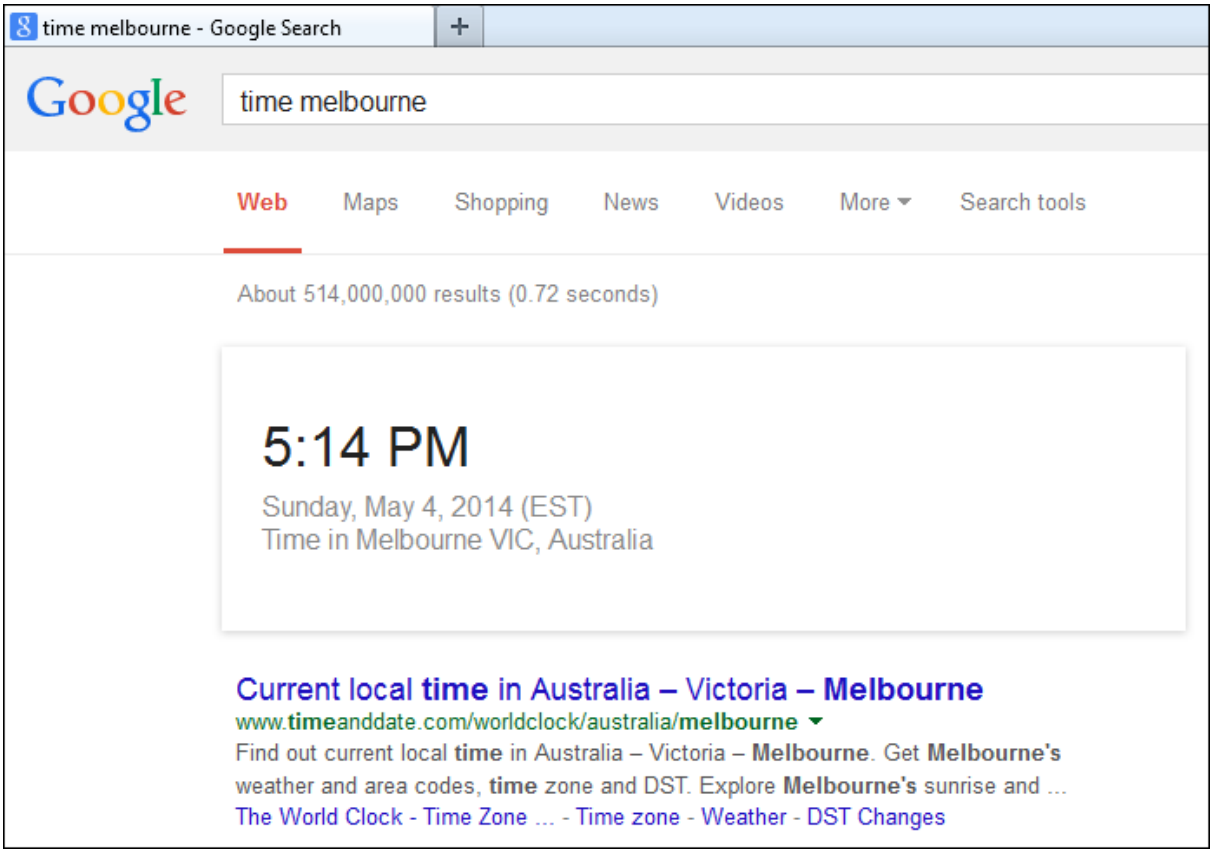

Figure 1.15: Local Time for Melbourne

- **Calculation Search**

Google can be used as a calculator to perform mathematical calculations. Even complex formulas can be entered to get the results within a few seconds. Care should be taken to provide the correct keyword query as even a small change would change the result. The user can use the search text box of Google for calculating values.

Perform the following steps to view the calculation:

1. Open a new **Google Chrome** window or tab.
2. Type **(14*(34+12)/78*(2 +79))** in the search text box.
3. Click **Search**.

Figure 1.16 shows the calculation search.

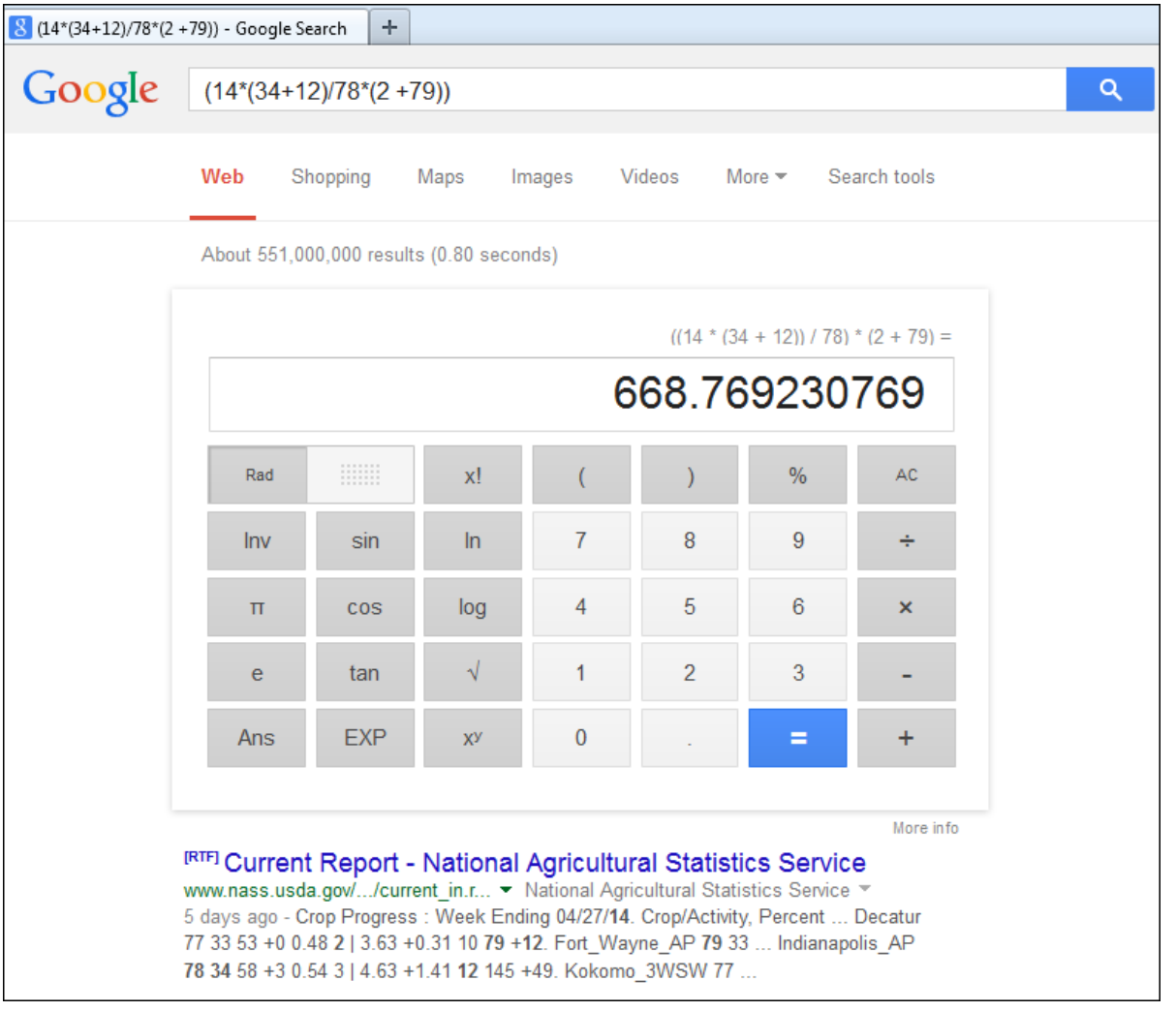

Figure 1.16: Calculation Search

- **Range Search**

Google enables users to search for a range using the .. symbol.

Perform the following steps to view the range results:

1. Open a new **Google Chrome** window or tab.
2. Type **ebook Reader $50..$80** in the search text box.
3. Click **Search**.

Figure 1.17 shows the range search results.

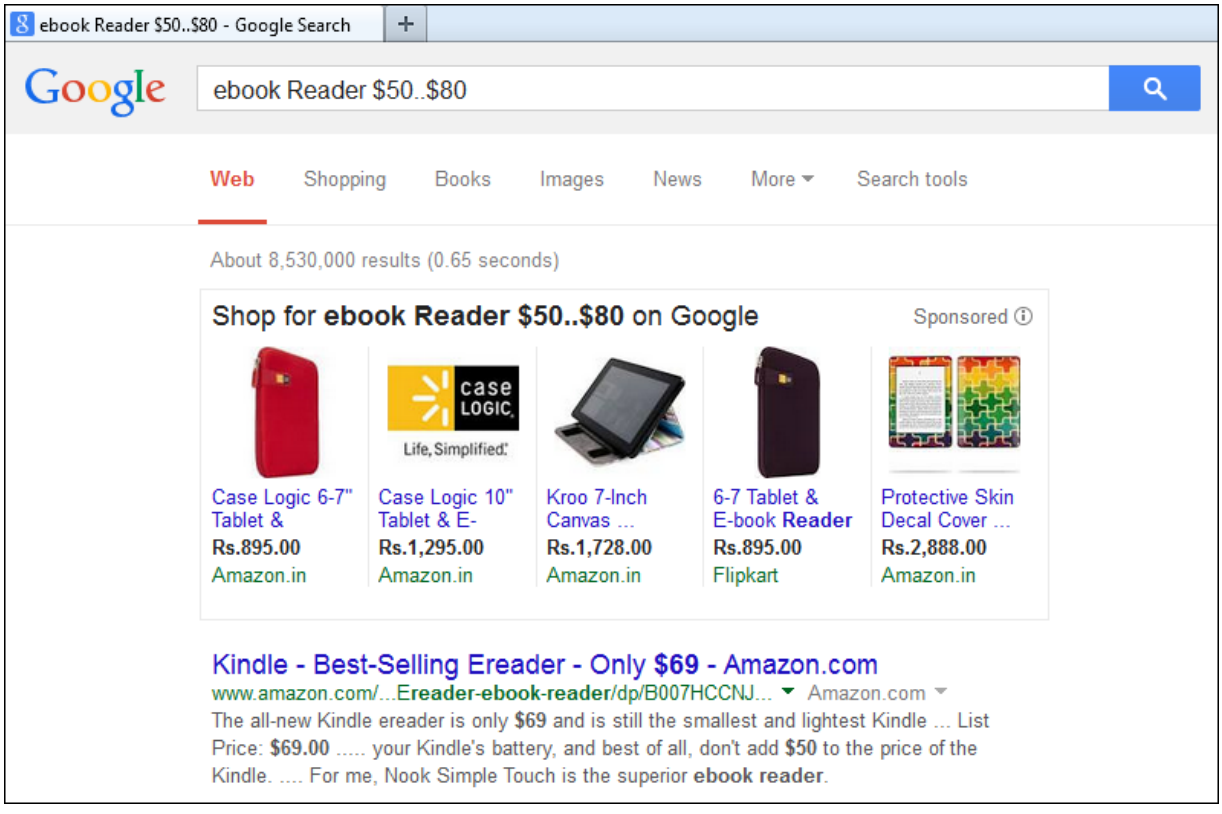

Figure 1.17: Range Search

This will show all the Web sites that feature ebook readers between 50 dollars to 80 dollars.

☐ Definition Search

Google enables users to search the definition using the **define** keyword.

Perform the following steps to view the definition search:

1. Open a new **Google Chrome** window or tab.
2. Type **define kilobyte** in the search text box.
3. Click **Search**.

Figure 1.18 shows the definition search.

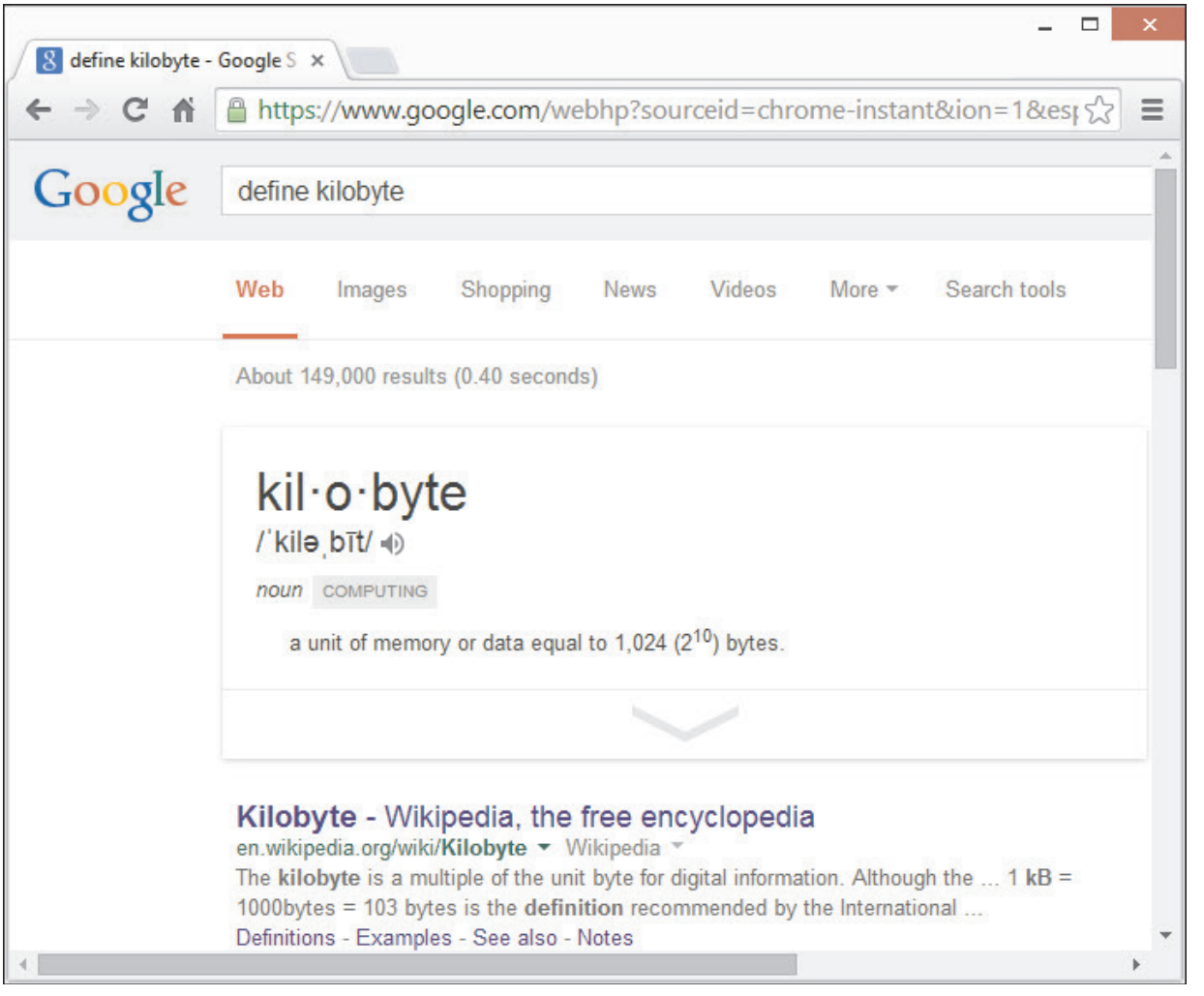

Figure 1.18: Definition Search

This will show all the Web sites that feature definition of kilobyte.

Check Your Progress

1. Which of the following is a network of computers connected globally?

(A)	Internet	(C)	Web Site
(B)	Web Page	(D)	Search Engine

2. Which of the following is a small piece of text that is exchanged between the client and the server?

(A)	Web Page	(C)	Web Browser
(B)	Web Site	(D)	Web Cookie

3. Which of the following is a collection of interconnected documents that can be accessed over the Internet?

(A)	Search Engine	(C)	URL
(B)	World Wide Web	(D)	Browser

4. Identify the software program which enables the user to search and view different types of information available on the Internet.

(A)	Search Engine	(C)	Browser
(B)	Internet	(D)	Cookie

5. Which of the following feature searches and retrieves information on WWW and other Web-based servers?

(A)	Web Search Engine	(C)	IP Address
(B)	URL	(D)	Internet Service Provider

Answer

1.	A
2.	D
3.	B
4.	C
5.	A

Introduction to Google Search

Summary

- The Internet is a network of computers connected globally.
- The WWW is a collection of interconnected documents that can be accessed over the Internet.
- A browser is a software program which facilitates the user to search and view different types of information available on the Internet.
- A Web page represents a document that resides at remote location.
- A collection of Web pages that are accessed over the Internet is called a Web site.
- A Web search engine searches and retrieves information on the WWW and other Web- based servers.
- Google is mainly used to search information on the Web.

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02 Exploring Google Search

Welcome to the Session, **Exploring Google Search**.

This session explains some advanced search techniques which can be used to improve the effectiveness of the Google searches. The session also explains the different types of advanced search techniques such as Google Instant, Google Voice search, and Google Image search.

In this Session, you will learn to:

- Describe advanced search techniques
- Describe Google Search settings or preferences
- Describe the different types of Google Search
- Explain the working of Google Search

Session
02

Exploring Google Search

2.1 Introduction

Google, the most popular search engine, is used to search for information from all over the world. This Information can include Web pages, images, videos, news, maps, and so on. Google Search enables to narrow down the searches.

This session explains some advanced search techniques which can be used to improve the effectiveness of the Google searches. The session also explains the different types of advanced search techniques such as Google Instant, Google Voice search, and Google Image search.

Search Techniques

A user can use various techniques and tools to search for relevant content on Google. In the basic search techniques, the user enters a keyword and a search result page is displayed. However, the advanced search option allows user to quickly refine the search by using Boolean logic fields such as AND, OR, and NOT. It also allows the user to narrow down the search results by defining the file formats, number of result pages, language preferences, and so on.

To open the **Advanced search** option, perform the following steps:

1. Open the **Google Chrome** browser.
2. Type **www.google.com** in the Address bar.

Click the **Settings** link at the bottom of the page and select **Advanced search** from the menu.

Figure 2.1 displays the **Advanced search** option.

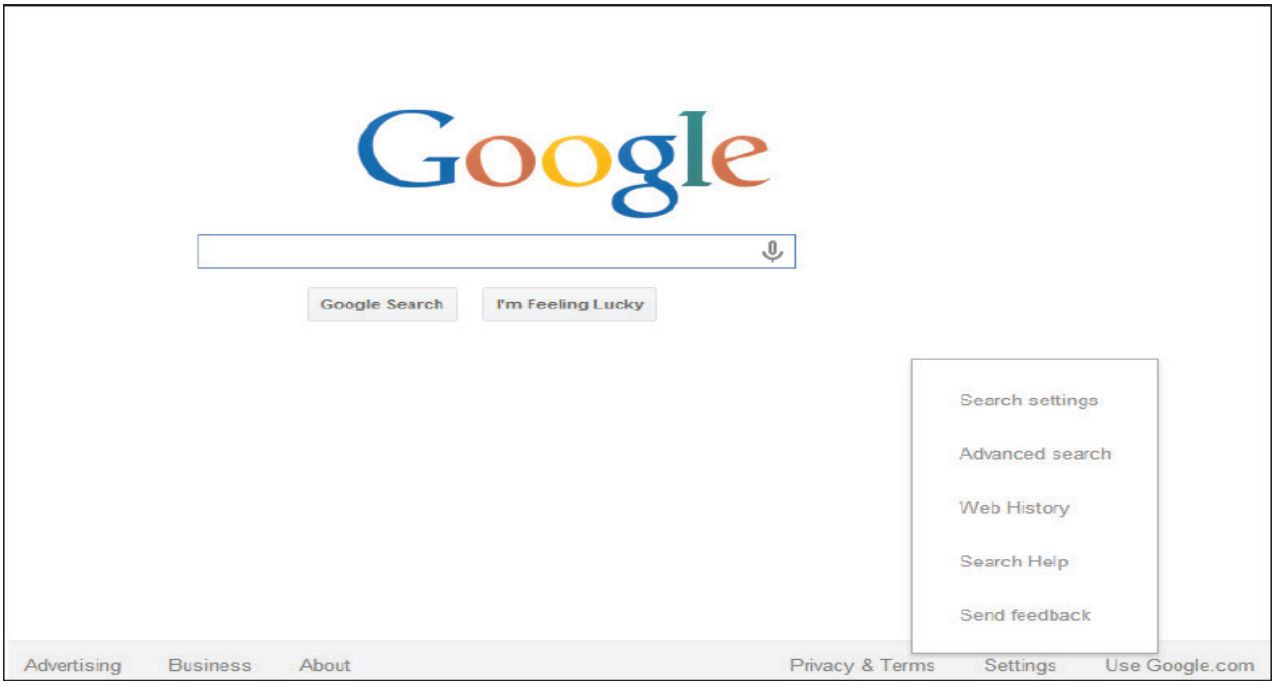


Figure 2.1: Advanced Search Option

The **Advanced Search** page appears with different parameters as shown in figure 2.2.

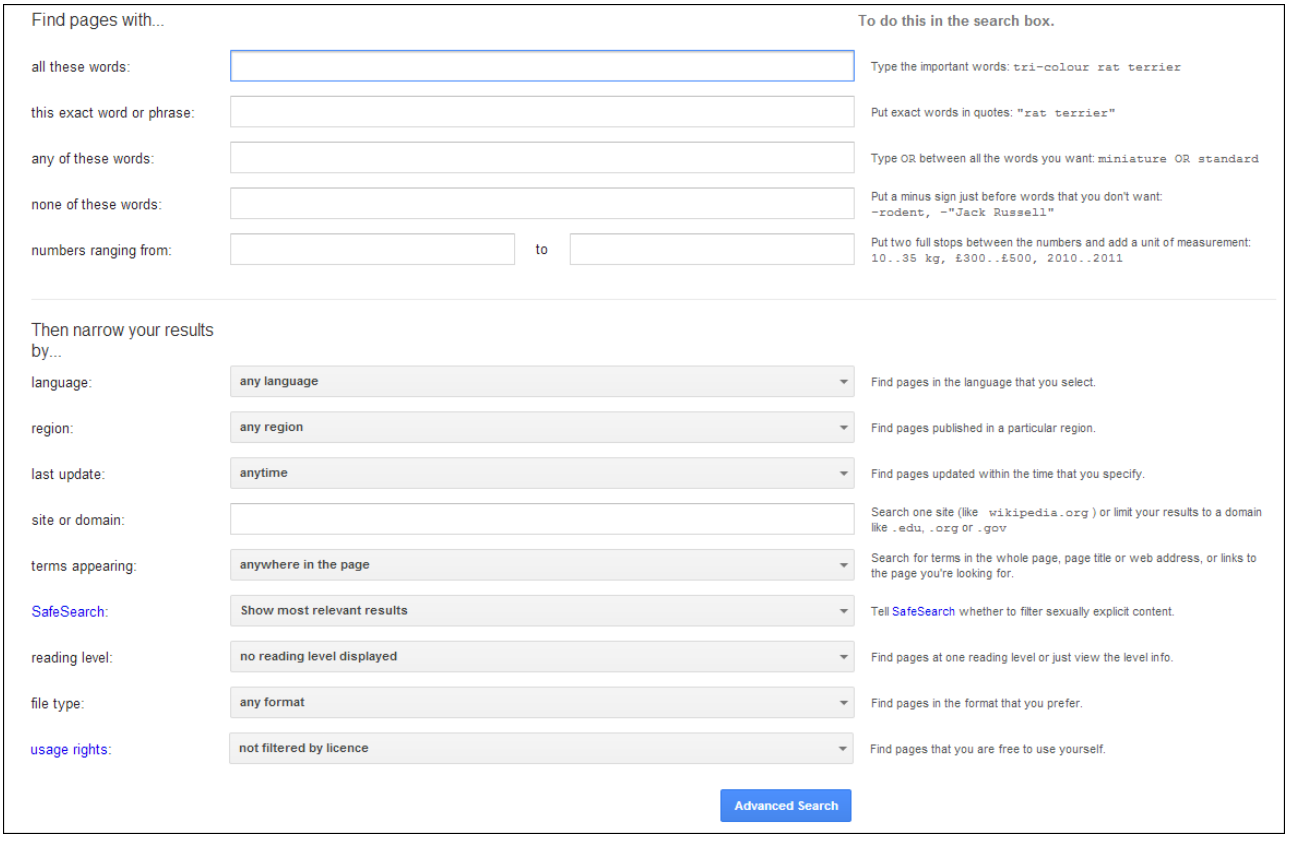


Figure 2.2: Advanced Search Page

Some of these parameters present in the Advanced Search page are explained as follows:

- **all these words** - Searches for all words typed in the search box.
- **this exact word or phrase** - Searches for all the words that must be present together in a Web site. This is similar to putting quotes ("") around the words. Can also include stop words such as the, a, and so on.
- **any of these words** - Searches for results that match either one or more of the words.
- **none of these words** - Eliminates unwanted words from the results page. It is similar to using minus (-) sign in basic search of Google.

You can also filter or narrow down your results based on various criteria. Some of these are as follows:

- **language** - Restrict results to pages written in a language that is selected from the language list. Figure 2.3 displays the **Language** list.

Figure 2.3: Language List

- **region** - Displays search result pages from a country which is specified in the region list. Figure 2.4 displays the **Region** list.

Figure 2.4: List of Regions

- **last update** - Find pages updated within the time that you specify.

- **site or domain** - Restricts search results to only a specific Web site such as, **wikipedia.org** or a domain say **.edu**, **.org**, or **.gov**.

- **terms appearing** - Specifies the location of the search term which can appear either in the page, or in the title, or in the text, or in the URL, or in links to the page. Figure 2.5 shows the available options.

Figure 2.5: Where Your Keywords Show Up

- **SafeSearch** - Filters out adult content and inappropriate graphics. It eliminates them from appearing in the search results. Figure 2.6 displays the various **SafeSearch** options.

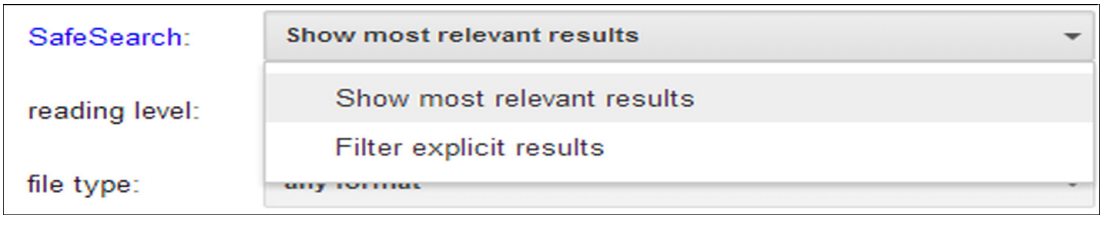


Figure 2.6: SafeSearch Option

- **reading level** - Restricts the search result page to any one of the three categories namely, basic, intermediate, and advanced. In other words, this option enables the user to limit the search results to a specific reading level based on the requirement of the user. Figure 2.7 displays the **Reading level** list.

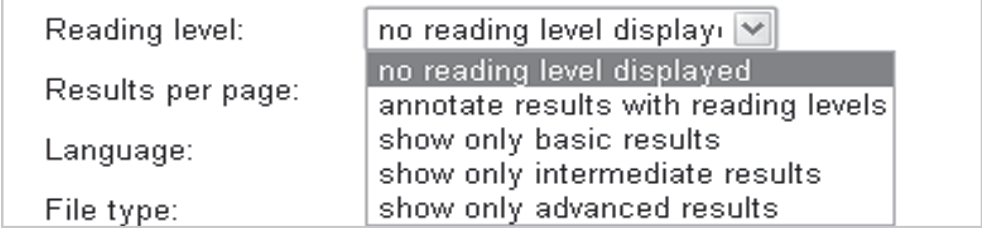


Figure 2.7: Reading Level List

- **file type** - Restricts search results to a file format specified from the file type list. Figure 2.8 displays the **file type** list.

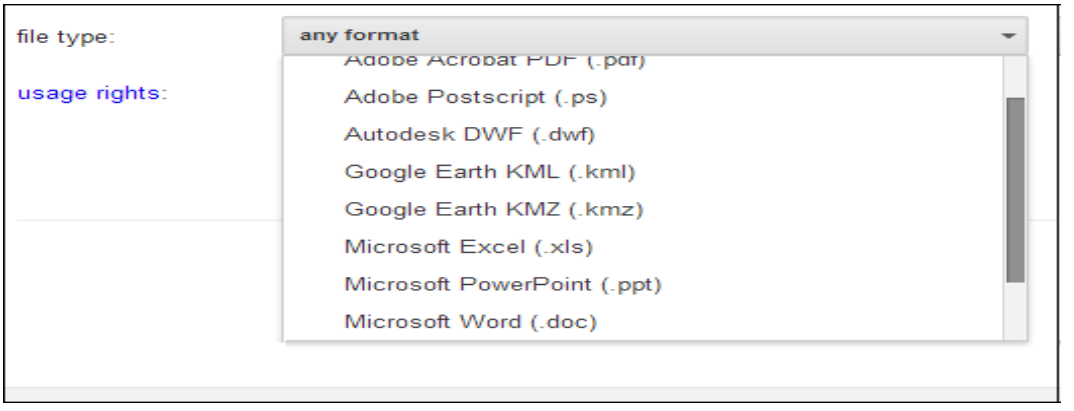


Figure 2.8: File Type List

- **usage rights**: Finds Web content such as images, text, and videos which can be reused, modified, or shared. The pages displayed by Google by default cannot be freely used. Hence, this option is useful to locate content that can be reused.

  Figure 2.9 displays the usage rights list.

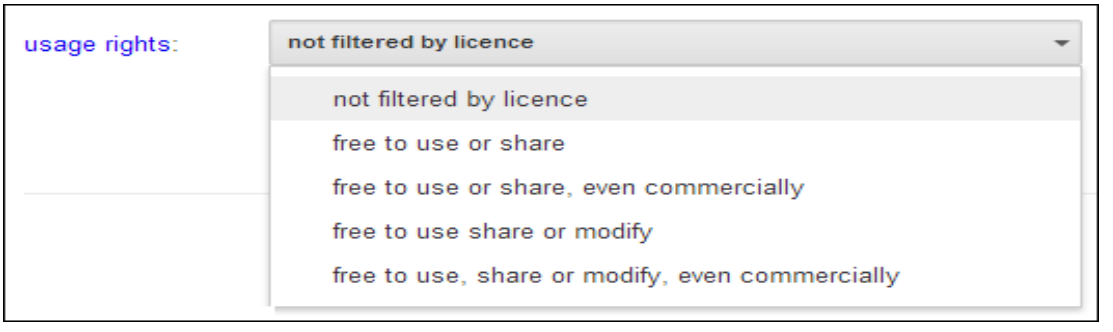


Figure 2.9: List of Usage Rights

2.2 Google Search Settings

Google Search is used to provide relevant and useful search results. These search results can also be customized to appear according to the user's requirements. This can be done using the Google Search settings.

To open the Google Search settings page, perform the following steps:

1. Open **Google Chrome** browser.
2. Type **www.google.com** in the Address bar.

   Click the **Settings** link at the bottom of the page and select **Search settings** from the menu. Figure 2.10 displays the **Search settings** option.

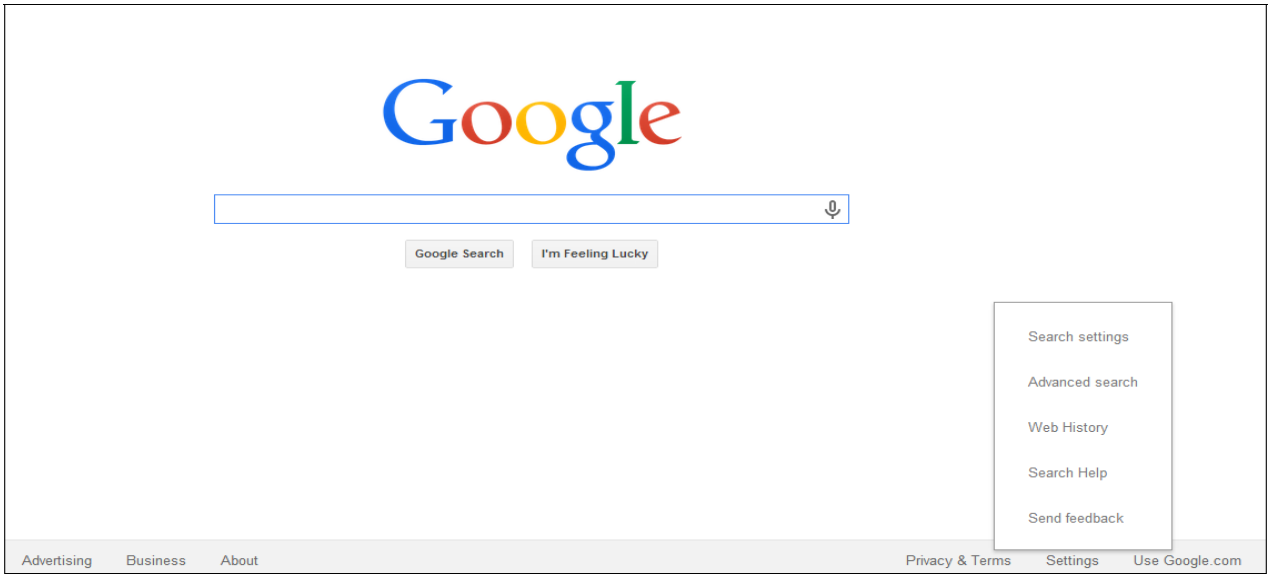


Figure 2.10: Search Settings Option

The **Search Settings** page is displayed in figure 2.11.

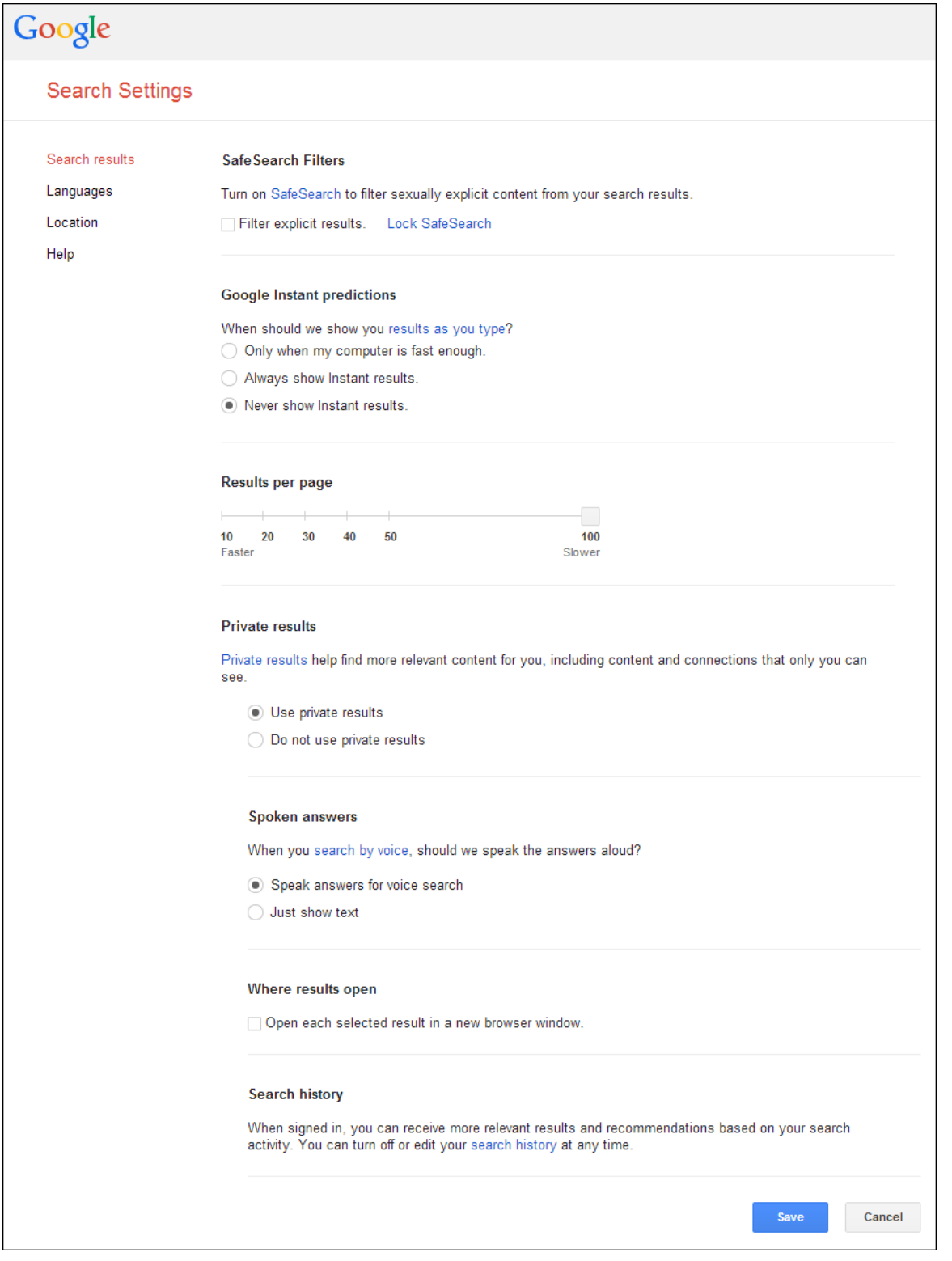


Figure 2.11: Search Settings Page

The **Search Settings** page can also be displayed by typing **http://www.google.com/preferences** in the Address bar of **Google Chrome** Web browser.

In the **Search Settings** page, users can modify **Search results** and **Languages**.

All these settings are saved in a small file called a cookie. This file is saved in the local computer. Hence, these settings will apply only to that specific computer and browser.

2.2.1 Search Results

Search results pane allows the user to customize and filter the search results. This pane displays the various options that are required to be set according to user requirement and these options are as follows:

- **Google Instant predictions** - This feature allows Google to display the search results as the user types the search term. In other words, when the user types the search term in the search box, **Google Instant** automatically displays results for a popular search that begin with the same letters. If the desired results are not found, the user must keep typing and the results would update dynamically on the result page.

  The search engine technology predicts the user's search query and displays text in grey in the search box even before the user finishes typing the search terms. For example, type the letter 'q' in the search window, **Google Instant** predicts the query to be **quotes**, **qvc**, and **quotes about life** as shown in figure 2.12.

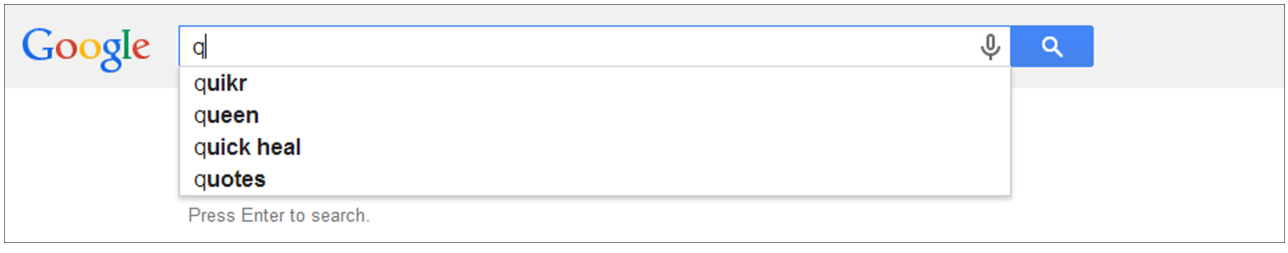


Figure 2.12: Displaying Results after Entering Letter 'q'

Users can disable the **Google Instant predictions** option by clicking the **Never show Instant results** button on the **Search results** pane of the **Search Settings** page.

Figure 2.13 displays the disabled **Google Instant predictions** option.

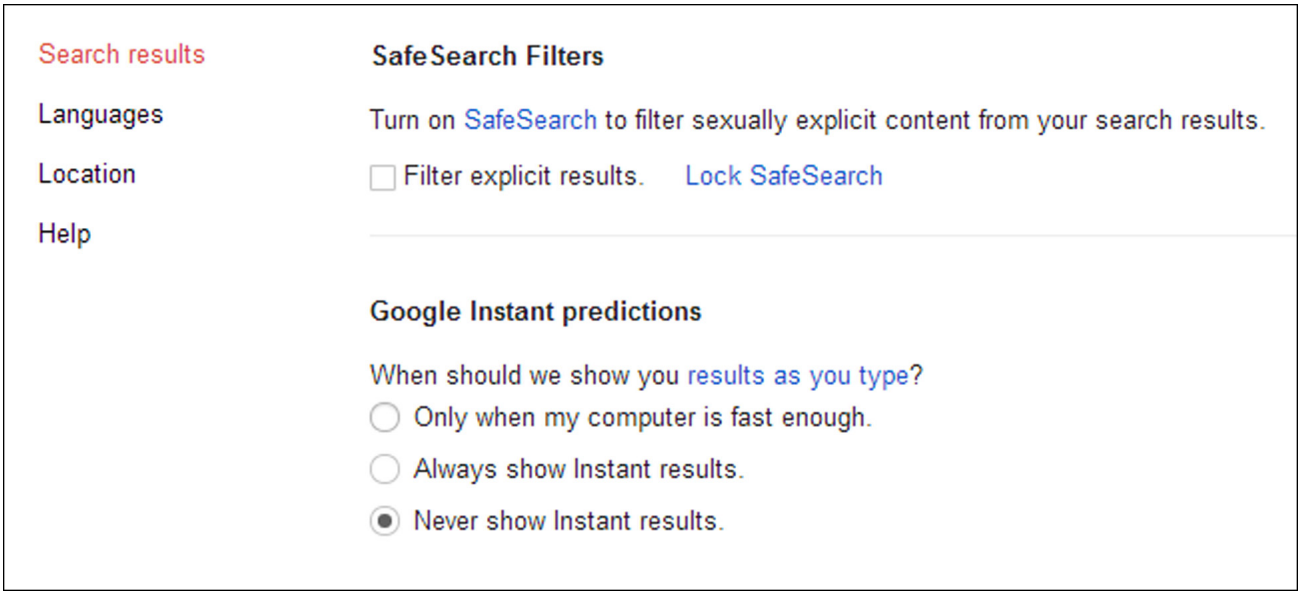


Figure 2.13: Disabling Google Instant Predictions Option

- **Location** - Selecting a Location helps Google find information relevant to the specified location when using Search, Maps, News, and other Google Apps. Figure 2.14 shows the Location feature.

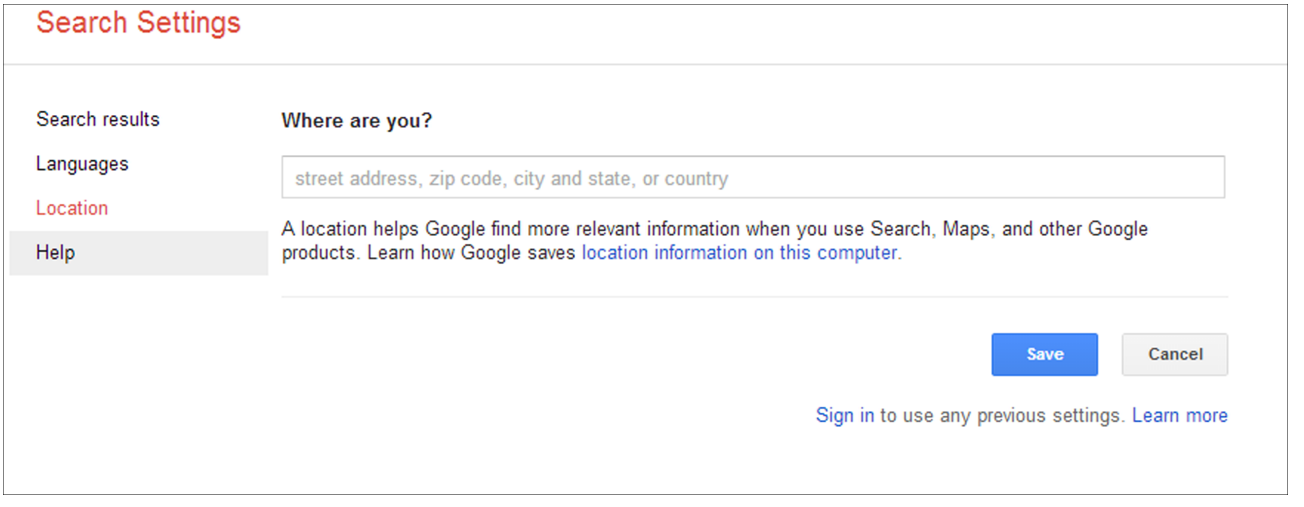


Figure 2.14: The Location Feature in Search Settings

- **SafeSearch**: The **SafeSearch** feature is designed to filter out explicit adult content from search results. It allows the users to change the browser settings and prevent adult content from appearing in the search results. Google's **SafeSearch** feature blocks objectionable content by using an automated method that identifies this content.

  To turn **SafeSearch** on or off, perform the following steps:

  1. Open the **Search Settings** page.

2. In the **SafeSearch filters** pane, select the **Filter explicit results** check box. When **SafeSearch** is on, sexually explicit videos and images will be filtered from Google Search result pages, along with results that might link to explicit content.

   To turn off **SafeSearch**, uncheck the **Filter explicit results** check box. The Google Search results page may now include explicit content.

3. Click the **Save** button at the bottom of the page.

The **SafeSearch** filters setting will be applied to any other users who may use the same computer and Web browser.

- **Lock SafeSearch** - The **Lock SafeSearch** option locks the **SafeSearch** filters settings so that other users cannot change the settings. The lock can only be removed only by the Google account that set it.

To turn on **Lock SafeSearch**, perform the following steps:

1. Open **Google Chrome** browser.
2. Type **www.google.com** in the Address bar.

   Click the **Settings** link at the bottom of the page and select **Search settings** from the menu.
3. In the **Safe Search Filters** pane, make sure SafeSearch is turned on by checking the Filter explicit results check box.
4. Click **Lock SafeSearch** as shown in figure 2.15.

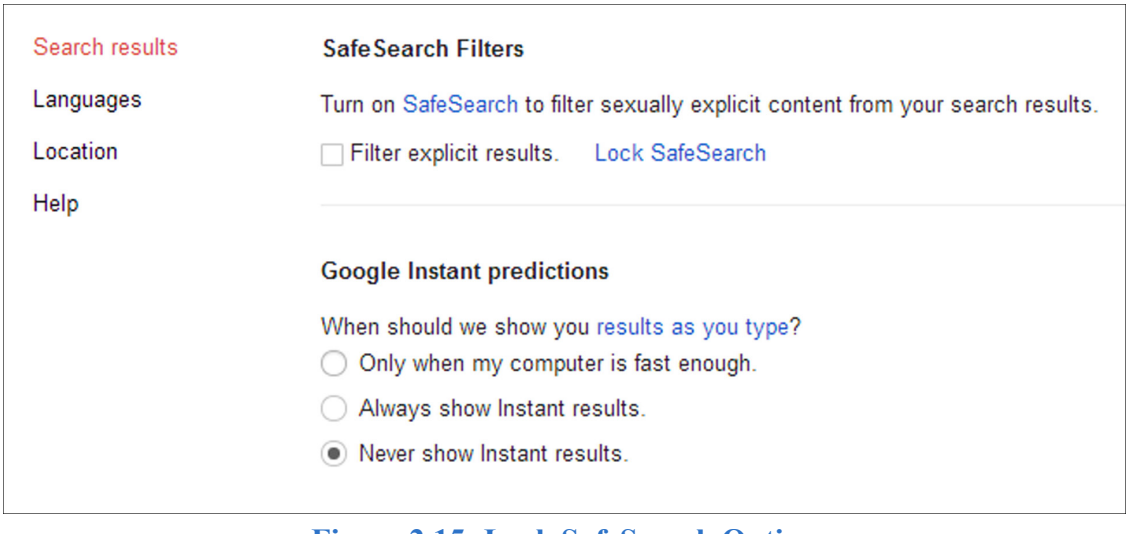


Figure 2.15: Lock SafeSearch Option

A sign in page is displayed, prompting the user to sign-in to their Google account. A message providing information about the **Lock SafeSearch** option is displayed.

5. Click **Lock SafeSearch**.

This process takes time. A percentage progress counter is displayed.

A confirmation page is displayed indicating that **SafeSearch** lock has been applied. Figure 2.16 displays the **SafeSearch Filtering** confirmation page.

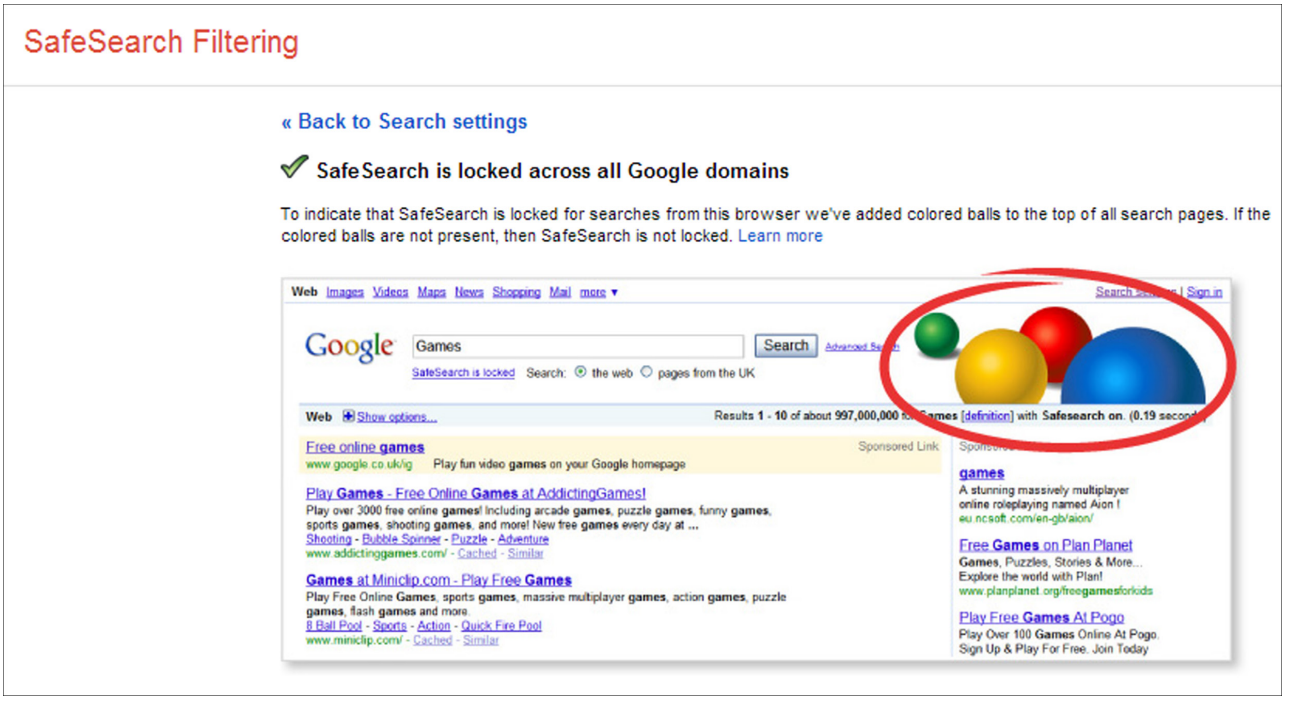


Figure 2.16: Lock SafeSearch Confirmation Page

To unlock SafeSearch, follow the same instructions and click **Unlock SafeSearch**.

Note

The two ways that enable the user to identify that the **SafeSearch Lock** feature is on, are as follows:

1. **SafeSearch locked** is displayed on the right of the search box as shown in figure 2.17.
2. Colored balls appear on the right of the Google Search result page as shown in figure 2.17.

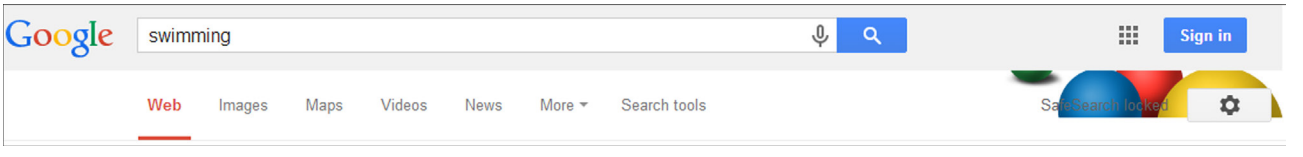


Figure 2.17: SafeSearch is Locked

- **Results per page** - By default, Google Instant displays 10 search results per page. The more results per page (such as 20, 30, 50, or 100) can be viewed by disabling the Google Instant feature.
- **Where results open** - By default, clicking any search result will open the page in the same window or tab.

To open the results in a separate window, select the **Open each selected result in a new browser window** check box.

- **Spoken answers** - This new feature of Google allows users to search by speaking into an inbuilt or an external microphone attached to the computer or mobile device. Users can choose to have their voice search results returned in voice by selecting **Speak answers for voice search** radio button or as text by clicking the **Just show text** radio button. **Spoken answers** are turned on by default, as shown in figure 2.18.

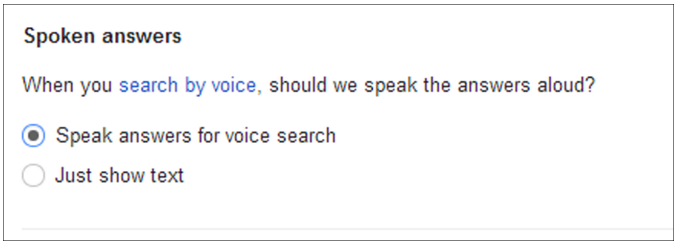


Figure 2.18: Spoken Answers

- **Search History** - Google tracks searches and visited pages, whenever user searches on the Web. Users can turn Web History ON to receive faster, more relevant results and smarter predictions based on their previous search activities. Users can turn off or edit your search history at any time.

To turn on and edit Web History, perform the following steps:

1. Open the **Search Settings** page and scroll down to the **Search History** pane.
2. Click the **Search History** link.
3. On the **Web History** page, click the **Turn Web History on** button. Figure 2.19 shows the button.

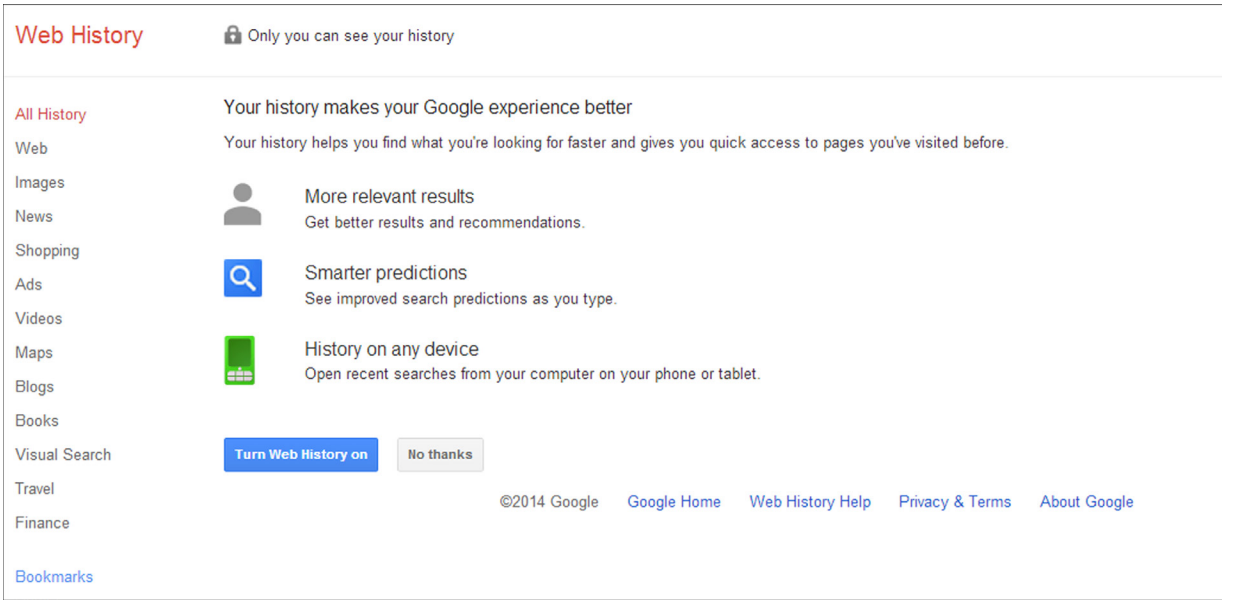


Figure 2.19: Turning Web History On

For security reasons, it is recommended to delete the Web history by signing in to the Google account.

To remove the searched sites from Google Web history, perform the following steps:

1. Open **Google Chrome** browser.
2. Type **www.google.com** in the Address bar.
3. Click **Sign in** button in the top right of the page. Enter Google account e-mail and password, and then click **Sign in**.
4. Click the **Settings** link at the bottom of the page and select **Search settings** from the menu. Scroll down on the **Search Settings** page and click **Search history** hyperlink.
5. Select individual entries from the list that need to be removed and click the **Remove items** button.
6. After deleting Web history, sign out of the Google account.

2.2.2 Languages

Users can customize the language of the Google home page. Users can select their interface and search result language. Click the **Languages** option on the left to display the **Languages** pane as shown in figure 2.20.

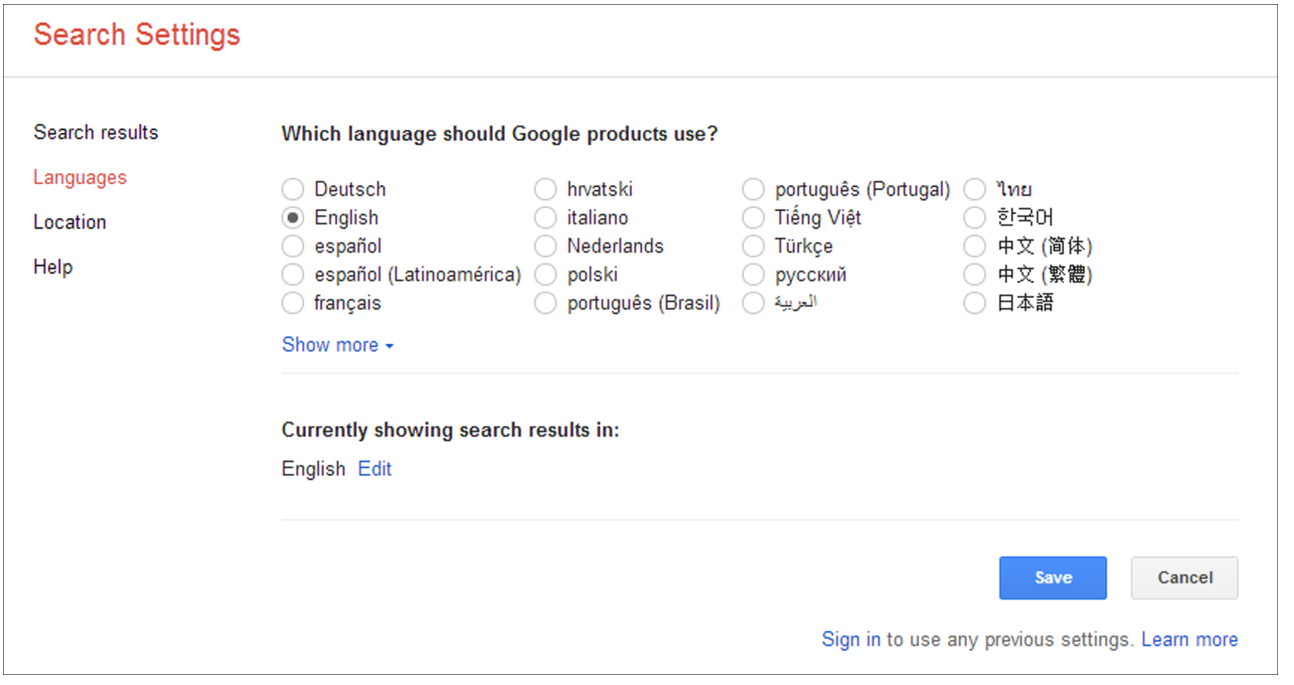


Figure 2.20: Languages Pane

2.3 Types of Google Search

Google Search helps by allowing different types of advanced searches such as images, videos, news, and so on. The newly added Google Search types are as follows:

1. Google Instant
2. Google Voice Search
3. Google Image Search

As mentioned earlier, **Google Instant** is a new feature added by Google. It displays the search results while the user types the search terms in the search box. It is faster because, before completing typing of the search term, it automatically displays results for a popular search that begin with the same letters. If the required results are not found, the user must continue typing and the results will be updated in the results page. Figure 2.21 displays the **Google Instant** search results while typing the search string.

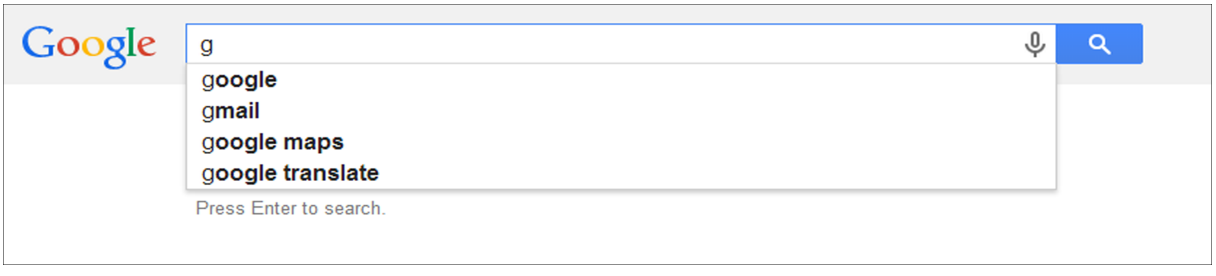


Figure 2.21: Results after Typing Letter 'g' in the Search Box

Figure 2.22 displays the changed search results page while some more characters of the search string have been typed.

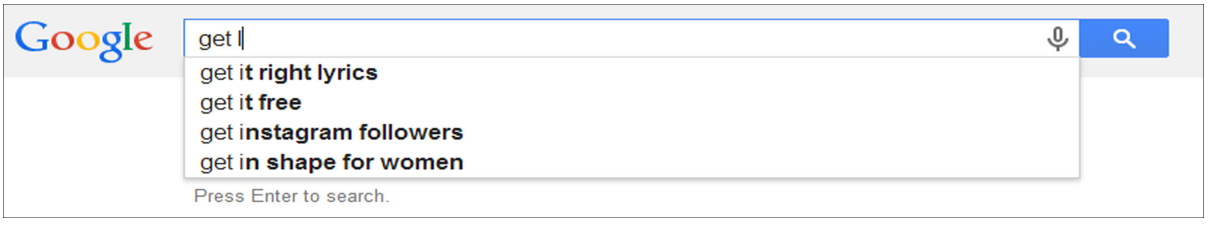


Figure 2.22: Results after Typing 'get l' in the Search Box

Benefits of the **Google Instant** are as follows:

1. **Saves Time** - It saves time because it shows relevant results while typing.
2. **Smarter Predictions** - It shows the dynamically generated results and users can stop typing as soon as they find the result. The top predictions are displayed in grey text directly in the search box.
3. **Search Smarter** - It shows instant results while typing in the search box. If the required results are found, add one or more words in the search box.

2.3.1 Google Voice Search

Google Voice search is a new feature added by Google to the desktop PCs. In **Google Voice search**, the user has to speak instead of typing the search query, to search on **google.com** site. The **Voice search** displays search results based on the user's audio input (voice). To search using Voice search, the user must have **Google Chrome** and a built-in or an externally attached microphone.

Google Voice search is available in 27 languages and dialects for mobiles and for desktop it is available in English language only.

The benefits of using **Google Voice search** are as follows:

- It saves time to write search query manually and repeatedly.
- It can search the difficult-to-spell words very quickly.
- It can search lengthy search queries.

To start search using **Google Voice search**, perform the following steps:

1. Open Google Chrome browser and type **www.google.com** in the Address bar. This opens the Google home page. The microphone icon ( ) is shown in the search box. Figure 2.23 displays the microphone icon in the search box.
2. Click the icon and speak the search terms when the **Speak now** tooltip is displayed.

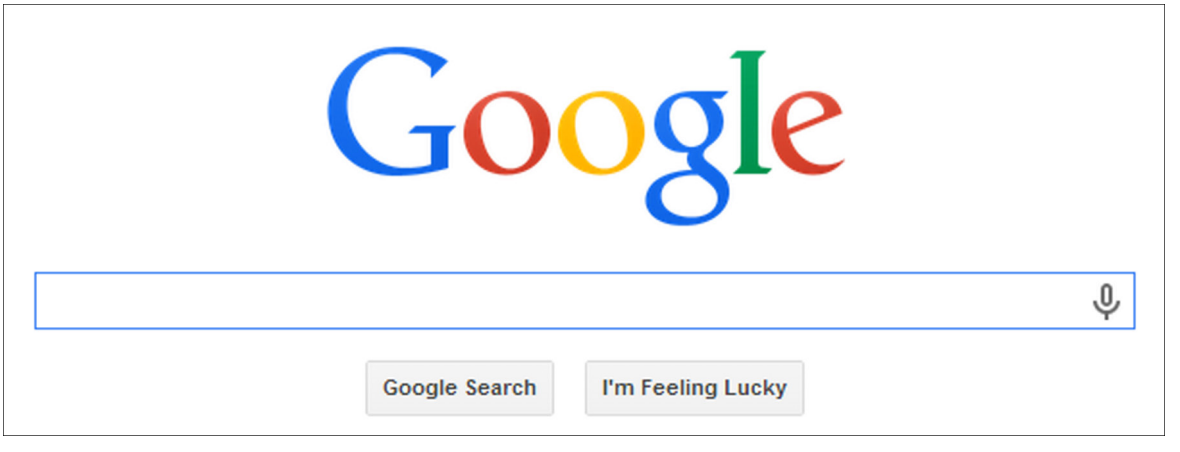


Figure 2.23: Microphone Icon

The search term will be displayed in the search box when it is recognized. It also displays results on the same page as shown in figure 2.24.

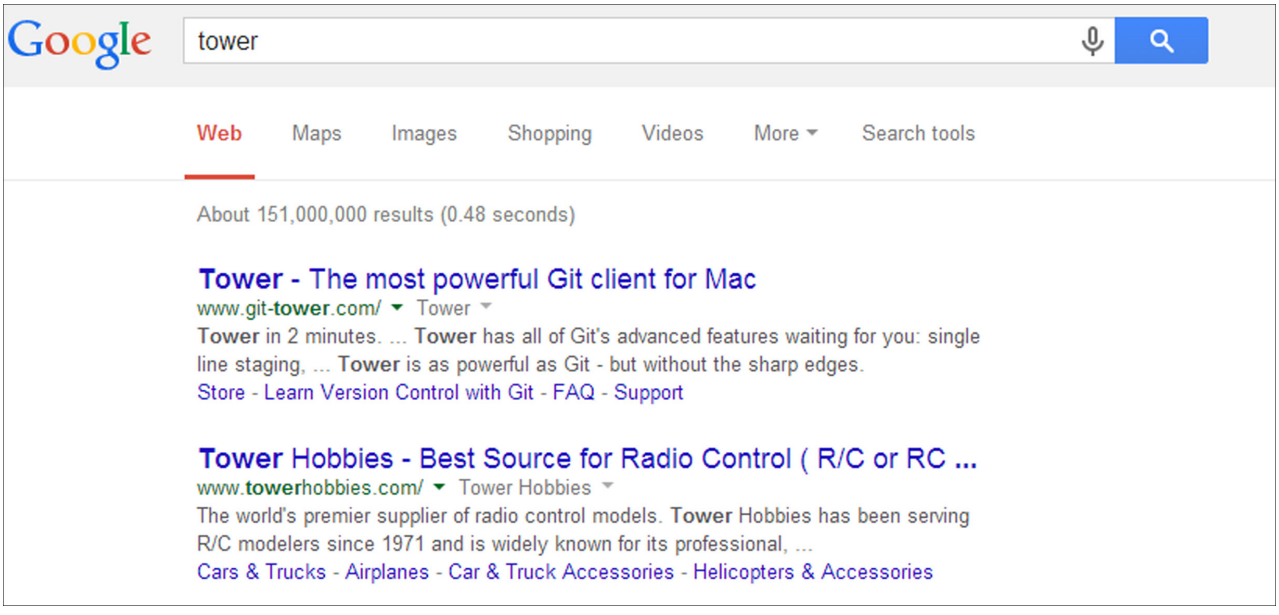
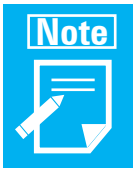


Figure 2.24: Displaying Searched Term

If Voice Search is unable to understand the user's speech, it lists a set of possible search terms that he/she might search for. The user can then click the required search term.

Users can use Voice Search to find answers to various questions by simply speaking them out. Some examples of such Voice search questions are as follows:

- Where is the nearest Italian restaurant?
- What time is it in Sydney?
- How many dollars is 200 pounds?
- What is the weather in Seattle?



Users can also set reminders using Google Voice Search.

2.3.2 Google Image Search

Google Image search is used to search for images on the Web. The keywords for the image search are based on the link text pointing to the image, the file name of the image or text adjacent to the image. A thumbnail for each image is displayed in the image search results.

To search an image, perform the following steps:

1. Open **Google Chrome** browser.
2. Type **www.google.com** in the Address bar.
3. Click the **Images** link in the top right corner of the page. The **Google images** page appears as shown in figure 2.25.

Figure 2.25: Google Images Page

4. Type **mango** in the search box and click Search as shown in figure 2.26.

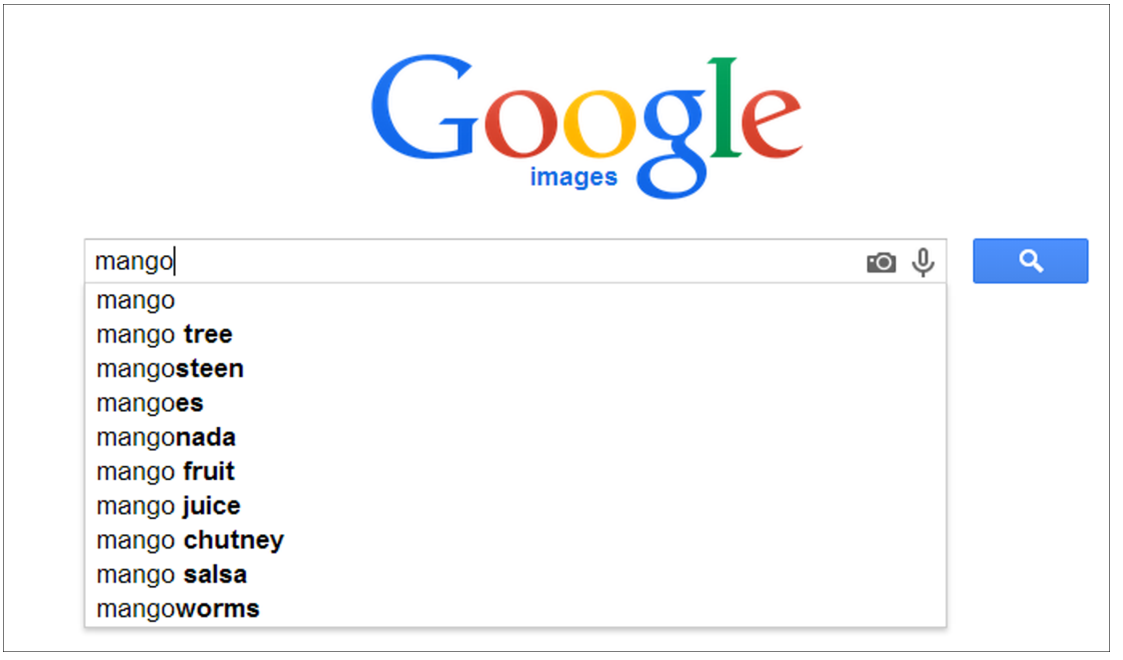


Figure 2.26: Typing 'mango' in the Search Box

The search result is displayed as shown in figure 2.27.

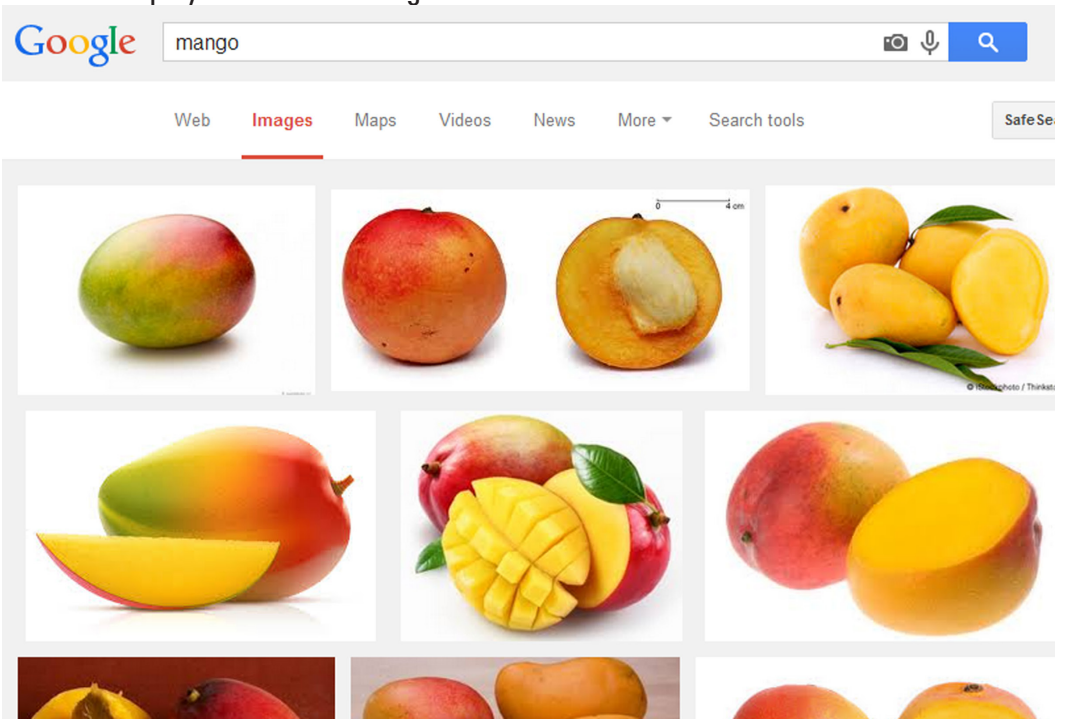


Figure 2.27: Mango Search Result

5. Click any thumbnail. The image is zoomed and displayed in a box along with the Web site address. Figure 2.28 displays details of the image.

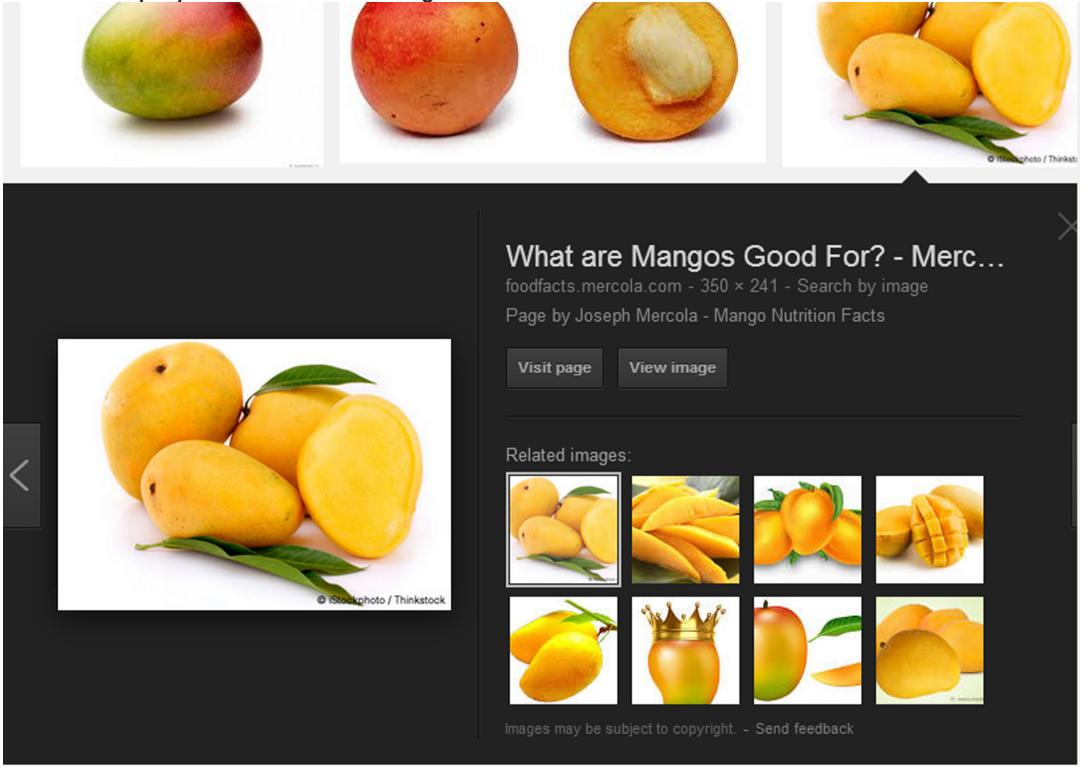


Figure 2.28: Details of the Image

Search by Image

Google has introduced a new feature for image search. This feature is called **Search by Image**. Here, the user does not have to enter or remember a word or a phrase. There are four ways to search by image and they are as follows:

1. Upload an image from the desktop
2. Copy and paste an image URL
3. Drag and drop an image
4. Right-click an image on the Web

The user can directly add specific image on the local system or on the Web and search for similar Web pages or images about that specific image.

Upload an Image from the Desktop

To search an image by uploading it from the desktop, perform the following steps:

1. Open **Google Images** page. In the search box, Camera (📷) icon is displayed as shown in figure 2.29.

Figure 2.29: Camera Icon

2. Click the Camera (📷) icon, a pop-up box shown in figure 2.30.

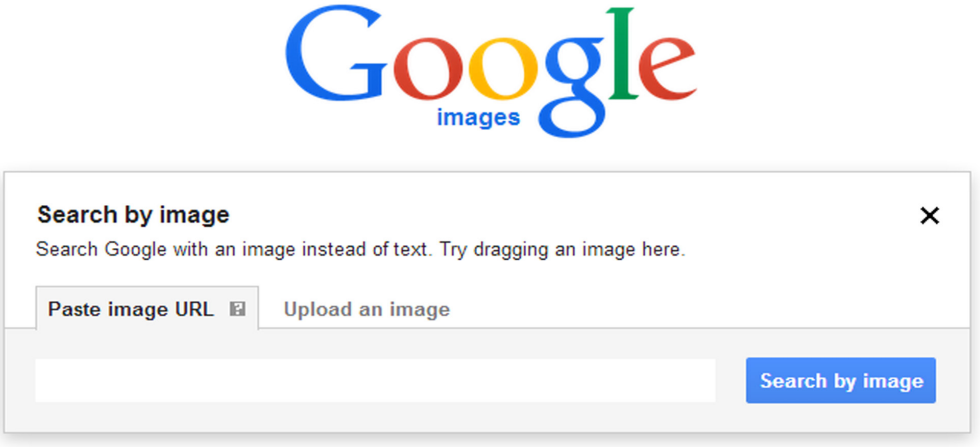


Figure 2.30: Clicking the Camera Icon

3. Click **Upload an image**. The screen appears as shown in figure 2.31.

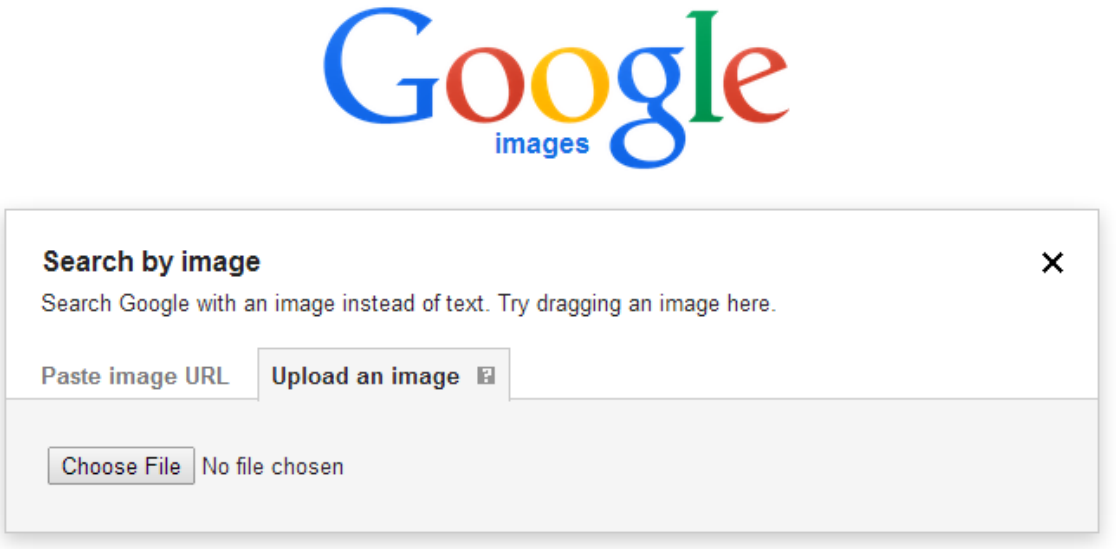


Figure 2.31: Clicking Upload an Image Tab

4. Click **Choose File**. It displays the **Open** dialog box. Select the file and click **Open** as shown in figure 2.32.

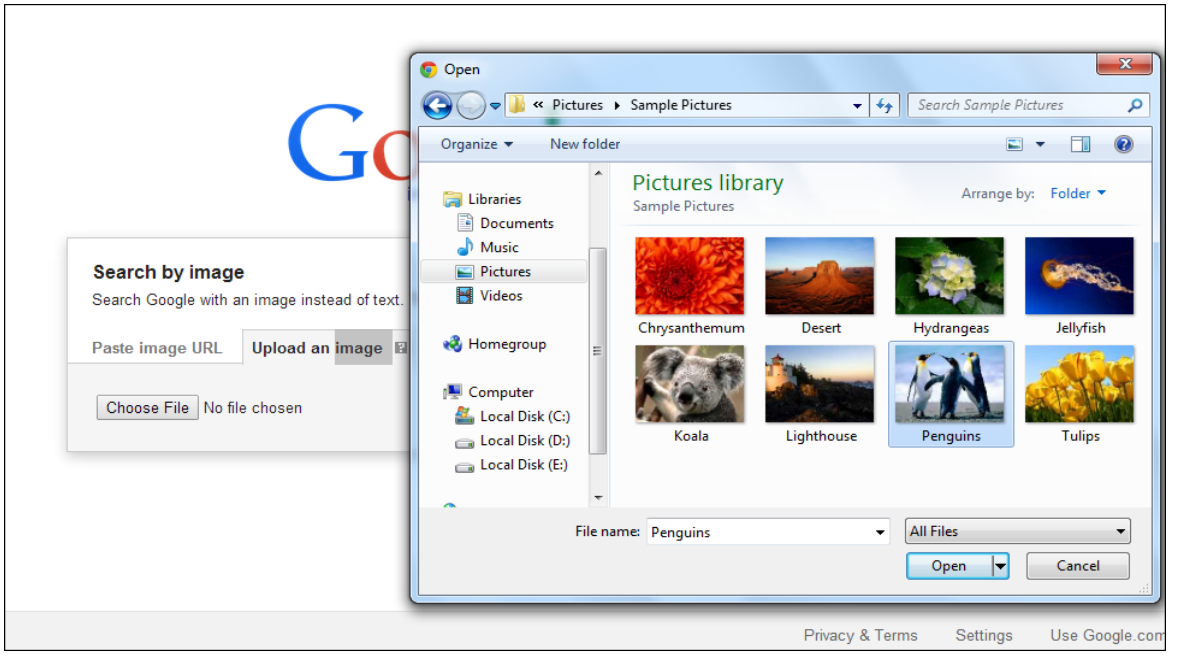


Figure 2.32: Selecting the Image

The image uploading process begins as shown in figure 2.33. It takes some time to upload an image.

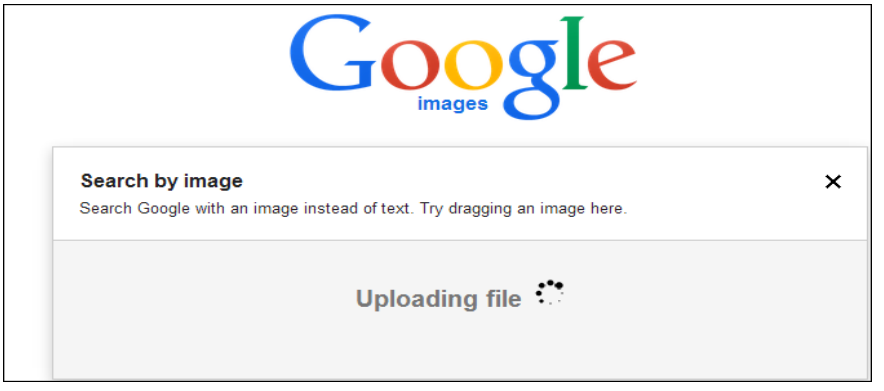


Figure 2.33: Uploading an Image

The result page appears after the image has been uploaded as shown in figure 2.34. The result contains image thumbnails similar to the uploaded image. It also displays Web site that has images similar to the uploaded image.

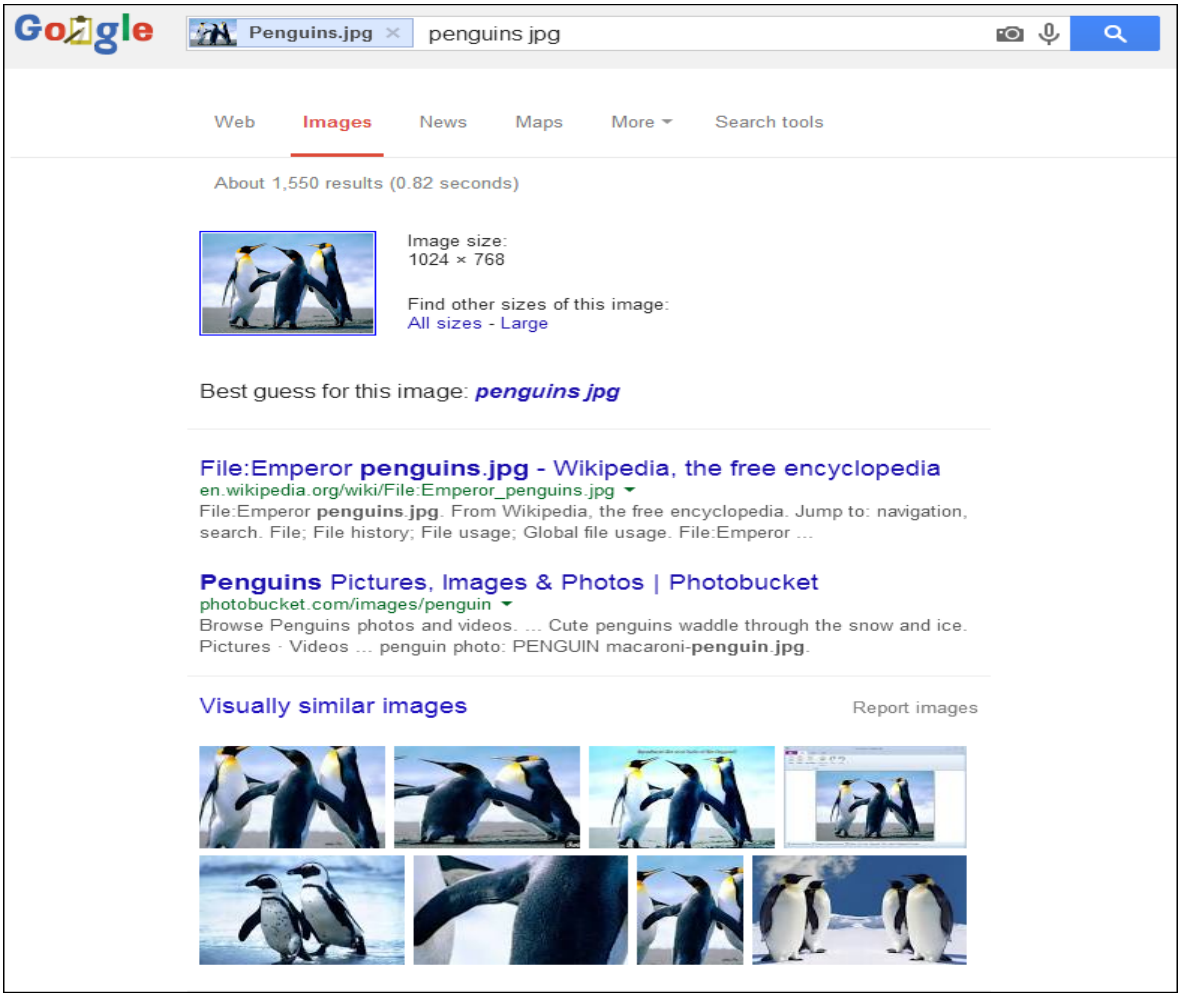


Figure 2.34: Image Result Page

Search Using an image from a Web site

To search an image by copying the URL, perform the following steps:

1. Open the **Google Chrome** browser and type **www.google.com** in the Address bar.
2. Type any search term in the search box, for example, **elephants**. The result page will be displayed.
3. Open any link from the results page. For example, **http://en.wikipedia.org/wiki/Elephant**
4. Right-click an image and select **Copy image URL** option as shown in figure 2.35.

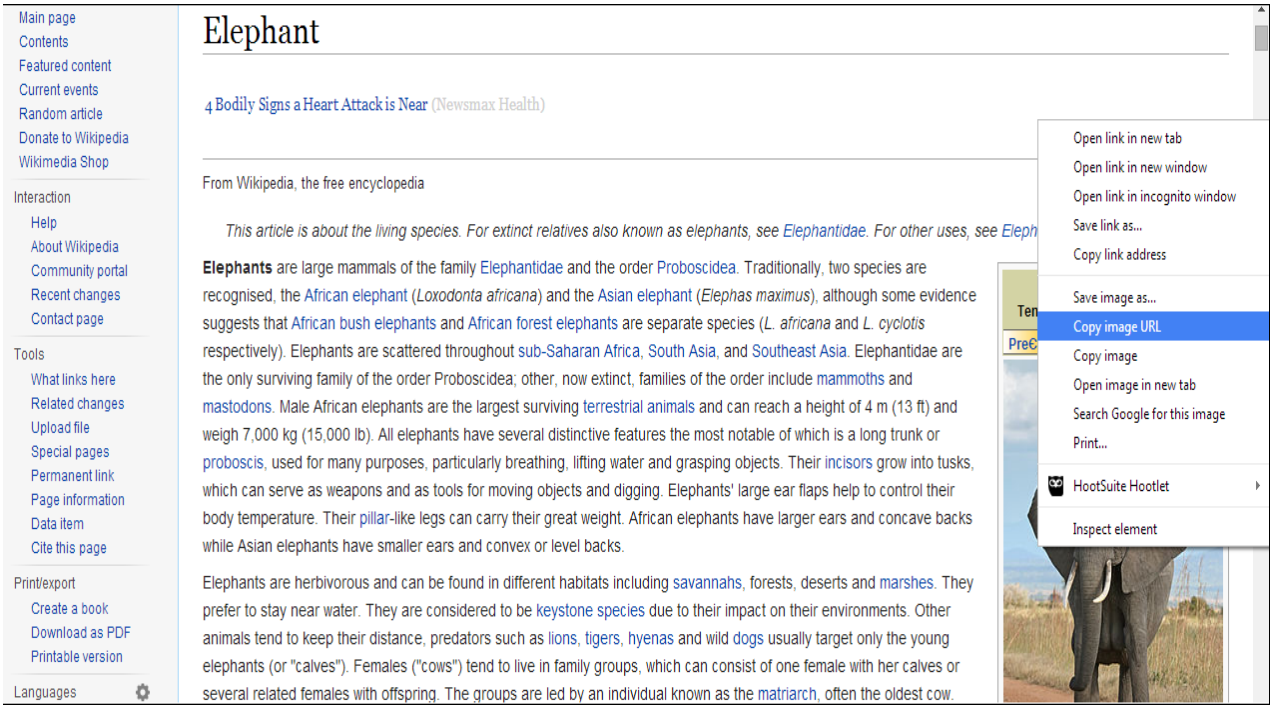


Figure 2.35: Selecting and Copying Image URL

5. Open **Google images** page. Click the ( ) icon in the search box. The **Search by image** prompt appears. Paste the image URL which has been copied in the search box as shown in figure 2.36.

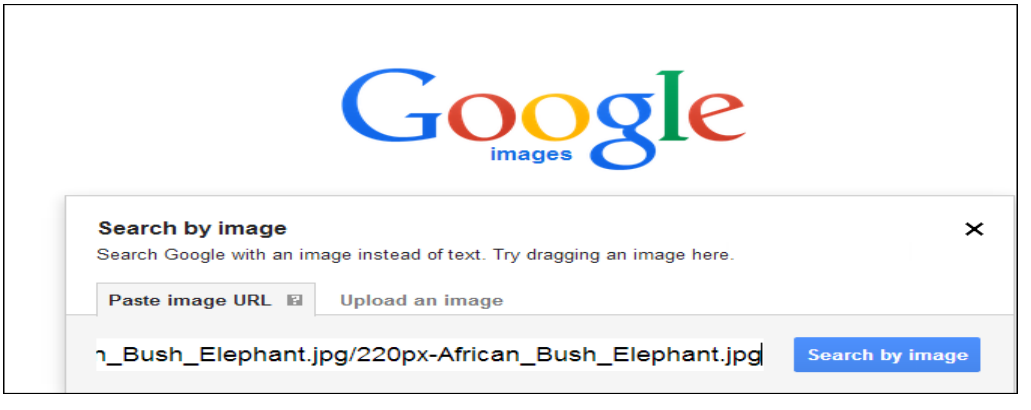


Figure 2.36: Pasting the Copied Image URL

6. Click **Search**. The search results will be displayed as shown in figure 2.37.

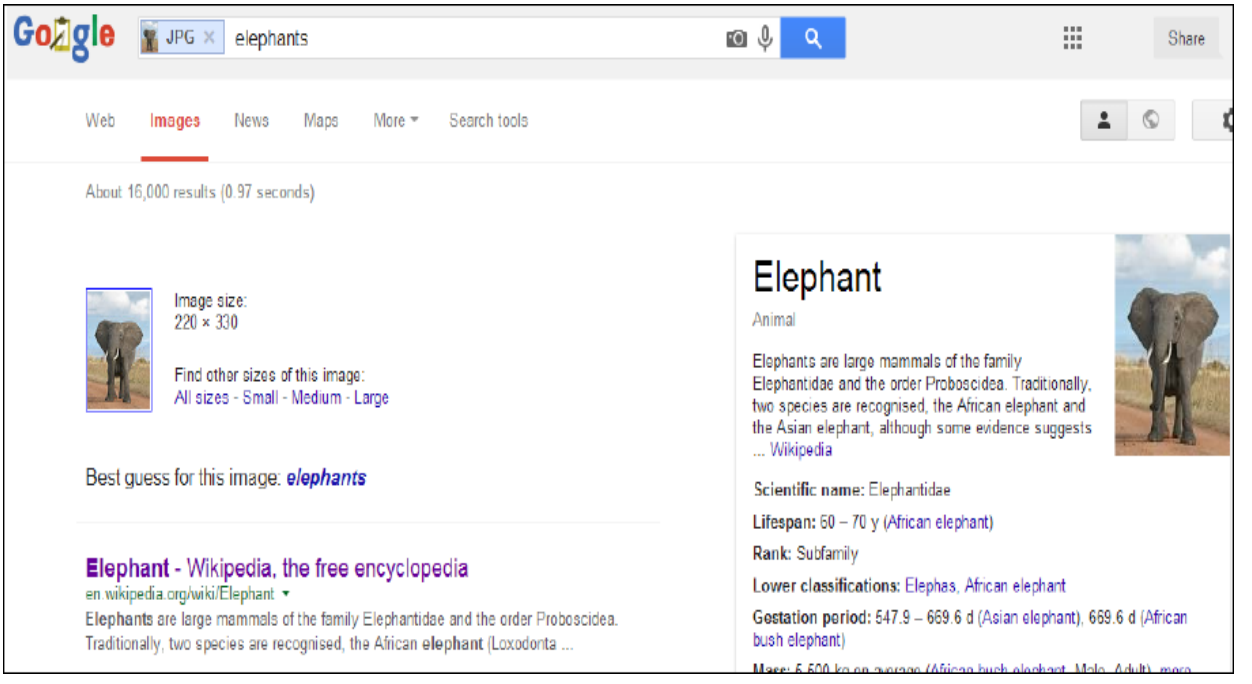


Figure 2.37: Result Page for Paste Image URL

Search by Dragging and Dropping an Image

To search an image by dragging it, perform the following steps:

1. Open **Google images** page.
2. Search for any image, for example, **Rose**.
3. Drag the required image from the Web and drop it to the Google image search box as shown in figure 2.38.

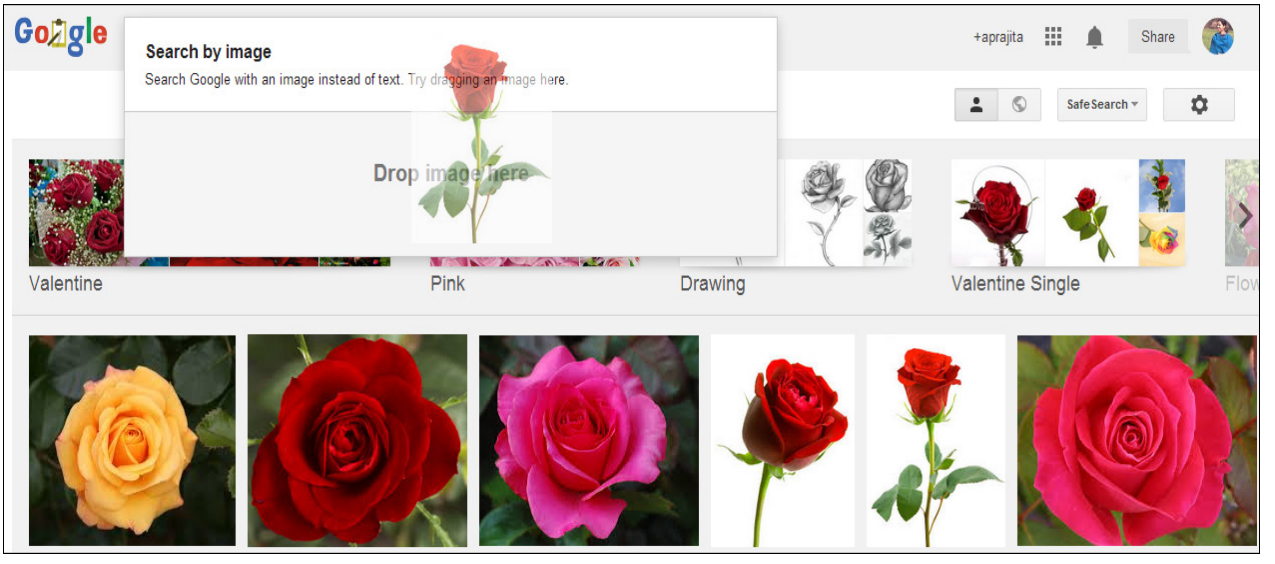


Figure 2.38: Dropping the Image

The uploaded image will be processed and compared to all the similar images on the Web and the relevant result will be displayed as shown in figure 2.39.

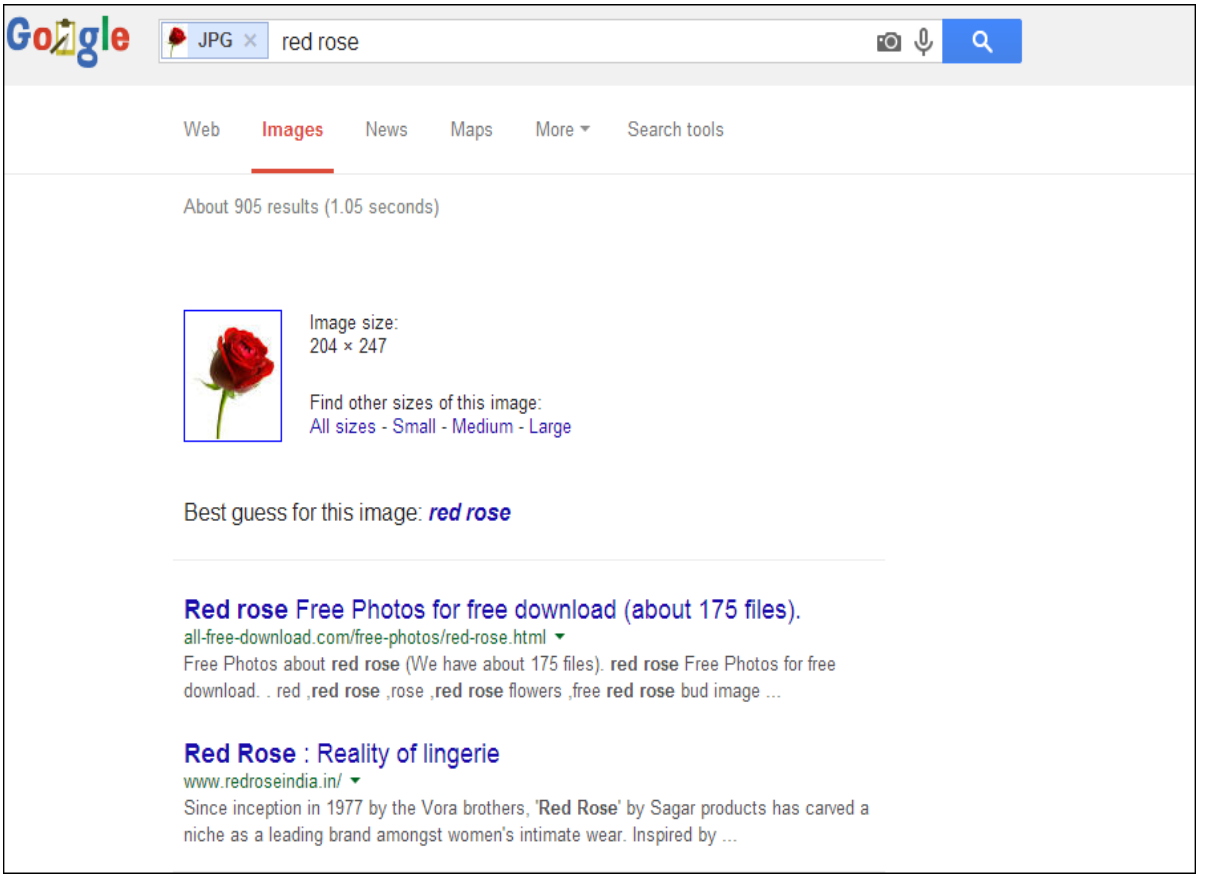


Figure 2.39: Result Page after Dropping an Image

Search Google with this image Option

Users must download the **Google Chrome** or **Firefox** browser extension to search image on the Web. This method does not require copying the URL or uploading the image.

To search by an image on the Web, perform the following steps:

1. Open **Google Chrome** browser.
2. Type the search term in the search box, for example, **rose**. The result page will be displayed.
3. Open any link from the result page. The Web page will be displayed.
4. Right-click required image on the Web page and select **Search Google with this image** option as shown in figure 2.40.

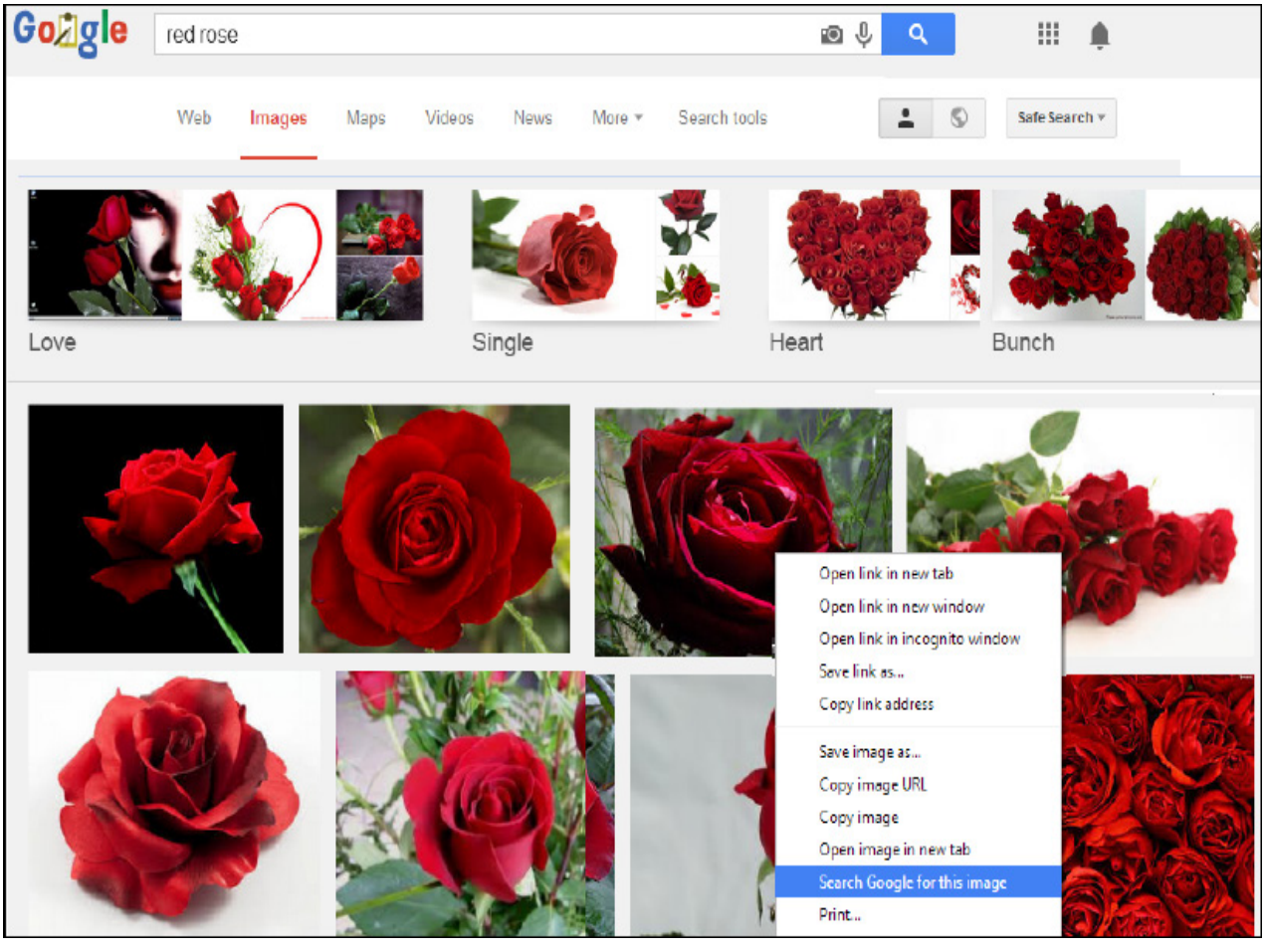


Figure 2.40: Selecting Search Google for This Image Option

The result page will be displayed as shown in figure 2.41.

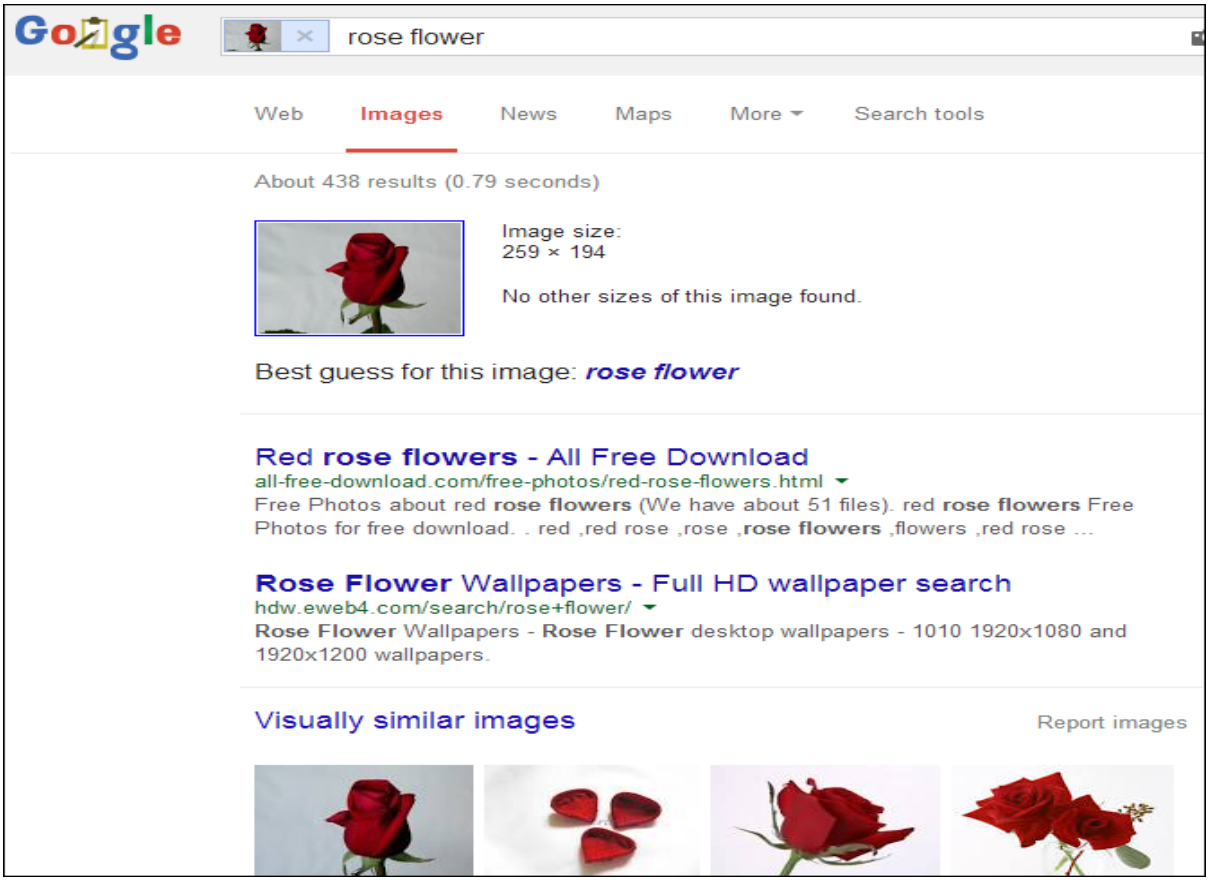


Figure 2.41: Search Result Page with Google with This Image

2.4 Working of Google Search

When the user types keywords as the search query, Google matches the search query with different Web sites and determines the order in which the search results would appear. An index is used to match the query. An index is a reference to all the sites that are present on the Internet. This index data is huge and is stored on the database servers by Google. The results that appear after indexing are of two types that are as follows:

- **Organic Search Results** - These are the search results that according to Google are the most relevant to the user. For example, if a user types the term 'greeting cards', Google Search will show Web sites which would be most relevant to the user. Hence, these search results are organic.

- **Sponsored Search Results** - These search results comprise advertisements that are used as a promotion for a certain organization, product, or service. They consist of links that are related to the keywords but may not be directly associated with them. For example, for the term 'greeting cards', advertisements of certain companies selling greeting cards may be shown under sponsored search results.

The background process of delivering a result page consists of three steps that are as follows:

- **Crawling** - A crawler is a program that collects information from different Web sites and stores the data in the database of the search engine. Google uses its own crawler called **Googlebot**. **Googlebot** collects information from the Web pages and also finds new and updated Web pages to be added to the Google database named Google index.

  The process that **Googlebot** uses to collect information from different Web sites is as follows:

  1. The **Googlebot** crawls through the Web and creates a list of Web sites in the Google index. This is a continuous process.
  2. The **Googlebot** also regularly crawls through the Web page URLs in the Google index and checks if there are any updates to the Site map data of the Web site.
  3. All the links and sub-links related to a particular site are added to the Google index.
  4. Also, changed links are updated and dead links are removed.

- **Indexing** - In this process, the Google index plays a major role. After the **Googlebot** collates and processes the data, it sends the data to the Google index which consists of all the words and their Web page location. These words act as keywords for the site. Though the **Googlebot** can collect information from all content types, it cannot process and classify all content types. Hence, for such types of content, Google uses the information received in the key content tags and attributes that are present in the `title` and `alt` attributes of the site.

- **Serving** - After a user types the search query in Google, the search engine searches the index to find the relevant pages and then lists these pages in the results page. To list a site in the Google Search results, the site must instruct the **Googlebot** to crawl and index the page.

  After the information is collected, a **PageRank** is generated for the site. A **PageRank** is determined based on the number of incoming links from different sites to the present site.

Some of the features provided by Google for better output and performance of the Google results page are as follows:

- Google Autocomplete
- Did you mean

Check Your Progress

1. Which of the following option collects information from different Web sites and stores the data in the index of the search engine?

(A)	Crawling	(C)	Surfing
(B)	Indexing	(D)	Serving

2. Which of the following option is enabled to filter out adult sites?

(A)	Usage rights	(C)	SafeSearch
(B)	Web history	(D)	LegalSearch

3. Which of the following option in search preferences allows user to search results as they type?

(A)	Google Instant	(C)	Google Voice search
(B)	Google Instant predictions	(D)	Google Video search

4. Which of the following is Google's default setting for the SafeSearch?

(A)	Advanced	(C)	Strict
(B)	No filtering	(D)	Moderate

5. Which of the following options enable users to understand that SafeSearch is locked?

(A)	'SafeSearch locked' is displayed on the right of the search box
(B)	The colored balls appear on the right of the Google search result page
(C)	Security icon is displayed in the Address bar
(D)	There is no way to find the SafeSearch is locked

(A)	A, B	(C)	C, D
(B)	B, C	(D)	B, C, D

Answer

1.	A
2.	C
3.	B
4.	D
5.	A

Summary

- Google Search is used to search information that includes Web pages, images, videos, and news.
- Using basic search techniques, a user enters a keyword and a search result is returned.
- In the advanced search option, users can define the file formats, results per pages, language preferences, and so on.
- The search results can be customized to appear according to the user's requirement using the Google Search settings.
- Newly added Google search types are Google Instant, Google Voice search, and Google Image search.
- Google Instant displays the search results while the user types the search query.
- Voice Search allows the user to quickly search the Web by speaking the search terms.
- Google Search classifies the search results as organic search results and sponsored search results.

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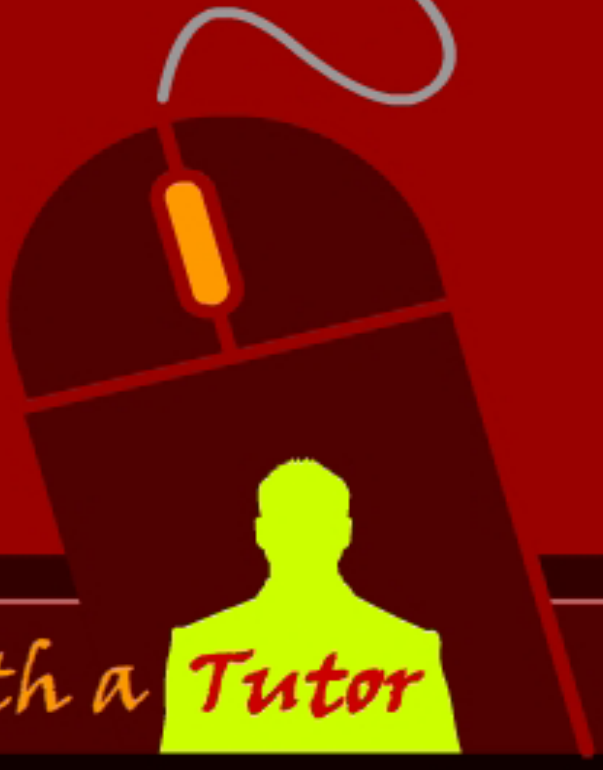

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03

Introduction to Search Engine Optimization

Welcome to the Session, **Introduction to Search Engine Optimization**.

This session begins with an introduction to SEO and continues to explain the basic guidelines of SEO. It also explains the roles and responsibilities of an SEO consultant.

In this Session, you will learn to:

- Define and describe SEO
- Explain the roles and responsibilities of SEO Consultant
- List the Google Webmaster guidelines for SEO
- Explain the basic techniques used to implement SEO
- Explain the process of submitting pages to Google
- List the causes for sites not performing well in search

Session
03

Introduction to Search Engine Optimization

3.1 Introduction

Consumers of Web site content are often visitors who depend on search engine results for their queries. Therefore, optimization of Web sites for a search engine becomes extremely important as the search engines result can drive traffic to the sites.

SEO is the process through which a Web site is listed in the top ranking results of search engines. SEO helps in creating a strong position for the Web site by using certain techniques to reach the top rank. It also improves the number of visits to the site. Good content and relevant keywords play an important role in SEO.

Following tasks are included in the process of SEO:

- Review of your site content or structure
- Technical advice on Web site development
- Content development
- Management of online business development campaigns
- Keyword research
- SEO training
- Expertise in specific markets and geographies

3.2 Introduction to SEO

A few decades ago, the number of people using Internet were very less, there were hardly any search engines, and therefore, very less competition in the Internet arena. Today, the number of Web sites across the world exceeds 200 million. On an average, there are over 60 billion search queries per month on Google alone. Given such statistics, it becomes crucial to utilize strategies and techniques that can help a business/ enterprise or an individual to improve traffic to their Web site.

Consider a scenario where a user types **computer education**. The organization would like Aptech Computer Education Web page to be displayed as one of the top search results.

Figure 3.1 shows the user query **computer education** in Google search engine.

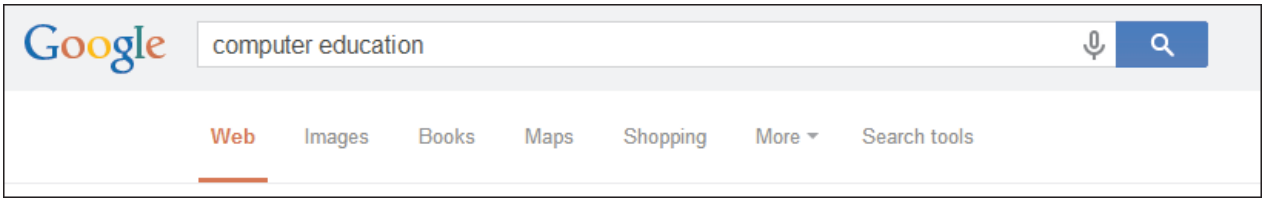


Figure 3.1: Searching for the Keyword

Figure 3.2 shows the desired outcome expected from the Web search for **computer education**.

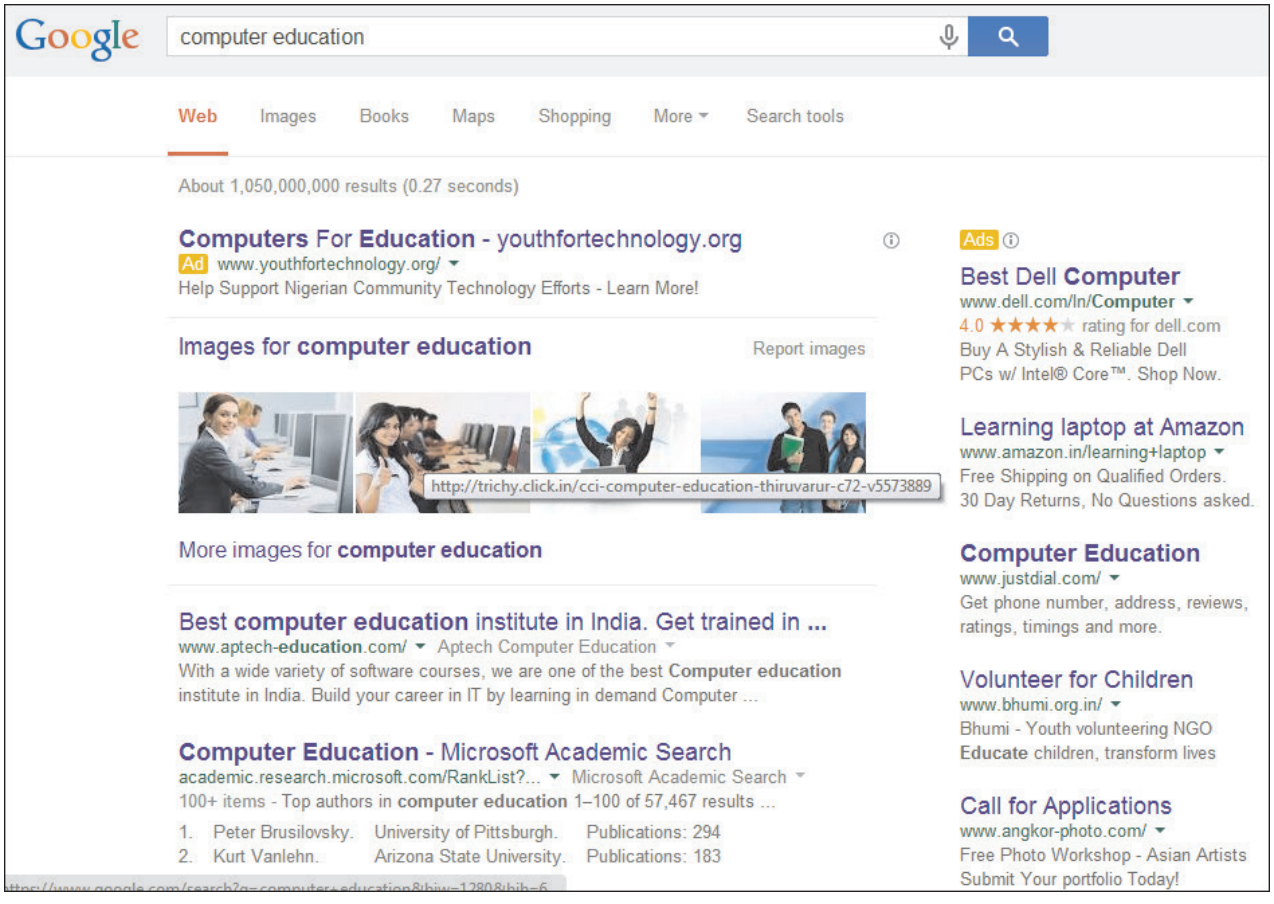


Figure 3.2: Search Results Displayed for the Keyword

SEO is the process that enhances the visibility of a Web site to the search engines query result. Typically, Internet users are most likely to select the Web site or Web page that appears frequently in search results list or is among the first ones listed. It is here that SEO can make a difference by helping search engines find and rank a Web site higher as compared to the other sites. In simple words, SEO is a technique that increases the number of visitors to a Web site.

Apart from text and content search, SEO can also search for videos, images, and news.

An important factor to note is that SEO achieves the goal of improving the visibility of a Web site through the natural or un-paid search results and not through sponsored links. These search results are also called organic or algorithmic search results. SEO considers how search engines work, what items people search for, the search queries entered in search engines, and the search engines preferred by users.

3.3 SEO Consultant

Key responsibilities of an SEO consultant include performing all the tasks in an SEO workflow as well as perform page optimization wherever required. In page optimization, the consultant checks the markup and code of a page and cleans it up wherever necessary.

This may include checking the page titles and other parts of a Web page, site navigation, user-friendliness, search engine friendliness, and so on. As most search engines including Google provide a high page rank based on keywords, a Web site should be targeted towards the use of several relevant and constructive keywords. An SEO consultant must help to select the correct keywords to target the desired audience.

Besides these tasks, an SEO consultant may also have to perform the following:

- Review site content or structure
- Provide technical advice on Web site development including hosting, redirects, and error pages
- Conduct SEO training

3.3.1 Considerations for Hiring an SEO Consultant

By hiring an SEO consultant, an organization or individual client can potentially improve the ranking of their site and save time, but there is also a good amount of risk involved. An irresponsible or unethical SEO can not only harm the site but also destroy the reputation of the site owner.

Following are some of the considerations that must be observed while hiring an SEO consultant:

- Avoid SEO firms and Web consultants or agencies who send spam or unsolicited e-mail offering their services. Such firms or agencies may not be genuine, therefore, it is best to steer clear of them.
- Avoid SEOs who claim to guarantee a number one ranking on Google.
- Watch out for SEOs who are secretive or do not clearly explain what they aim to do.
- Watch out for SEOs who create suspicious, misleading, or false content on the behalf of the site owner just to draw more traffic, because the site could be removed from Google's index.
- Watch out for SEOs who claim to guarantee rankings on the basis of special relationship with Google or priority submit to Google.

There is no feature called priority submit for Google. The only method to submit a site to Google is through the Add URL page or by submitting a Site map. A Site map is a file specifically created to provide links to different pages of the Web site. This is one of the most important file that must be created by the Web site developer for SEO purposes.

Note that even after using a Site map, there is no guarantee that Google will crawl or index all the URLs of the Web site.

3.4 Google Webmaster Guidelines

Google provides a set of guidelines for enhancing SEO for a Web site. These guidelines help to index and rank the Web site on the Google search results page. It also helps to maintain the quality and avoid spamming. The Google Webmaster guidelines can be classified into the following three categories:

- Design and content guidelines
- Technical guidelines
- Quality guidelines

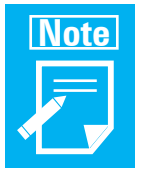
Design and Content Guidelines

Some of the design and content guidelines for Webmasters specified by Google are as follows:

- Submit the Web site to Google.
- Submit the Site map to Google.
- Submit the Web site to other search engines and related Web sites.
- Provide hyperlinks connecting all the pages on the site.
- Provide a Site map to the users for easier navigation.
- Limit the number of links on a single Web page.
- Ensure that quality of the content inside a Web page is informative and specific in nature.
- Ensure that the content includes keywords that would be affiliated to the business or service provided by the Web site.
- Ensure that names of all the images match with their description. For example, if a logo is used, the image should be named as 'logo' so that search engines can pick this up easily.
- Check that all the links on the Web site are functional.
- Allow crawlers of different search engines to crawl on the Web site.
- Monitor the site performance by using tools such as Google Analytics.
- Do not use irrelevant keywords.
- Avoid information that may lead to phishing, spamming, or viruses.

Technical Guidelines

- **Test using text browsers** - Before submitting the Web site, it is recommended to test it using a text browser such as Lynx. Text browsers display the content as viewed by search engine spiders. All important content and links in the Web site should be accessible and readable in such browsers and not hidden behind technologies such as JavaScript, Flash, DHTML, and frames.
- **Keep search bots free from session IDs and arguments** - Session IDs and arguments should not block the path of search bots. Else, they might not be able to do proper indexing of the Web site.
- **Ensure support for If-Modified-Since HTTP header** - Web sites are dynamic and keep changing. Webmasters need to ensure that the Web server hosting the Web site supports the If-Modified-Since HTTP header, so that the Web server can communicate to Google whether the Web site content has changed since it was last crawled.



The If-Modified-Since HTTP header saves the Web server bandwidth and overhead.

- **Regularly update robots.txt** - Web crawlers gather information about which Web site directories can be crawled and which cannot. Web masters must regularly update the **robots.txt** file on the server and check that the Googlebot crawler is not accidentally blocked. The Google Webmaster Tools provides an analysis tool that can be used to test the **robots.txt** file for accuracy.
- **Ensure that content management system pages do not block crawlers** - If using a content management system to create and manage the Web site, make sure that the pages and links created are accessible to search engine crawlers.

- **Test Web site in multiple browsers** – Different users might prefer using different browsers for accessing Web sites. It is therefore important to test all the content and links in the Web site for accurate visibility and accessibility in different browsers.
- **Test site speed and performance** - It is strongly recommended that Webmasters regularly rest their Web sites for optimal performance and speed using various tools such as **Page Speed**, **YSlow**, and **WebPagetest**. Google provides the **SitePerformance** tool which shows the speed of a Web site as experienced by users around the world.

Quality Guidelines

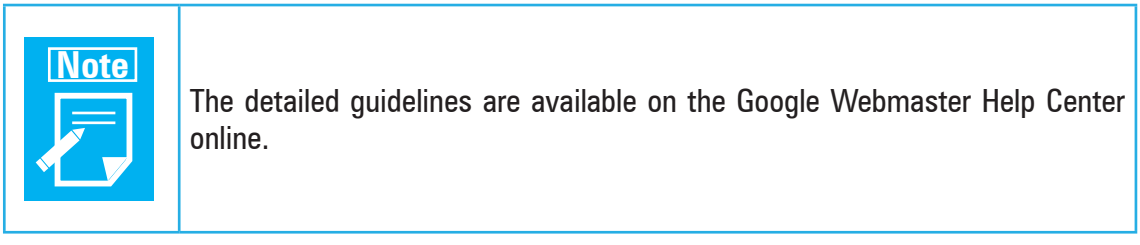
Web sites must always be designed keeping the users in mind. Web masters must present content such that it is easily accessible to users who need the information it provides. Concentrate on providing information that is valuable and engaging to users. Never create Web sites just to attract search engines and increase the page rankings.

Webmasters should engage in good practices such as regularly monitoring Web sites to prevent and remove spam and to protect Web sites from hacking and remove hacked content at the earliest.

Following are some of the quality guidelines for Webmasters when developing Web sites:

- Avoid using automatically or programmatically generated content.
- Avoid participating in link schemes as it is a violation of Google's Webmaster Guidelines.
- Do not display different content and URLs to users and search engines. This is called cloaking.
- Do not program sneaky redirects where the search engine is presenting a certain kind of information, while on visiting the Web site, the user is redirected to a page containing completely different information.
- Avoid placing hidden text or links using techniques such as white text on white background, text behind images, placing text off-screen using CSS, or setting font size to zero.
- Do not include doorway pages which are pages that do not contain any useful information for users, but are optimized for specific keywords of phrases that attract search engines and increase page rankings.
- Never use scraped content which is content lifted from other popular Web sites just to increase the number of pages in the Web site.
- Do not create pages with malicious behavior that promote phishing, install viruses, Trojans, or other malware.
- Do not program Web sites to send automated queries to Google.

Violating any of the quality guidelines may result in Google taking action against the Web site.



There are various methods that can be used by a Web user that helps to enhance the SEO experience. Some of these methods for better SEO results are shown in figure 3.3.

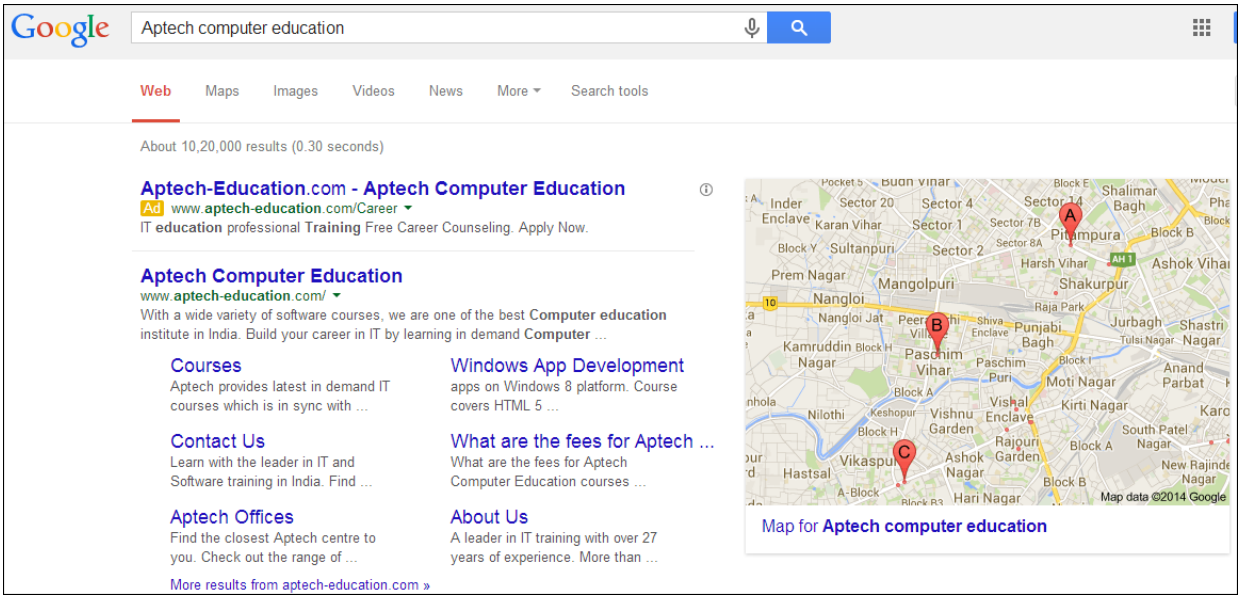


Figure 3.3: SEO Methods

The methods depicted in figure 3.3 are as follows:

- **Keywords** - Keywords are popular terms that are attributed to a certain Web site. The right keywords must be selected to direct the user to accurate and useful results. This can be done by selecting specific and focused keywords instead of generic keywords. For example, the keywords used in figure 3.3 are **Aptech**, **Computer**, and **Education**.
- **Title** - The title of the page can also be used to search for a specific term or Web site. The title must be short and instructive. For example, the title specified in figure 3.3 is **Aptech Computer Education: Pioneer in IT education | Home**.
- **URL** - A user may either know only a part of the URL or may be unsure of the URL spelling. In such cases, the user can type part of the URL in the Google search engine. Google would provide suggestions by providing results that are closest to the URL searched for.
- **Description** - Each Web site must specify a description that would include the keywords for the Web site. A user can also search for a part of the description to receive the required results. The description for Aptech Computer Education Web site is **Aptech offers IT courses for engineers, for 12th pass/college students, graduates and working people. Aptech provides most- wanted IT courses, specially trained...**
- **Related Links** - Hyperlinks that can lead to the main Web site also help in SEO. Also, the main Web site result can lead to a specific Web page that may be required by a Web user. In figure 3.3, the related links are related to the short-term courses, certification courses, and so on. Here, if the user had to see only the certification courses, it can be accessed directly through the related links.
- **Web site Content and Images** - The Web site content must contain the keywords that are attributed to the Web site. All the images in the Web site must have a description and an `alt` tag.

To view the Web site of Aptech Worldwide, the user can use any of the following details to search for the site and reach the required search result:

- **Keywords** - Aptech, Worldwide, Training, Content Development, Assessment Solutions, Education, Aviation and Hospitality, Hardware and Networking, Animation and Multimedia, and so on.
- **Title** - Aptech Global Learning Solutions: Retail & Corporate education
- **URL** - www.aptech-worldwide.com/
- **Description** -IT, animation & multimedia, hardware & networking, aviation & hospitality education, content development, assessment solutions & training for corporates.
- **Related Links** - Contact Us, Careers with Aptech, Current vacancies, IT education, Investors, and About us.
- **Web site Content and Images** - Aptech is an ISO 9001:2000 organization that commenced its education and training business in 1986 and has globally trained over 6.4 million students. The organization is headquartered in Mumbai, India, and has presence across 40+ countries.

Here, if the Web user knows any one of the details mentioned, SEO would help to find the site.

Consider an example where a user searches by using keywords. Type some of the keywords related to the Aptech such as **Training, Content Development, Assessment Solutions**. The results displayed by searching using keywords would have Aptech Worldwide as a top ranking result. Refer to figure 3.4.

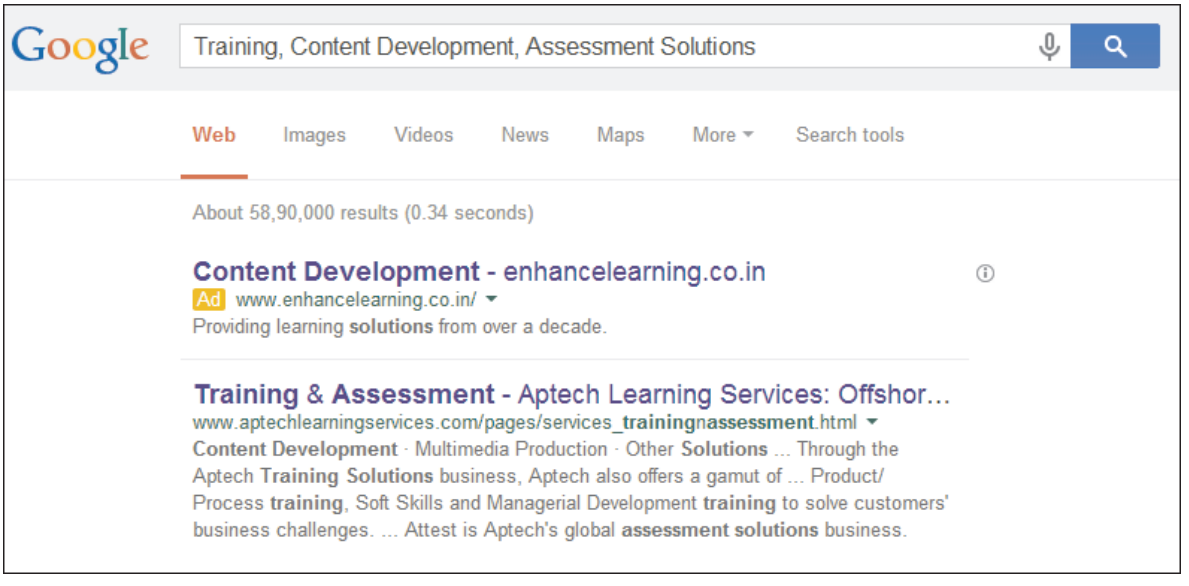


Figure 3.4: Results by Searching Using Keywords

3.5 Submitting Pages to Google

Googlebot, the Google crawler crawls through almost all the Web sites that are available on the Internet.

These sites are automatically added to the Google index. For faster crawling and indexing, a user can submit the Web site to Google after it has been developed. There is no additional cost involved to submit the site to Google.

A user can also re-submit an already developed site to Google if there are any changes to the Web site.

This helps to update the information that is present in the Google index and replace it with the new information. To confirm if a site is indexed in the Google index, type the following in the Google search box:

site:www.sitename.com

The steps to submit a Web site to Google are as follows:

1. Type the link **www.google.com/addurl/** on the Web browser.
2. Log in using a Google account.
3. Type the URL name in the URL box. Refer to figure 3.5.

Figure 3.5: URL Box

4. Type the **CAPTCHA**.

> **Note**
>
> CAPTCHA is a program that generates and grades tests that only humans can pass but not computer programs. It generates random words or letters which are displayed in a distorted fashion.

5. Click **Submit Request** to submit the Web site.

There are numerous sites that are being created on the Internet. The crawler Googlebot can miss out crawling through certain sites. Some of the reasons for this can be recent launch of the site, errors on the site, bad design of the site, and non-connected links. It is therefore better to submit the Web site directly to Google. Also, making a Web site crawler friendly helps to ensure that the site is available in the search results. An additional support can be provided by submitting the Site map along with the Web site. This would help Google to crawl through all the pages of the Web site.

3.6 Submitting Sitemaps to Google

Once a Site map is created using the Site map protocol, it must be submitted. To submit a Site map to Google, the developer must use the Google Webmaster Tools service. This service helps to receive strategic statistical information about the Web site.

Before a Site map is submitted, the following sites should be added and verified using the account created in Webmaster Tools service. They are as follows:

- The site containing the Site map location
- The URLs of sites referred in Site map

The steps to submit a Site map to Google are as follows:

1. Create and upload the Site map of a Web site.
2. Log in to the Webmaster Tools home page and click the verified site for which the Site map has to be submitted.
3. Under **Site configuration**, click **Sitemaps** as shown in figure 3.6.

Figure 3.6: Sitemaps Option

4. Click **Submit a Sitemap**.
5. In the text box, type the path of the Site map as shown in figure 3.7.

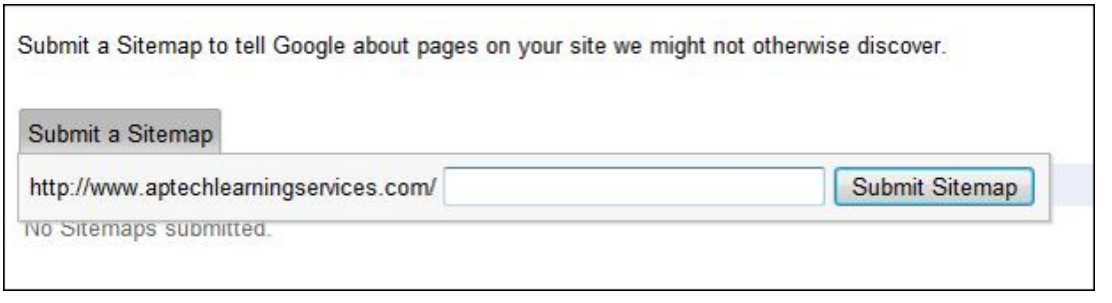


Figure 3.7: Submit a Sitemap Text Box

6. Click **Submit Sitemap**.

A Site map can also be submitted to Google and other search engines using **robots.txt** file. The **robots.txt** file can be edited to allow crawlers to access the Site map. To achieve this, the following line must be added to the **robots.txt** file:

```
Sitemap: http://example.com/sitemap_location.xml
```

3.7 Causes for Sites Not Performing Well in Search

The main aim of a search engine is to return relevant Web sites for each and every search query. Search index which is responsible for returning the search results is constantly being updated as sites are added and modified. At times, even after submitting a Web site to Google, it does not appear in the search results.

Some of the causes for sites not performing well in search are as follows:

- The Web site design is not crawler friendly and the content cannot be crawled upon by Googlebot.
- The Web site violates the guidelines specified by Webmaster.
- The hyperlinks of the site are not connected and are on standalone basis, thus they do not lead to other pages.
- The site structure is not easily accessible.
- The site has been recently launched and Googlebot has not yet crawled upon it.
- There are errors in the site or its syntax.
- Keywords, title, or description may not be present.
- The content of the site is not specific and it does not support the keywords.
- The site popularity is low and hence, it does not appear on results page.
- New content has been added but the updated site has not been submitted to Google.

Following are the steps that Webmasters can perform to identify and fix potential causes of their Web sites not performing well:

- Verify that the Web site is listed in the Google index.
- Check if the Web site has suffered from a manual spam action.
- Test and check if Google is able to find and crawl the Web site and index it.
- Make sure that the content on the Web site is useful and relevant for users.

Check Your Progress

1. Which of the following statements about black hat SEO techniques are true?

(A)	Black hat SEO techniques generate results that are long lasting.
(B)	Black hat SEO comprises techniques that search engines do not approve.
(C)	With black hat SEO techniques, the risk of a penalty or removal from a search engine database is very high.
(D)	Black hat techniques believe in placing users first and does not exclusively target search engines.

(A)	B, C	(C)	A, C, D
(B)	C, D	(D)	A, B, D

2. Which of the following guidelines should be followed while writing descriptive titles for a Web page?

(A)	Titles used in the page must be short and instructive
(B)	Titles must contain all the content given in the page
(C)	Titles must contain some unrelated keywords
(D)	Titles must be same for all the pages

(A)	A	(C)	A, C, D
(B)	B, D	(D)	A, B, D

3. Which of the following activities is typically not performed by an SEO consultant?

(A)	Check the page titles and other parts of a Web page and optimize them
(B)	Improve site navigation
(C)	Design the layout of the Web site
(D)	Implement search engine friendliness for the Web site

Check Your Progress

(A)	B, D	(C)	C
(B)	C, D	(D)	A, B, D

4. SEO achieves the goals through the _____ search results.

(A)	Natural
(B)	Unnatural
(C)	Sponsored
(D)	Paid

(A)	A	(C)	C
(B)	B	(D)	D

5. Which of the following URL is used to submit a Web site to Google?

(A)	www.google.com/urlsubmit
(B)	www.google.com/urladd
(C)	www.google.com/addurl
(D)	www.google.com/submiturl

(A)	A	(C)	C
(B)	B	(D)	D

Answer

1.	A
2.	A
3.	C
4.	A
5.	C

Summary

- SEO is the process that enhances the visibility of a Web site to the search engines.
- An SEO consultant is a person who plans and implements SEO for an organization or individual's Web sites.
- Google provides a set of guidelines for indexing and ranking a Web site.
- The methods used for SEO include keywords, title, URL, description, related links and Web site content and images.
- A user can submit the Web site to Google after it has been developed.
- A user can also resubmit an already developed site to Google if there are any changes to the Web site.
- An additional support to crawlers can be provided by submitting the Site map along with the Web site.

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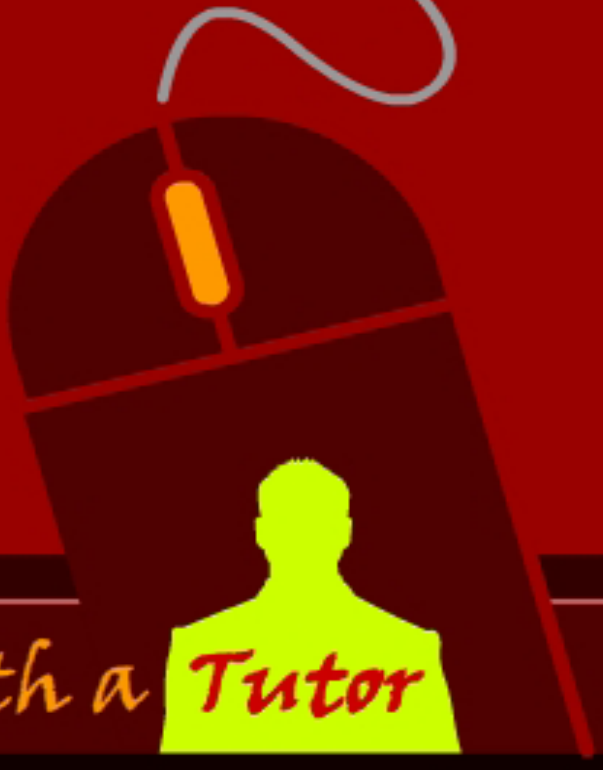

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04 On-Page Optimization

Welcome to the Session, **On-Page Optimization**.

This session describes on-page optimization and various techniques related to it.

In this Session, you will learn to:

- Explain on-page optimization and its need in SEO
- Describe different techniques for achieving on-page optimization
- Explain the importance of keywords while developing Web site
- Explain the selection of keywords on the Web pages
- Explain the optimization of site structure
- Explain the significance of developing quality content
- Explain the navigation structure for a Web site
- Explain the optimization of essential files such as robots.txt and sitemap.xml

Session
04

On-Page Optimization

4.1 Introduction

SEO is basically the science of designing a Web site which can achieve maximum ranking by the search engines. Ranking is related to the listing and placement of a Web page in the search result pages for a user search query. The more the pages are optimized, greater are the chances to achieve higher ranking in the search results.

Today, most organizations use Google as first choice of search engine for their Web site advertisements. It is one of the most well-known and widely used search engines which directs almost 70% of search engine traffic to the Web site. Thus, an SEO consultant has to adopt different strategies to optimize the Web site which can improve the ranking on the Google search result pages.

There are two different design strategies which can be followed by an SEO consultant to achieve maximum optimization of the Web site. They are as follows:

- **On-Page Optimization** - Refers to selection of proper keywords, proper site structure, and quality content through which Web site is easily recognizable by the search engine.
- **Off-Page Optimization** - Refers to building links from other Web sites to your site through directories, articles, blog postings, and so forth.

4.1.1 Overview of On-Page Optimization

On-page optimization is concerned about the various elements provided on the Web pages of a Web site. It is a process to optimize the Web sites depending on their nature and business. This includes use of proper keywords in the different elements of the Web page and their relevance with the overall content. It also deals with the internal linking of the pages and better Web site structure that can help in achieving better ranking and increased visitor satisfaction.

The different techniques used for achieving on-page optimization on the Web site are depicted in figure 4.1.

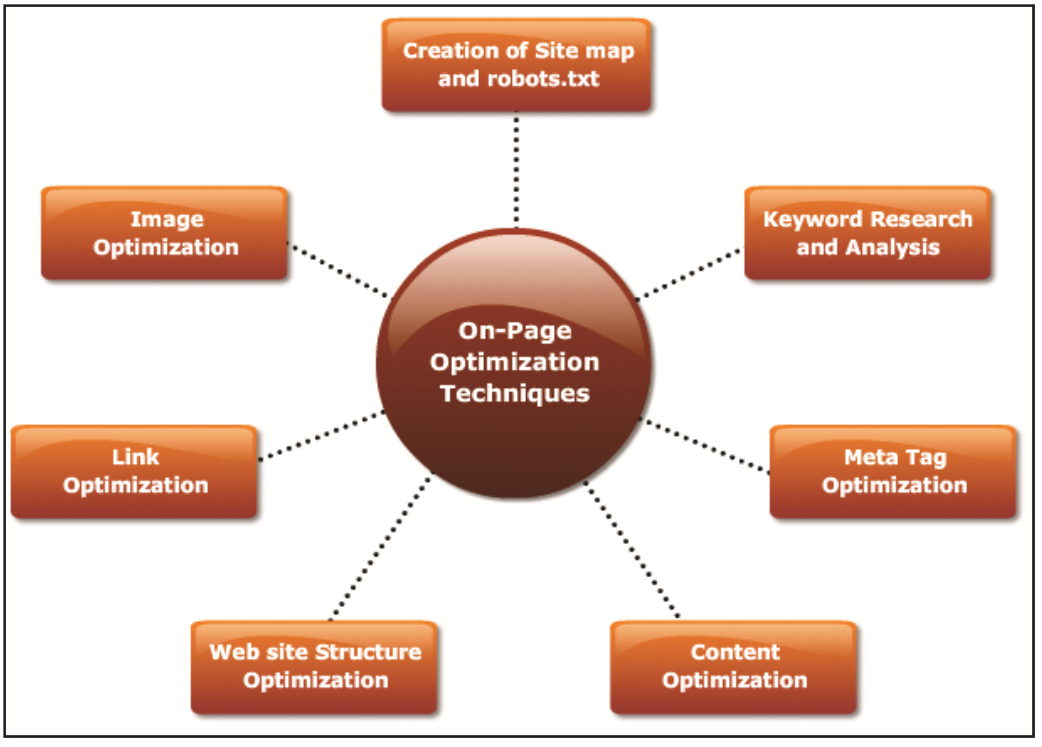


Figure 4.1: Different Techniques for On-Page Optimization

4.1.2 Characteristics of an Optimized Page

Following are the characteristics of an optimized page:

- **Unique content**

  An optimized page is one that provides unique content that cannot be found phrased similarly, on any other Web site on the Internet. It uses certain keywords phrases that help the right users to get to this Web site and no other Web site uses those keyword phrases. In the unique content layout, an optimized page provides useful and valuable content to the visitors.

- **Good User Experience**

  One of the goals Web page optimization is higher visitor counts. To ensure that visitors return to the Web site multiple times, it must provide a great user experience. Web pages should contain easy to understand content, easy to navigate and find required information, be attractive and pleasing to the eye, load quickly even on slow connections, be compatible across all types and versions of browsers currently in use by a majority of Internet users.

Web Crawler/Web bot Accessible

Search engine crawlers and Web bots crawl the pages of Web sites and index them to make them available in their search engine results. Optimizing Web pages involves building pages such that they are accessible for engines to crawl, index, and rank the Web pages accurately.

To ensure that Web pages are accessible to these crawlers and Web bots:

- Ensure that content on one page is not repeated on other pages in the Web site. Any pages with duplicate content must be redirected back to the original page by using the `rel=canonical` attribute in the `<link>` tag.
- Ensure that the Web site implements the best practices with respect to the balance between the use of static and dynamic pages, Really Simple Syndication (RSS) feeds, and XML site maps.
- When using **robots.txt** and meta robots to limit what engines see, make sure that it does not prevent the search engine crawlers from crawling and indexing the Web page content.
- If for some reason, a specific Web page is temporarily down, place a 503 status code instead of a 401 code, to let the search engine crawlers understand why the Web page is inaccessible. Similarly, if users need to be redirect from a specific page to a new location, place a 301 status code with a permanent redirect. Do not use multiple redirect chains or 30x status codes.

4.2 Keyword Analysis

Keywords are the cornerstone of SEO and play an important role in optimization. Keywords are words reflecting a potential visitor's thought while typing in the search box. Keywords can be a single word or phrases.

Phrases are the long tail keywords which combine two to five words that are expected to be typed by visitors to search specific information. A long phrase search reflects better targeting and increased search traffic to the Web site.

An SEO consultant must take care to determine and short-list keywords that are most relevant to the products or services offered by the Web site. Single word keywords are difficult to search as they are not well targeted for audience. Longer keywords are easier to rank well and have better adaptability among users.

Figure 4.2 shows some of the terms used in phrases by the visitors to perform search for specific information on the **www** containing the resources.

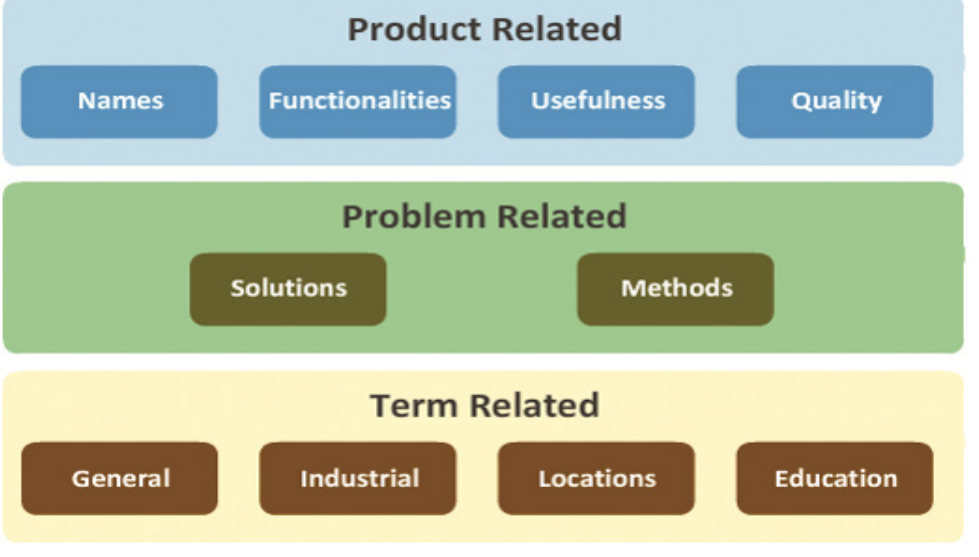


Figure 4.2: Terms Related to Long Phrases

Consider a scenario where a user searches for information related to services. The user types the word services in the search box on the Google search engine page, as depicted in figure 4.3.

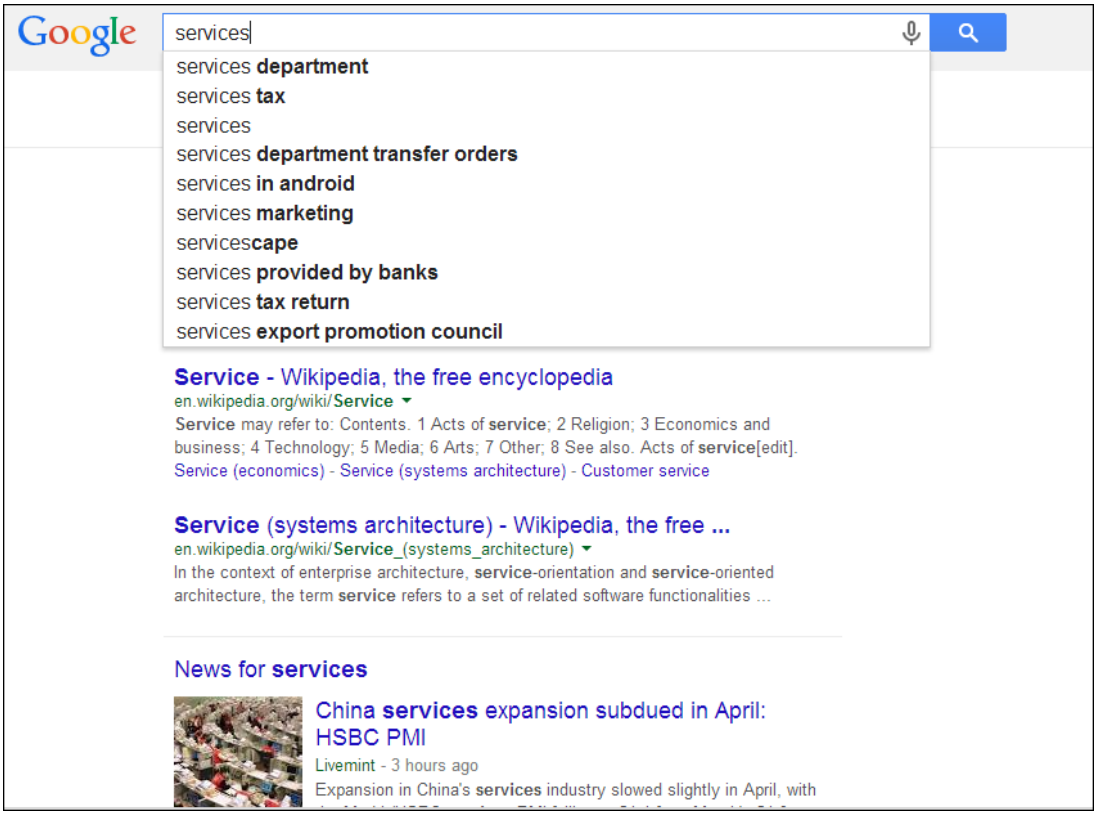


Figure 4.3: Search for Information Using Single Keyword

As shown in figure 4.3, while the user types in the search box, Google search engine's autocomplete algorithm offers keywords similar to the user's search words. It also examines the corresponding indexed pages of the Web sites and displays pages of different sites that have used the keyword services to describe their offerings.

To narrow the search to services in education field, the user types learning services in the Google search box as depicted in figure 4.4.

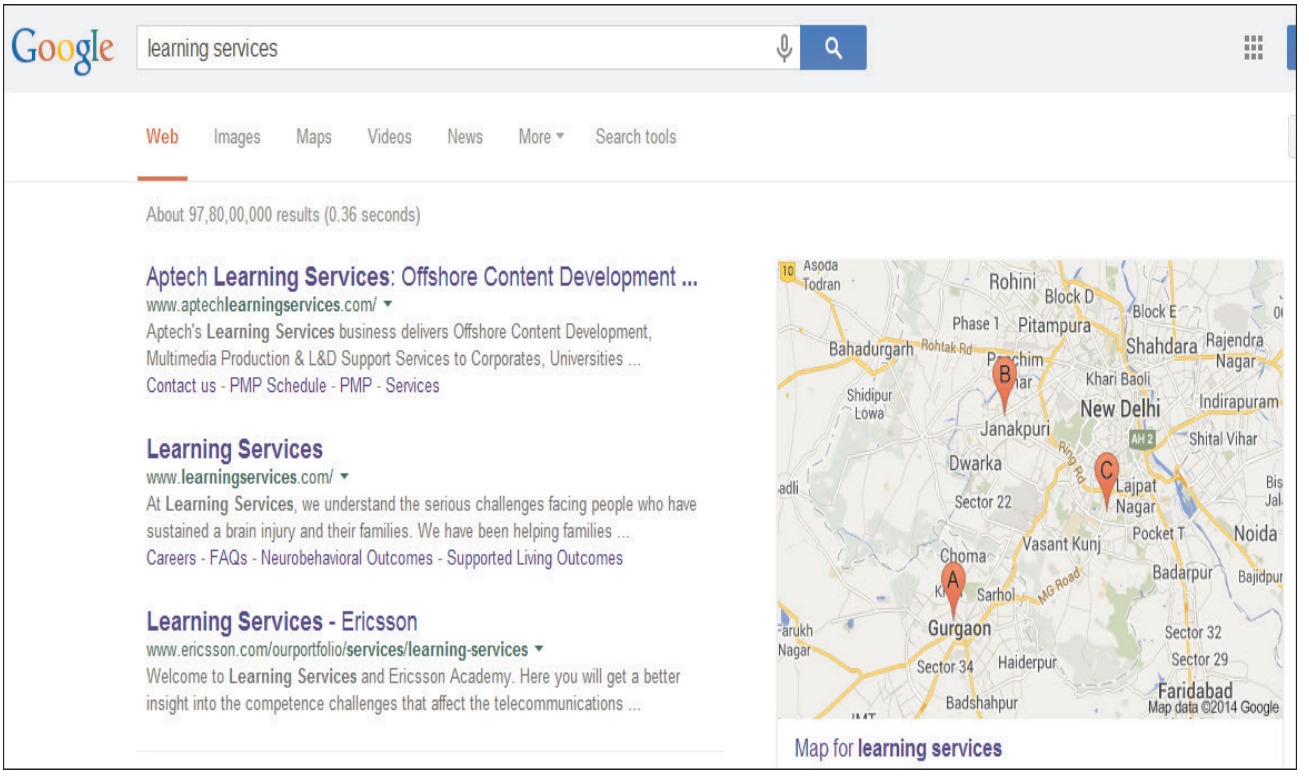


Figure 4.4: Narrowing the Search Using Phrases

As shown in figure 4.4, use of two or more words narrow the search to display the Web pages of highest ranking Web sites offering services related to learning.

4.2.1 Research on Keywords

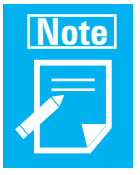
The core of any SEO strategy is the research on keywords. More targeted and related keywords used on the Web pages increase the possibilities of getting more customers or high sales on the Web site. An SEO consultant should research for keywords that clearly represent the services, products, or information provided by the Web site.

Some of the guidelines which can help SEO consultant in selecting effective and informative keywords are as follows:

- **Knowledge of your product or service** - The selection of the keywords should relate users with the characteristics of the products or services provided on the Web site. Thus, spend time to know your products or services to be marketed on the Web site.

- **Customers** - Customers are the people who can help in providing best keywords while searching for products or services on the Web.

- **Competition analysis** - Gaining knowledge about the strategies adopted by the competitors to rank high on the keywords.



Usage of keyword suggestion tools help SEO consultants to find relevant keywords or phrases for the Web site. The tools such as **KeywordDiscovery** and **WordTracker** help to find all permutations of search keywords including synonyms and wrong spellings used by visitors to find the similar sites.

4.2.2 Selection of Keywords

Web site can be targeted by search engines for different keyword phrases. Thus, an SEO consultant should choose keywords that can target search traffic to the Web site.

The guidelines for optimizing keywords for the Web pages are as follows:

- A Web page containing repeated keywords in keyword phrases led to overlapping. So, the page needs to be optimized frequently for overlapped words used in the keyword phrases.
- Try not to use stop keywords such as 'a', 'an', 'the', 'when', 'and', 'or', and 'but' on the Web page. These keywords are not considered by search engines while searching.
- Most of the search engines such as Google are not case-sensitive. Thus, search for keyword 'aptech' or 'Aptech ' is treated as similar words for them.
- Most search engines treat hyphen as space. Try to put hyphen between two words as this ranks the page better.

4.3 Placement of Keywords on the Web Pages

Once the selection of right keywords are completed, an SEO consultant can use them to gain the value in the search results. Placing keywords at the right place helps search engines to notice and interpret them properly. Some of the elements where keywords are placed on the Web page are namely, page titles, Meta tags, headings, text links, and images.

4.3.1 Page Titles

Page titles inform the users and search engines about the type of the information contained on the page. They optimize the capabilities of a search engine. They are created using the `<title>` tag that is ideally unique for each page on a Web site. The `<title>` tag is placed in the `<head>` section of the HTML Web page.

An SEO consultant can place the keywords in the beginning of the `<title>` tags which helps in improving the effectiveness of the tags. Page title contents are the first information about Web sites, displayed in search results.

The keywords in the <title> tag can include the following information:

- Name of the Web site or business name such as 'Aptech Learning Services' or 'Aptech Computer Education'
- Geographical location such as 'China', 'USA', or 'South Africa'
- Main offerings or focus areas such as 'offshore content development' or 'Pioneer in IT education'

Consider a scenario, where an SEO consultant wants to inform visitors about the Aptech Learning Services Web site. To achieve this, an SEO consultant researches for the keywords which should be informative as well as speak about the services offered by Aptech.

As a result of intensive search on Aptech services and products, an SEO consultant selected certain keywords to be used in the <title> tag as shown in Code Snippet 1.

Code Snippet 1:

```
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title>
Aptech Learning Services: Offshore Content Development Services
</title>
</head>
...
<body>
...
</body>
</html>
```

The significance of the <title> tag depicted here is that, it represents the business name of the Aptech Web site and its main offerings in the learning industry.

As a visitor, when a user enters the keywords 'aptech learning services' in the search box, it displays the result pages for the query as shown in figure 4.5.

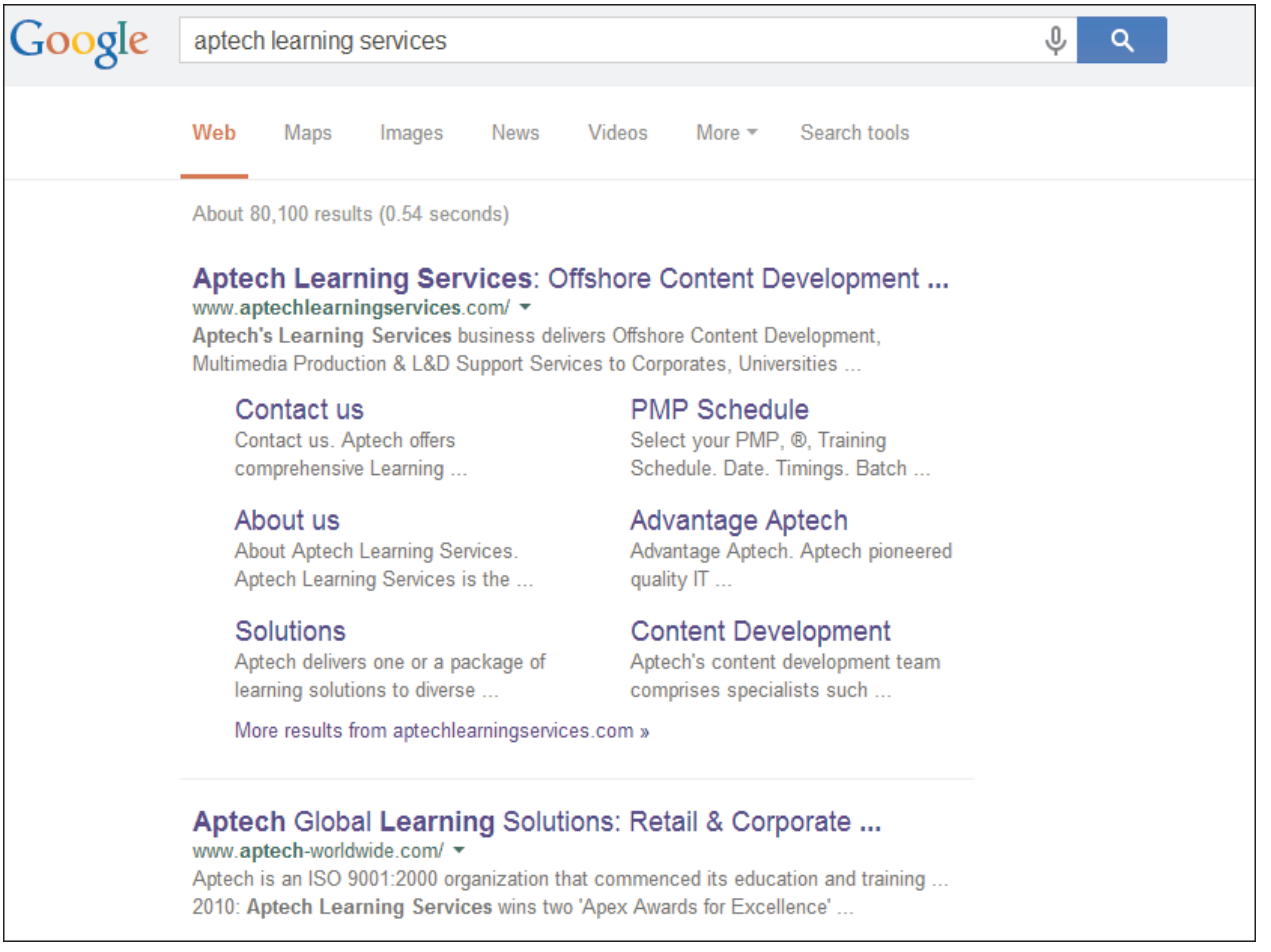


Figure 4.5: Content of <title> Tag in Search Results

Following points are visible in figure 4.5:

- Content of the title tag appear in the first line of the result displayed for the user's search query.
- Words of the title appear bold which helps users to decide whether the page is relevant to their search.
- On clicking the displayed result, title appears as page title of the Web page displayed in the browser.

The Aptech Learning Services home page with the page title is shown in figure 4.6.

Figure 4.6: Page Title Appearing in the Browser

The guidelines to be followed while defining page titles for the Web pages are as follows:

- **Accurate page content description** - The selected title should effectively communicate with the information contained in the page. Google crawlers uses title tags as the main source for determining the page topic.

- **Unique titles** - Every page on the Web site should have unique content in the title tag. This helps Google to distinguish between the pages on the Web site. Take care not to include titles that do not match the content on the page.

- **Descriptive titles** - Titles used for pages must be short and instructive. If the content in the title are too long, then Google search result will display only a portion of the title. Google allows up to 66 characters including spaces to be used in the `<title>` tag.

Note

Following are the points to be taken care of while defining the page title for the Web page:

- Avoid using keywords such as 'Untitled', 'New Page 1', or irrelevant keywords in the title.
- Avoid selection of a single title tag for all pages of the Web site or for collection of pages.
- Avoid specifying lengthy keywords in the title as they are of not much help to the users.

4.3.2 Meta Tags

Meta tags play a vital role in search engine optimization. An SEO can use Meta tags to provide information about their sites to the search engines. These Meta tags are written in the `<head>` section of the page.

Meta tags supported by the Google search engine are depicted in figure 4.7.

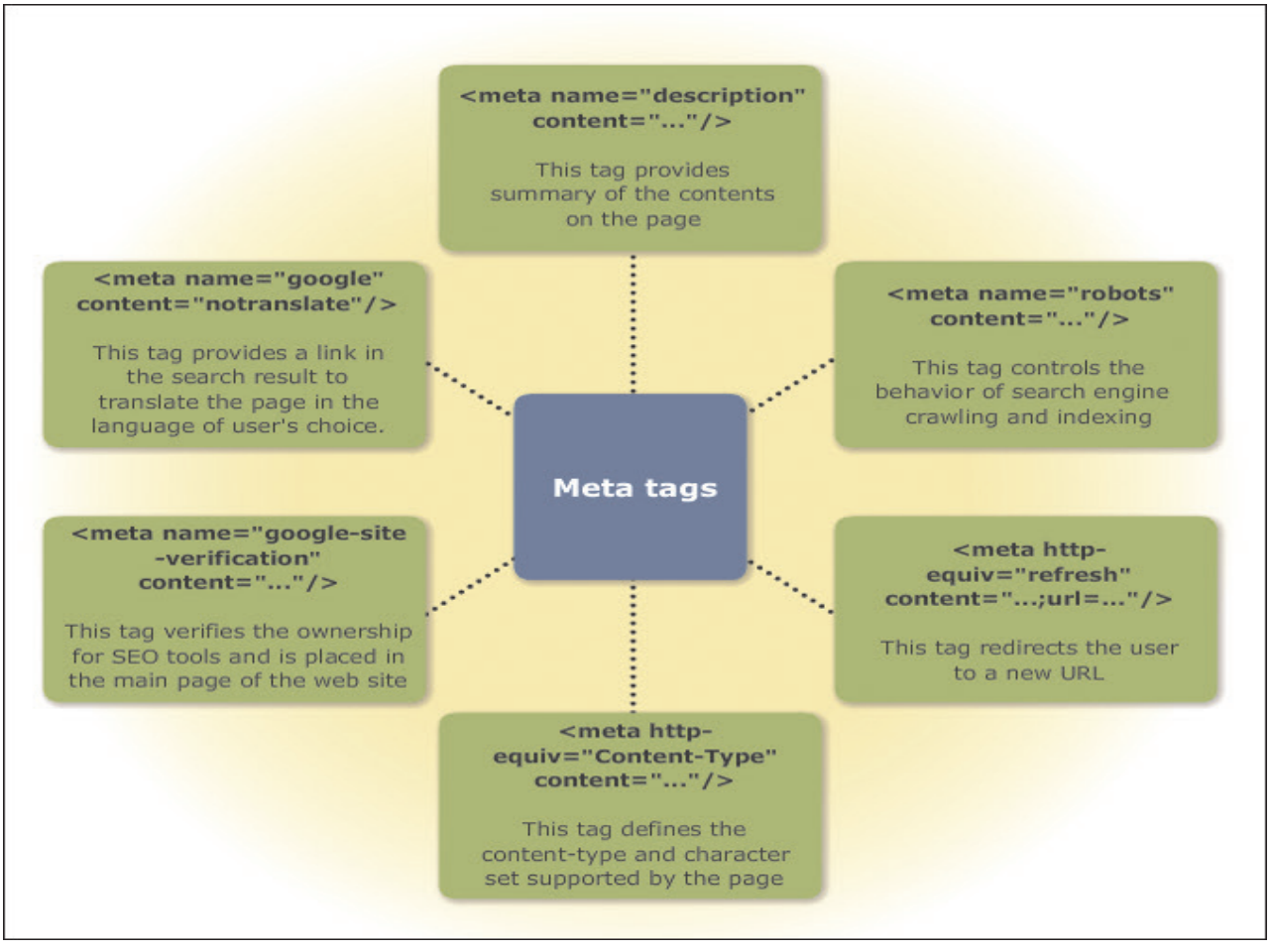


Figure 4.7: Meta Tags Supported by Google Search Engine

Some examples of Meta tags are as follows:

- `<meta http-equiv="Content-Type" content="text/html; charset=utf-8" />`
- `<meta name="description" content="Aptech's Learning Services business delivers Offshore Content Development, Multimedia Production & L&D Support Services to Corporates, Universities & Others"/>`
- `<meta name="robots" content="noindex,nofollow">`

4.3.3 Meta Description Tag

Meta descriptions are HTML attributes that provide summary of content on the page. They are commonly used by Google and other search engines to display a snippet of the page in the search results. When compared to page titles, descriptions are of few sentences or short paragraph.

Apart from Meta descriptions, Google may also use other approaches to describe pages in the search results.

These approaches are as follows:

- Might use text in relevant section of the page for matching up the user's query.
- May use a site's description from the Open Directory Project (ODP) in case the site is listed in it. ODP is a well-known directory on the Internet used to list the Web site.

Consider a scenario in which an SEO consultant wants to include the overview of Aptech offerings on the home page of the Web site.

To accomplish the task, an SEO consultant includes `<meta>` tag within the `<head>` section of the page as shown in Code Snippet 2.

Code Snippet 2:

```
...
<head>
<meta name="title" content="Aptech Learning Services: Offshore Content
Development Services" />
<meta name="description" content="Aptech's Learning Services business
delivers Offshore Content Development, Multimedia Production & L&D
Support Services to Corporates, Universities & Others" />
</head>...
```

The significance of the description Meta tag here is that, it outlines various offerings of the services provided by the organization.

As a visitor, when a user enters the keywords 'aptech learning services' in the Google search box, then the search result pages displayed for the query as shown in figure 4.8.

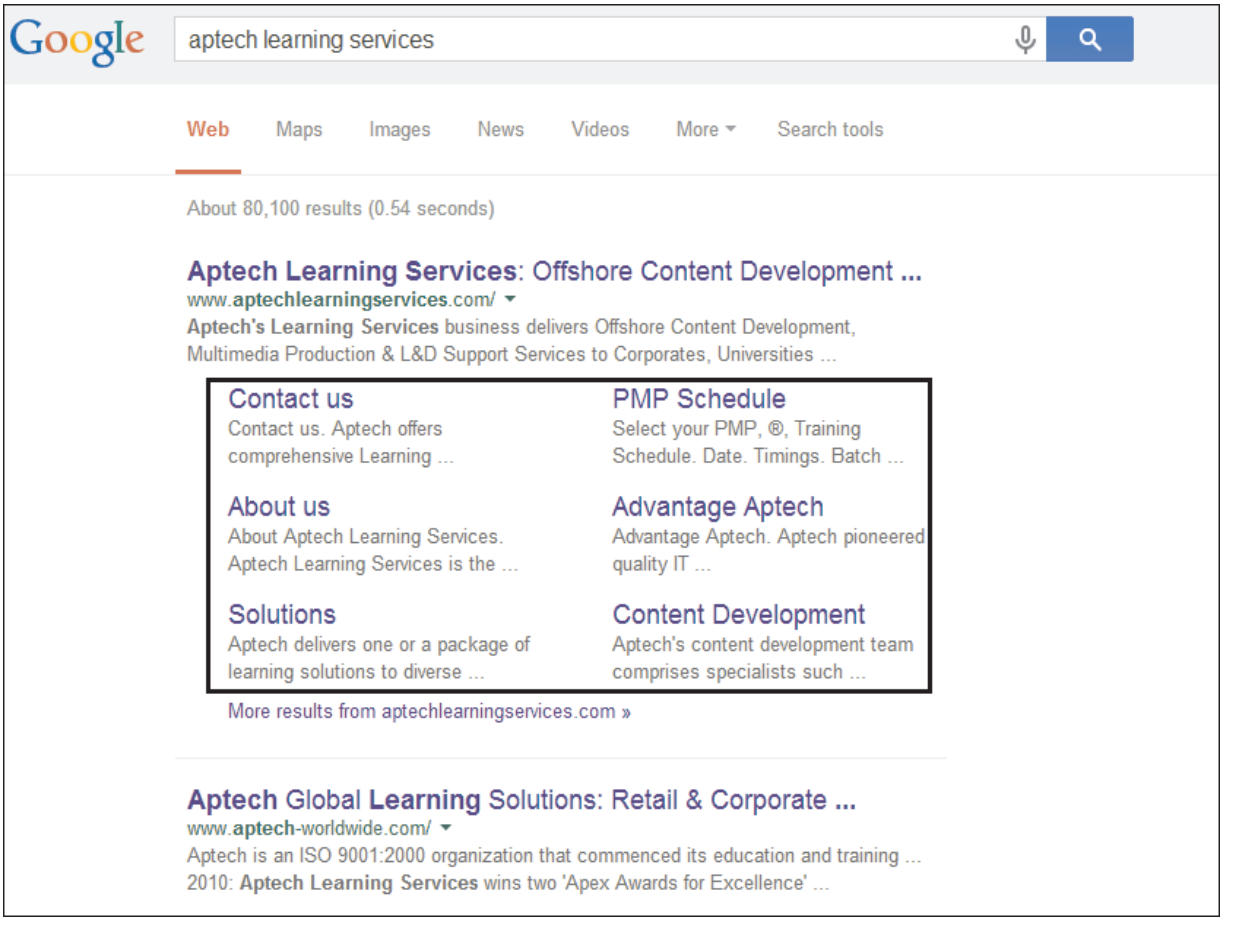


Figure 4.8: Snippets in the Search Results

The guidelines for using the Meta description on a Web page are as follows:

- **Summarize page content accurately** - Users should write informative descriptions about the page which is of great significance for the users.
- **Unique descriptions for pages** - Users should provide different description Meta tags for each page of the Web site. This helps Google to provide multiple pages in the search results available on the Web site. If the Web site contains thousands of pages, then description Meta tags can be automatically generated based on the content.

Note

Following are the points to be taken care of while defining the descriptions for the Web page:

- Avoid descriptions that are not relevant to the page content.
- Do not copy-paste entire contents of a Web page into the description Meta tag.
- Avoid generic phrases such as 'This is a Web page' or 'Page about services'.
- Avoid using only keywords or entire content of the page in the description.

4.3.4 Heading Tags and Cascading Style Sheets

The heading tags are embedded in the `<body>` section and are used to emphasize the important text on the page. They enable the users to read quickly about the page and search the information they want.

Heading tags allow you to place important keywords that are contextually appropriate with the rest of the page. There are six sizes of heading tags that can be used to present text structure on the page. These headings are depicted in figure 4.9.

Figure 4.9: Sizes of Heading Tag

As shown in figure 4.9, different heading tags can be used for the important text on the Web page as it help users to understand the type of content included under them.

Consider a scenario, where an SEO consultant wants to highlight the different types of services offered by the Aptech Learning Service on the service page. Thus, the use of heading tags in this context will make the text larger than its normal size that will attract the user's attention.

Code Snippet 3 shows the heading tags applied by an SEO consultant on the Web page content.

Code Snippet 3:

```
...
<body>
<h1>Services</h1><!--Page Topic-->
...
<h2>Consulting</h2><!--Main Topic-->
...
<h3>...</h3><!--Sub Topic-->
<h2>Managed Services</h2>
...
<h2>PMP Training</h2>
</body>...
```

As shown in the code, applying heading tags in proper structure makes easier for the users to navigate within the page.

Figure 4.10 shows the effect of the heading tags as viewed by users on the Web page.

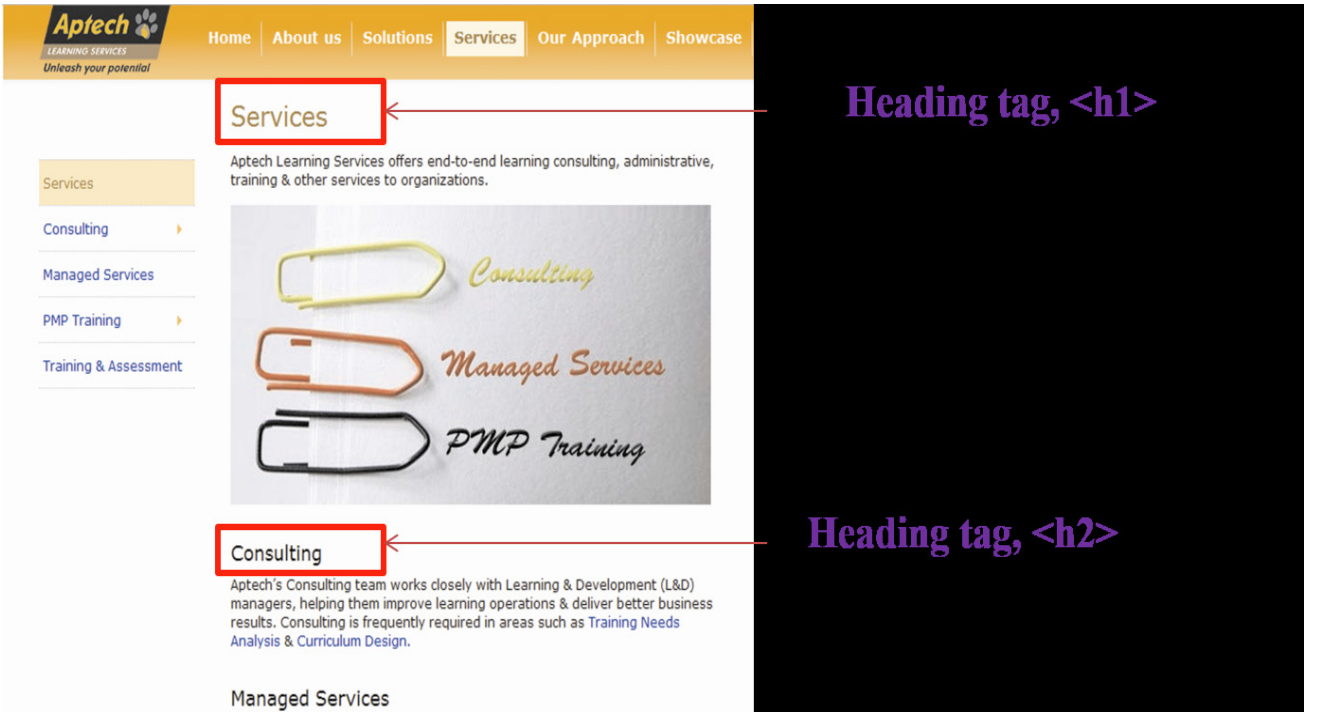
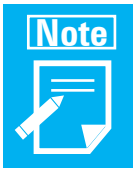


Figure 4.10: Heading Tags

The guidelines for providing the headings in the content of the Web page are as follows:

- **Outline the points** - Before applying the heading to the text, decide on the content of main point and sub-points on the page and accordingly apply the heading tags.
- **Limit the use of headings across the page** - Excessive use of heading tags on a page make it difficult for the users to find the required information on the page.



Following are the guidelines to be avoided for heading tags:

- Avoid unrelated text in heading tags as it does not define the structure of the page correctly.
- Avoid replacement of `<em>` and `<strong>` tags with heading tags.
- Avoid excessive usage of heading tags on the page.
- Avoid placing all the text of the page in the heading tags.
- Avoid using heading tags as presentation of text, instead use them for presenting the structure of text on the page.
- Avoid random use of heading tag sizes.

To increase the readability of the tags by the search engines, they need to be formatted properly. The design elements which can help in formatting are Cascading Style Sheets (CSS). While other tags help in increase the ranking of the page, CSS is strictly restricted to the formatting of the tags.

The use of CSS ensures that Web site pages are consistent. Thus, they can be applied on most of the elements to ensure their consistency on the pages.

CSS files can be created separately and then linked on the Web pages as shown in Code Snippet 4.

Code Snippet 4:

```
...
<head>
...
<link rel="stylesheet" href="style.css" />
</head>...
```

Consider a scenario, where an SEO consultant wants to achieve consistency in terms of styling on the Aptech Learning Services Web site.

Thus, the CSS file with font and styling information can be created as shown in Code Snippet 5.

Code Snippet 5:

```
...
body {
font-size:12px;
padding:10px;
margin:0 auto; color:#000000;
font-family:Verdana, Arial, Helvetica, sans-serif
}
a:link, a:visited, a:hover{ color:#0066CC }...
```

4.3.5 Working with Anchor Text

One of the most important uses of keywords in a Web page is the anchor text. These are text that appears as links (hyperlinks) on the page and which can be clicked further to move to the result of link. In other words, it is a clickable text on the page that is also referred as linked text.

The anchor text is placed within the anchor tag, `<a href=". . .">anchor text</a>` on the page.

Keywords used in the anchor text should match with the keywords on the page that is linked by it.

The guidelines to select keywords for the anchor text are as follows:

- **Short and descriptive keyword** - An SEO consultant should select short and self-explanatory keywords used in the anchor text. This provides a basic idea to users about the type of information contained in the linked page.

- **Formatted links** - An SEO consultant must distinguish anchor text from the normal content on the page by formatting them. Formatting can be done using different text-styles or CSS.

- **Pay attention to internal links** - An SEO consultant should give equal importance to anchor text used for both internal links and external links. This helps users and search engines in better navigation of the site.

Figure 4.11 shows Aptech Learning Services Web site containing anchor text that help users and Google search engine in understanding about the page linked by them.

Figure 4.11: Anchor Text in the Web Page

Better and descriptive anchor text help users to navigate site. They also helps search engine crawlers to understand about the information contained on the linked pages.

Links pointed by the anchor text can be of the following types:

- **Internal links** - Refers to those links that are pointing to other pages on the same Web site.
- **External links** - Refers to those links that are pointing to content on some other Web site.

Note

Following are the points to be avoided while selecting anchor text on the Web page:

- Avoid specifying generic keywords such as 'page', 'click here', or 'article' to be used as anchor text.
- Avoid selecting keywords that have no relevance with the content pointed by the anchor text link.
- Avoid using URLs as anchor text.
- Avoid writing paragraphs, lengthy sentences, or excessive keywords as anchor text.
- Avoid using CSS or text styling that make links look just like regular text.
- Avoid creating unnecessary links that do not help with the user's navigation of the site.

4.3.6 Working with Images

Images are the fundamental components included while designing the Web pages. An SEO consultant should take efforts to optimize them based on their usage on the page. The element used to display image on the page is `<img>` tag. The `<img>` tag contains the attributes such as file name, display size of the image, and the alternative text that are depicted in figure 4.12.

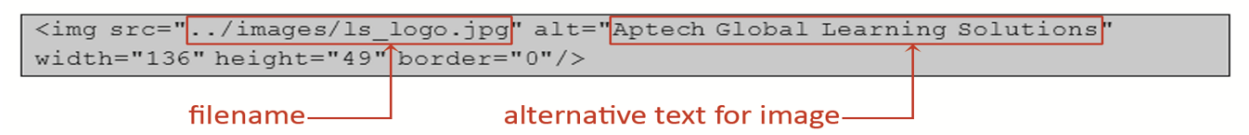


Figure 4.12: <img> Tag

The text in the `alt` attribute is used as a replacement text for the image when it is neither supported nor displayed by the browser. This attribute tells the user about the type of information that might have been depicted by the image on the Web page.

The guidelines for using the images on a Web page are as follows:

- **Path of the image** - An SEO consultant should consolidate all the images in a single directory, instead of storing them in multiple directories. For example, an SEO consultant creates `/images` directory to store all images of the Web site.
- **Descriptive names** - Use short and explanatory text for file name and `alt` attribute of the image.
- **Alt text for images** - An SEO consultant should mention text for `alt` attribute of an image used as link. The `alt` attribute of an image is similar to the anchor text provided for the text on the page.

Note

Following are the points to be avoided for images provided on the Web page:

- Avoid specifying generic file names such as **picture.gif**, **a.jpg**, or **1.jpg** for images
- Avoid selecting lengthy file names and long text description in `alt` attributes for images
- Avoid using images as links, as Google cannot see the graphic-based links
- Avoid stuffing keywords or copy-pasting entire sentences into `alt` text

It is useful to include an image site map file on the Web server. Googlebot can use these image site maps to gather more information about the images on a Web site and in turn about the content of the Web site. The structure of image site maps is similar to an XML Site map file for Web pages.

Alt stands for alternative and is used with images and graphs provided on the Web page. The `alt` attribute serves as a replacement for tables, pictures, or other elements of a Web page that are not displayed properly on the visitor browser.

These attributes are specifically designed for visitors and not for the search engines.

The purpose of using `alt` attribute is to provide the same information which might have been depicted by an image, a link, or an element to the user.

Consider a scenario, where an SEO consultant wants to specify keywords for an image as shown in figure 4.13. This image will be used to support the text on various deliverables provided for content development on the Aptech Learning Services Web site.

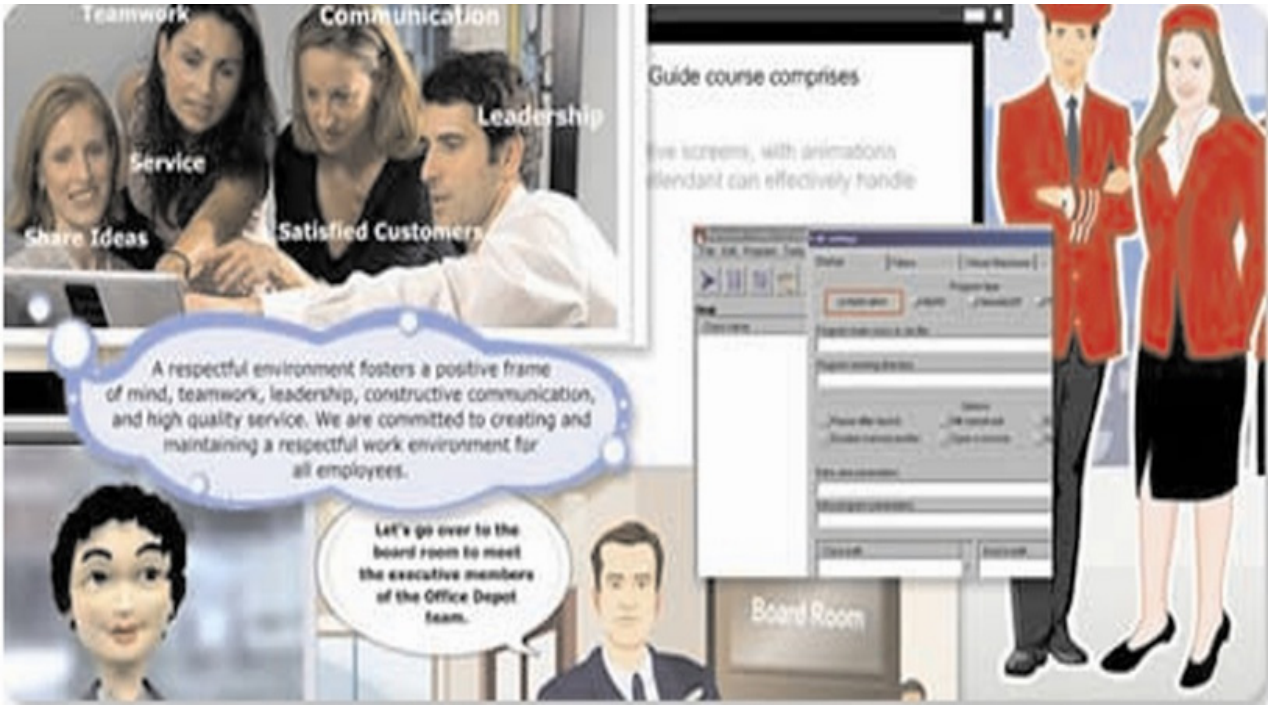


Figure 4.13: alt Text for the Image

The `alt` attribute used here for describing the image can be `alt="Content Development"`, `alt="Customize Content Development"`, or `alt="Custom Content Development"`.

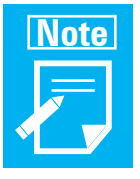
The `img` tag used for displaying the content development graphics on the page with its corresponding `alt` attribute is shown in Code Snippet 6.

Code Snippet 6:

```
<body>
<img src="../images/ContentDevelopment.jpg" alt="Custom Content
Development" width="472" height="218" /><br />
...
</body>...
```

The guidelines for defining the `alt` attribute for the images or other elements on the Web page are as follows:

- Length of keywords used in the `alt` attribute must be between 50 and 125 characters.
- Keywords used in the `alt` attribute must convey the information to the visitors and their connectivity with the rest of the content on the page. To increase the effectiveness, specify `alt` attribute for every image provided on the Web site.



Alt tags or Alt attributes are important when visitors accessing the Web site use only text browsers, or have turned-off the graphical capabilities of their browsers, or are reading Web pages using screen readers device.

4.3.7 Metadata and Snippets for Additional SEO Markups

Following is a list of a few Metadata attributes and snippets that Webmasters can include in Web pages for optimization:

- `Rel=author` and `rel=publisher` attributes can be included as Metadata in the link tags to display the brand image of the Web site along with its details in Google search results.
- Bing provides a linked pages feature that allows users to have pages linked to their public profiles or brand profiles. These linked pages show up alongside search results for the profiles of a person or a brand.
- Use schemas to mark up the HTML content of Web sites to help search engines understand the content of Web pages better and provide greater visibility of the Web pages in search results. Google provides a rich snippets testing tool that can be used to test the markup schemas implemented in a Web site and identify errors, if any.

4.4 Optimization of the Web Site Structure

SEO is not just about placing the right keywords in the elements provided on the Web pages.

It is also about the way a Web site is built. The first thing which attracts a search engine crawler to your Web site is the structure of the Web site. In other words, it is the actual design of the Web site.

An SEO consultant can enhance the structure of the Web site by providing easy to understand URLs, proper file names, better quality content, and proper navigation structure for the Web pages.

4.4.1 Structure of URLs

URLs are the address of the Web pages and other resources on the WWW. They are displayed as part of the search result below the document title in bold.

An SEO consultant should keep the URL structure of the site simple, user-friendly, and understandable.

The user-friendly URLs helps to convey the information easily contained on the page. Simple URLs are helpful to users as well as to the search engines. The long and cryptic URLs with few keywords may confuse the visitors while searching. For example, an URL such as, **directory25/189765/num/1099872b.html** can be confusing and can give a hard time while delivering the URL from memory or creating a link.

Consider a scenario, where a user wants to know about the career opportunities at Aptech. In this context, he or she searches for the keywords 'career with aptech' in Google search engine page, as shown in figure 4.14.

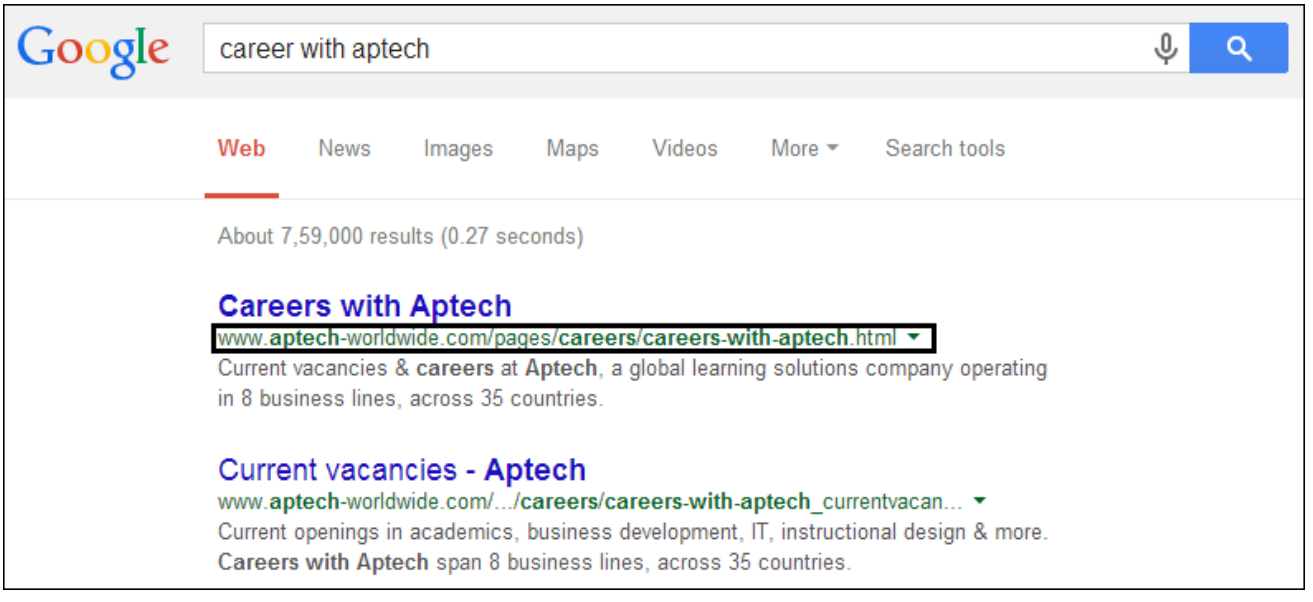


Figure 4.14: URL of the Web Page

The information on content development is displayed with a URL, **www.aptech-worldwide.com/pages/careers/careers-with-aptech.html** in the search result. The URL is in human-readable words that help the users in deciding whether the information on the page is relevant to their search.

Google search engine supports the usage of hyphens (-) in URLs than underscores (_). For example, a URL represented as **www.aptech-worldwide.com/pages/careers/careers-with-aptech**. HTML is more understandable by users than represented as, **www.aptech-worldwide.com/pages/careers/careerswithaptech.html**.

The guidelines to improve the site structure using URLs for the Web pages are as follows:

- **Usage of words in URLs** - Words used in forming the URLs must be related to the content and structure of the document. This helps in providing a friendlier approach while navigating the site and help visitors to remember them. URLs must also be named in lowercase for better recollection.
- **Simple directory structure** - Users can organize the content in a directory structure. This helps the visitors to know exactly where they are while navigating in the site. The directory structure should be mapped with the type of content displayed at that URL.
- **Single URL for a document** - Users should focus on providing and referring to a single URL in the structure. If the visitors are accessing the same content through different URLs, then redirecting from non-preferred URLs to the main URL can be done.

Following are some of the points to be avoided for URLs:

- Avoid selection of long URLs with parameters, session IDs, generic names such as **page1.html**, or keywords.
- Avoid nesting of subdirectories at deeper level or selecting directory names that are not related to the content.
- Avoid pages from different directories accessing the same content.

4.4.2 Creating Quality Content

Content is the most important aspect of Web site design from the user and Webmaster perspective.

An SEO consultant should spend lot of time in studying and creating quality content, as it compels the visitors to come back to the site. Frequent visits to the pages increases the site ranking and search engine crawlers register them as a valid and in-demand site.

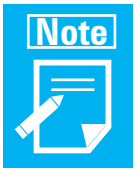
An SEO consultant must target the audience and accordingly create content strategy plan which can impart right type of content to the users. If users find quality content, then they direct other users to the site through blogs, social media services, e-mail messages, forums, and so forth.

Google provides tools such as AdWords that helps Webmasters identify keywords for their Web sites. Google AdWords can be used to input keyword variations and check the search volume for each keyword. Similarly, Webmasters can use Google Webmaster Tools to identify the top search queries for which a Web site shows up in search results. Webmasters can accordingly modify or retain the right combination of keywords for the Web site.

The guidelines for an SEO consultant when developing content for Web pages are as follows:

- **Use of keywords** - Use of right keywords and phrases is necessary in content. Selection of keywords should not be more than three on any Web page of the Web site. Too many keywords can reduce the effectiveness of the content and search engines can exclude the pages in the indexes.
- **Write easy and understandable text** - An SEO consultant should write content that is readable and is easily understood by the users.
- **Organize the topics** - An SEO consultant should divide the content logically, as this helps the users to understand the start and end of the topic in the content.
- **Unique content** - An SEO should always update and write high quality unique content on the Web site. This helps to sustain existing users as well as attracts the new users.
- **Prime focus of content** - An SEO consultant should design the site content according to the user's

need. The proper usage of keywords increases the accessibility of the Web sites in the search engine results.



Following are some of the points to be avoided while designing content for the Web site:

- Avoid writing large amount of unrelated text on the page.
- Avoid spelling and grammar mistakes.
- Avoid embedding of text in images as search engines may not read it.
- Avoid writing text without paragraphs, sub headings, and layouts.
- Avoid duplication of content across the site.
- Avoid text that is hidden from users but visible to search engines.

Figure 4.15 shows some of the recommended pages to be included on the Web site.

Figure 4.15: Web Pages on the Web Site

4.4.3 Use of Programming Languages

One of the aspects of Web site designing is the use of programming languages. These languages are not similar to HTML and require different SEO strategy to deal with them.

Some of the languages used in the Web page designing are as follows:

- **JavaScript**

  JavaScript is used to create dynamic content on the page but is not well supported by search engines. Search engine such as Google has set a time limit to crawl in the Web site. If the Web site contains large amount of code compared to visible content, then it may not crawl to all the pages in one pass. This may result in lower search engine ranking of a Web site.

  Following are guidelines to minimize such problems:

  - Code used in the page should be lean and clean that helps to increase the download speed of the page. It also improves the user's experience on visiting the site.

- JavaScript code should run as a file from some external location. To achieve this, code should be separated in a .js file and a link to it must appear in the <head> section of the page as shown in Code Snippet 7.

Code Snippet 7:

```
...
<head>
...
<link rel="stylesheet" href="style.css" />
</head>...
```

- **Frames**

  Frames provide navigational capabilities to the Web site, but search engines normally cannot crawl through such sites. Secondly, users cannot bookmark any pages of the framed site or link to them. Thus, the use of frames should be avoided.

- **Flash**

  Flash is another technology occupying more resources. A page containing Flash might slower down the loading of the page and even do not allow a user to move ahead until its execution is completed. It might also block crawler from tracking the site. This results in not indexing the site and crawler moves to the next site displayed in the searched result. Best practice is to either avoid it or retain it to some location on the page or provide a link to it.

4.4.4 Use of Keywords in File Names

Search engines such as Google give consideration to keywords used in the file names of the Web pages.

The words in the file name must be separated by hyphen (-), otherwise, Google may treat it as a single word. For example, the URL of Aptech Web site informing about career opportunities is shown in figure 4.16.

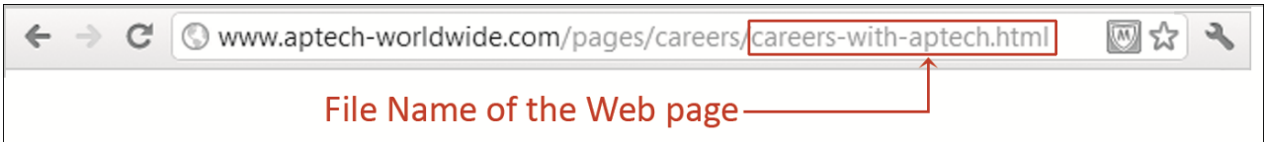


Figure 4.16: File Name of the Web Page

4.4.5 Knowledge on Navigation

Navigation means moving from one page to another page on the Web site using links displayed on the Web pages. Search engines use links to understand what content are important according to an SEO consultant. The proper navigation structure of a Web site also helps visitors to move in the Web site and find the relevant content as per their need.

Navigation is of two types that are as follows:

- **Internal navigation** - It involve links which make the user move from one page to the other page on the same Web site.
- **External navigation** - It involve links that move the user from page to some another Web site.

The guidelines to be considered by an SEO consultant while planning navigation for the Web site are as follows:

- **Navigation based on home page**

  The start page of all Web sites is the 'home' or 'root' page. This page is the most frequently visited page by the visitors and is considered as the starting point for planning the navigation of the Web site.

  An SEO consultant should take into consideration that how a user will move from a root page to a page containing content on specific topic.

  For example, if a Web site contains pages on specific topic, then the path of navigation will be: **home page → page listing related topics → page containing specific topic**.

Figure 4.17 shows a directory structure of the Aptech Learning Services Web site that helps the SEO consultant to decide the navigation path from the home page.

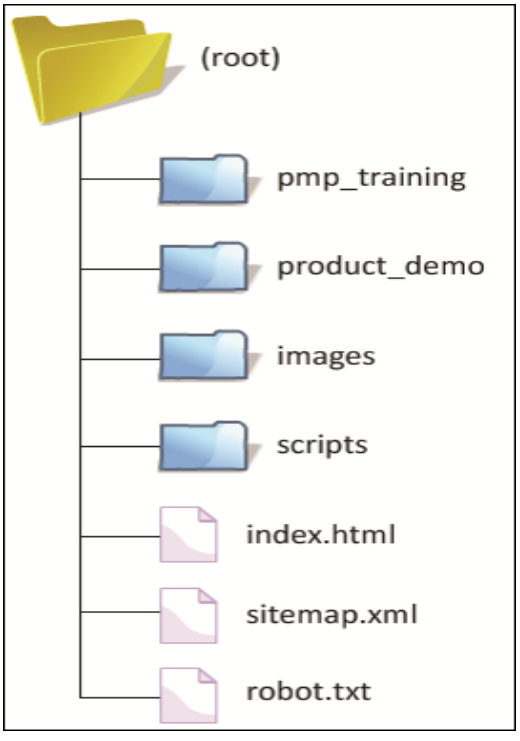


Figure 4.17: Directory Structure of Aptech Learning Services

- **'breadcrumb' link navigation**

For effective navigation of the pages, an SEO consultant can create a row of internal links at the top or bottom of the page. These internal links are called as 'breadcrumb' and they help user in moving from a specific page to previous page or home page. For example, a breadcrumb list created by an SEO consultant for Aptech Web site is depicted in figure 4.18.

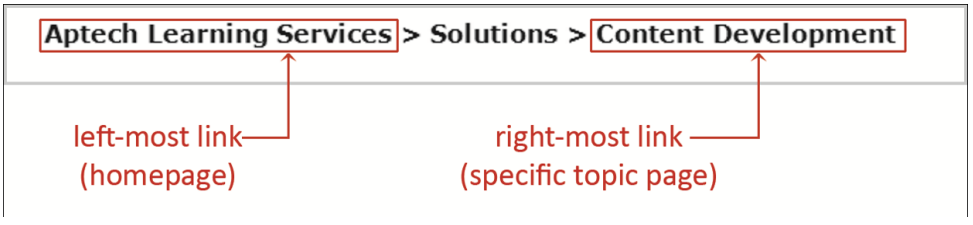


Figure 4.18: Breadcrumb List

- **Text-based navigation**

The navigation in the site should be based on text links as it helps the search engine crawler to understand the site structure.

Preparing HTML site map

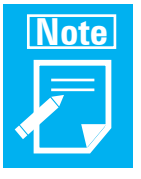
A site map is a simple page of a Web site. It displays the structure of the Web site. It consists of hierarchical listing of the pages in the site.

Following are the benefits of using site map for different stack holders:

- **Visitors** - If visitors have problems finding pages on the site, then they can visit the site map page.
- **Search engines** - Search engines can crawl into this page and understand about the pages provided on the site.

Creating a page handling 404 error

Sometimes, a user may reach on a page that is either not existing or whose link is broken and which takes them to an error page with the HTTP status code 404. To avoid this, an SEO consultant can customize a 404 page which can guide user back to the home page. It can also provide some popular links or related topics which increase the accessibility of the site. The design of the error page should be consistent with the rest of the pages on the site.



Following are some of the points to be avoided while planning the navigation for the Web site:

- Avoid linking every page on the Web site to the other page, as it makes the navigation complex.
- Avoid planning navigation on images, animations, and drop-down menus.
- Avoid out dated pages in HTML site map.
- Avoid text that is hidden from users but visible to search engines.
- Avoid messages such as 'error page' or '404' on the error page.

4.4.6 Understanding Site Map

A site map is an overview of the navigational structure of the Web site. It is a text-based page containing links of all pages in the Web site. The data in the site map helps the search engine crawlers to crawl all the linked pages from it. All the pages once crawled are included in the search engine index and finally appear in the search results for a user's search query.

Figure 4.19 shows a site map created by the Google Web site.

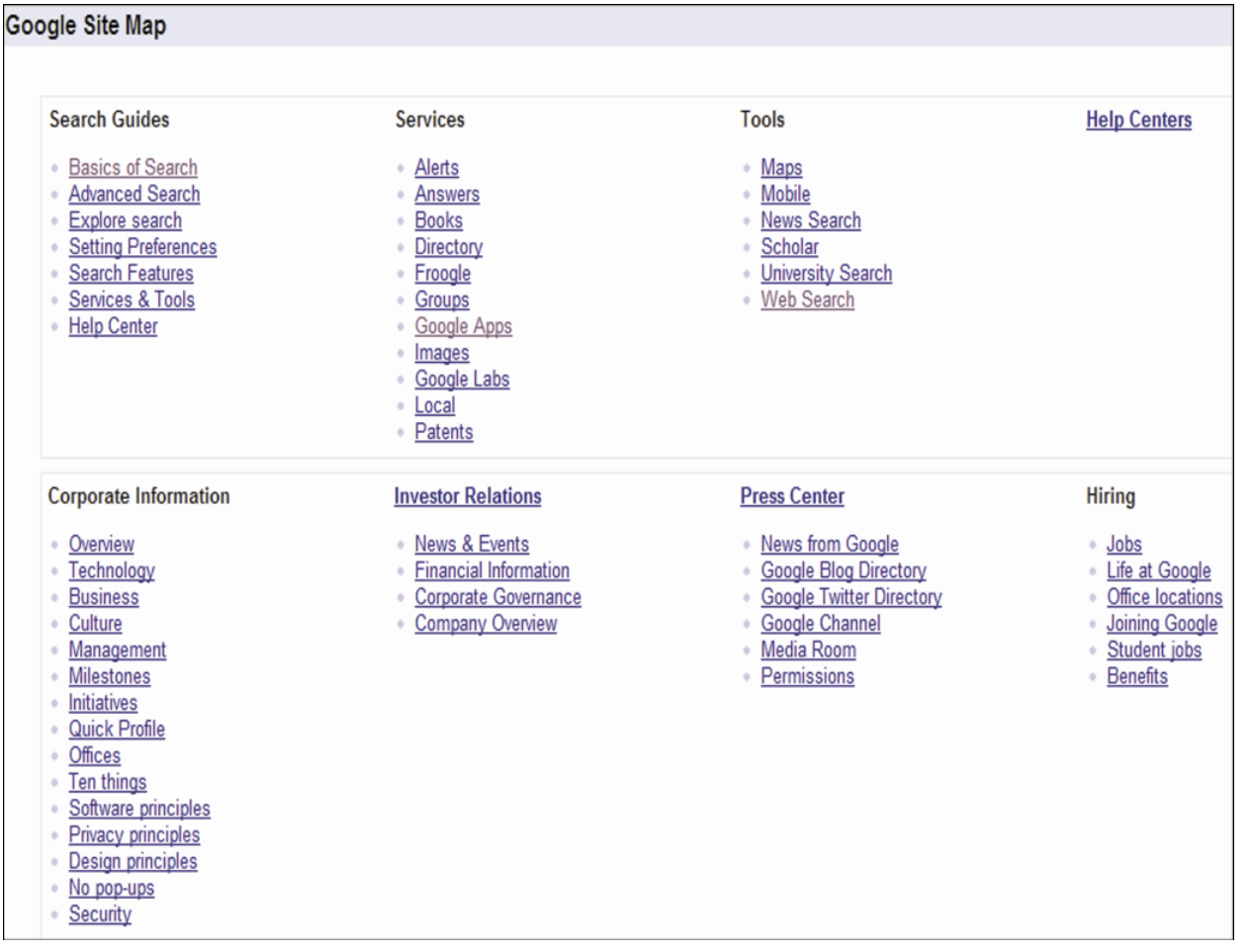


Figure 4.19: Google Site Map Page

4.5 Optimization of Essential Files

The only part of the search engine that is visible to the users is a query interface and the search result pages. The other part of the search engine working behind the scenes are crawlers, spiders, and robots that determine how the Web site can achieve a better ranking in the search results.

Crawlers, spiders, and robots are programs that crawl from one page to other page following the links in the Web site. The work of these programs is to collect information such as text and link from the Web site, index them, and save in the database. This further helps in searching and ranking the Web site for user's search queries.

Figure 4.20 shows the path taken by a crawler on the Web.

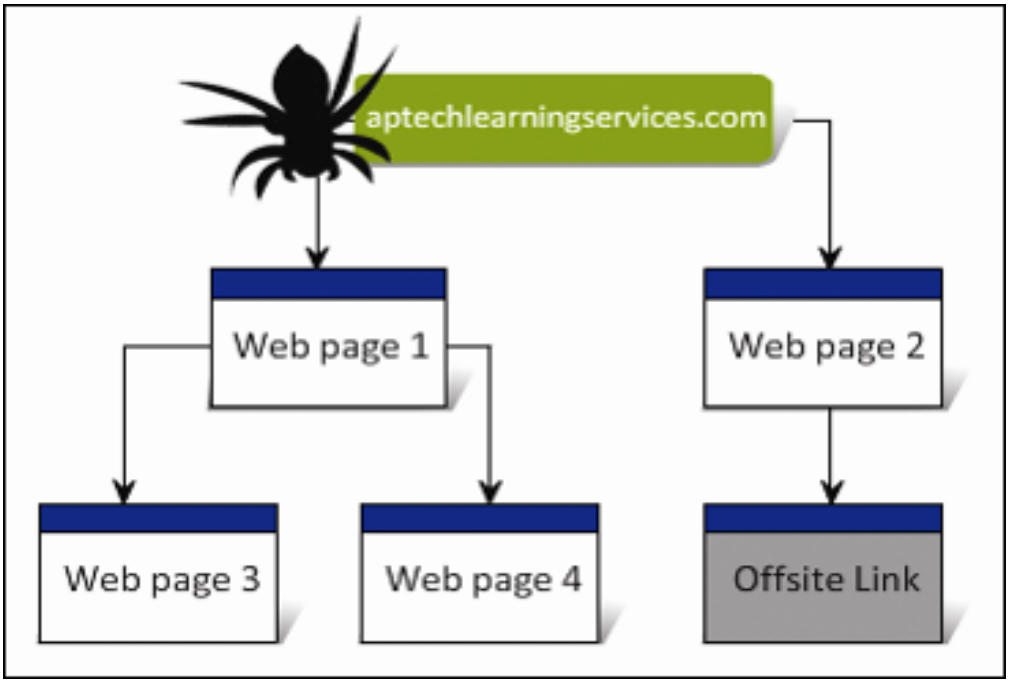


Figure 4.20: Web Crawler Path

There are possibilities that crawlers can cause destruction on a Web site. Thus, to control their behavior, certain guidelines are to be followed. These guidelines are also called as Robot Exclusion standard or **robots.txt**.

4.5.1 The robots.txt File

The **robots.txt** file is a text document that is placed in the root directory of the Web site. They contain instructions that inform the search engines about what they are allowed and not allowed to access while crawling in the Web site. These files are created by using a friendly **robots.txt** generator which is available in Google Webmaster Tools.

A typical **robots.txt** file can contain the following code:

```
User-agent: *
Disallow: /images
Disallow: /download
```

The essential parts of the file are as follows:

- **User-agent indicator** - Tells which user agent or crawler is commanding. The * indicates that all crawlers can command.

- **Disallows indicator** - Tells what a crawler is not allowed to access. The / indicates that crawlers cannot access any content stored under /**images** directories or any URL that begins with /**download**.

Crawlers read **robots.txt** file from top to bottom. If any guidelines are applicable on them, they leave the search result page and move to next one on the site.

Consider a scenario where an SEO consultant does not want a crawler to access images, JavaScript, or access any URL whose path begins with /**search** for Aptech Learning Services Web site. In this context, an SEO either generates or creates a **robots.txt** file using a text editor such as Notepad.

This file will be stored in the root directory of the site and the name of the file will be in **robots.txt** (lower case letters). It provides information as shown in Code Snippet 8.

Code Snippet 8:

```
User-agent: *
Disallow: /cache/
Disallow: /download/
Disallow: /FCKeditor/
Disallow: /giaitri/
Disallow: /images/
Disallow: /includes/
Disallow: /javascript/
Disallow: /languages/
Disallow: /modules/
Disallow: /register/
Disallow: /sitemapgen/
Disallow: /themes/
Disallow: /upload/
Disallow: /vuoncao_hna/
Disallow: /news/article/
Disallow: /news/chuongtrinh/
Disallow: /news/index.php
Disallow: /search/
Disallow: /news/danh-sach-cac-cong-ty.hna
```

The **robots.txt** file that guides the crawlers for the Aptech Learning Services Web site will be accessed on the URL, **http://www.aptechlearningservices.com/robots.txt**.

4.5.2 Inclusion with XML Site Mapping

A site map, as discussed earlier in the session, outlines the navigational structure of the Web site. There exists another site map which is different in functionality from site map and is referred as Extensible Markup Language (XML) Site map. The XML Site map is an XML file that lists the URLs for a Web site. It is used by crawlers to index the Web site. The XML Site map file can be placed in **robots.txt** or are directly submitted to the search engines.

Code Snippet 9 shows the **sitemap.xml** file created for Aptech Learning Services Web site.

Code Snippet 9:

```
<urlset xmlns="http://www.google.com/schemas/sitemap/0.84">
<url>
<loc>http://www.aptechlearningservices.com/</loc>
<lastmod>2008-12-12</lastmod>
<changefreq>weekly</changefreq>
<priority>0.5</priority>
</url>
<url>
<loc>
http://www.aptechlearningservices.com/pages/services_
pmptraining.html
</loc>
<lastmod>2008-10-23</lastmod>
      <changefreq>hourly</changefreq>
      <priority>0.5</priority>
</url>
<url>
<loc>
http://www.aptechlearningservices.com/pages/showcase.html
</loc>
<lastmod>2008-10-23</lastmod>
   <changefreq>hourly</changefreq>
  <priority>0.5</priority>
```

```
</url>
...
```

As shown in the code, the <urlset> references the protocol standard and the <url> tag contains the address of the Web page. Each <url> tag is the parent tag containing child tags such as <lastmod> which represents the last modification date of the file. <changefreq> indicates when the search engine will crawl in the page.

Webmasters therefore, are recommended to prepare two site maps, one for users and one for search engines. While the former is for users if they have trouble finding the required pages or content on a Web site, the latter is for search engines for easier crawling and gathering information about the pages on the Web site. Search engine crawlers also use the regular site map for users to gather information.

Following are the ways to submit the Site map to Google:

- **Submitting using robots.txt** - Add a directive in the **robots.txt** file with the complete path that will be auto discovered by the crawlers. For example, Site map: **http://example.com/ sitemap _ location.xml**
- **Submitting using search engines** - In this, an account needs to be created in Google which is linked Webmaster Tools account. Using this interface, submit the Site map.

4.6 Common Mistakes in On-Page Optimization

Some of the deceptive techniques which should be avoided while optimizing the pages or structuring the Web sites are as follows:

- **Duplication of content**

  Duplicate content are identical content on a page similar to those in a page either on the same Web site or to some other Web site. This content can be completely similar or presented with some minor changes.

  The different kinds of duplicate content are described in table 4.1.

Function	Description
Reprints	These are content published on multiple sites with copyright holder's permission. This results in lower ranking of the Web site in the search results.
Site mirroring	This is maintaining multiple copies of the Web site and publishing them at different places on the Internet. It is done to avoid downtime when one site is crashed. This kind of duplication which can lead to delisting from the search engines.
Content scraping	This is reusing the content of one site at another Web site with little modifications such as formatting the text. This kind of duplication can lead to legal action for violation of copyright.
Same-site duplication	Duplicating content from your site in pages is also considered as duplication.

Table 4.1: Different Kinds of Duplicate Content

Google search engine keeps on indexing the pages. So, if many versions of the same content are found and neither of them is blocked, then only one version is listed in the index. This can also result in lower page ranking.

To avoid this, use `noindex` tag on the header of the page that is not required to be indexed. The syntax of `noindex` tag that instructs the Google engine not to index the page is written as: `<meta name="robots" content="noindex,follow">`

- **Using 301 redirect**

When a Web site is moved to a new domain then a transition is required in the URL. In this case, a 301 server-side redirect is required. The 301 is a status code which means a page is redirected permanently to a new location. This will help the search engine to redirect the search traffic to the correct page on the Web site.

- **Consistent of internal linking**

If URL addresses are given in the linked text on the Web pages, then they should be consistent at all places. For example, avoid specify URL as, **http://www.aptechlearningservices.com at** one place and **http://www.aptechlearningservices.com/index.htm** at other.

- **Keyword stuffing**

Keyword stuffing is a practice to load the pages with maximum keywords. This is mainly done to achieve maximum ranking in search results. This contradiction can lead to poor search traffic to the Web site and can also affect ranking in the search results. An SEO consultant should use informative and descriptive keywords that are relevant to the content on the page.

- **Preferred domain**

A Web site can be accessed through multiple URLs. For example, Aptech Learning Services home page can be accessed using **http://www.aptechlearningservices.com**,

http://aptechlearnigservices.com, or http://www.aptechlearningservices.com/index.html. Using www and non-www Web sites can be considered as duplicate content by search engines and may not index those sites. To avoid this, select the preferred URL also called as canonical destination to index the Web site pages.

For example, whenever a user types **http://www.aptechlearningservices.com** or **http://aptechlearningservices.com**, the search engine always redirects him/her to **http://www.aptechlearningservices.com**.

- **Doorway pages**

These are pages with poor-quality design, where each page is optimized with a keyword or phrase. This is done to achieve ranking for particular keyword or phrases which can take the search traffic to the same Web site while searching.

The results of doing so can make users to move out of the site on finding pages of no relevance to them.

Google search engine recognizes doorway pages and directs the users to the sites that are relevant to their search, instead of moving to sites which they selected. Sometimes, it removes such Web sites from Google index.

Cloaking

Cloaking is a practice of showing different pages based on the request. For example, if a visitor clicks the search result and moves to a page, then he/she may view only a single page. However, if search engine crawls in the Web site for indexing, then it may see many optimized and relevant pages. Search engines such as Google dislikes this practice. If caught, it may penalize on the Web site or may even completely remove it from listing.

Check Your Progress

1. Which of the following plays an important role in optimizing the Web pages?

(A)	Page styling	(C)	Adding maximum pages
(B)	Keyword research	(D)	Dynamic content

2. Which of the following tag is used by Google search engine to display snippets in the results?

(A)	Anchor tag	(C)	Meta Description tag
(B)	Title tag	(D)	Header tag

3. Which of the following links are used to point other pages on the same Web site?

(A)	Internal Links	(C)	Back Links
(B)	External Links	(D)	Image Links

4. Which of the following points should be avoided for URLs?

(A)	URL must be related to the content and structure of the document	(C)	Naming URLs in lower case
(B)	Accessing different content from the same URL	(D)	Selecting long URLs with parameters, session IDs, and generic name

5. Which of the following document contains the instruction that inform the search engine crawlers about what not to be accessed on the Web site?

(A)	sitemap.xml	(C)	index.html
(B)	robots.txt	(D)	Styles.css

6. Which of the following tag is used to exclude the page from indexing?

(A)	`<meta name="robots" content="noindex, follow">`	(C)	`<meta name="keywords" content="noindex, follow">`
(B)	`<meta name="description" content="noindex, follow">`	(D)	`<meta name="refresh" content="noindex, follow">`

Check Your Progress

7. Which of the following option shows a proper breadcrumb list?

(A)	page listing related topics > specific topic > home page	(C)	home page > page containing specific topic
(B)	home page > page listing related topics > page containing specific topic	(D)	page listing related topics > home page > specific topic

Answer

1.	B
2.	C
3.	A
4.	B
5.	D
6.	A
7.	B

Summary

- SEO is basically the science of designing Web site.
- An SEO consultant adopts different optimization strategies such as on-page and off-page to achieve maximum ranking in the search engine results.
- Keywords are the cornerstone of SEO and play an important role in optimization.
- Some of the elements where keywords are placed on the Web page are Page titles, Meta tags, Headings, Text Links, and Images.
- The site structure can be enhanced by providing easy to understand URLs, proper file names, and better quality content on the page.
- Proper navigation structure help visitors to move on the Web site and find relevant content as per their need.
- Crawlers, spiders, and robots are programs that crawl from one page to other page following the links displayed on the Web site.

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05 Off-Page Optimization

Welcome to the Session, **Off-Page Optimization**.

This session explains off-page optimization, various services that can be used to implement it, and guidelines to avoid mistakes.

In this Session, you will learn to:

- Describe off-page optimization techniques
- Explain different off-page optimization services
- List the guidelines for avoiding common off-page SEO mistakes

Session
05

Off-Page Optimization

5.1 Introduction

SEO is performed to generate traffic to Web sites. On-page and off-page optimization are a part of the SEO process used to increase the Web site traffic. On-page and off-page optimization always work in tandem with each other.

On-page optimization denotes factors that have an effect on a Web site or Web page listing in search results. Since these factors are off-site, they are controlled neither by the user nor by the codes present on the page. On-page optimization factors include HTML code, Meta tags, keyword placement, keyword density, and so on. On the other hand, off-page optimizations are not controlled by the user. To get a good ranking, both on-page and off-page optimization should be configured correctly.

5.2 Off-Page Optimization

In SEO, off-page optimization plays an important role in the ranking of the Web site or Web page. Off-page optimization factors are those that are not controlled by the user or the codes present on the Web page. These factors might affect the Web site or Web page ranking in the search results page. Off-page optimization technique improves the site's performance by increasing the exposure, traffic, and revenue of the Web site.

Off-page optimization is associated with the site's internal linking structure and it also relates to the factors that influence a Web site's ranking on the SERP. The main point to be considered in off-page optimization is to increase the Web site ranking by allowing the user to exchange links with competitor's Web site. In other words, the Web sites should obtain links from those Web sites that are ranked higher than the user's Web sites. The point to be considered while doing this is that the user's Web sites should be relevant to the business performed by the high-ranking Web sites. Web sites will rank better when there are more number of quality links from popular Web sites, newspapers, blogs, and so on. Search engines can recognize the difference between relevant and irrelevant links, so the inbound links should be from the Web sites with similar keywords.

The factors that are to be considered for establishing quality links are as follows:

- Check if the linked site or page is relevant
- Check if the linked site or page is temporary or permanent
- Check if the linked site or page is linking to any other low quality topic or content sites
- Check if the linked site or page ranks well in the search engines
- Check if the linked site has multiple links from other Web sites

Some of the off-page optimization techniques that can be adopted are as follows:

- Link Popularity
- Link Building
- Directory Submission
- Document Sharing
- Article Submission
- Social Bookmarking
- Classified Submission

- Forum Posting
- Press Release Submission
- Blogging
- Video Submission
- Deep Linking
- Infographics

Figure 5.1 displays the different off-page optimization techniques.

Figure 5.1: Off-Page Optimization Techniques

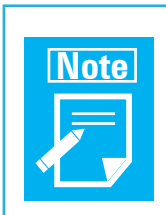
5.2.1 Link Popularity

Link popularity is an important factor used in determining the ranking of Web sites by the search engines. Link popularity evaluates the popularity of the Web site on the Internet and is determined by the number of links that other sites place in a user's/organization's Web site. These links are also called in-bound links; it allows other users to visit the Web site. Higher the number of inbound-links, higher will be the ranking of the Web site by the search engine. Thus, it can be said that link popularity depends on the quality and quantity of inbound links. The quality of a Web site depends on the Google Page Rank (PR) value that varies from 0 to 10 with 10 being the highest PR value.

The link popularity details for each Web site is displayed alongside the search results.

There are four components to it:

- **Link quantity**: This indicates the number of links pointing to a Web page.
- **Link quality**: Link quality is determined by analyzing the number of backlinks that a Web site has pointing to itself from respected or high quality Web sites. Google uses a factor called Page Rank to indicate the link quality of a Web site. The higher the page rank the better the link quality.
- **Anchor text**: This is the hyperlink text that is displayed in the results. Clicking the Anchor text will take the users to a page on the Web site. The Anchor text comprises of hyperlinked keyword phrases and help users decide whether the Web site might contain information they are looking for.
- **Relevance**: Relevance if a Web site is determined based on the topics that it covers or the geographical are to which it caters. When the topics match the search query or match the geographical region from which the user is placing the query, the relevance of the Web site is higher.



Page Rank is Google's system of counting link votes and determining which pages are most important. These scores are then used along with other factors to determine if a page will rank well in a search.

Link popularity also checks the online visibility of the Web site. There are many tools that can check the link popularity of the Web site. It is necessary to keep a record of the link popularity index so that the user has knowledge of the Web site ranking on search engines. However, to develop link popularity or obtain links from other Web sites is a difficult and time-consuming process. Before approaching the Web administrator/ owner for obtaining links from other sites, the user should ensure that the Web site is functional and allows links from other pages. The user should first place a link to the other Web site and then send a mail briefly explaining about the Web site and requesting them to link back to the user's Web site.

The factors to be considered while developing quality link are as follows:

- Page Rank of the site that provides the inbound link
- Content of the linking site should be relevant with the content of the Web site
- Words used in the anchor text of the links should be relevant to the content of the lined page
- Number of other links present on the linking page

5.2.2 Link Building

Link building is the process of obtaining links from other Web sites. It is an SEO technique that helps in getting quality traffic to the Web site and thus, enhances the popularity of the Web site. One must ensure that sites ranking high in search engine results should be used for building links. For instance, a link from a popular Web site as The New York Times will help you get a good page ranking as compared to a link from a newly established unknown Web site. In link building, the user receives inbound links for the Web site that increases the link popularity and this puts the Web site on the top of the search result. It helps to increase the ranking in major search engine result pages for any target keywords. The more the number of links from high quality sites, the better the page ranking of your Web site.

The advantages of link building are as follows:

- Link building helps in receiving quality traffic from relevant sites, which will increase the page rank of the site in the search engine.
- It creates awareness, credibility, and visibility of the Web site.
- Link building is an effective way for the site to get indexed by different search engines.

5.2.3 Link Building Strategies

Following are some link building strategies that will improve the page rankings of the Web site:

- **Content creation and promotion**: Create content that is attractive and interesting enough to compel visitors to link to it and share with people.
- **Submissions**: Submit events and happenings on your Web site to press releases or site directories.
- **Reviews and Mentions**: Network with influential bloggers or Web site owners to review, reference, or mention your products and services. Have friends and partners also refer and discuss the Web site products and services.

5.2.4 Guidelines to Appropriate Linking Methods

Link building if not done correctly might be detrimental to the Web site page rankings and the placement of the Web site in search results. Following are a few guidelines to appropriate methods of building links:

- Select sites that are most relevant to your Web site content for building links.
- Build creative content and include useful information that will compel visitors to read the site content, revisit it, and share it with others.
- Do not build link farms. A link farm is a Web page containing a list of unrelated links to other sites simply to increase incoming links to a site. Search engines discourage sites with link farm pages and consider them as spam. Search engines penalize site containing link farms and hence do no good to the site's page ranking.
- Do not aggressively purchase links. Some Web sites offer to sponsor or promote links to your Web site for a price. Be careful when investing in such purchases as they can have negative impacts for a Web site. If the Web sites sponsor or promote links incorrectly or place them in their link farm, they are likely to be penalized or banned. Also, too many links either to or from a site are also considered by search engines as akin to link farms.

5.2.5 One-Way Inbound Link

There are two types of links namely, Inbound and Outbound links. Outbound links start from a user's/organization's site and lead to an external site, while inbound links or backlinks come from an external site to a user's/organization's site. For example, if ABC.com links to XYZ.com domain, the link from ABC.com is an inbound link for XYZ.com. On the other hand, the link from ABC.com to XYZ.com is an outbound link for ABC.com.

One-way link helps in building links. It links a Web site to another Web site of interest and does not hyperlink back to that Web site. Inbound links are also defined as links found on another Web site on the Internet that direct users to a user's Web site only. One-way Inbound link is the most important factor for link popularity development. One-way inbound links are the hardest to acquire, as the Web site needs to create a reputation in market regarding its usage. Secondly, there is no hyperlink back so the Web site placing the link of the other site and thus will not have any benefit from the other site. Inbound links are preferred types of links by search engines and have more weightage.

For example, A, B, C, and D, are four Web sites, if A is the Web site that needs one-way inbound links then B, C, and D will send one-way inbound links to A.

Figure 5.2 displays Web site B, C, and D sending one-way inbound links to A.

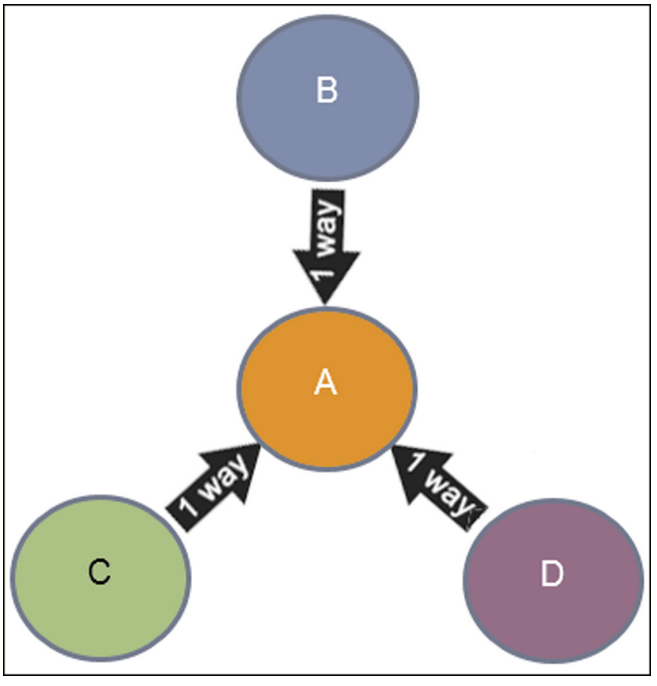


Figure 5.2: One-way Inbound Links

80% of the searches are performed on the Internet using Google search engine. These searches use inbound links, which is a factor in determination of Google PR.

Inbound links are also necessary if the Web site is looking to reach a large number of Internet users. In addition, one-way inbound links helps in increasing the traffic for the Web site.

Links can be either text-based or graphic-based links. However, text-based links are more effective in increasing the traffic and PR. In text-based links, keywords should be used as hyperlinks for inbound links as it will have an effect on the rank and position of search engines.

5.2.6 Directory Submission

Directory submission is an important SEO technique. It is the process of adding a Web site to a Web directory. Directory submission involves submitting Web site URL and other details of the Web site to a proper category. Using an appropriate anchor text (inbound linking) in the directory submission will increase the search engine ranking. User can submit to paid or free directories. The charges for directory submission for a Web site URL for paid Web directories depend on the number of directories.

Having many links to an organization's Web site is like having signboards on the road. The greater the number of links, the greater is the chance of a human or a search bot to notice the Web site.

Directories are typically available as paid services. Depending on the price paid for subscribing to the services, various directories provide some or all of the features listed as follows:

- Directories may charge either a one-time fee or a recurring fee for using their services. Some directories do not levy any charge for reviewing and listing of the site.
- Some directories make it compulsory for the submitted Web site to add a link back to the directory Web site. Others may make adding link back to the directory Web site optional.
- Some directories require the Web sites to bid for being placed on it. The order in which the sites are placed for reviewing and listing is based on the bid amount.
- Some directories include the feature of affiliate links where the directory charges a commission for each customer that visits the Web site from its listed link on the directory site.
- Some directories, depending on the amount paid for the service, place the Web site link in premium locations such as the home page or in a highlighted box. These are called sponsored listings.
- The links listed on directories typically have a `rel="nofollow"` attribute. This means any outbound links from the links will not be counted by search engines towards the increase in hits or page rank of the linked site.

The steps for directory submission are as follows:

1. Preparation of title and description of Web site
2. Open a separate e-mail account for submissions
3. Make a directory list for submission
4. Choose a proper category
5. Submit detail in directory submission form
6. Confirmation of mail acceptance
7. Preparation of the report of complete work

Figure 5.3 displays the directory submission process.

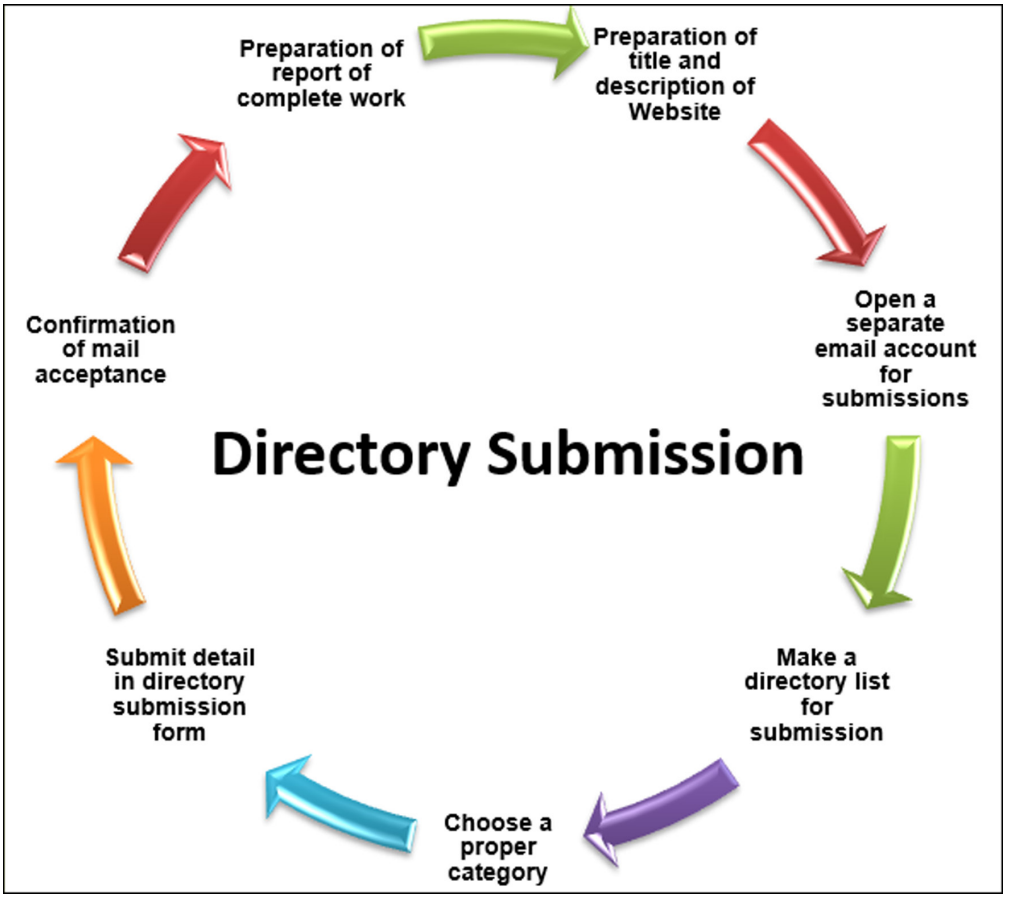


Figure 5.3: Directory Submission Process

Some of the benefits of directory submissions are as follows:

- Increases link popularity
- Improves search engine indexing
- Increases exposure
- Builds a brand

5.2.7 Document Sharing

Document sharing is a new SEO technique in which users can upload content as various types of documents such as Word documents, PowerPoint slides, PDF files, Open Office documents, and so on. There are a number of document sharing sites that provide this service.

Documents submitted to the document sharing sites get indexed with search engines as quickly as 24 hours. Also, there is no approval period before the documents are available for sharing. Thus, making the documents available in search results very fast.

Scribd, DocStoc, and SlideShare are some examples of document sharing Web sites.

How to Create a Document

Following are the guidelines and steps for creating an SEO document for document sharing:

1. Prepare a list of keywords that are related to the Web site content.
2. Write an article containing at least 400 words, using the keywords judiciously. The article may be a Word, PDF, Open Office, or any such document format.
3. Remember to include links to related information throughout the article.
4. Include descriptive images to the article that add value to the content. Hyperlink the images wherever required and make sure to use proper text wrapping.
5. Use a legible font with appropriate formatting to make it easy and attractive to read.
6. At the end of the document, include a resource box listing the links related to the article.
7. Save and upload the document to a document sharing site.

Following are the benefits of implementing the document sharing technique:

- Web sites and articles get indexed in Google very quickly.
- Hyperlinked images can be included in the shared documents, which can help attract users and higher page hits.
- A single document can be uploaded to multiple document sharing sites to reach to a wider range of Internet users and thus attract more page hits.
- Add affiliate links to related articles and Web sites to attract more users.

Figure 5.4 shows an example of document sharing.

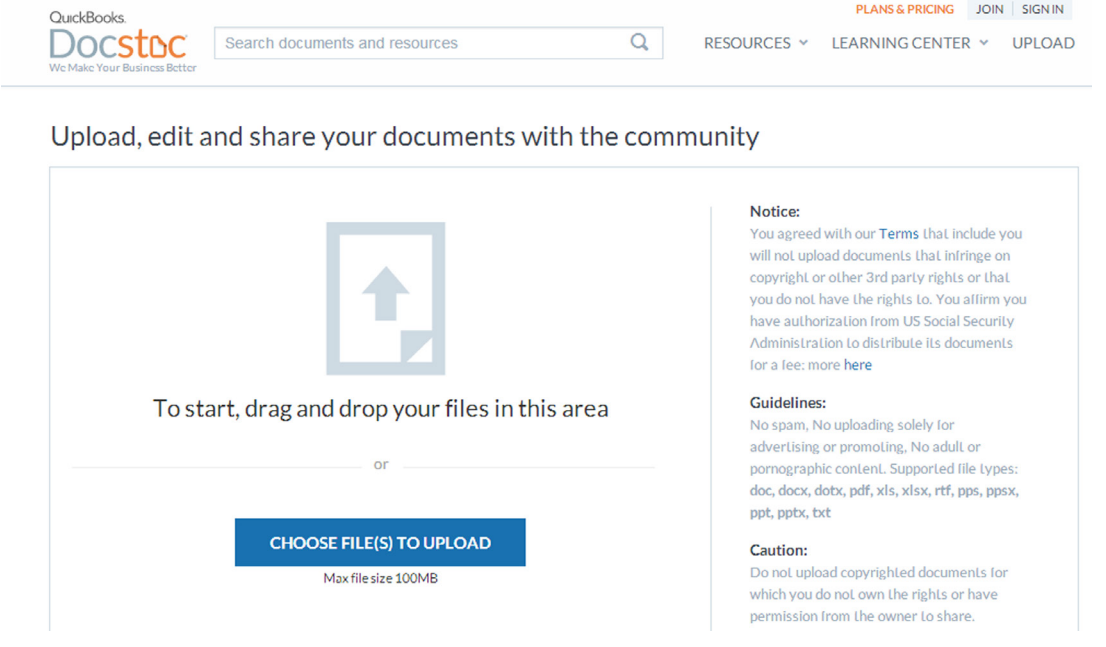


Figure 5.4: Document Sharing on Docstoc

Following are the document sharing tips for attracting more traffic to Web sites:

- Include the document file name as a keyword in the document.
- Keep trying uploading the documents in different formats on different document sharing sites. On some document sharing sites, .doc files might be ranked better than .pdf files or the other way round.
- Make sure to use a catching, descriptive but short title for the document that describes the content of the Web site accurately. This will help attract the right crowd in larger numbers and multiple times.
- It is recommended to use a larger font of at least 16 point for writing the documents as the documents display in a much smaller font when viewed by users online.

Some of the top document sharing sites are Docstoc, Zoho Viewer, TwitDoc, Calameo, and Google Docs.

5.2.8 Article Submission

Article submission is a popular and oldest standard in Web site SEO strategies. Article submission is a method of submitting articles to article directories or article submission pages by Web site administrators. The article directory in turn provides inbound links to author's Web site. The articles need to be relevant, not only to the Web site content but also to the article site where it is being submitted. These article submission pages can be blogs, forums, and so on.

Some of the benefits of article submission service are as follows:

- Ensures quality one-way links through submissions in high PR article directories
- Ensures quality traffic to the Web site
- Reaches maximum number of readers through interesting content

5.2.9 Social Bookmarking

Social bookmarking is a new way of marketing on Internet. Internet users use this method to organize, store, manage, and search for bookmarks of resources online. In other words, the user bookmarks the Web site with those Web sites that receives high traffic and thus, enables the user's Web sites to be exposed to thousands of users. A user generally bookmarks all the Web sites that have high traffic flow. This ensures that the Web site of an organization is exposed to other sites.

Figure 5.5 displays social bookmarking in a Web page.

Figure 5.5: Social Bookmarking in a Web Page

Some of the key features of social bookmarking are as follows:

- Is done specifically on direct links
- Provides a permanent one-way backlinks from Web sites with a high PR
- Submits these links exclusively to the public social bookmarking Web sites
- Submits all links from the user's personal profiles
- Helps the Web site to get easily indexed by Google

Few benefits of social bookmarking are as follows:

- An experienced editor can manually submit the bookmarks
- It increases the link ability of the Web site
- All top social networking Web sites can be bookmarked
- It increases PR value of a Web site

Figure 5.6 displays some social bookmarking icons.

Figure 5.6: Social Bookmarking Icons

5.2.10 Classified Submission

Classified submission is an important SEO tool that helps to make a Web site popular and obtain high ranking in search results. It is the cheapest and most effective way of advertising the product online. Classified submission can be used to display the product features or specialization to the Web users.

The two ways in which classified submission helps are as follows:

- Helps in marketing an organization's product or services
- Helps in making the Web site visible by posting links under classified submission

Figure 5.7 displays classified submission on Web.

Figure 5.7: Classified Submissions on Web

The few important things to be considered for obtaining results through classified submission are as follows:

- The classified should be attractive and be able to deliver the proper message.
- The classified should be posted on correct sites, where traffic is good.

5.2.11 Forum Posting

Forum posting is an easy method to receive multiple one-way links to a Web site. In other words, it is a link building method. In forum posting, users can quickly and easily post their opinion in a forum. Once certain amount of posts has been added, user can add signatures. A signature is a sign off appearing at the bottom of the posts containing relevant links to an organization's Web site. These posts have anchor text in the signature line of the advertisement.

Figure 5.8 displays a forum post.

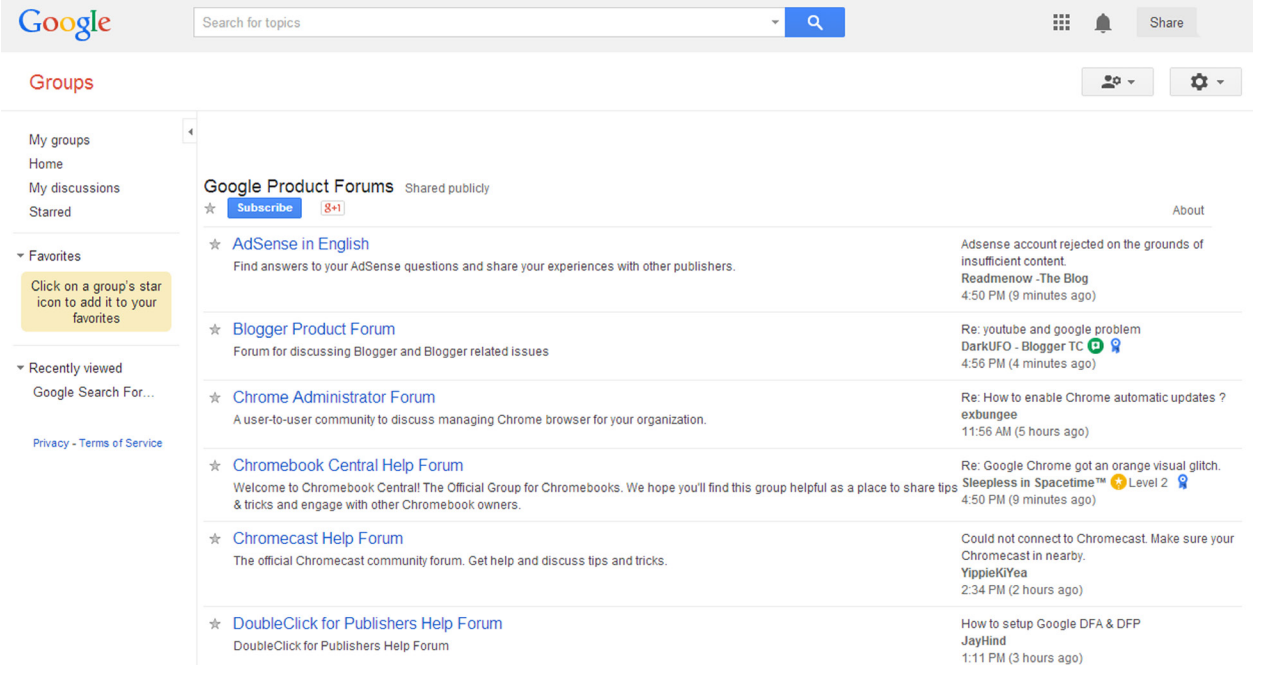


Figure 5.8: Forum Posts

Some of the advantages of keywords in signature lines are as follows:

- It represents the important content of the site
- It improves the traffic leading to the site
- It helps to achieve high rank on search engines results page

Hence, SEO manual submission is an effective way for visitors to choose a site for more information.

An important function in forum posting is to direct traffic to the Web site. Users will read messages in the forum, and if they find it interesting or if they want additional information, then they will visit the Web site. If a forum is monitored, has good reviews, and flow of traffic from the forum post are very high then the forum is a success.

5.2.12 Press Release

A press release is a written statement that is used to broadcast a range of news such as scheduled events, awards, personnel promotion, sales accomplishments, new products, and services to the media. Generally, they are couriered, faxed, or e-mailed to the Public Relation Officer (PRO). They can be editors of newspapers, magazines, and other media. PRO's goal is to attract favorable media attention and to provide publicity for the products, services, or events.

Figure 5.9 displays a press release Web site.

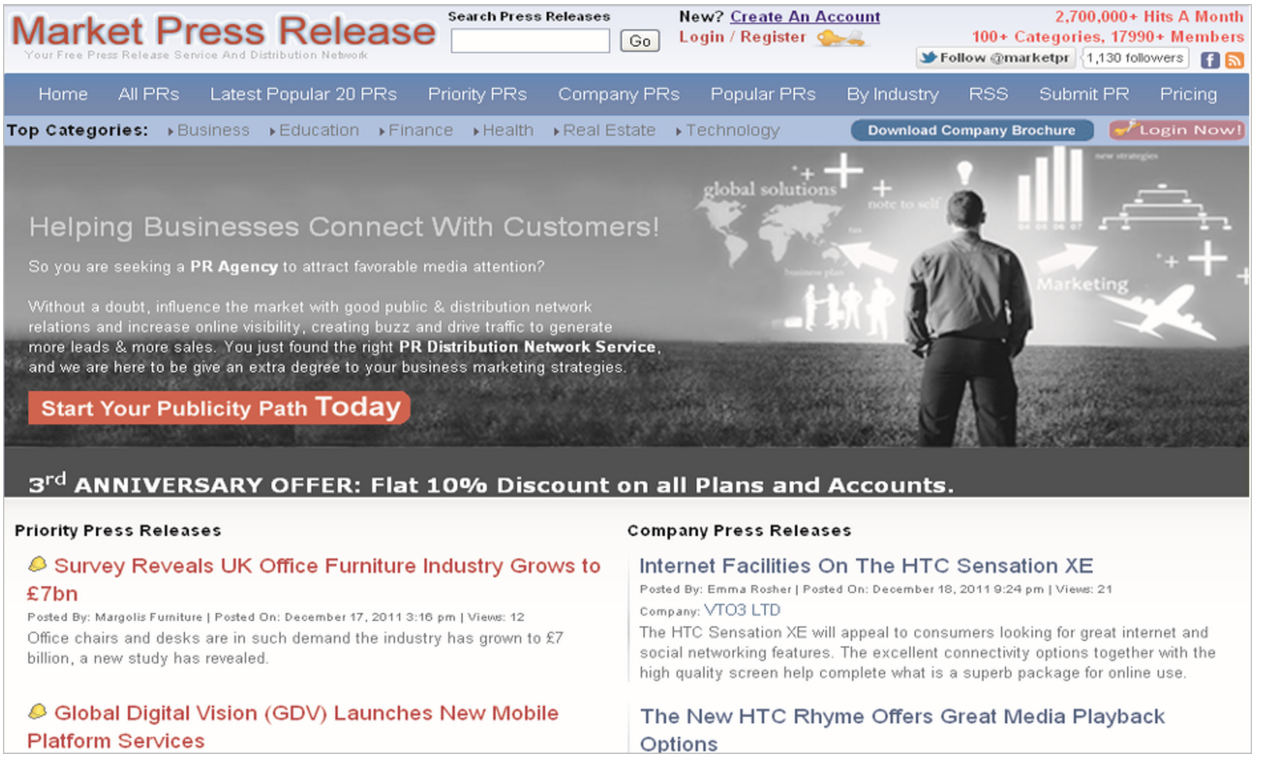


Figure 5.9: Press Release Web site

The benefits of SEO Press release are as follows:

- Provides backlinks from trusted news sites that will increase the ranking of a Web site or blog
- Improves the visibility of the Web site in search engines
- Ensures good ranking by using keywords in Google News and Yahoo News
- Promotes headlines across various blogs, Web sites, and news sites based on Really Simple Syndication (RSS)
- Introduces prospective improvement in media coverage

5.2.13 Blogging

A Web site needs to be optimized and that is achieved by blog posting and creation. It makes the Web site visitor friendly. A blog is a Web site in which items are posted on a regular basis and displayed in reverse chronological order. The term blog is a shortened form of Weblog or Web log. Authoring a blog, maintaining a blog, or adding an article to an existing blog is called blogging. Individual articles on a blog are called blog posts or entries. A person who posts these entries is called a blogger. A blog comprises text, hypertext, images, and links to other Web pages with video, audio, and other files. In blogs, the blogger uses a conversational style of documentation for dynamic entries and debating on the post. Microblogging is another form of blogging that features very short posts.

Regular blogging is an important activity associated with blog promotion. Once the readability of the blog increases, the blogger needs to keep the readers interested by publishing more content. This will keep readers from not losing interest in blogger's activities. Thus, the ranking of the Web site will increase.

Figure 5.10 displays a Blog page.

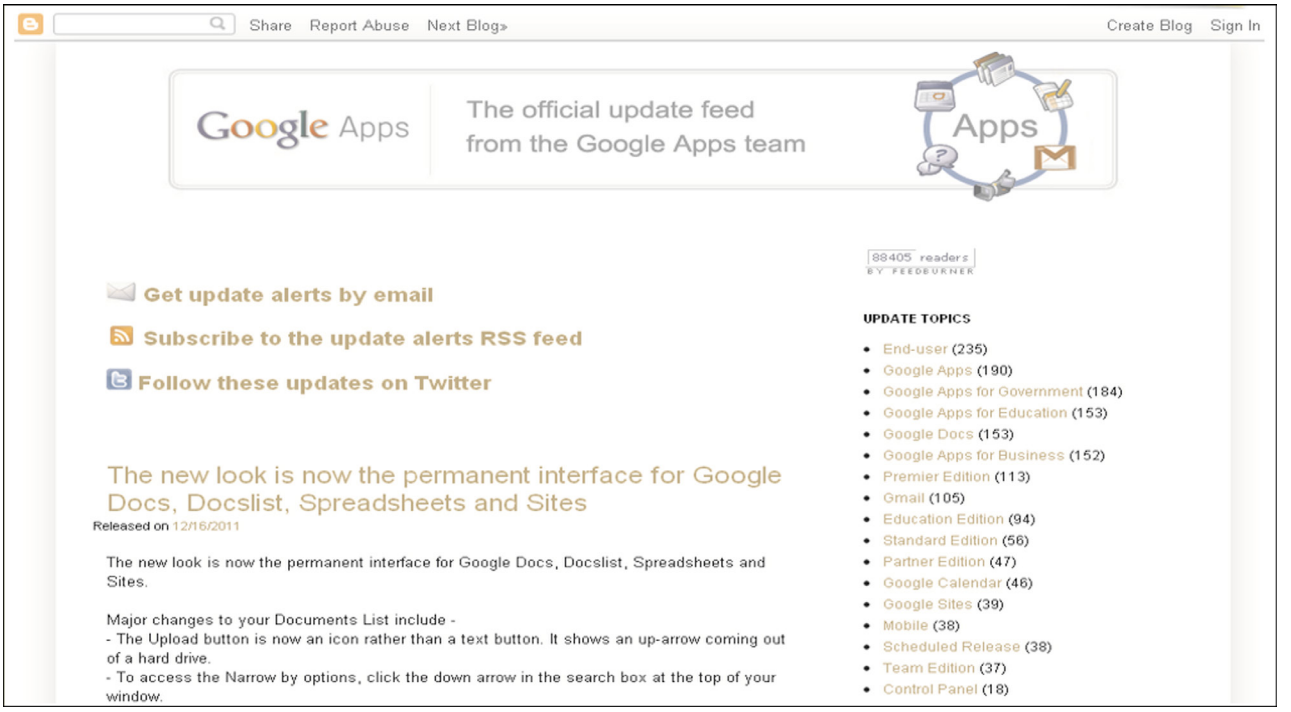


Figure 5.10: A Blog Page

5.2.14 Video Submission

In today's world, there is lot of competition for visibility. Nowadays, video submissions have become a successful marketing strategy for online products. A video submission also needs to go through the process of SEO as any other file or Web site. For this, a title and description of a video must be added, which will act as the Meta title tag and Meta tag description of the page. Viral marketing is a promotional strategy where an organization can create a video that is promoted on social networking sites. If the video is appreciated and watched by many people, they will surely visit the Web site for more information. There are many companies that specifically perform video promotions and distributions. By this method, the organizations can guarantee for their online product marketing.

Normally, all the video promotions are done through YouTube Web site. Figure 5.11 displays the YouTube page.

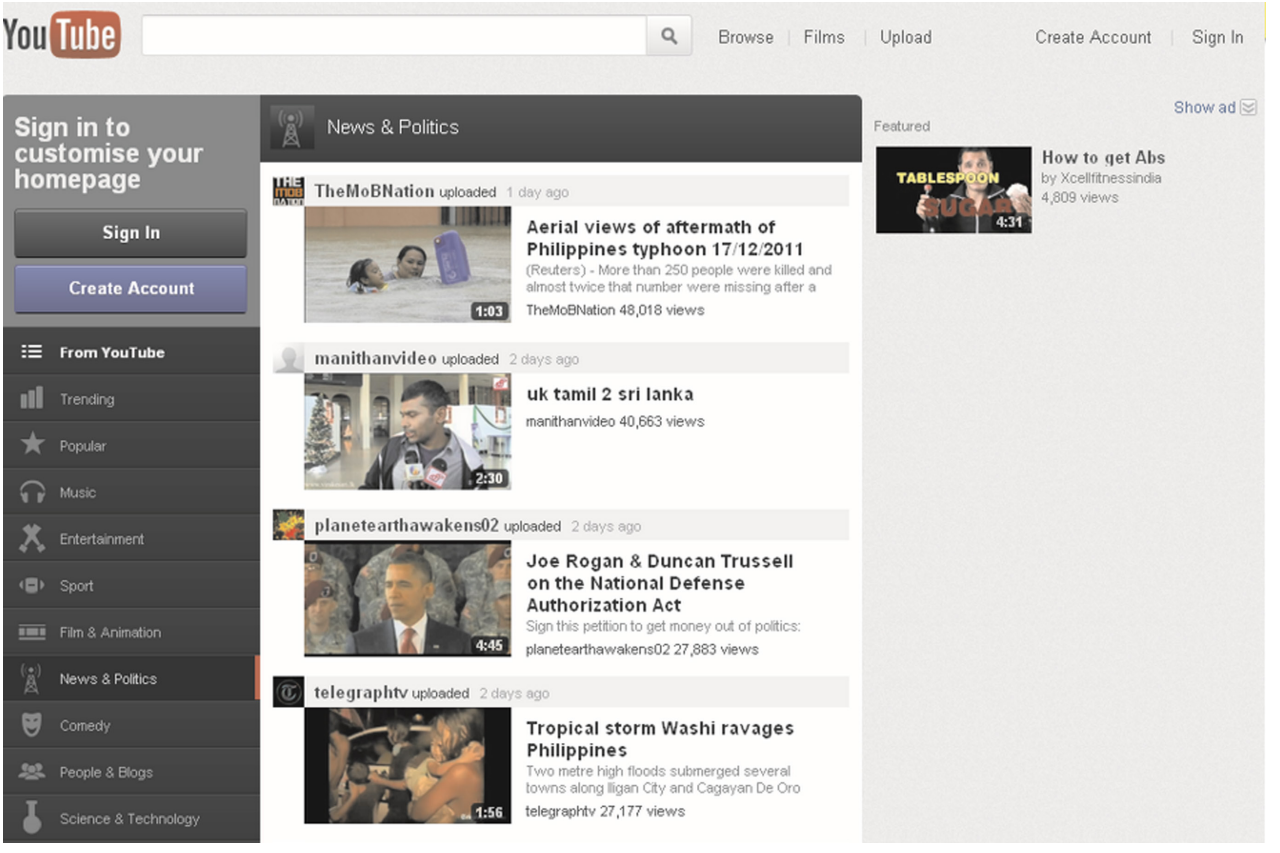


Figure 5.11: YouTube Web Site

5.2.15 Deep Linking

Deep linking on the WWW is a tool used by Web site developers. It is used for creating a hyperlink that guides a user directly to a specific page or image on the same or another Web site, instead of the Web site's main or home page. Deep linking allows users to move directly to the information they require, which is available on a certain page of the same or another Web site. For example, the link **http://en.wikipedia.org/wiki/Deep_linking** is a Web site that gives information on deep linking. The URL contains all the information required to point to a particular item. By deep linking, users go directly to the English Wikipedia article on deep linking.

Figure 5.12 displays deep linking on Wikipedia, instead of the Wikipedia home page at **http://www.wikipedia.org/**.

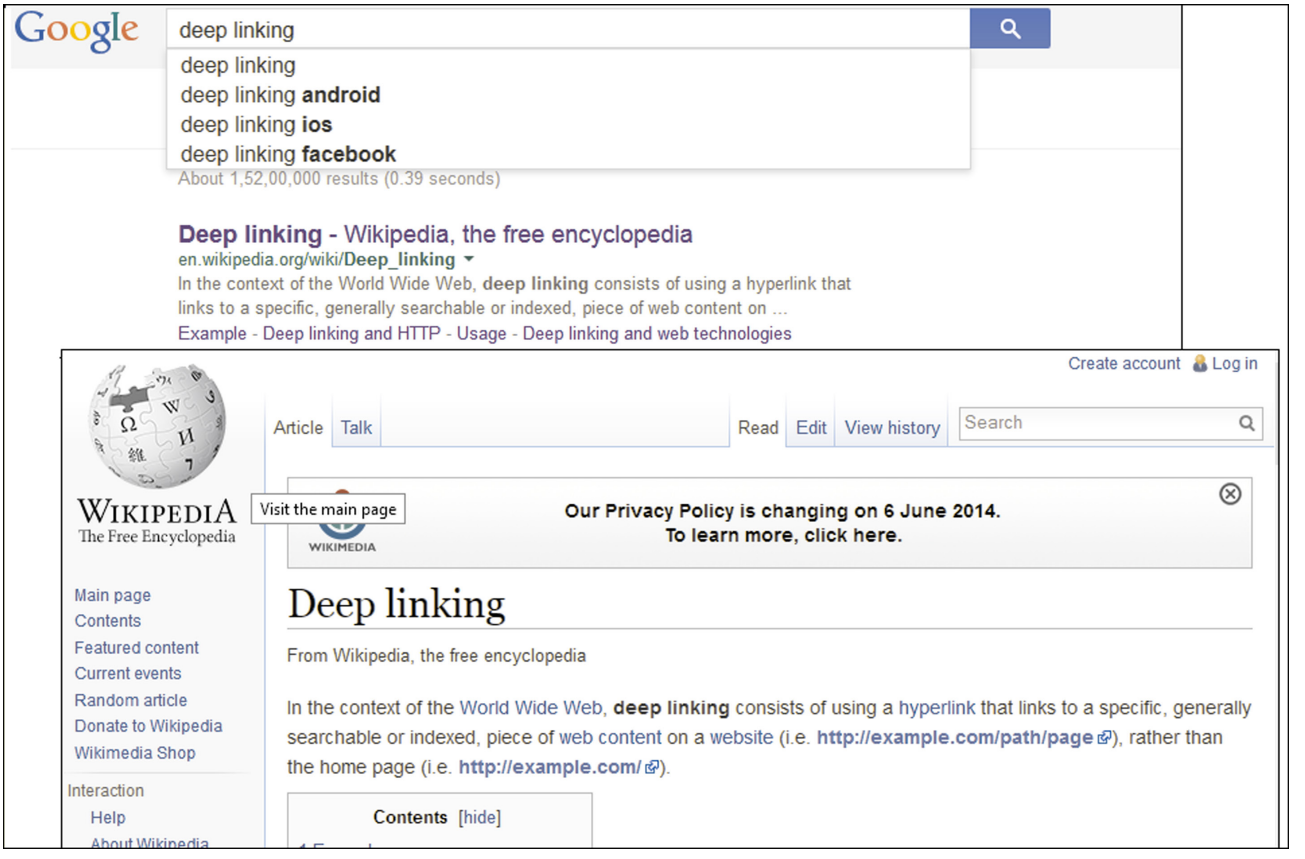


Figure 5.12: Deep Linking on Wikipedia

Web sites developed using Web technologies such as Adobe Flash, Asynchronous JavaScript, and XML (AJAX) do not support deep linking, as it creates usability problem for users visiting these sites. The users of these Web sites face problems such as unable to save bookmarks to individual pages, disabled forward and back buttons of the browser, return of the initial page when using browser's refresh or reload button. However, these are fundamental limitation of this technology. Tools, techniques, and libraries such as SWFAddress and History Keeper are used to provide deep linking to pages within their sites.

5.2.16 Infographics

Infographics are the latest and one of the most effective ways of communicating information. An infographic or information graphic is a visual representation of information through a combination of images, text, graphs, and numbers. An Infographic can be used as an SEO technique by including links and keywords within it.

Following are the benefits of using infographics as SEO tools:

- Infographics are graphical, eye-catching, and attract Internet users easily to click and learn more.
- They appeal to the Internet audience who are used to skimming through articles and grasping information as opposed to actually reading large amount of text.

- Infographics can be easily shared and linked across various domains on the Internet including mailers, social networks, and related Web sites.
- Infographics can be used as marketing and branding tools.

Figure 5.13 displays an infographic of the components of SEO best practices (source: http://masterinbound.com/item/the-ingredients-to-a-tasty-seo-sandwich-infographic.html).

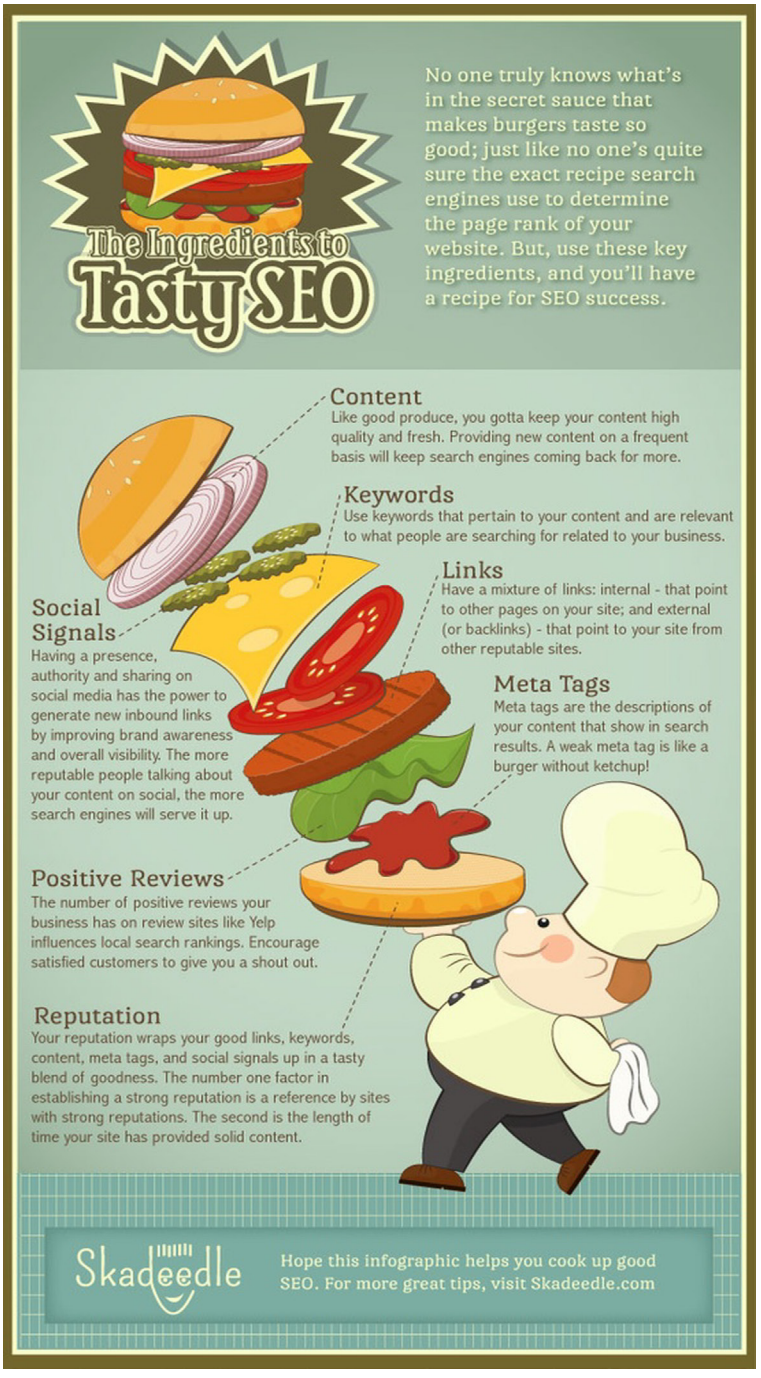


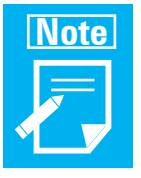
Figure 5.13: Infographic Depicting Components of SEO Best Practices

5.3 Guidelines for Avoiding Common Off-Page SEO Mistakes

In the off-page optimization techniques, inbound links plays an important role in enabling the Web site to obtain a high PR value. In the process of acquiring high PR value, the Webmaster may make mistakes, which can produce bad results later.

Few off-page SEO guidelines that are required to be followed to avoid such mistakes are as follows:

- Avoid using duplicate keywords in link advertisements
- Use less site-wide links causing link growth spikes
- Avoid using on-page SEO to do the work of specialist off-page SEO
- Avoid placing random links without keywords near your link advertisement



A link growth spike is a sudden jump in the rate of backlink growth to a site, followed by a sharp decrease in the rate of growth of links to the site.

Check Your Progress

1. In off-page optimization, to increase the Web site ranking, the user can exchange the links with _____ Web site.

(A)	Developer's	(C)	Competitor's
(B)	Webmaster's	(D)	Organization's

2. Which of the following options are factors for off-page optimization services?

(A)	Article Submission
(B)	Social Bookmarking
(C)	Classified Submission
(D)	Particle Submission

(A)	A, B, C	(C)	C
(B)	C, D	(D)	A, B, D

3. _____ percent of searches done on the Internet are by using the Google search engine.

(A)	70%	(C)	66%
(B)	85%	(D)	80%

4. Article submission is a method of receiving quality _____ to a Web site and it is a standard in Web site search engine optimization strategies.

(A)	Link building	(C)	Backlinks
(B)	Outgoing links	(D)	One-way inbound links

Check Your Progress

5. Which of the following are advantages of keywords in signature lines?

(A)	It improves the traffic leading to the site
(B)	It improves the Web page presentation
(C)	It helps to achieve top ranking on search engines results page
(D)	It helps to reduce the budget of the site

(A)	B, D	(C)	C
(B)	A, C	(D)	A, B, D

Answer

1.	C
2.	A
3.	D
4.	D
5.	B

Summary

- Off-page optimization is associated with the site's internal linking structure and it also relates to the factors that influence a Web site's ranking on the SERP.
- Higher the number of inbound-links, higher is the link popularity.
- Inbound links are defined as links found on another Web site on the Internet that direct users to user's site only.
- Some of the off-page optimization techniques that can be adopted are as follows:
  - Link Popularity
  - Link Building
  - Directory Submission
  - Article Submission
  - Social Bookmarking
  - Classified Submission
  - Forum Posting
- Page rank is Google's system of analysing the popularity of a Web site based on the number of hits and the number of back links it has from other quality Web sites.
- Documents submitted to the document sharing sites get indexed with search engines as quickly as 24 hours thus enabling almost instant visibility on the Internet.
- Infographic is a visual representation of information through a combination of images, text, graphs, and numbers.
- Viral marketing is a promotional strategy where an organization can create a video that would be promoted on social networking sites.

06 Customization, Personalization, and Other Features

Welcome to the Session, **Customization, Personalization, and Other Features**.

This session describes how to customize and personalize Web site SEO through localization and internationalization.

In this Session, you will learn to:

- Describe SEO localization
- Describe SEO internationalization
- Describe Social Media Optimization

Session
06

Customization, Personalization, and Other Features

6.1 Introduction

During the recession that occurred a few years ago, businesses reacted to different economic conditions by proactively searching for new opportunities. By marketing unconditionally, the small as well as large organizations were able to enter into the international market.

The desire to grow in testing times requires to be backed by very effective tools. Before investing money, organizations should study the international market trend and then apply the required strategy.

The Web is now the first choice for most consumers when shopping for a new service or product, so the key to enhancing the business internationally is shown through its Web site.

SEO customization, localization, and internationalization are cost-effective way of entering into new markets, with a simple text-based Web site costing very minimal to set up in a foreign language.

6.2 Introduction to SEO Localization

SEO localization is the process of optimizing the Web pages based on its geographical location where it is operated. The Web page can be optimized globally or specific country-wise.

Generally, a small organization that does business in their country or state optimizes their Web site in their country itself. In such cases, SEO localization is helpful.

The ways in which SEO localization can be applied to Web sites are as follows:

- After uploading the Web site in Google Webmaster Tools, optimize the pages in specific geographical location. Figure 6.1 shows the Google Webmaster Tools.

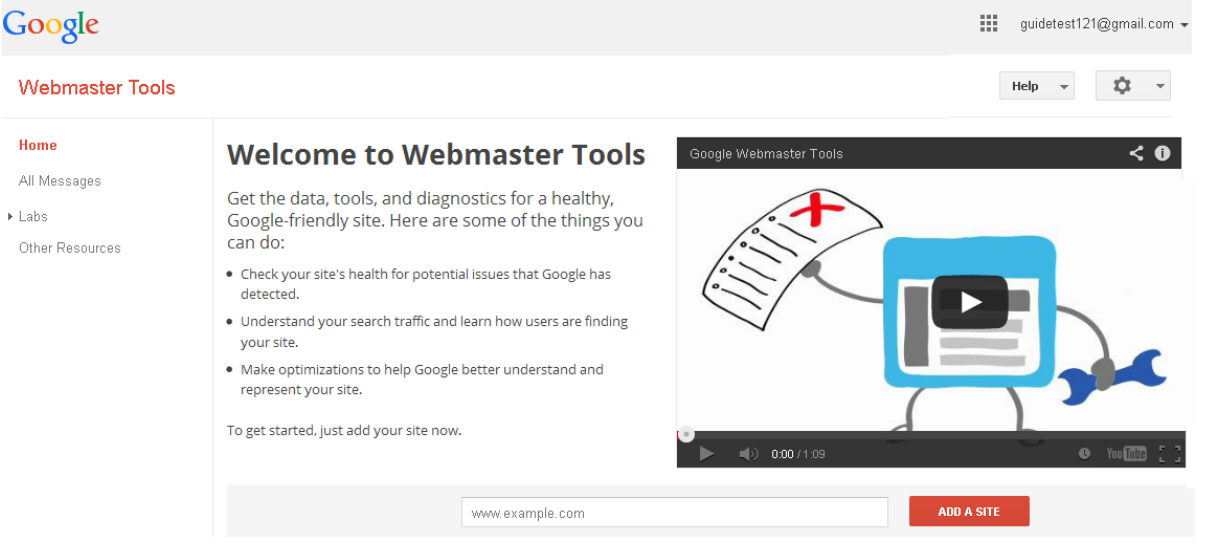


Figure 6.1: Google Webmaster Tools

This option is present in **Webmaster Tools - Site Configuration - Settings**.

- Add country name or area name to the search keywords wherever applicable. For example, add US Embassy, Australia or UN headquarters, Thailand.

  Generally, in the Google search, the users use those keywords that are specifically for the services offered in their location. So including the city and country names along with the search keywords helps in generating more search results.

  Before using these types of keywords, search once with the **Google AdWords Keyword Planner** tool as shown in figure 6.2. The **Google AdWords Keyword Planner** tool is used to identify whether a keyword has a good search volume. If the keywords have good search volumes, optimize it with the Web page.

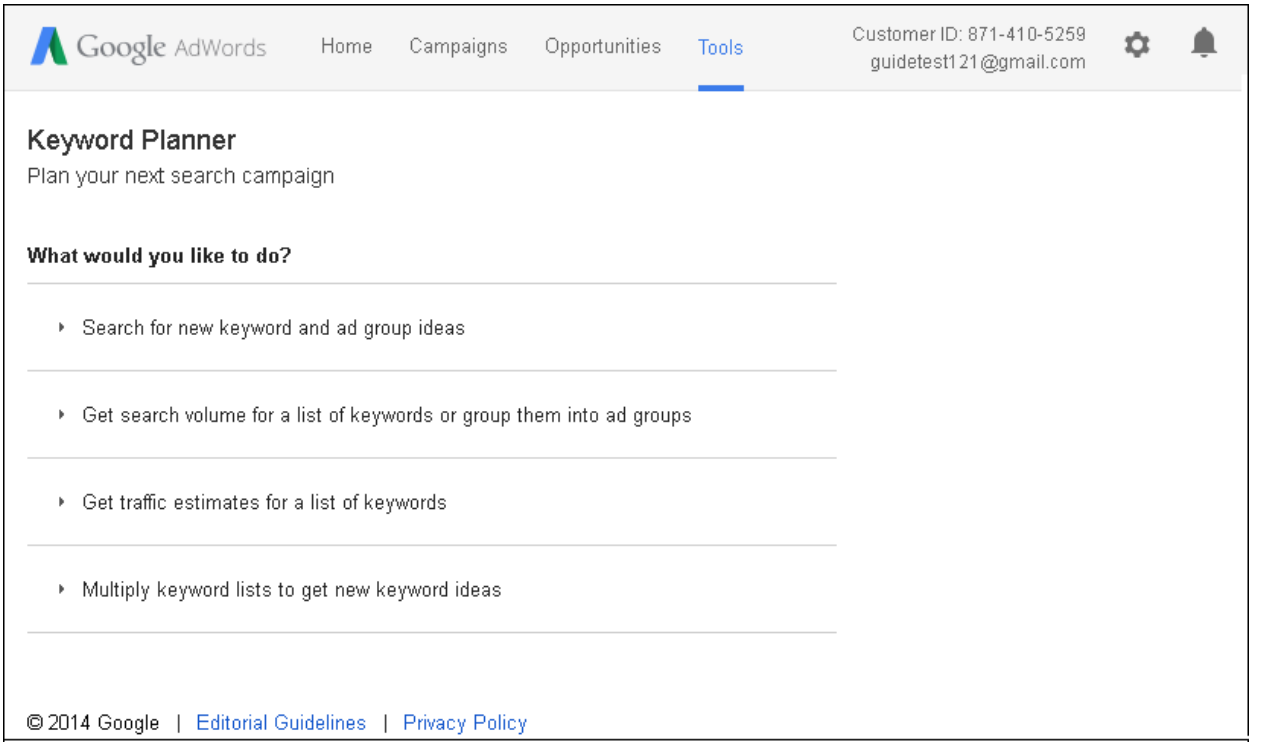


Figure 6.2: Google Keyword Planner Tool

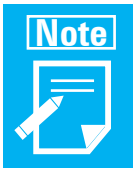
- Users can perform local optimization, when registering the domain name. If the domain name is registered with country's Top Level Domain (TLD), the SEO will be optimized as the country-level search engine. For example, the TLD of Google Germany is **google.de**.

6.2.1 Research and Analysis for SEO Localization

For better reach into the market, products and services are typically customized to the local preferences and cultures. Web site content too needs to be modified to ensure that it caters to the preferences of the target audience. SEO localization involves adapting the content to suit the language, culture, and geography of the audience in the target market.

Following are the various types of research and analysis that Webmasters need to perform to arrive at a content plan for SEO localization of the Web site:

- **Keyword Research** – Users in different regions might use different phrasings for certain topics or words. Also, the topics of interest might vary from region to region. Keyword research identifies the keyword phrasings that users of the target region might use to find the content on your Web site. It helps identify relevant keywords which when placed in Web page content, can generate maximum search demand in the target region.
- **Competitive Analysis** – When marketing products and services in a market, it is essential to analyze the competitors in the market. Webmasters need to collect information and identify the languages they use, the search engines they target, the keywords they focus on, and the Web sites from which they get their backlinks. This information will help them draw the content plan for their Web sites.
- **Culturally-Sound Adaptation** - When targeting specific regions for marketing products and services, it is essential to understand the culture of the region and ensure that the marketing strategies used are culturally sensitive. When building the content plan for Web pages targeting specific regions, it is important to ensure that the content does not contain language, or depict emotions that are offensive or considered a taboo in the region. The tone and language used should be acceptable to the audience of the region and respect their emotions and culture.



While looking into the cultural adaptation of the content, it is important that the content does not lose its selling point and the adapted content remains as compelling as its original Web site.

- **Content Plan** - After keyword research and analysis of the competitor Web sites, Webmasters can focus on designing the content plan. When drawing up a content plan, determine the keywords that are already a part of the Web site content and those that need to be added. Check if the existing content can be modified to include the new content or does it require creating new content from scratch. The changes can be in the phrasing of words, placement of content, or even the format of content (text vs images). Modify the content to map the preferred format if required.

6.2.2 Using Keyword Planner for Keyword Research

Using the Google Keyword Planner is a fourfold process. These processes are grouped under the following categories:

- Search for new keyword and ad group ideas.
- Get search volume for a list of keywords or group them into ad groups.
- Get traffic estimates for a list of keywords.
- Multiply keyword lists to get new keyword ideas.

Search for New Keyword and Ad Group Ideas

This option helps users derive a list of new keywords and ad groups most relevant to the products or services described on their Web site. To use this option, users need to enter relevant information in one or more of the text boxes that lie under **Enter one or more of the following**.

- **Your product or service**: Enter words or phrases that describe the contents of the Web site.
- **Your landing page**: Enter the URL of the main page of the Web site.
- **Your product category**: Click in the box to display a hierarchical list of categories and select the categories that match the product or service offered by the Web site. If none of the categories provided match, enter a new string of relevant categories.

To filter larger sets of keywords and group ideas, fill up both **Your product or service** and **Your landing page** boxes.

Figure 6.3 displays the searching for new keyword and group ideas section of the Google Planner.

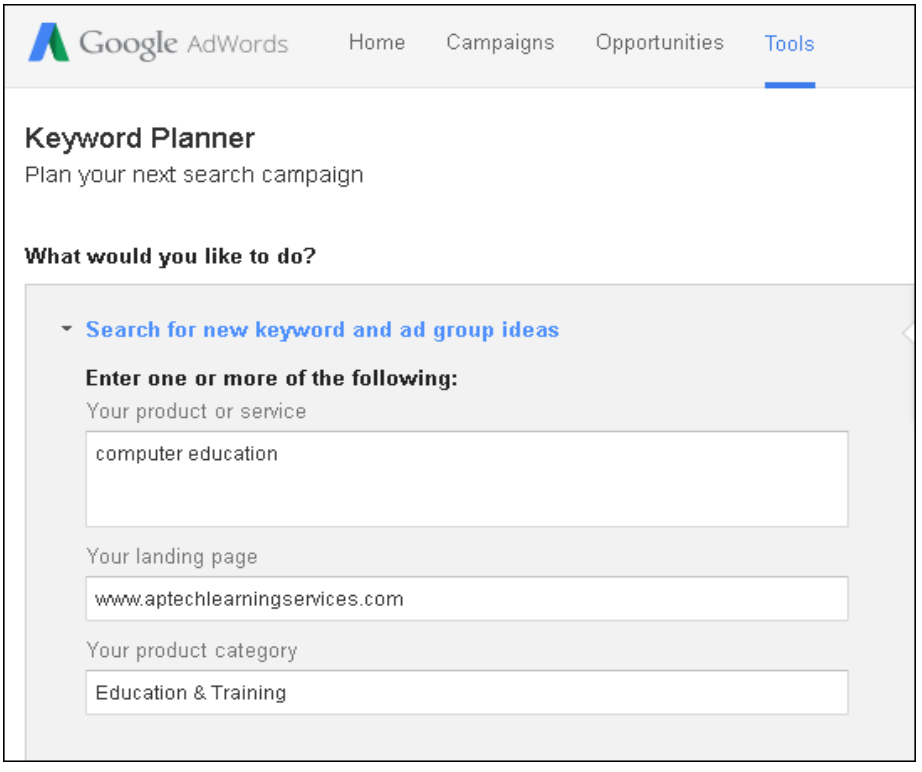


Figure 6.3: Searching for New Keyword and Group Ideas

Get search volume for a list of keywords or group them into ad groups

This option is useful for Webmasters after they have arrived at a list of keywords, to get historical statistics such as search volume of the keywords. It requires them to enter the list of keywords manually or as a Comma Separated Values (CSV) file.

Perform the following steps to use this option:

1. Click the **Get search volume for a list of keywords or group them into ad groups** link.
2. In the **Option 1: Enter keywords** text area, enter a list of keywords, one per line or separated by commas. Alternatively, click the **Browse** button under **Option 2: Upload file** and upload a CSV file containing the list of keywords.

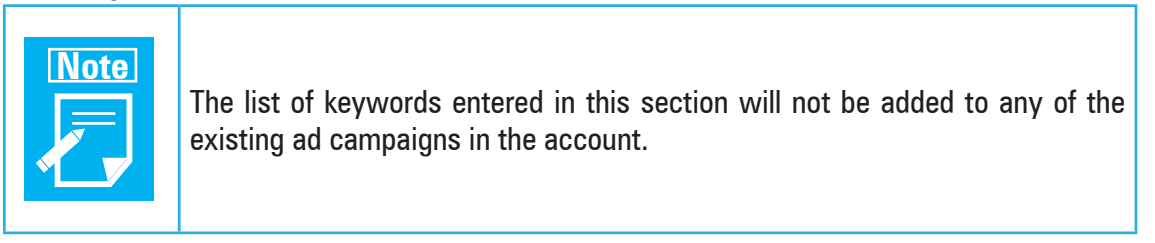


Figure 6.4 displays the **Get search volume for a list of keywords or group them into ad groups** section.

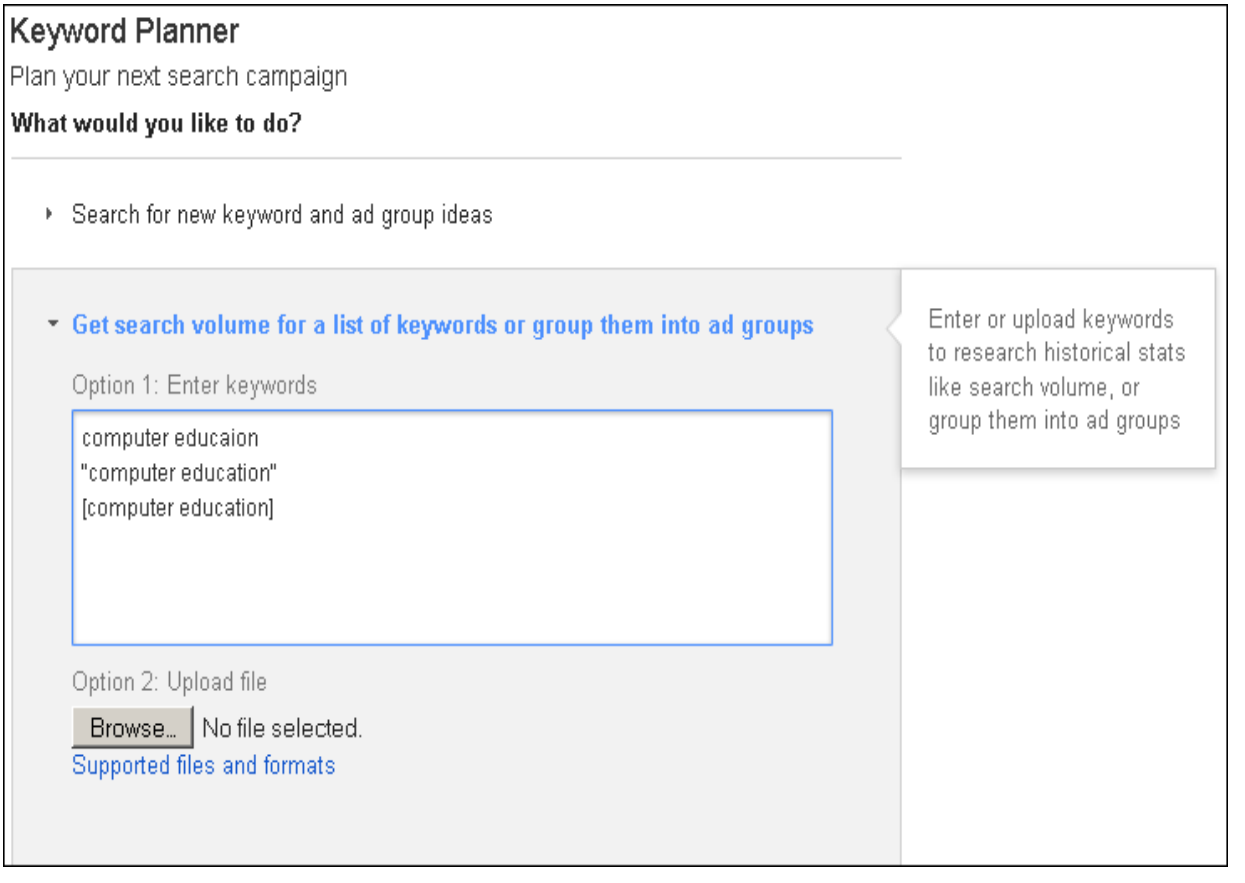


Figure 6.4: Get Search Volume for a List of Keywords

3. Under T**argeting**, **there** are three options for filtering the search volume results:

   - The search results by default targets for all locations. If the Web site is targeting a specific geographic location to market their products or services, click **All locations** and enter in the text box, the target country, city, or region.
   - Click **Google** and specify whether search volume results should show only for Google search engine or for Google as well as its partner search engines.
   - Under **Data range**, select the period for which the search volume results need to be displayed. To view how search volume results for the selected period compare to a previous period, change the **Compare** slider to **ON** and select **Previous period, Same period last year**, and **Custom**. Selecting **Custom** allows specifying a specific backdated period to compare.

4. Click **Get search volume**.

Get traffic estimates for a list of keywords

Users can also retrieve an estimate of number of clicks and cost estimates for a list of keywords. Users can enter the list of keywords manually or upload them as a CSV file. The estimates are provided keeping in view, the AdWords account and its selected bid.

Perform the following steps to use this option:

1. Click **Get traffic estimates for a list of keywords**.
2. Under **Option 1: Enter keywords** text area, enter a list of keywords, one per line or separated by commas. Alternatively, click the **Browse…** button and upload a CSV file containing the keywords.

The list of keywords entered here will not be added to any existing campaigns on the account.

Figure 6.5 shows how to get traffic estimates.

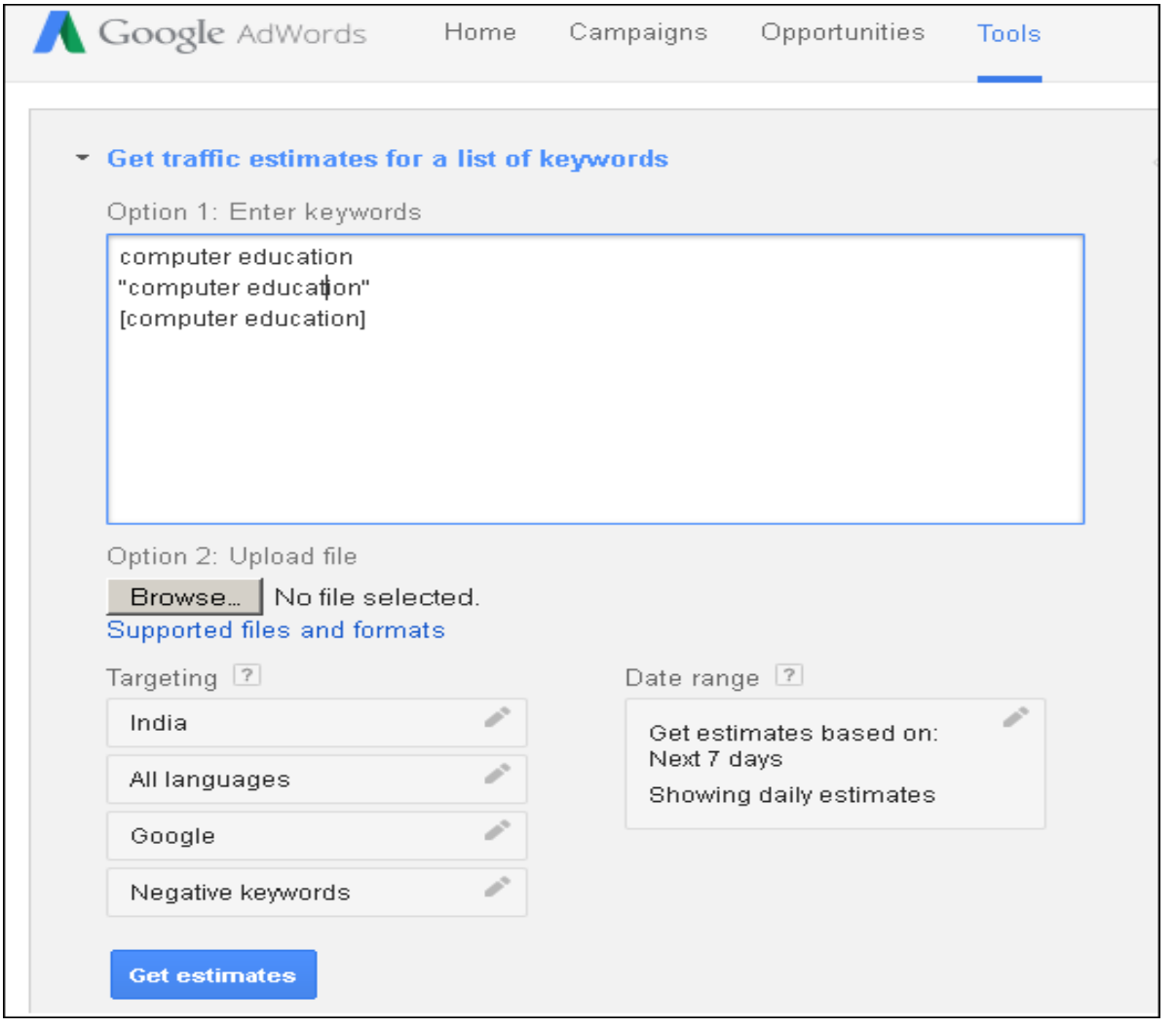


Figure 6.5: Get Traffic Estimates for a List of Keywords Section

3. Here too, users can filter the results by target location, preferred language, target search engine, and the keywords to be eliminated.

4. Finally, specify the period for which the traffic estimates need to be displayed and click **Get estimates**.

Multiply keyword lists

This option enables users to combine multiple lists of keywords into a single list and get search volume results or traffic estimates for them.

1. Click **Multiply keyword lists to get new keyword ideas**.

2. In the **List 1** text area, enter a list of keywords, one per line or separated by commas.

3. Similarly, in **List 2** box, enter a different list of keywords to be combined with the keywords in List.

4. To include a third keyword list to merge, click the faded X next to the **List 2** text areas. This will display a new text area, **List 3**.

Figure 6.6 shows how to use multiple keyword lists at once for search volume and traffic estimates.

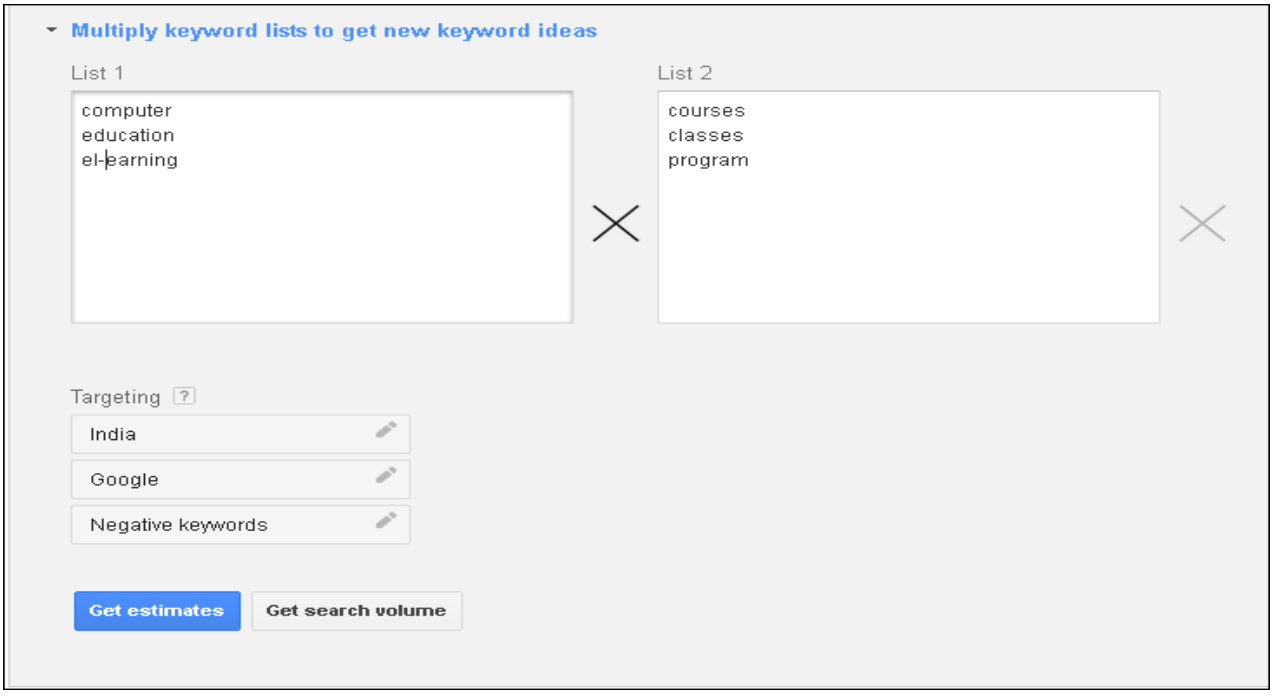


Figure 6.6: Multiplying Keyword Lists

5. This section too allows filtering the search volume and traffic results by target location, search engine, and keywords that need to be eliminated.

6. Click **Get estimates** to retrieve traffic estimates.

7. Clicking **Get search volume** displays search volume results.

6.2.3 Optimizing Multilingual Search

An SEO localization consultant can optimize the multilingual side of search. Consider a scenario, where the French, Spanish, or German customer searches for the work that an organization has done to produce and communicate through the Web site. In such a situation, SEO optimization with multilingual option can push the organization's Web site to the top of the search results.

Using the right keyword for each language is necessary and will have a huge impact on the natural ranking and Pay-Per-Click (PPC) Advertising results in local markets for the organization's Web sites. For multilingual options, using direct translation of the English keywords will not state the correct set of keywords for local market.

Most of the international business happens due to search engines, so the organization endorsing its product through Web sites has to optimize it as per the market.

Words have multiple meaning, so do the product names and service descriptions. If the word-to-word translation of any language is done, the meaning will be misinterpreted.

For example, if the user types a keyword 'Windshield repair' from the English language Web site, the direct translation into French would be 'réparation pare-brise'. However, this keyword would be incorrect if it is a French site. By using the Google Keyword Suggestion tool its understood that direct translation has a lower local search volume than some alternate suggestions such as, 'auto pare brise' or 'moto pare brise'.

For the UK English version, the keyword will be 'Windscreen Repair' instead of 'Windshield Repair', because terms for car parts, have different English names in the UK and US markets. This will be displayed in the suggestion if the multilingual option is optimized in an SEO Localization.

6.2.4 Difference between Localization and SEO Translation

Localization and translation does similar work or to be precise a good translator can do both, but majority of them do only localization.

The difference between localization and translation is as follows:

- **Localization** - Translating the Web pages for a different culture or human beings who reside in a different society is called localization. A good brand name in one language will be a joke in another. For example, an Iranian soap product called 'Barf', the actual meaning in Persian is 'snow', but in English, it means 'vomiting'. Similarly, the Rolls Royce 'Mist' means, 'manure', or 'dung' in German.

  An excellent marketing slogan in one language can be a total disaster in another. For example, an American airlines slogan for first class seats was 'fly in leather' when directly translated to Spanish 'vuela en cuero', which in Spanish means 'fly naked'. A good translator will know the language as well as its culture and will adjust according to expression and situation. Therefore, localization is the process of translating text to adapt to the foreign language and making it familiar and attractive.
- **SEO Translation** - Keywords, expressions, titles, tags, anchor texts, script messages, and every single attribute on a Web page is translated to the target language to make the page understanding. Therefore, when a client searches in foreign language the SEO translated site pops up as the first result in the search engine.

  This helps in getting more business with increase in customers and sales. An organization can develop best-localized Web site but if the translated page does not appear in the search engine result then the effort is useless.

  For an organization, both localization and translation are important as they both together can make a Web site the best-searched site. Rather, if the translation or localization is imperfect then it could be a laughing stock for few because of the direct translation of words to different languages.

6.3 Internationalization

Web users prefer local and regional language-based Web sites as compared to global Web sites. Hence, an organization targets the requirements of regional audiences, even though the Internet is global. For this, translation to regional languages is recommended. So, when implementing this, there are some requirements that need to be checked.

Some of the requirements for translating into specific languages are as follows:

- **One ccTLD for Every Language** - All Web sites receives a Country Code Top Level Domain (ccTLD) that belongs to the language. For example, Germany gets **example.** , United Arab Emirates (UAE) **example.ae**, and so on, where **<example>** is a domain name. For English content, it is mainly **.com** domain. Even the sub domains can be made with language preference such as **de.example.org** or **ae.example.org**.

- **Language Header** - Ideally, the Web server should send Content-Language HTTP Header and the HTML document that follows will be in the designated language. In practice, this could create a problem because many Webmasters are unaware of or unfamiliar with this header. Also, there are meta-headers which are set in contradiction to the HTTP header. This will challenge the search engine and it will have to know the language using some other characteristics, which will take time. Therefore, Webmasters should use the HTTP Header viewer to check if the language header was sent, if it is the same as the language in the content. The viewer should also check if the meta-header is correct. Search engines also try to detect the language through the text, so use only one language per URL.

- **Links from Same Language Sites** - There are different ways in which the Google search engine tries to find the language of a Web site. One way is through incoming links. If the ratio of incoming links of English and local language Web sites to the submitted Web site is unfavorable, then the site will fall in English language index, even though the language is different.

- **Local IP Address** - Normally search engines such as Google not only try to know the language of the Web site, but also its origin. This technique is used to check whether the sites are localized in the country, of which the language is detected. The origin of the Web site is detected by the IP address of its server on which the Web site is hosted. Hence, it is helpful for an organization to get an IP address associated with the country whose language is used.

- **No Auto Translation** - This does not directly affect the SEO, but is still recommended to use a native speaker to translate the content of the organization's Web site. Text translated through a program cannot be used for long term, since it is not validated. So, even if the content is great and if it is not understood in the target language, it is useless.

6.3.1 Challenges and Tips

Any organization with international Web presence will have some challenges. Some suggestions to avoid challenges and create a good Web presence are as follows:

- **Domains** - Selecting a domain is very important for good SEO. Specifically for international SEO, a Top Level Domain (TLD) for each targeted country is recommended. However, many countries insist on having a physical presence of the organization before obtaining a TLD. For example, to get a German (.de) TLD, the organization must have an office in Germany.

In the case of search engines, the organization tries to rank the most relevant search results for the Web user. If a user is searching when he or she is in Germany, on **google.de** as shown in figure 6.7, Google will give search results with **.de** for better outcome.

Figure 6.7: www.google.de

For example, the top ranking Web sites for 'travel' in **google.de** are **statravel.de** as depicted in figure 6.8, **statravel.com** and **wiki travels**.

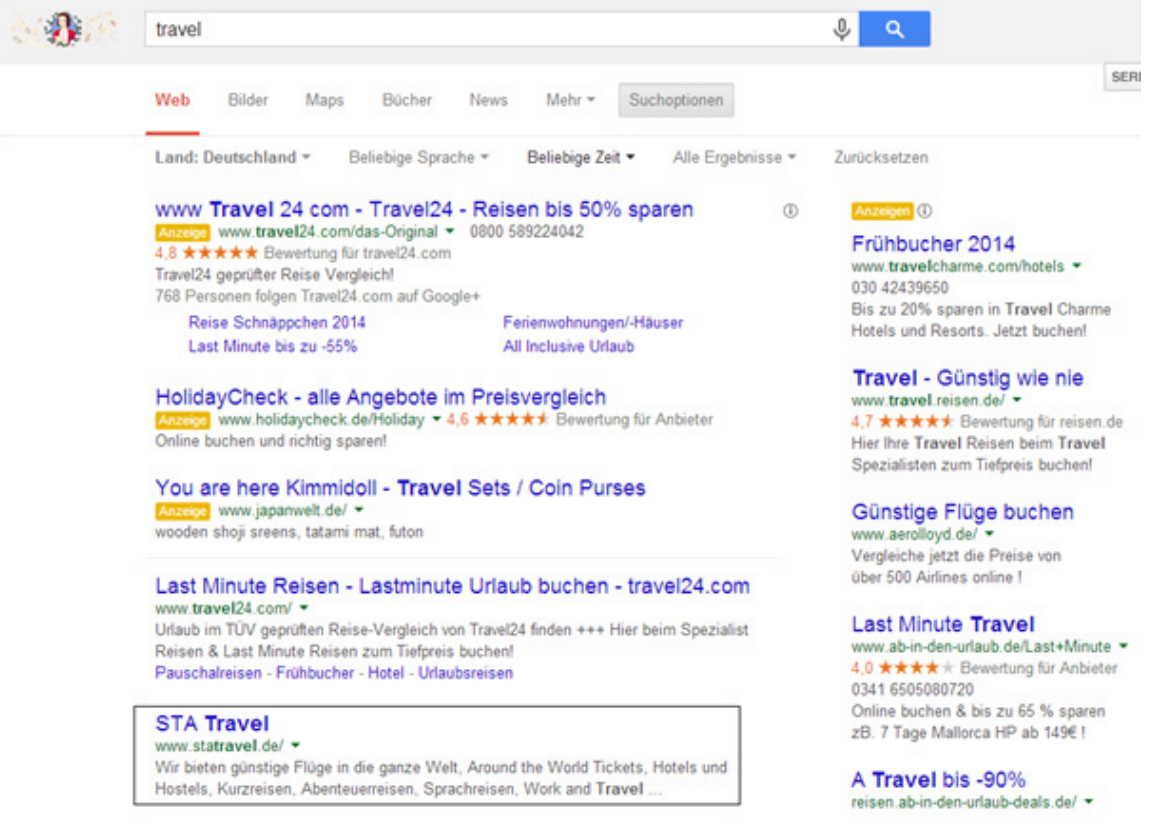


Figure 6.8: Searching www.google.de

Interestingly, in this case, **statravel.de** ranks better than **statravel.com**. When analyzing the number of pages of **statravel.de** indexed on **Google.de**, it is found to be 17,100 pages, whereas, for **statravel.com** is 27,500 pages. Based on these values, it can be expected that **statravel.com** outranks **statravel.de**. However, in this case, localized domain matters.

- **Hosting** - Generally, it is mandatory that all large organizations have their Web sites hosted in one facility, across the globe. Search engines normally look at the host location (IP) of Web sites, so that they can deliver relevant search results to Web users. For getting an organization's Web site in the top ranked searches, they have to host their unique TLDs in the country. If the organization is targeting Germany, then they require .de domain Web site and host the domain within Germany.

  Similarly, if they are operating from different countries, follow the same principle of hosting the domain within the country.
- **Content** - As the organization localizes the domain and the SEO for each country, hence, for obtaining optimized search results, it is also necessary to localize the content within each country.

6.4 Social Media Optimization

The integration of SEO and social media marketing has gained popularity among online marketers globally. Distinctively, SEO and social media functions are not effective enough but work better when integrated. Considering consumers information discovery behavior, the integration of SEO and social media is successful.

There are two areas to focus on for the future of search and social media market. The first area is the technology that powers the search and social media experience for Web users. The second area is the logistics of implementing integrated search and social media campaigns within a company.

Technologically, search has become increasingly influenced by social media. For example, Google gets feed data from Twitter and Facebook. If a user is logged on to Google, the search results can have comments updated by the user's friends. Due to social media networks, searching has become very creative and helps to connect to a large audience with great inputs.

Social Media Optimization (SMO) uses viral publicity through multiple social media outlets and communities to increase the awareness of a product, brand, or event.

SMO includes using social news and bookmarking sites, social media sites, and video and blogging sites.

SMO is a technique of viral marketing where word of mouth is done by social networking and by the use of video and photo sharing Web sites. SMO is an integral part of Online Reputation Management (ORM) or Search Engine Reputation Management (SERM) strategy for organizations. SMO was first used and described by Rohit Bhargava .

Bhargava's original five rules for conducting social media optimization are as follows:

1. Increase your linkability
2. Make tagging and bookmarking easy
3. Reward inbound links
4. Help your content travel

5. Encourage the mashup

After the initial five rules, four years later, Bhargava posted an updated set of five new rules and they are follows:

1. **Create shareable content**: This rule has replaced the earlier rule of 'Increase your linkability' because sharing has become more popular in recent times.
2. **Make sharing easy**: This rule has replaced the earlier rule of 'Make tagging and bookmarking easy'.
3. **Reward engagement**: This rule has replaced the earlier rule of 'Reward inbound links'.
4. **Proactively share content**: This rule has replaced the earlier rule of 'Help your content travel'.
5. **Encourage the mashup**: This rule is retained intact as mashups are still relevant today.

During the evolution of search and social media, many other authors updated these basic rules and now there are 17 rules that are translated into French, Dutch, Italian, Spanish, German, Japanese, Greek, Portuguese, Russian, Hebrew, and Thai.

There are two types of SMO methods and they are as follows:

- Social media features are added to the content such as RSS feeds, social news and bookmarking sites, user rating and polling tolls, and videos.
- Social media has promotional activities such as blogging, commenting on other blogs, discussion group participation, and posting status updates on social network profiles.

Check Your Progress

1. Global optimization of the Web pages takes longer time as compared to _____ optimization.

(A)	Basic	(C)	Local
(B)	Master	(D)	Metropolitan

2. What is full form for ccTLD?

(A)	Custom code Top Level Data	(C)	Country Call Top Level Data
(B)	Country Code Top Level Domain	(D)	None of these

3. Social Media Optimization (SMO) uses _____ through multiple social media outlets and communities to increase the awareness of a product, brand, or event.

(A)	Social publicity	(C)	Online publicity
(B)	Web publicity	(D)	Viral publicity

4. How is the German domain represented in URL?

(A)	.fr	(C)	.de
(B)	.ae	(D)	.uk

Answer

1.	C
2.	B
3.	D
4.	C

Summary

- There are different ways in which SEO localization can be applied to Web sites.
- SEO localization users can optimize the multilingual side of search.
- SEO optimization with multilingual option can make the organization's Web site at the top of the search results.
- Web users prefer local and regional language-based Web sites as compared to global Web sites.
- SEO and social media functions are not effective unless they are integrated.
- Bhargava's five rules are used for conducting social media optimization.

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07

SEO Tools

Welcome to the Session, **SEO Tools**.

This session introduces various SEO tools and describes them.

In this Session, you will learn to:

- Describe the need for Search Engine Optimization (SEO) tools
- Describe various SEO tools and their functions
- Compare between various SEO tools
- Explain how to analyze using different Google Optimization tools
- Describe the usage of various Google SEO tools

Session
07

SEO Tools

7.1 Introduction

SEO requires various methods, techniques, and strategies to be utilized by Webmasters and optimization experts who monitor search engine marketing efforts. The chances to draw huge volumes of traffic to a Web site are greater if Webmasters make use of the search engine tools.

SEO tools are various programs, assessment methodologies, and data collection techniques that work together to maximize the impact of a Web site on search engine listings and thereby boost the Web site's business popularity in a fluid and dynamic marketplace.

A professional SEO organization always has a full range of SEO tools. These tools are specifically designed for analyzing content, keyword density, choice, and links (both inbound and outbound). Search engine optimization is all about SEO tools and their effective usage.

7.2 Need for SEO Tools

Every business related Web site aims to reach the top-most position in search engine results to get noticed by worldwide Internet users. Performing SEO using various search engine tools is the best way to succeed against the tough competition in the Internet world.

The traditional SEO techniques cannot provide all the inputs for a Web site's optimization. One must also use some powerful and effective SEO tools to draw large volumes of traffic to a Web site.

Following sections describe a few popularly used Google SEO tools that help to get a Web site ranked quickly and correctly, and thus increase its traffic.

There are also some 3rd party SEO tools available, such as Rank Checker, BuzzStream, and OpenSite Explorer.

7.3 Google Webmaster Tools

The **Google Webmaster Tools** is a free service by Google that gives Webmasters valuable Google search rank information and enables them to monitor and maintain their site's presence in Google search results.

Google Webmaster Tools enables Webmasters to:

- Identify potential health issues in their Web site. It enables to monitor and resolve any malware or spam issues with the Web site and ensure that it is clean.
- Study the traffic flow and visits to the pages on the Web site. Some search queries might generate more traffic to the Web site.

- Identify possible optimizations and ensure that Google can crawl and index all content to help Google index and represent the Web site better.
- Monitor a site's performance in Google search results.
- Find top search queries that list the Web site as a result.
- Locate all Web sites that link to the blog or Web site.
- Associate the Web site with the most common keywords.

7.3.1 Google Site Verification

Before Google can crawl and index a Web site and display its search results, it needs to verify that the site exists and that the Webmaster owns it.

Webmasters can verify their Web site with Google using the following methods:

- By uploading an HTML file
- Via their domain provider
- By adding a Meta tag
- By using the Google Analytics tracking code
- Via the Google Tag Manager

These are described in detail as follows:

- **Verifying a Web Site by Uploading an HTML File**

Verification by uploading an HTML verification file is the Google-recommended method for Web site verification. Following are the steps to verify a Web site by uploading an HTML file:

1. Open the Google Webmaster Tools home page at **www.google.com/Webmasters/tools**.
2. Click the **Manage Site** button next to the name of the Web site that needs to be verified and then, select the **Verify this site** option from the drop-down menu.
3. Download the HTML verification file from the link as displayed on the **Recommended method** tab.

Figure 7.1 shows an example of a HTML verification file that can be downloaded.

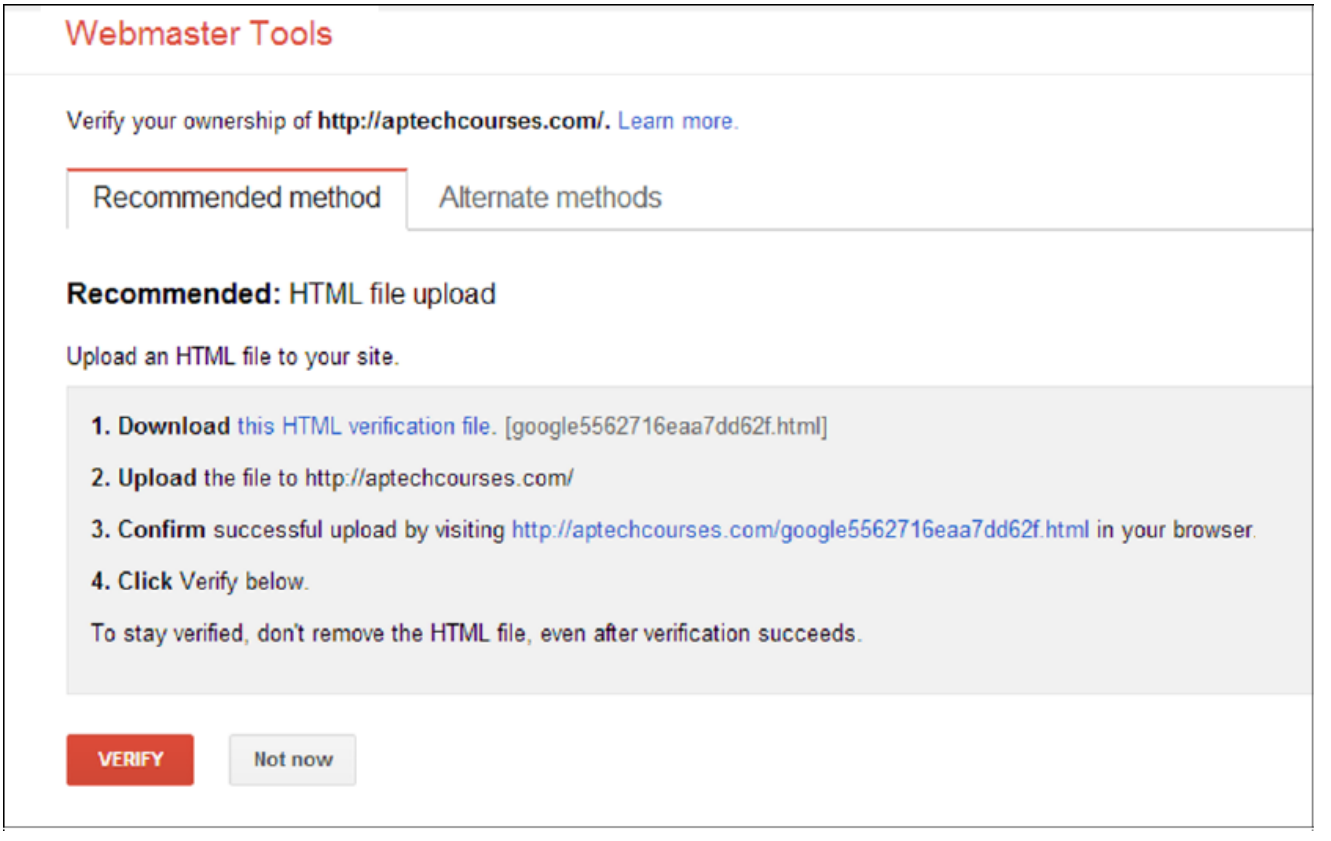


Figure 7.1: Verifying Web Site By Uploading HTML Verification File

4. Upload the downloaded verification file to the root of the Web site folder on the Web server hosting it. To do this, you may need to use an File Transfer Protocol (FTP) client.

5. Once the HTML file has been uploaded, click the **VERIFY** button at the bottom of the Google Webmaster Tools page.

Verification via the Domain Name Provider, by adding a Meta tag, by using the Google Analytics tracking code, or via the Google Tag Manager are the alternate methods of Web site verification. These methods are listed on the **Alternate methods** tab adjacent to the **Recommended method** tab.

Figure 7.2 displays the **Alternate methods** tab.

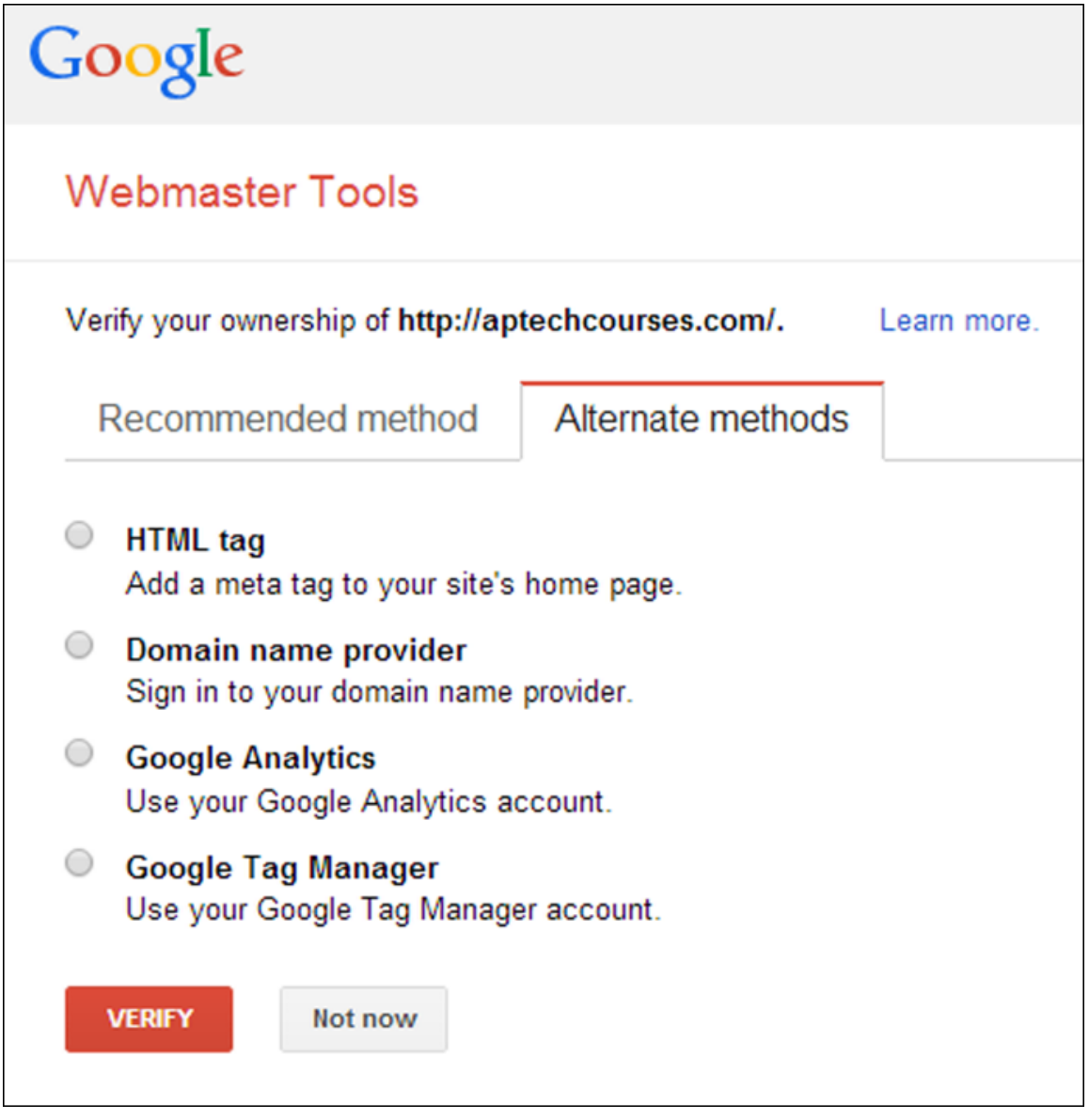


Figure 7.2: Alternate Methods for Web Site Verification

- **Verifying Web Site via the Domain Name Provider**

Following are the steps for verifying Web site via the domain name provider:

1. On the Webmaster Tools home page, click the **Manage Site** button next to the name of the Web site to be verified.
2. Click the **Verify this site** link from the drop-down menu.
3. Click the **Alternate methods** tab.

4. Select the **Domain name provider** option.
5. From the drop-down list that is displayed, select the Web site's domain registrar or service provider. This requires signing into the domain service provider's server from within Google Webmaster tools Web site. If the Web site domain name provider is not listed, select **Other** in the list. This will display a DNS TXT record, for the Web site. Copy this record text and add it to the DNS configuration for the Web site on the Web server hosting it.
6. Click the **VERIFY** button.

Verify Web Site by Adding a Meta Tag

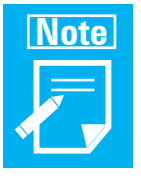
Webmasters can also have their Web site verified by adding a Meta tag to their Web site. To do so, select the **HTML tag** radio button on the **Alternate methods** tab. Copy the Meta tag that is displayed and paste between in `<head>` section of the Web site's home page.

Verifying Web Site Using Google Analytics Tracking Code

Webmasters using the Google Analytics tool can have their Web sites verified using a Google Analytics tracking code. However, before using this verification option, ensure that the Google Analytics tracking code is appropriately placed between the `<head>` tags in the Web site home page. This tracking code can be generated from their Google Analytics Web page.

Verifying Web Site Using Google Tag Manager

Webmasters using Google Tag Manager for managing their Web site can have the Web site verified by adding a Google Tag Manager snippet code to their Web site home page. Before using this verification option, ensure that the Google Tag Manager container snippet code is appropriately placed in the Web site home page. The Webmaster needs to have View, Edit, and Manage permission for their Google Tag Manager account.



Google Tag Manager is a free tool that provides Webmasters a user-friendly interface for updating the Web site tags and mobile applications easily without the need for intervention of the software programmers.

Once submitted though, Google will automatically upload updates from the site map file without any further work.

Once loaded, the number of URLs submitted by the site map and the number of URLs currently in Google's search index can be seen.

If the site is new, it may take a while for Google to index the pages. As a result, the number of URLs indexed may be 0 or very low. Once Google starts to index your pages, this number will automatically rise.

7.3.2 Changing Verification Method

It is possible to change the verification method at a later stage. Following are the steps to do so:

1. On the Webmaster Tools home page, click the Web site link for which the verification method needs to be changed.
2. Click the gear icon at the top corner of the Web page and select **Verification details** from the drop-down menu.
3. The page displays the verification method currently used. Click the **Verify using a different method** link displayed under it.
4. Select the required verification method and follow the new verification process.

Sometimes, the verification process might fail and give some error. Depending on the type of verification method used, Webmasters may face errors such as:

- General verification errors
- HTML verification file errors
- Meta tag verification errors
- DNS verification errors

These are described in detail as follows:

- **General verification errors**: These include errors in verification due to server timeout, a busy or down server, a DNS error due to a down server, or a DNS routing problem, or problems with password authentication.

  When faced with such general verification errors, check that the server is not down and that the domain address is being resolved correctly.
- **HTML file verification errors**: These include errors such as verification file not found, verification file as the wrong content, or verification file is empty. When faced with such errors, download the HTML verification file from the Google Webmaster tools Web site and upload to the correct location on the Web server hosting the Web site. Do not make any changes to the file name or contents of the file while doing so, as then Google will not be able to verify the Web site ownership.

  HTML file verification error might also occur if the verification file on the Web site returns an HTTP status code of XXX instead of 200 (which indicates OK). The Google Webmaster Tools will then not be able to verify the Web site.
- **Meta tag verification errors**: These errors are typically caused by missing Meta tag code or incorrect placement of Meta tag code. Copy the Meta tag from the Google Webmaster Tools Web site and make sure that it is pasted within the `<head>` section after `<title>` on the Web site's home page.

  Meta tag verification error might also occur if the Web site home page returns an HTTP status code of XXX instead of 200 (which indicates OK).

HTML file verification does not work for Web sites that redirect to other Web sites or to a disallowed location. It is recommended to use the Meta tag verification method for such Web sites.

- **DNS verification errors**: These include DNS not found, DNS wrong TXT, DNS resolve permanent error, and DNS resolve errors. These errors are typically caused if the DNS TXT record was not found on the Web site, the DNS TXT record contains the wrong content, or if Google encounters an error while looking up the DNS TXT record.

7.3.3 Google Webmaster Tools Dashboard

Figure 7.3 displays start page of Google Webmaster Tools.

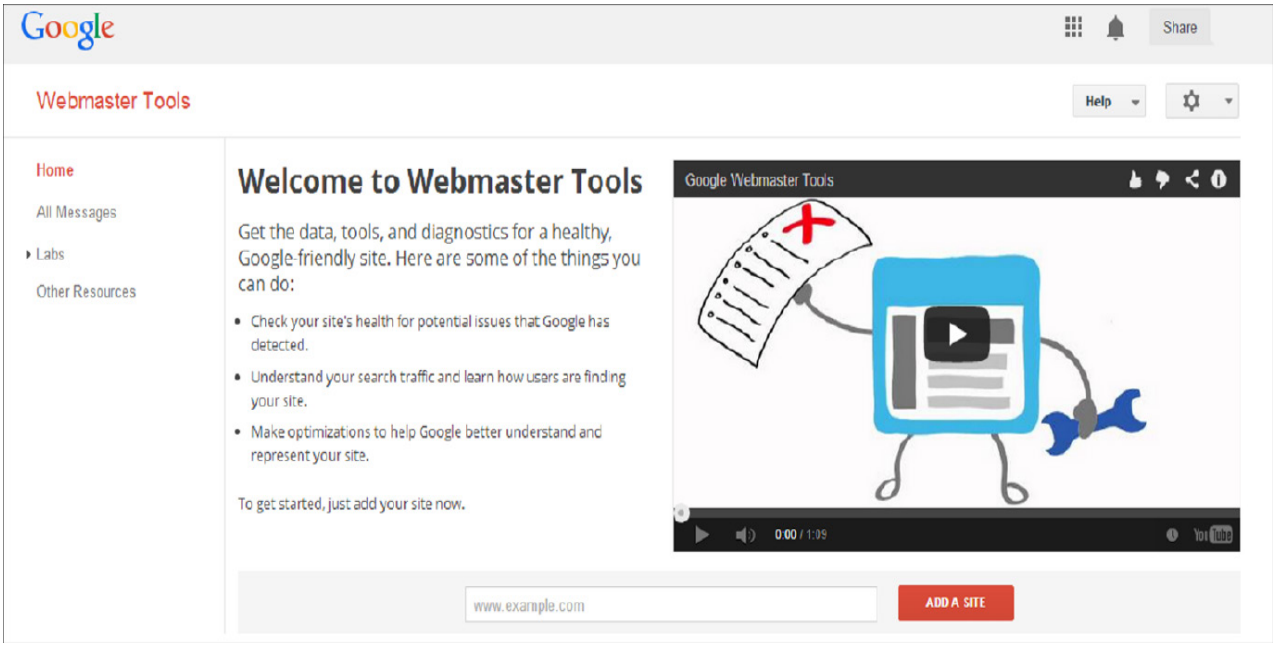


Figure 7.3: Google Webmaster Tools

Once you have successfully added a site and verified it, you can see the dashboard.

Following are the different tabs available on the dashboard of the Google Webmaster Tools:

- **Overview** - The Overview tab gives a quick look at:
  - When the Web site was last crawled
  - Whether any of its pages are included in Google's index
  - Summary of Web crawl errors in the Web site such as broken links, unreachable pages, or server problems

- **Diagnostics** - The Diagnostics tab lists problems encountered during its latest crawl of the site such as:
  - **Web crawl** - lists any pages or URLs that had errors, including URLs not followed, not found, restricted, or unreachable. This information helps to fix any errors with the Web site
  - **Mobile crawl** - lists the problems encountered on a mobile Web site. In order to view this, include the Web site in the mobile index and submit a Mobile Site map
  - **Content analysis** - helps to find problems such as missing or duplicate title tags, and Meta descriptions
  - **Non-indexed content** - helps to review Web pages containing non-indexable content such as Flash files or images that may have trouble being crawled
- **Statistics** - The Statistic tab helps to find:
  - **Top search queries** - gives an idea about the top searched keywords used by visitors to find the Web site
  - **How Google sees your content** - lists the anchor text used in links to your site
  - **Crawl stats** - helps to know the estimated importance of the Web page
- **Links** - The Links tab provides the information about:
  - Where your traffic is coming from
  - What kind of sites are interested in your site
  - All the internal links Google has found on your site
- **Sitemaps** - The Sitemaps tab displays information about:
  - Site map status
  - The date it was last downloaded by Google
  - Errors visitors may have encountered while crawling the sites listed in your Site maps
- **Tools** - The Tools tab offers the following functionalities:
  - **Analyze robots.txt** - Helps to show the content to Google and also ensure to keep blocked pages out of the results
  - **Manage site verification** - Helps to control who has access to detailed information about the Web site
  - **Set crawl rate** - Helps to request Google for change in frequency it crawls the Web site
  - **Set geographic target** - Helps to target users in a specific region

- **Set preferred domain** - Helps to ensure the domains entered in the dashboard is the chosen preferred domain
- **Enable enhanced image search** - Helps to allow Google to include the images in detailed image search
- **Remove URLs** - Helps to remove pages from the index

7.3.4 SEO Best Practices When Using Google Webmaster Tools

Following are the best practices to make the most of Google Webmaster Tools as an SEO tool:

- Sign up for e-mail forwarding
- Check out search queries
- Use keyword to target content
- Reduce duplicates through HTML suggestions and URL parameter handling
- Diagnose crawl errors
- Prioritize content through internal links
- Verify access through Fetch as Googlebot
- Use site performance to improve speed

7.4 Google KeywordPlanner Tool

Keywords are absolutely fundamental to SEO strategy. It is a very powerful tool that helps build new search and ad campaigns for a Web site as well as optimize existing campaigns. Webmasters can check which combination of keywords and ad group ideas give best search results. The most popular keyword research tool is Google Keyword Tool, which helps to:

- Discover targeted keywords proposition
- Show the level of competition for the chosen keywords
- Determine anticipated traffic volumes
- Suggest in-trend keywords

Following are the steps to use Google Keyword tool:

1. Sign in to AdWords account. A Google AdWords account is required for this.

    - If you already have a Google AdWords account, just sign in to the account.
    - If a Google account already exists and the same login information is to be used for the AdWords account, sign in to the Google account, and follow the instructions.
    - If a Google account already exists but you would not like to use the same login information for the AdWords account, you can create a new AdWords account. The procedure to do this is as follows:

        i. Click **Sign up now**.

ii. On the **Create Google Account** window, select '**I have an e-mail address and password I already use with Google services like AdSense, Gmail, Orkut, or iGoogle**' option.

iii. Choose whether you like to have a single account for all Google services.

iv. Select the first option to use the existing Google account for AdWords.

v. Then, type the Google account login e-mail and password in the fields provided, and then click **Sign in**. This step is shown in figure 7.4.

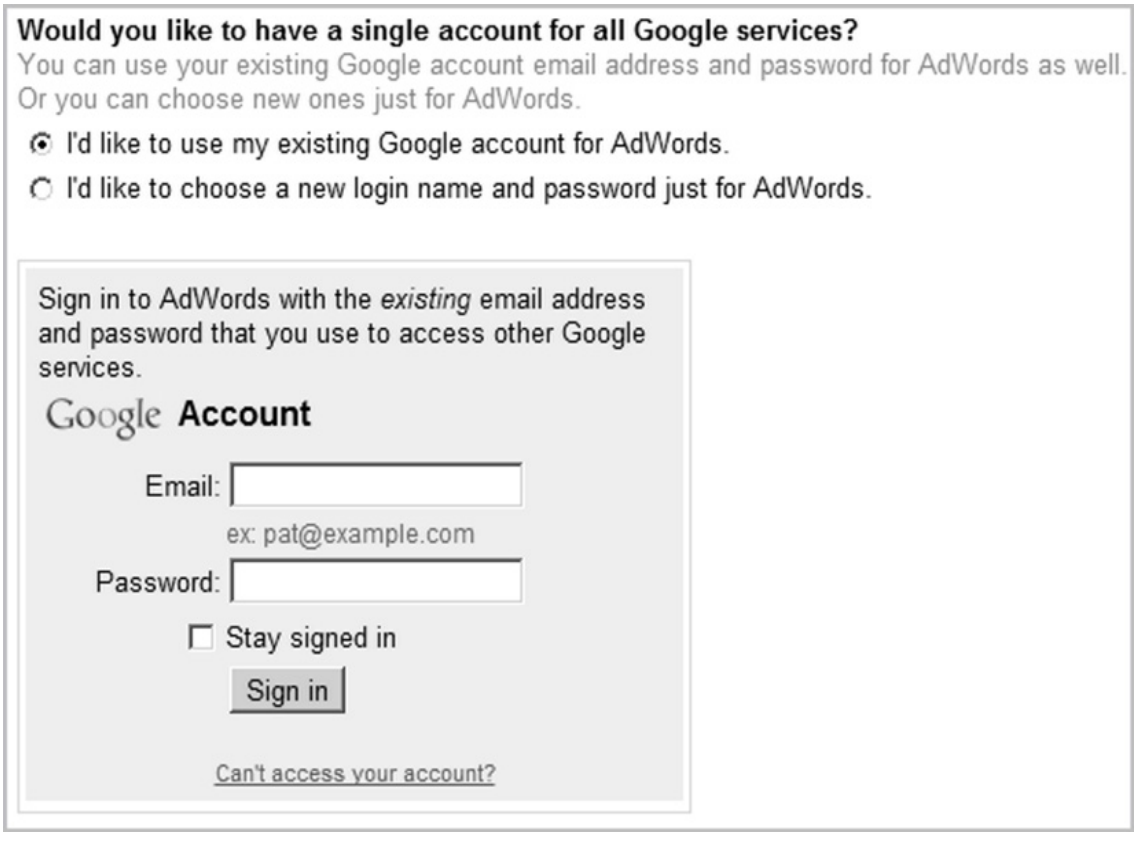


Figure 7.4: Google Account Sign Up for AdWords

vi. Select the second option, if you prefer to use different login information just for AdWords, as shown in figure 7.5.

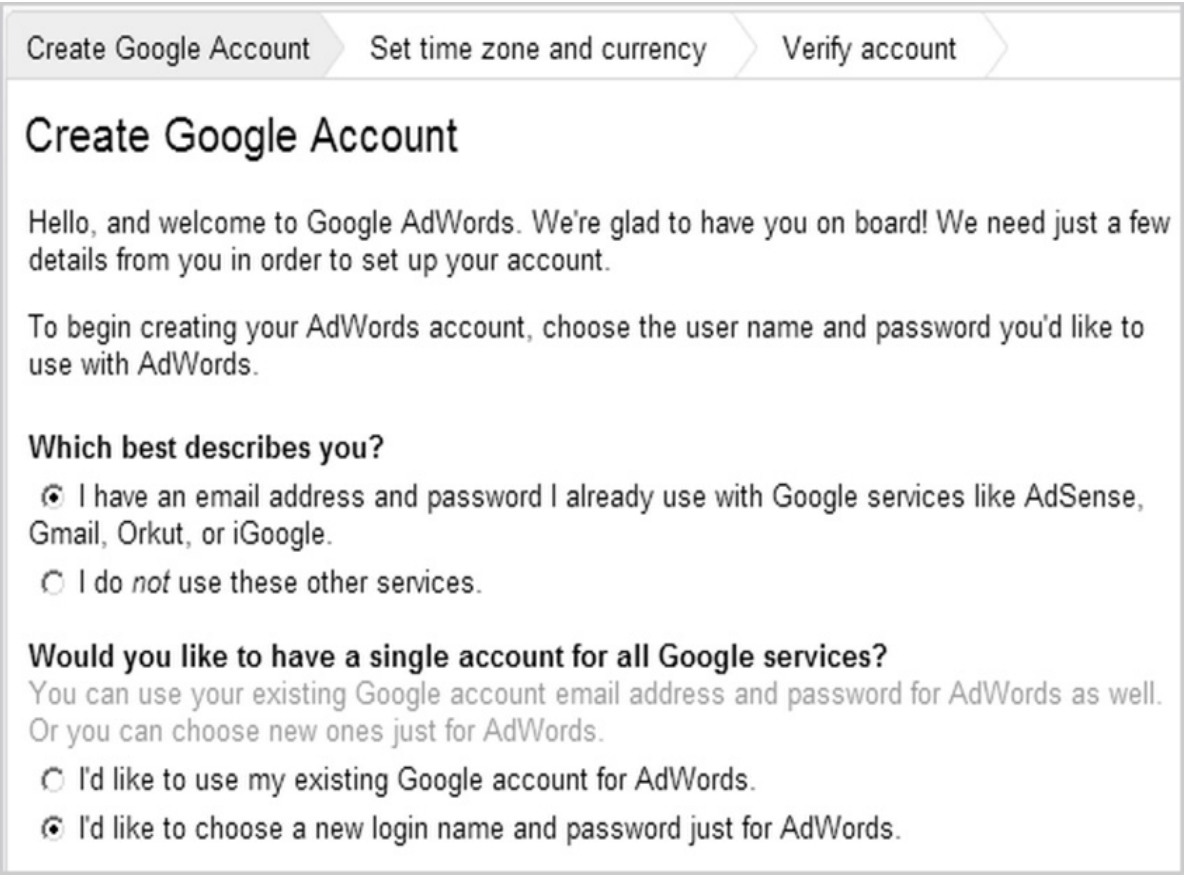


Figure 7.5: Google Account with a New Login and Password for AdWords

vii. Type the e-mail address and the password you will use for the AdWords account, and then click **Create Account**. You will receive an e-mail asking to click a link to verify the e-mail address. Click the link in the message and then sign in to your account with the e-mail address and password submitted on the registration form.

- If you do not have a Google account or Google AdWords account, the procedure to create the account is as follows:

  i. Click **Sign up now**.

  ii. On the **Create Google Account** window, select '**I do *not* use these other services**' option.

  iii. Type the e-mail address and the password for the new AdWords account, and then click **Create Account**. An e-mail asking you to verify the AdWords account will be received on the e-mail address provided on the registration form.

2. To start the Keyword tool, select **Keyword Planner** from the Tools menu. Figure 7.6 depicts the Google Keyword Planner Tool.

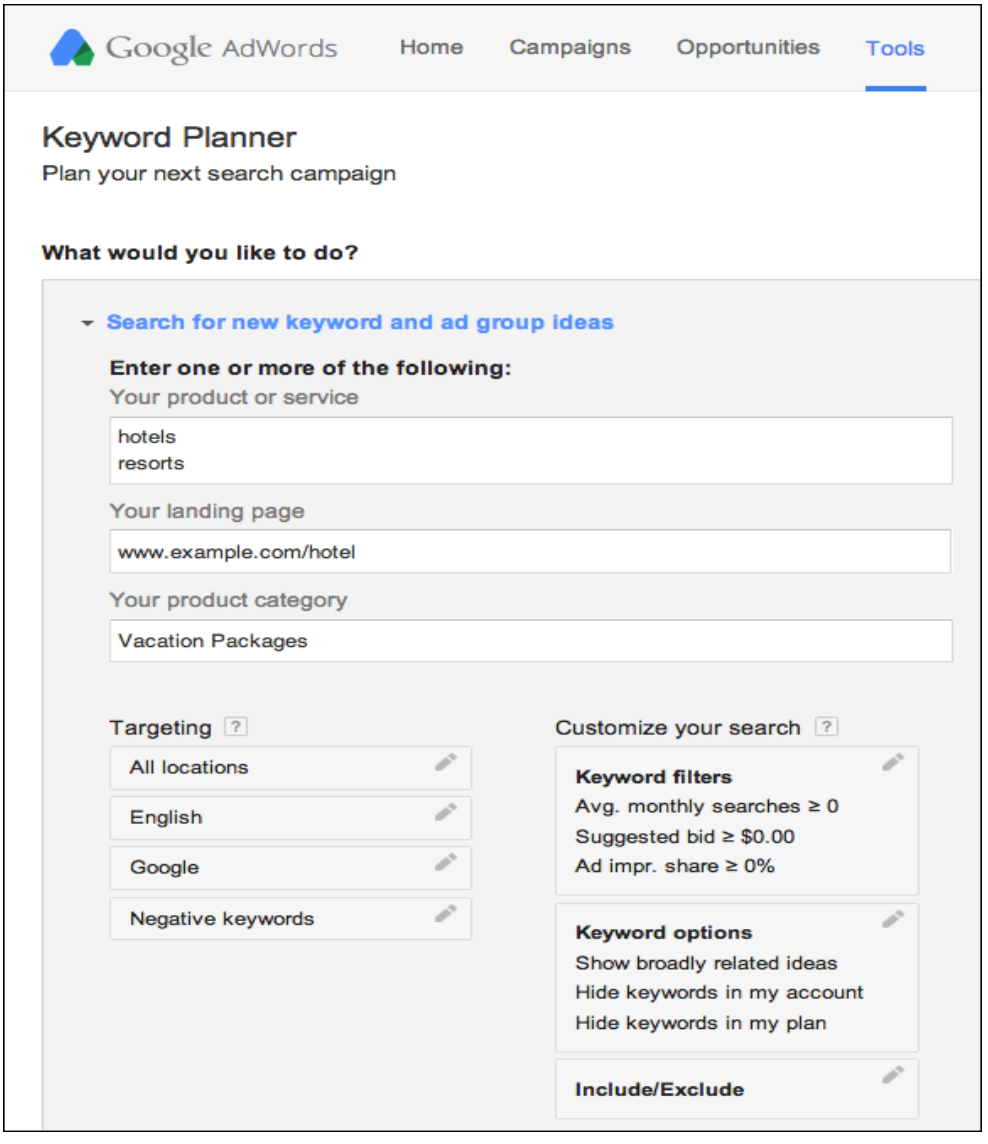


Figure 7.6: Google Keyword Planner Tool

3. Click the **Search for new keyword and ad group ideas** link and type a phrase or a group of words in the **Your product or service** text box.

4. Enter the Web site address in the **Your landing page** text box.

5. In the **Your product category** text box, type or select the relevant category. Click the **Get Ideas** button to display the search results. To further refine the search results, specify the target location, and search engine. The search results can be further customized under the Keyword filters, **keyword options**, and **Include/Exclude** sections.

7.5 Google Insights for Search Tool

Google Insights for Search is a tool that provides robust information and can be used for market intelligence and research for free. It shows what people are searching for, where they are searching, and when.

This tool categorizes search results based on location, time frames, and properties.

From a marketing point of view, Google Insights can be used for:

- Choosing advertising messages
- Examining seasonality
- Creating brand associations
- Entering new markets

The **Google Insights for Search** tool can be used as follows:

1. Go to the **Google Insights for Search** page (**http://www.google.com/insights/search**). By default, the **Search terms** option will be selected in the **Compare by** section. The options mentioned can be used to:

   - Enter the terms to view the keyword statistics
   - Compare the search results using the **Locations** option
   - View the trends within a time frame

   The default page of Google Insights for Search tool is shown in figure 7.7.

Figure 7.7: Google Insights for Search Tool

2. Compare two terms or products by using the **Add search term** link below the first text box. The search result helps to choose which one of the two can fetch more traffic and money.

   The search terms results displays the top searched keywords or terms as well as the keywords for which the search have increased within the time frame. This information can be used to estimate the number of clicks for the keywords or terms.

3. Set any filters that are required. By default, **Filter** category shows Web Search, Worldwide, 2004 - present, and All Categories. An advertiser or a marketer can filter the search results based on a category or country.

4. Once all the parameters are set, click **Search**. The interest for the search terms over the years or the keyword trending will be displayed through a graph.

Following are the best practices to make the most of Google Insights for Search as an SEO tool:

- **Targeting Seasonal Search Terms**: Online marketers can plan and deploy new landing pages specific to seasonal search terms.
- **Combining Insights for Search and the External Keyword Tool**: Search terms section of the Google Insights dashboard provides an insight into the most common related search terms. Analyze these terms using the Google Keyword Tool to identify the traffic volume as an actual number for specific search terms. Conduct keyword research with much greater confidence and efficiency by combining these two tools.
- **Leveraging Search Volume Spikes Using Google Trends**: Google Insights for Search displays a list of top fastest rising search queries with an option to drill-down to find specific information related to each keyword such as peak search time, search related keyword, top Web results, new result, and blog results. Use Google Trends to quickly identify keywords and topics, which can leverage search volume spikes by publishing relevant information on the Web site.

7.6 Google Trends Tool

Google Trends is a free tool to view the kind of content people search. This tool helps to make a decision whether or not to enter a certain market and compares the volume of searches between two or more markets/ items. This type of market research is very beneficial to predict, by looking at the way a certain market is trending over a period of time. Figure 7.8 shows the Google Trends tool.

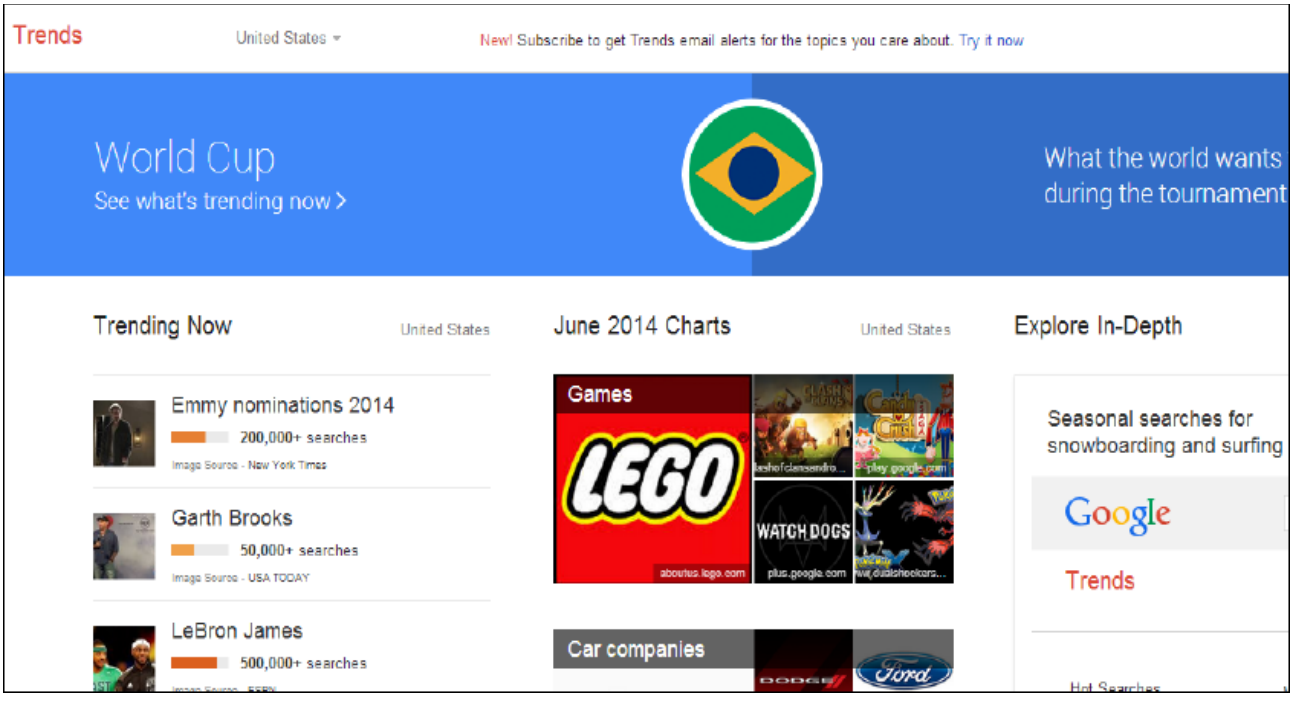


Figure 7.8: Google Trends Tool

Google Trends data can be used for SEO in the following ways:

- **Analyze Crowd Behavior**: Use the Google Trends graphs to build an accurate time line of popular events around a phenomenon.
- **Analyze Event Sequence and Significance**: Google Trends gives insight into most popular historical news stories that helps in finding more targeted keywords.

The key difference between Google Trends and Google Insights for Search are as follows:

- Google Trends only shows the trends and the areas where people are searching. While, Google Insights shows not only the same information but also the keywords, and most importantly the rising searches from within that topic.
- Google Insights for Search is more beneficial for the advertisers and researchers.

7.7 Google Analytics Tool

Google Analytics is a free statistics tracking and analysis service that tracks:

- Where visitors came from
- What they did on a site
- E-commerce data
- Conversion information

This information is presented in an easy-to-read and comprehensive manner, through intuitive visual reports.

This tool helps the Web site to improve return on investment, increase conversions, and make more money on the Web. This is specifically beneficial to e-commerce Web sites.

Following are the benefits of using Google Analytics:

- **Track multiple Web sites**: Google Analytics allows tracking any number of Web sites from a single account. The reports of each domain and sub-domain can be viewed separately.
- **Track blogs, WordPress, or Facebook pages**: Google Analytics also allows tracking and monitoring activities on social network media pages such as Facebook or WordPress.
- **Track traffic from RSS feeds**: Google Analytics can be used to track Web site traffic from RSS feeds. However, this data is not accurate and complete. Google Analytics cannot count page views that are loaded through RSS readers. This is because most RSS readers cannot execute the Google Analytics tracking code, which is in JavaScript format.

Figure 7.9 illustrates the Google Analytics tool.

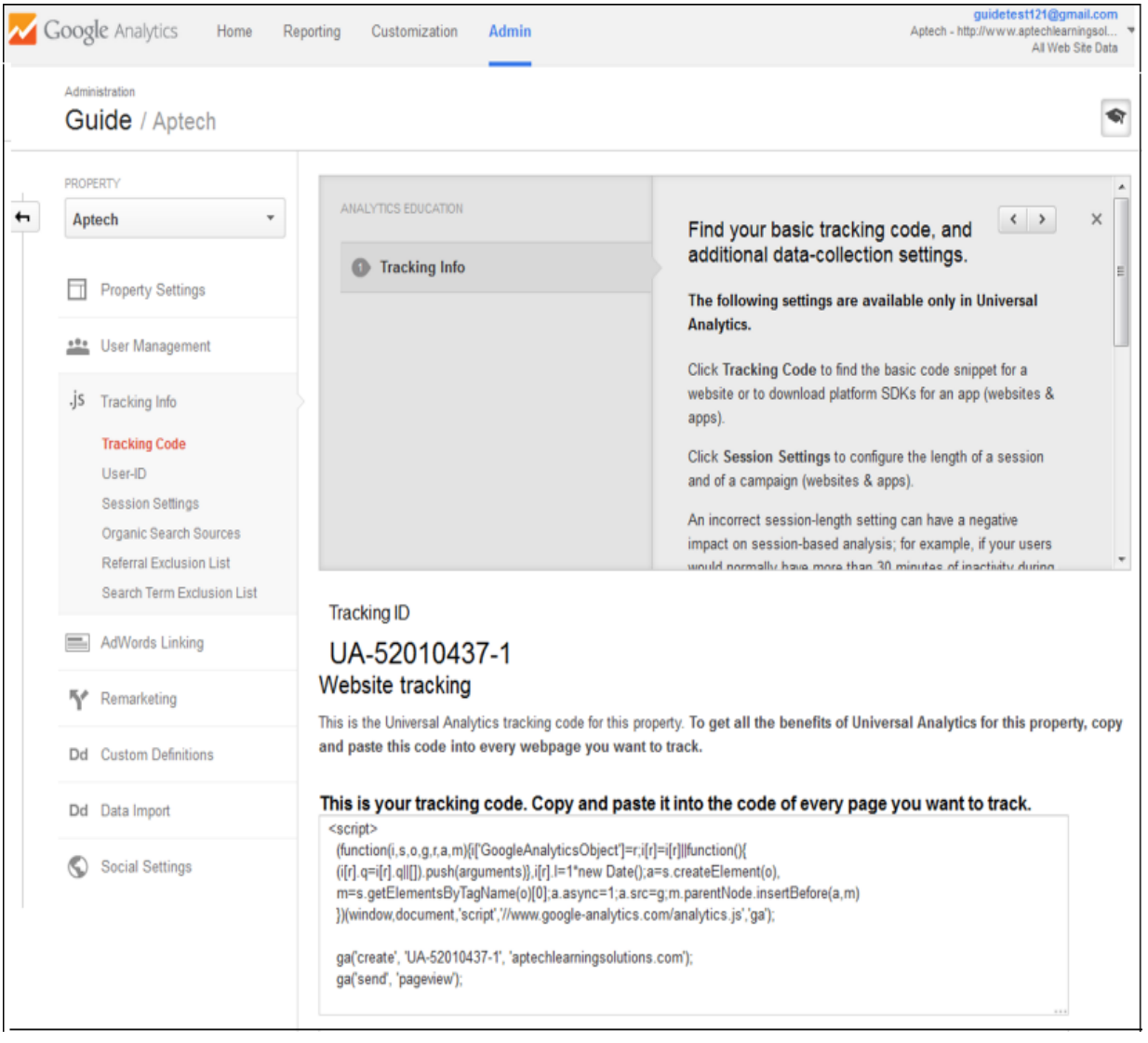


Figure 7.9: Google Analytics Dashboard

Following are the steps to use Google Analytics tool:

1. **Set up a Google Analytics account** - This will provide a tracking ID to add to the Web site. Only the Web site owner gets access to the information about the Web site or, to those special permission is given.

   The procedure to do this is as follows:

   i. Go to the Google Analytics Web site: **http://www.google.com/analytics/**

      You need to have a Google account to sign in. If you do not have a Google account, create one by clicking the **Sign Up Now** button and following the instructions given.

   ii. If a Google account exists, enter the e-mail address and password, and then click **Sign In**.

   iii. Enter an account name that will be used to track Web sites.

   iv. Enter a name for the Web site that needs to be tracked by Google Analytics.

   v. Enter the URL of the site to be analyzed.

   vi. Select the country along with the appropriate time zone.

   vii. Click the **Get Tracking ID** button.

   viii. Click the **I Accept** button to accept the terms of service. A Google Analytics tracking code snippet is displayed.

   ix. Copy the block of code and insert it into every Web page of the Web site that needs to be tracked.

Once a Web site is added to the Google Analytics account and the tracking code is in place on the Web site pages, users can start tracking the Web site performance.

The Google Analytics home page displays an expandable list of accounts and Web site properties set up to be tracked and columns to display information such as Visits, Average Visit Duration, Bounce Rate, and Ecommerce Conversion Rate.

There are four links displayed at the top of the Google Analytics Web page, as follows:

- **Home**: Displays the home page of the Google Analytics Web site for the account.
- **Reporting**: Displays various types of reports and sub reports including intelligence reports related to the Web sites being tracked. This page allows viewing numeric data as well as graphic and visual representation of data gathered by Google Analytics tracking.
- **Customization**: Allows creation of custom reports and analysis categories for the Web sites being tracked.
- **Admin**: Allows creating new properties and specifying Web site data to be tracked.

7.8 Google Analytics Content Experiments

Changes and improvements to Web sites is a constant process. Based on the Google Analytics reports, Webmasters try to make changes to the Web site content and design for increased visibility and search engine optimization of the Web site.

Google Analytics Content Experiments is a tool that is available as an extension of Google Analytics. It allows Webmasters to test the Web site changes and view its impact on SEO. It helps define optimization goals for the Web site and test which designs, layouts, and content are providing the most effective results in achieving these goals.

Google Content Experiments allows Webmasters to:

- Compare the performance of various Web pages or app screens using random samples of users
- Define the percentage of users that should be included in the user sample for experiment
- Define the optimization goals that need to be tested
- Set up to receive experiment result updates by e-mail

Google previously provided a feature called Google Web site Optimizer tags. This feature is no longer supported. Therefore, it is recommended to remove all Google Web site Optimizer tags that might be included in the Web site.

- **Experiments with Page Elements**

  Webmasters can create different versions of Web pages and test them using Google Analytics Experiments tool. They can test page elements such as headlines and headers, images and icons, text, calls to action, and page layouts.

- **Experiments with Analytics Goals**

  Webmasters can setup various types of optimization goals and test them in Google Analytics Experiments. This involves testing Web site design and content placement strategies that help pull maximum users to view specific pages of the Web site. Some of these goals are as follows:

  - **URL destination goals**: The goals typically target to get maximum traffic to pages that sell products or display the business location details, or advertise specific products or services.
  - **Event goals**: These goals target getting maximum users to perform actions such as signing up for newsletters, viewing videos, or buying products.
  - **Session duration goals**: These goals target to get visitors to spend at least a specified amount of time on the Web site so as to learn enough about the products and services offered.
  - **Pages per session goals**: These goals are to ensure that visitors are navigating through the maximum number of pages on the Web site and gathering enough information from it so as to sell the maximum amount of products or services offered.

7.8.1 Working with Google Analytics Content Experiments

To access Google Analytics Content Experiments, sign in to the Google Analytics Web site. Click the **Reporting** link at the top of the page to display Google Analytics Reports. Scroll down in the left pane and click **Experiments** under **Behavior**.

Following are the steps to create an experiment:

1. Click the **Create experiment** button. This displays the **Content Experiments – Create a new experiment** page. Figure 7.10 displays the **Choose an experiment objective** section on the **Google Content Experiments** page.

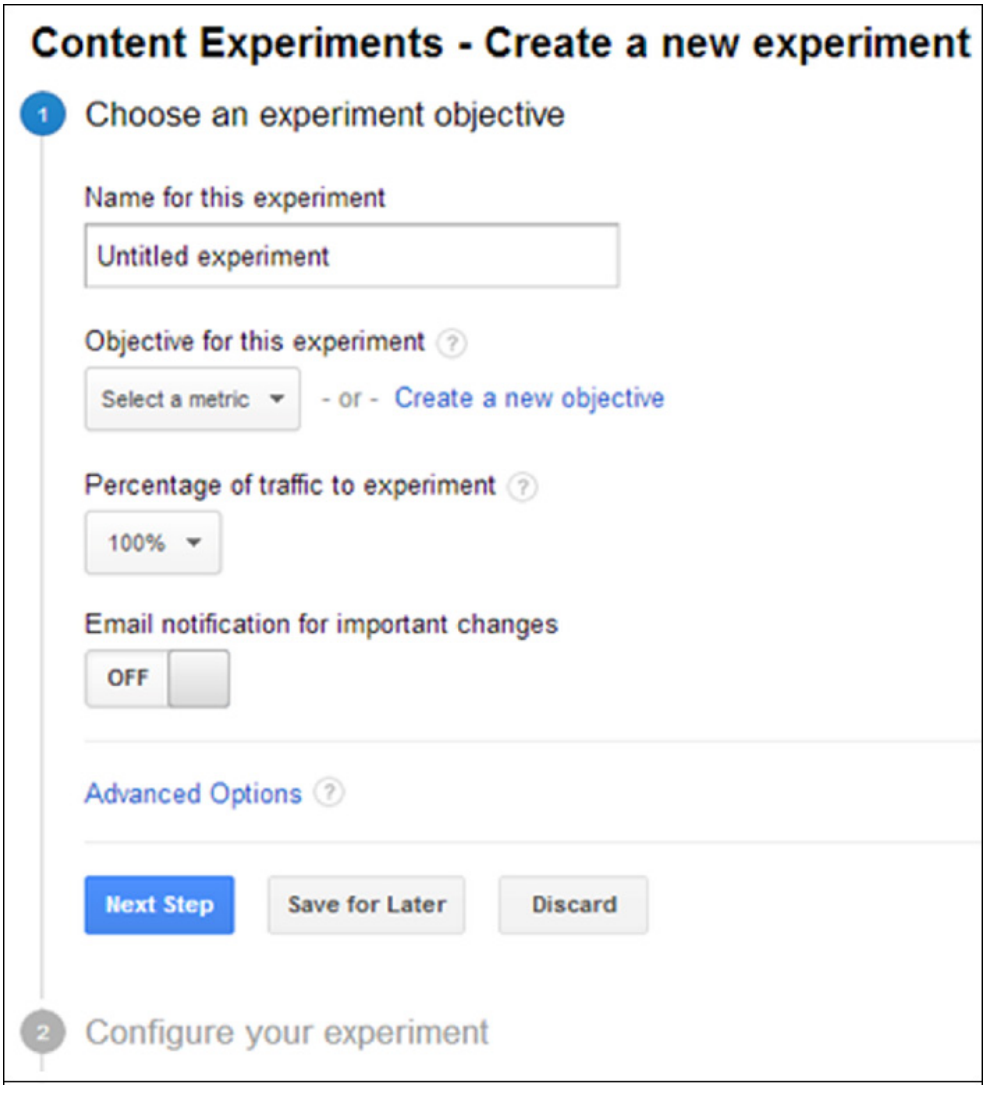


Figure 7.10: Google Analytics Content Experiment –Choose an Experiment Objective

2. In the **Choose an experiment objective** section, specify a title for the experiment in the **Name for this experiment** text box.

3. Under **Objective for this experiment**, click the list box to display a set of options for the goals that need to be measured and improved as part of the experiment, such as Ecommerce transactions or revenue or session duration and page views.

4. From the **Percentage of traffic to experiment** drop-down list, select the percentage of visitors to the Web site whose activities should be monitored as part of the experiment.

5. Specify whether e-mail notification should be sent for important changes to the experiment. If yes, specify the e-mail address to which the notifications should be sent.

6. Click the **Advanced Options** link to configure the following options for the experiment:

   - Whether or not the visitor traffic should be distributed evenly across all variations.
   - Specify the duration for which the experiment needs to be run.
   - Specify the confidence threshold for Google Analytics at which it should declare a winner. A higher confidence threshold might result in Google Analytics taking longer in concluding a winner.

7. Click the **Next Step** button to display the **Configure your experiment** section. Figure 7.11 displays the **Configure your experiment** section of the **Google Content Experiments** page.

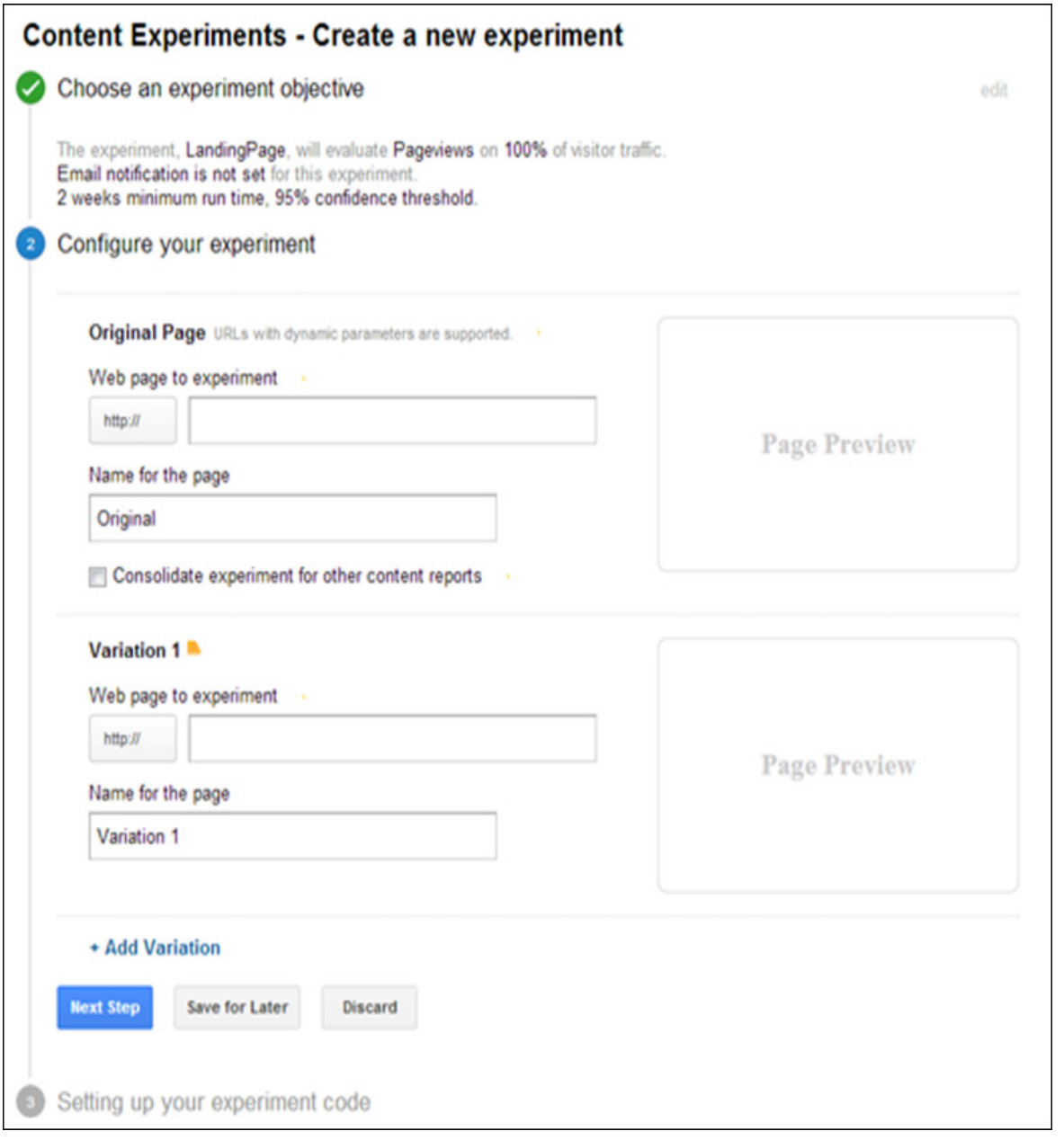


Figure 7.11: Google Analytics Content Experiment – Configure Your Experiment

8. In the **Web page to experiment** text box, enter the URL of the original Web page that needs to be tested as part of the experiment. A preview of the specified Web page is displayed to the right of the text box.

9. Specify a name for this original Web page in the **Name for the page** text box. The default name assigned is **Original**.

10. Under **Variation 1** section, in the **Web page to experiment** text box, enter the URL of the page with a variation of the original Web page. A preview of variation Web page is also displayed to the right of the text box.

11. Specify a name for the variation Web page in the **Name for the page** text box. By default, it is called **Variation 1**.

12. Click the **Add Variation** link if there are more variations of the original page that need to be tested. It is possible to add up to 10 variations of a Web page.

13. Click the **Next Step** button to display the **Setting up your experiment code** section.

14. Here, select an option and specify to insert the experiment code manually or send the code to Webmaster. Figure 7.12 displays the **Setting up your experiment code** section of the **Google Analytics Content Experiment** page.

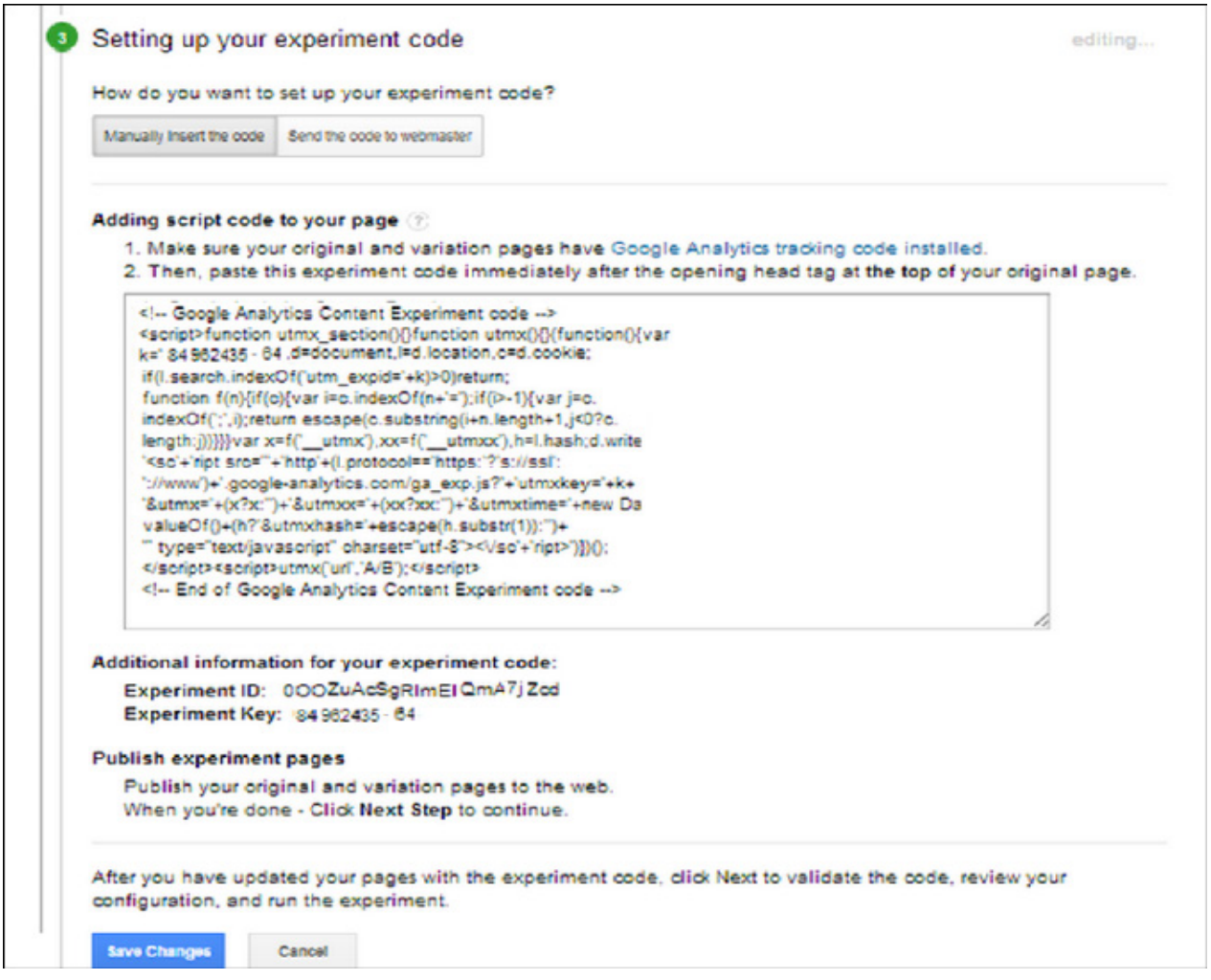


Figure 7.12: Google Analytics Content Experiment – Setting Up Your Experiment Code

Selecting the **Manually insert the code** button will display a code snippet that needs to be copied and pasted within the `<head>` tag of the Web page that is being tested. Google Content Experiments also assigns an Experiment ID and Experiment key for the experiment. Ensure that the Google Analytics tracking code is in place in the Web page. Also, ensure that the original page as well as the variation pages are all published after inclusion of the experiment code.

15. Finally, click the **Next Step** button to review and start the experiment. Figure 7.13 displays Google Analytics Content Experiment – Review and start section.

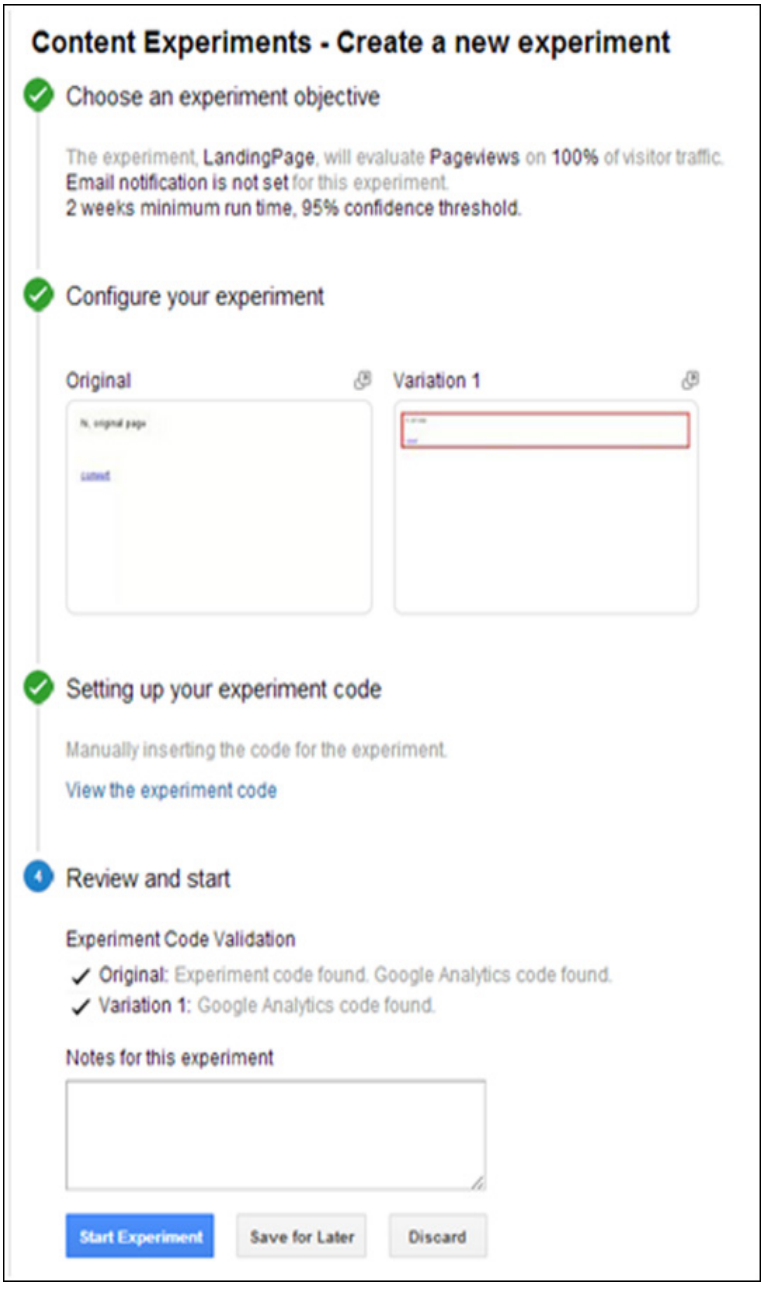


Figure 7.13: Google Analytics Content Experiment – Review and Start

Google Content Experiments validates the experiment code by checking the original Web page for the experiment code and Google Analytics tracking code. It also checks for the presence of the Google Analytics Web-property ID in the variation Web page. In case of error in validation, it displays an error. Ensure that they are all in place and click the **Start Experiment** button.

Once an experiment is created, it is displayed on the Content Experiments main page. It provides an overview of the various experiments with details such as important notifications related to the experiments, the names of the experiments, the start and end dates of each experiment, and the experiments that are currently running. It is also possible to create additional new experiments from this page.

Figure 7.14 shows the Google Analytics Content Experiment – Standard Reporting page.

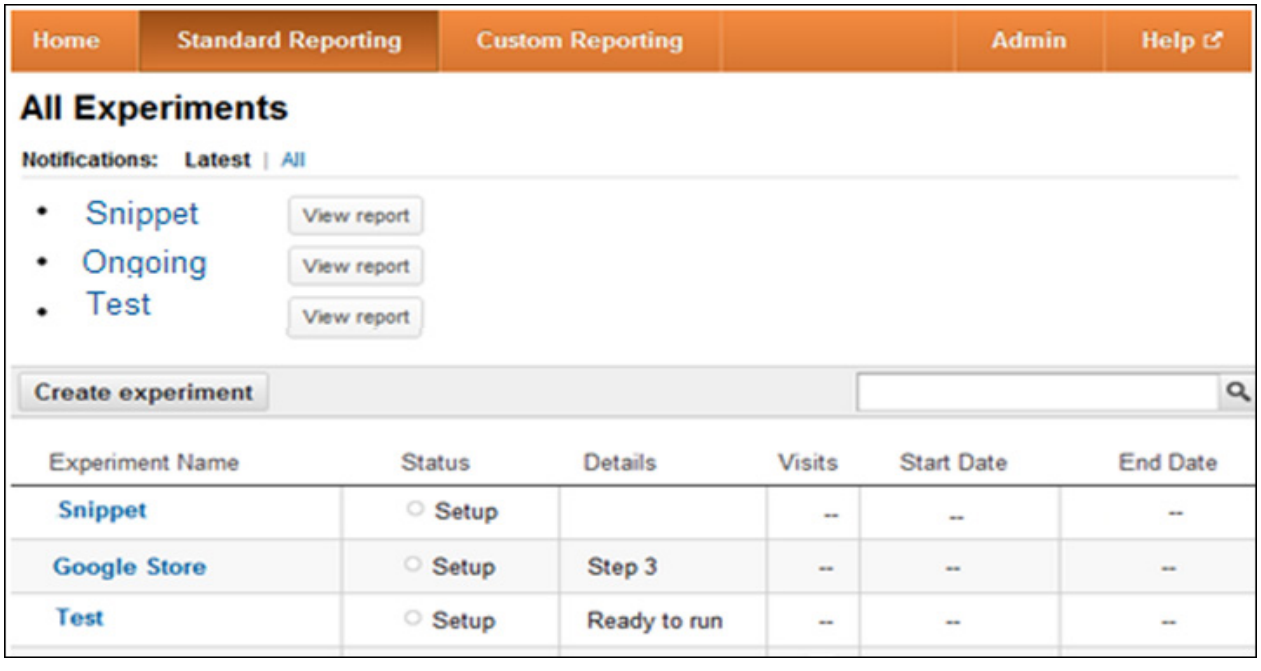


Figure 7.14: Google Analytics Content Experiment – Standard Reporting Page

Clicking an experiment displays details about the experiment as shown in figure 7.15.

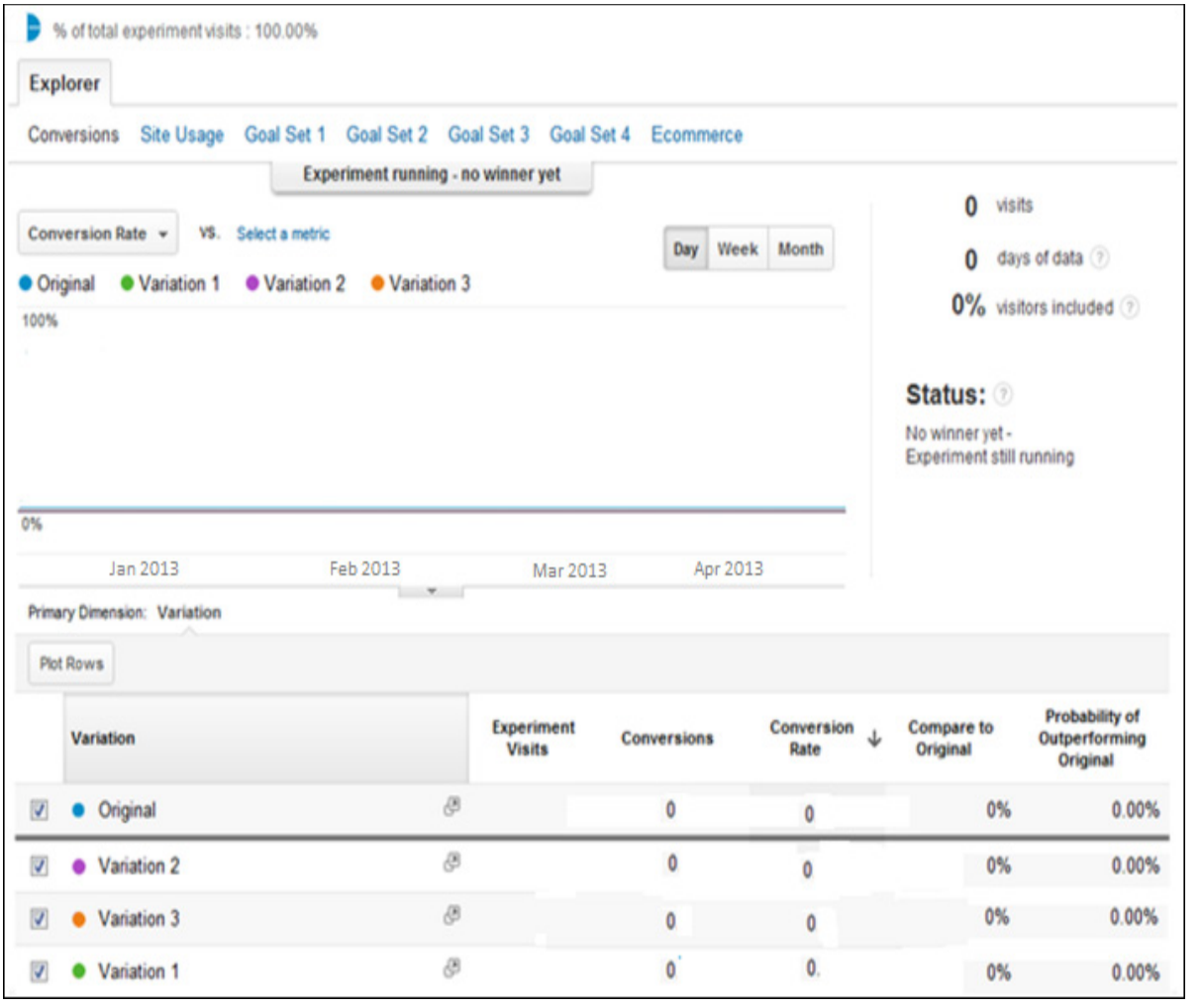


Figure 7.15: Google Analytics Content Experiment – Individual Experiment Details

Users can stop an experiment as well as modify the configurations of the experiment, such as changing the percentage of traffic to experiment and e-mail notifications about change to experiment.

7.9 Experiments and SEO Ranking

It is a wrong SEO practice to display multiple pages with same content on a Web site. Doing this might have a negative impact on the Web site's SEO rankings. To overcome this, insert a `<link>` tag with the `rel="canonical"` attribute in the variation pages. When search engine crawlers see the `rel="canonical"` attribute, it indicates to them that these pages are a variation of the original content for an experiment. It also indicates that only the original page should be indexed and displayed in search engine results.

Users who were part of the experiment and who visited the variation pages might sometimes bookmark them for later reference. After the experiment has ended, Webmasters might decide to leave those pages on the server for such users and redirect them to the original page on visit. For best SEO practices, Web masters can implement two types of redirects:

- **Server-side 301 redirect**: This redirect indicates to search engines that only the original page needs to be indexed and it may forget about the variation pages.
- **Server-side 302 redirect**: This indicates that this is a temporary redirect from the variation pages. While the original page needs to be indexed, the search engine crawlers need to keep checking whether the variation pages have been made live and need to be indexed in future.

7.10 Google Zeitgeist

'Zeitgeist' means 'the spirit of the times'. Google Zeitgeist lists most popular, fastest rising, and fastest falling search terms and keywords for the past years. The annual Zeitgeist report reveals what captured the world's attention in the past year— tend to be a good reflection of what is on everybody's minds as seen through search.

The Google Zeitgeist is calculated by looking at the aggregation of the most popular and fast-rising search queries submitted to Google by users. It includes those search terms that grew tremendously in the past years— the explosive queries, the topics that everyone obsessed over. To come up with this list, several thousand of most popular searches being ranked based on how much their popularity increased compared to previous years.

Using Google Zeitgeist, SEO professionals can look at the fastest rising queries as well as the fastest falling, letting them know what to avoid writing about. The standard SEO practice is to writing quality content on the latest hot topics to drive traffic naturally.

Figure 7.16 depicts the Google Zeitgeist tool.

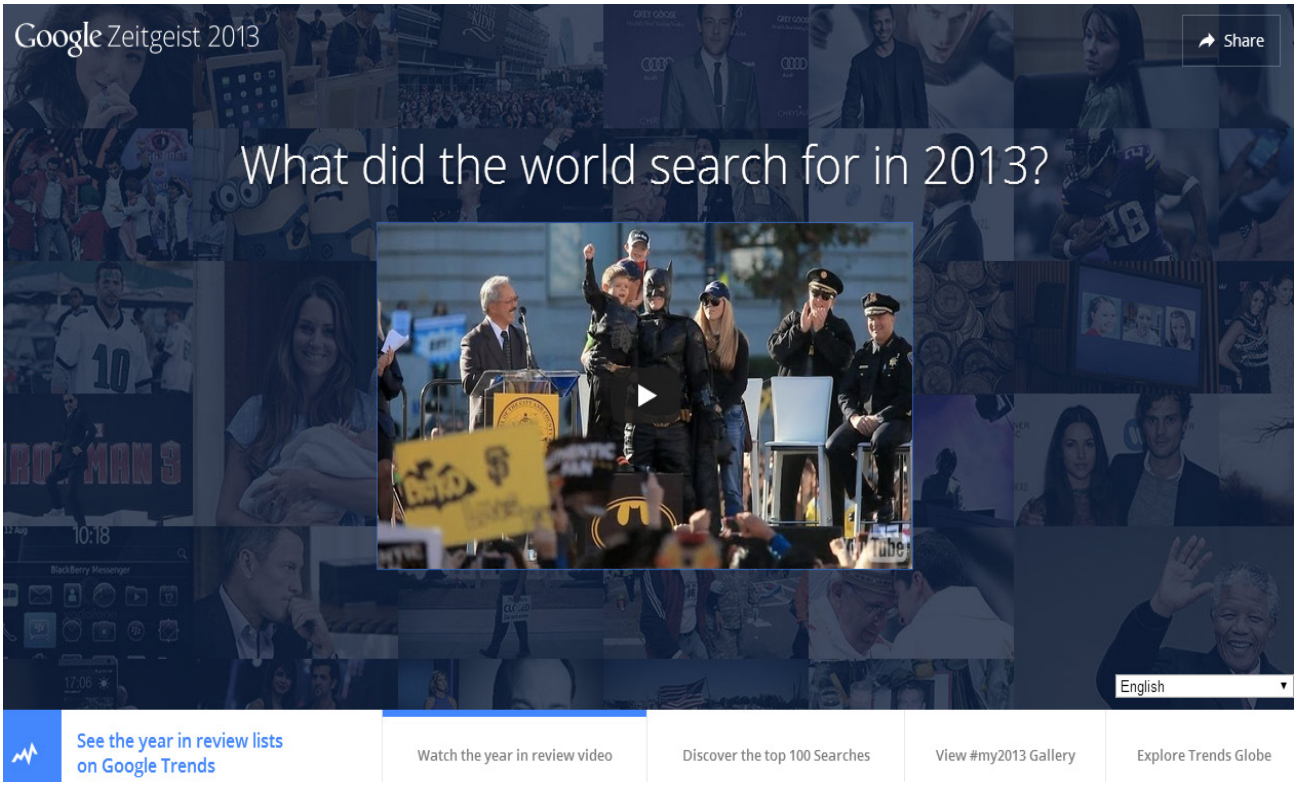


Figure 7.16: Google Zeitgeist Tool

7.11 Google Display Planner

Previously, Google provided an Ad Planner Tool that was a research and medial planning tool to help both advertisers and publishers in their campaign planning drive.

The Ad Planner was designed to address the three key challenges in campaign planning, namely, reach, relevance, and insights. Relevance is being confident that the sites targeting are appropriate to the audience. Insight is where the audience is going on the Internet beyond the obvious sites. Ad Planner addresses all the three challenges.

The Ad Planner has been replaced by another tool called the Display Planner. The Display Planner provides all the features of the Ad Planner plus an integrated Contextual Targeting Tool and the Placement Tool. Besides, since it is built into AdWords, it makes it easier for Webmasters to implement them in their SEO plans.

7.11.1 Benefits of Using Display Planner

Following are the benefits of using a Display Planner to help create a campaign plan:

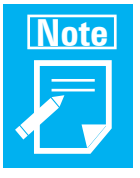
- Get ideas for target keywords, placements, and display media for your campaign.

- Get impression estimates and historical costs for various combinations of campaign ideas to help decide the best plan within your budget.
- Share plans with the team and clients for collaborative work.

7.11.2 Planning a Display Campaign Using Display Planner

Following steps describe how to use the Google Display Planner to plan a display campaign:

1. Open Google AdWords by opening **adwords.google.com** and signing into it. Create an AdWords account if you do not have one.



You may skip the billing process if all you do not intend to use AdWords and only require working with Display Planner.

2. On the AdWords home page, click **Tools** at the top of the page and select **Display Planner** from the drop-down menu.
3. In the **Your customers are interested in** text box of the **Get ideas and estimates section**, enter the list of keywords and topics that might interest visitors to the Web site.
4. In the **Your landing page** text box, enter the URL of the landing page of the Web site.
5. Under **Campaign targeting**, there is a series of drop-down lists in which you can specify the target location for the campaign, the language of the target audience for the campaign, and specify whether it is a direct campaign or a branding campaign.
6. Finally, click the **Get placement ideas** or the **Get ad group ideas** button.

Figure 7.17 shows the **Display Planner**.

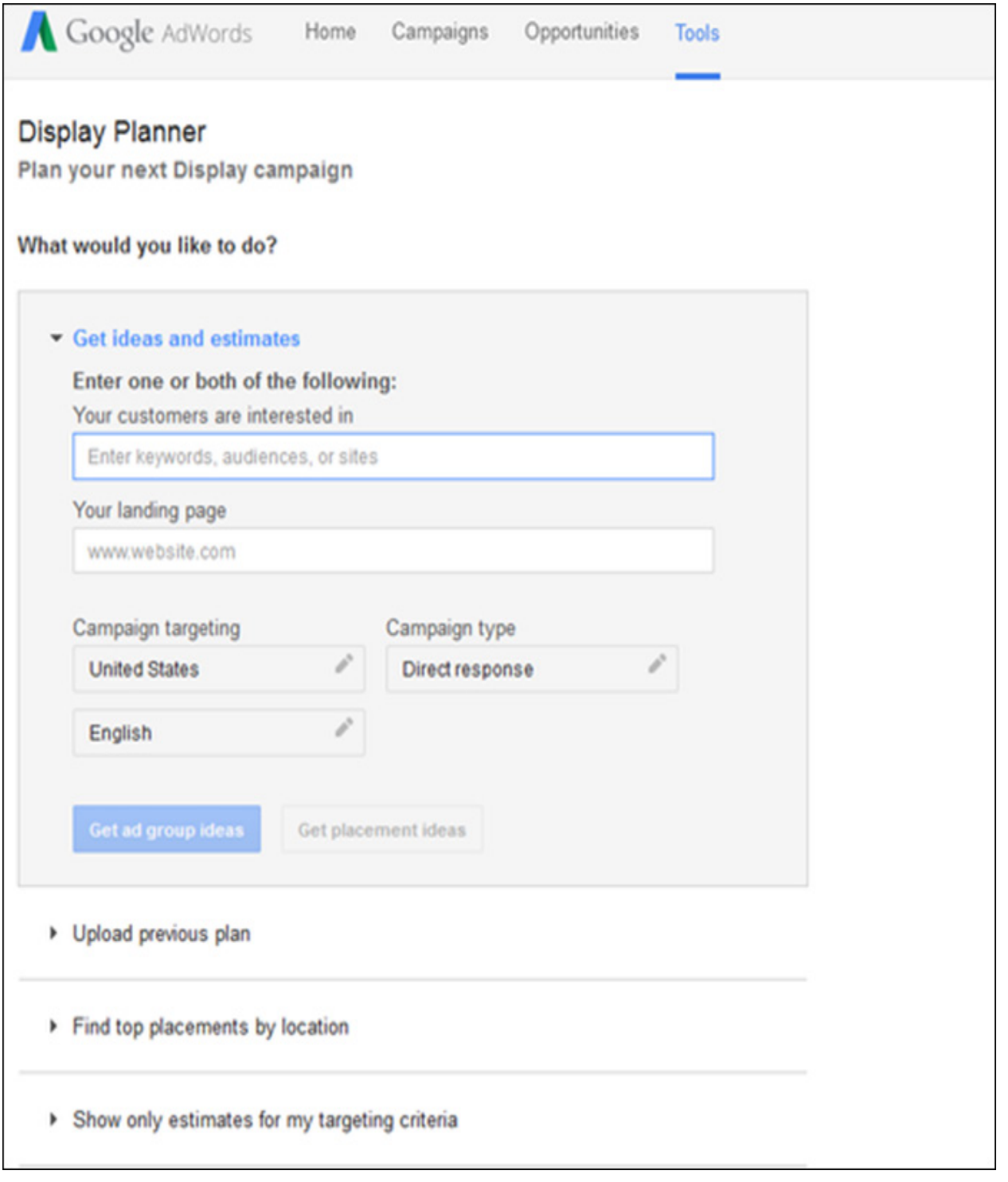


Figure 7.17: Google Display Planner – Planning a Display Campaign

Apart from creating a new plan, Webmasters can browse a previous plan created using Ad Planner and saved as a CSV file, in the **Upload previous plan** section.

The **Find top placements by location** section allows viewing placement results customized by target location for the campaign, the language of the target audience for the campaign, and specify whether it is a direct campaign or a branding campaign. Placements can also be classified by topic.

7.11.3 Using the Display Planner to Create a Campaign Plan

After entering the desired topics of interest, goals, and targets, clicking **Get placement ideas** displays the Display Planner page with an exhaustive list of placement ideas. It also displays network inventory data categorized by age and gender demographics and device users, for the listed topics of interest.

Click the **Individual ideas** tab to view estimated cookies per week, impressions per week, and historical costs for the placement ideas. The target ideas contain exhaustive lists categorized by keywords, user interests, topics, placements, and demographics to match your target goals.

Review each category and select target ideas to narrow down and customize your ad groups to meet your target goals. Clicking the chevron next to the target ideas in each group adds them to the **Your plan** panel on the right. These form your campaign plan.

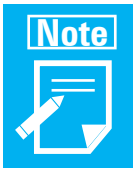
As each target idea is added to the plan, the expected network inventory for the plan is displayed in the **Your plan** panel. The expected network inventory details can be viewed for individual ad groups or for the entire plan.

The left panel on the Display Planner provides options to filter the placement ideas, ad formats, ad sizes, and mobile OS for apps.

7.11.4 Reviewing and Saving Campaign Plans

Once a campaign plan is created, click the **Review plan** button to study the detailed plan and check if the plan meets the Web site goals. Click the **Back to ideas** button at the top right corner to add/modify ideas to the plan. After arriving at a final campaign plan, click the **Save to account** button in the top right corner of the page to save the plan to the AdWords account.

To download a CSV file version of the plan to the desktop, click the **Download plan** button.



Users can also save the generated Web site placement ideas by clicking the **Download Web site placements** button at the far right corner above the **Individual target ideas** tab.

Clicking the **Save to account** button displays three options: to save the plan as a new campaign, to save the plan to an existing campaign, or add to an existing ad group. Selecting the **Create a new campaign** option requires entering a name for the campaign, a per day budget for the campaign, and a default maximum CPC bid for the campaign. Click the **Save** and **Continue** buttons.

Once the campaign plan is added to the AdWords account, set up the ad creative for the campaign.

Check Your Progress

1. Which of the following are features of Google Webmaster tools?

(A)	monitor and resolve any malware or spam issues with the Web site	(C)	build new search and ad campaigns for the Web site
(B)	study the traffic flow and visits to the pages on the Web site	(D)	provide market intelligence and research information

2. Google Display Planner contains an integrated:

(A)	Keyword Planner Tool	(C)	Contextual Targeting Tool
(B)	AdWords Tool	(D)	Google Content Experiments Tool

3. A manually created .CSV keywords file:

(A)	Can be uploaded as a .zip file	(C)	Can contain budget details
(B)	Can contain keywords and their definitions	(D)	Can only contain keywords listed in a single column

4. Which of the following sections in the Google Keyword Planner helps create from scratch, a fresh list of keywords and a campaign plan?

(A)	Get search volume for a list of keywords or group them into ad groups	(C)	Search for new keyword and ad group ideas
(B)	Get traffic estimates for a list of keywords	(D)	Multiply Keyword Lists

5. Which of the following Google SEO tools helps compare the volume of searches between two or more markets/items and make a decision whether or not to enter a certain market?

(A)	Google Keyword Planner	(C)	Google Trends
(B)	Google Analytics	(D)	Google Analytics Content Experiments

Answer

1.	D
2.	B
3.	C
4.	C
5.	B

Summary

- SEO tools are programs, assessment methodologies, and data collection techniques to maximize the impact of a Web site on search engine listings.
- Traditional SEO techniques cannot provide all the inputs for a Web site's optimization.
- Google provides powerful SEO tools such as Google Webmaster Tools, Google Keyword Planner tool, Google Analytics tool, Google Analytics Content Experiments, and Google Display Planner.
- Google Webmaster tools provides Webmasters valuable information regarding Google search rank of the site, traffic flow and visits to pages on the site, and enables them to monitor and maintain the site's performance.
- Google Keyword Planner tool is a powerful tool that helps Webmasters build new search and ad campaigns for a Web site as well as optimize existing campaigns.
- Google Analytics is a free statistics tracking and analysis service that helps Webmasters track where visitors came from, what they did on a site, e-commerce data, and conversion information.
- Google Analytics Content Experiments is a very useful tool that allows Webmasters to make changes and improvements to Web sites and fully test the changes before incorporating them permanently into the site.
- Google Display Planner is a tool integrated into Google AdWords and also includes and integrated Contextual Targeting tool that can be used along with the Ad Planner features create ad campaigns.

08 Search Engine Marketing

Welcome to the Session, **Search Engine Marketing**.

This session introduces Search Engine Marketing (SEM). It describes the various types of search engine marketing and Google-based tools that can be used for it. It also explores the various SEM strategies.

In this Session, you will learn to:

- Describe search engine marketing
- Compare types of search engine marketing
- Explain Google affiliate programs
- Compare Google AdWords and AdSense
- Explain search engine marketing strategies

Session
08

Search Engine Marketing

8.1 Introduction

Search Engine Marketing (SEM), is a process of improving a Web site's visibility in search engine rankings. An efficient SEM campaign along with appealing Web site design can be highly effective in driving targeted traffic to a company's Web site.

In other words, an effective SEM campaign can support the Web site in the following approaches:

- It can support the Web site in accomplishing high rankings on the search results page
- It can increase traffic to an Internet site
- It can maximize Return On Investment (ROI)
- It can make use of newest advertising strategies for Web site updates and the site will be able to harness the power of these technologies

8.2 Difference between SEO and SEM

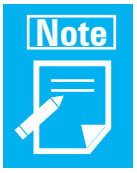
Search Engine Optimization (SEO) involves modifying the Web site content and design so as to improve its visibility in search engine results and attract more visitors to the Web site.

Search Engine Marketing (SEM) on the other hand is a medium for marketing products and services offered by a Web site, on the Internet. When Internet users search for keywords that match the contents on a Web site, advertisements of the products and services offered by related Web sites are displayed adjacent to the search results.

SEM does not help improve the search engine rankings of a Web site.

Following points highlight the difference between SEO and SEM:

- A key difference between how SEO and SEM affect Webmasters is that when Internet users click the search engine results to view a Web site, there is no charge to the Web site owner. However, for each click by users on an advertisement displayed alongside search engine results to view a Web site, the Web site owner incurs a charge.
- SEO requires technical expertise with understanding the combination of keywords and placement of content that helps achieve better visibility and page rankings for a Web site. Setting up ads for SEM is much simpler and does not required technical knowledge.
- For SEO, the Webmasters cannot specify how visible the Web site should be in search results, or where the Web site should be displayed in search results. They can only manipulate the content of the Web site and depend on Web site crawlers to index the Web site for optimized results. For SEM however, the Webmasters have complete control on determining where the ads should appear in search engine results, what type of ads should appear, the text contained in the ads, the budget for the amount they are willing to pay per click for the ads, and so on.



For best visibility on the Internet, it is recommended to use both SEO and SEM for Web sites, as they both act complementary to each other.

8.3 Understanding SEM

The goal of SEM is to drive prospective visitors and turn them to customers. SEM helps in increasing sales, increase the ROI, brand awareness, and improve visibility in SERPs.

SEM is an umbrella term that covers two broad areas:

- Gaining traffic through free SEO efforts
- Gaining traffic through paid search engine advertising

SEM is a continuous work, which involves:

- Monitoring the traffic
- Analyzing the traffic
- SEO campaigning to rank higher among specific keywords
- Driving traffic using Pay-Per-Click (PPC)
- Increasing the number of links to the Web site through link campaign
- Converting the traffic to actual goals

8.3.1 Need for SEM

Basic SEO tasks have become common and easy to replicate. A Web site business may face severe competition these days on the Internet. To compete with rival organizations and elevate the Web site's visibility, SEM tactics are required along with the traditional search engine optimization.

Most people use search engine on a regular basis and do not look past the top 30 search results. Therefore, to be considered competitive and to expose the business to potential customers via search engines, the Web site needs to appear higher in the search engine listings. SEM is a mix of various online marketing strategies that help to increase the exposure of the Web site on the Web.

8.4 Types of SEM

To be successful with an online business, one must know how to market the Web sites and the products or services. There are two types of SEM to reach and attract more customers:

1. Organic Search Engine Marketing: SEO
2. Paid Search Engine Marketing: PPC

A sample SERP is shown in figure 8.1.

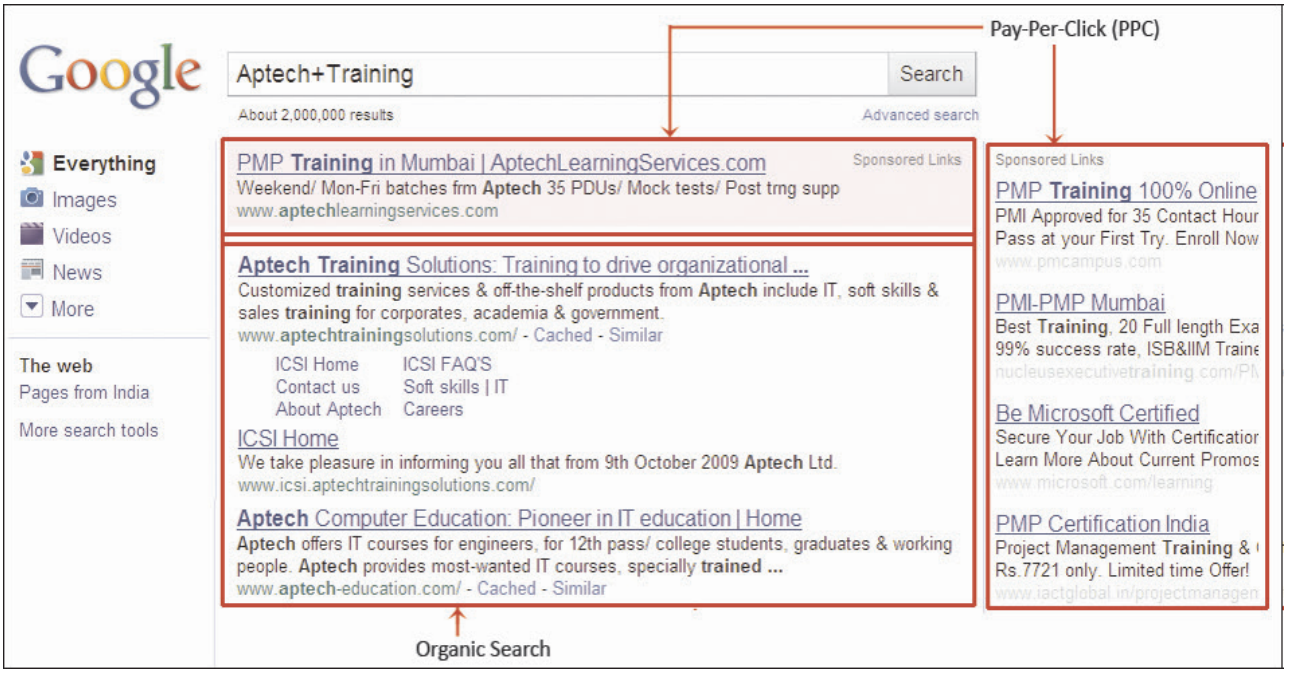


Figure 8.1: Organic Search and PPC

8.4.1 Organic Search Engine Marketing: SEO

Organic search engine marketing is a great way of marketing because it brings traffic for free. It consists of getting the Web site to rank higher in the organic or non-sponsored results using various SEO techniques.

SEO is important because, for many search terms, the top organic search results tend to get most of the traffic and have more implicit trust in the eyes of customers than sponsored results.

Additionally, Google often uses organic search result ranking to help decide which businesses to include in the Local Business Results section. Following are some best practices to rank the Web page high in popular search engines through search engine optimization:

- Create a Site map
- Product high quality content
- Maintain keyword density
- Include internal links

8.4.2 Paid Search Engine Marketing: PPC

Paid search engine marketing or Pay-Per-Click marketing is a marketing method that gives the opportunity to buy search engine placement of a Web site by bidding on different keywords or keyword phrases.

In a PPC program, Web site owners pay an amount each time a Web viewer chooses the ad listing link and clicks into the Web site. PPC applies to human-based search engines, known as directories. A successful directory campaign depends on the title, description, and the category selection.

Pay-Per-Click (also known as Pay-Per-Ranking, Cost-Per-Click, Pay-Per-Placement, or Pay-Per-Position) targets specific audience reaching up to 85% of all Internet users in a cost effective and immediate way, along with tracking the effectiveness of the marketing campaign.

There are two major types of PPC marketing. In the first PPC type, users pay a fee for an actual search engine placement of their Web site, and sometimes, they also pay per-click fee. This means, more the payment, the higher is the page rank results.

The second type of PPC marketing involves bidding on relevant keywords or phrases that appear in the text ads. For example, Google AdWords is a true example of this type of PPC marketing.

The steps involved in an effective PPC campaign are as follows:

- **Google Account Setup** - A Google account is required to start the PPC campaign.
- **Advanced Keyword Research** - Choosing the most relevant keywords or keyword phrases is the most crucial factor to make the PPC campaign successful.
- **Creating PPC Ads** - Attractive and descriptive ads are required to depict the client's business and convert the visit into a sale.
- **Bid and Campaign Management** - Ensures that the bid amount should not exceed the defined budget.
- **Monitoring of Campaigns** - Consistent site traffic monitoring is also an important aspect of PPC campaign. It helps in determining the target audience that visits the site.
- **Campaign ROI Analysis** – The Web site's conversions need to be analyzed to maximize the ROI.

Figure 8.2 shows the different stages of Pay-Per-Click.

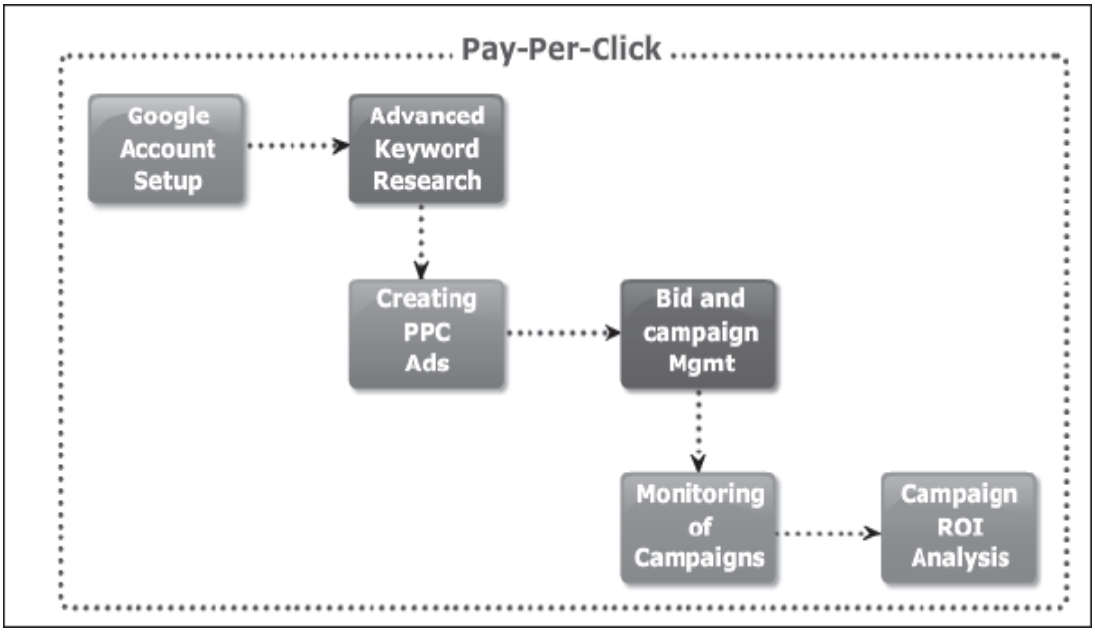


Figure 8.2: Stages of Pay-Per-Click

Following are the most important factors to drive a PPC program:

Select which PPC agent to use - The most widely known PPC agent on the Web is Google AdWords.

Google AdWords have the ability to start and stop PPC marketing campaign easily.

Additionally, Google AdWords outperform other competitors when it comes to generating conversions and return on investment.

Select the target keyword or phrases - The PPC ads are triggered by the keywords that are tied to the ads and not the Web site's content. You can get better results by increasing the density of the selected keywords in the Web site's HTML code, the Web site's description, and the Web site title.

Select the monthly budget - Decide on the maximum amount that will be paid each month for click to the Web site chosen, and then select how much to pay by keyword. The targeted traffic to the Web site will halt automatically once the maximum budgeted amount is reached.

8.4.3 Comparison between SEO and PPC

The goal of both SEO (non-sponsored search engine listing) and PPC (sponsored search advertisements) is to drive targeted traffic to the Web site. These two are complementary strategies and can be used together to get improved results.

The difference between SEO and PPC is summarized in figure 8.3.

Figure 8.3: Difference Between SEO and PPC

Some key differences between SEO and PPC are as follows:

1. **Displayed on the SERPs** - On Google SERPs, the PPC search results or the sponsored results are displayed as the top two or three listings as well as on the right column. The organic listings or the non-sponsored listings are considered as the regular listings and are showed up on the left and below the sponsored links.

2. **Traffic potential** - The possibility of customer to choose the organic listing is more than the paid listing on Google SERPs. Traffic generation effects of SEO are long lasting, while in case of PPC once the campaign is stopped the traffic dries up entirely.

3. **Cost of ranking** - SEO sometimes referred to as free traffic as the cost is simply much less tangible. In PPC strategy, the Web site owner has to pay for the clicks the ads.

4. **One-time investment versus long-term** - In bringing relevant traffic to the Web site, PPC is one-time investment and it is much faster to obtain paid listing while SEO is a long-term effort.

5. **Promotional versus Traditional** - SEO is a promotional strategy, while PPC is like traditional advertising.

6. **Time Taken for Results** - Organic search results take six months or more to show up, while PPC results can start showing up within 48 hours.

8.4.4 Google AdWords

Google AdWords Tool is a useful SEO tool assist the Web site owner in marketing products or services as a sponsored link through the Google Search Engine. This is a type of PPC marketing. AdWords ads are displayed in the side column of a search page, under 'Sponsored links' along with the positions above the free search results. The format of the ad includes a title, a URL link to a Web site, and two lines to describe the product or service.

Figure 8.4 displays an example of Google AdWords on SERPs.

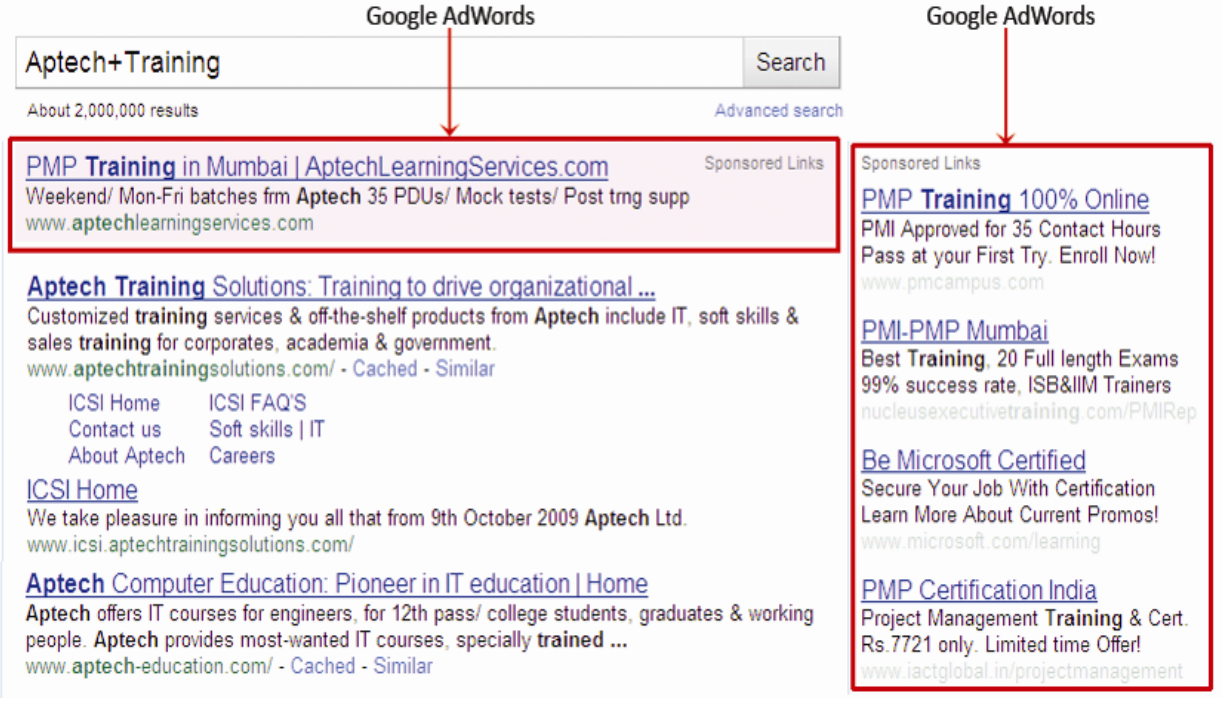


Figure 8.4: Google AdWords

The Web site owner can pitch where the ad will appear through bidding for a series of phrases. However, the Web site owner pays the amount only if Web viewer clicks the ad during a Web search. The higher the Web site owner bid for the phrases, the higher the ad appears on Google SERPs.

Google AdWords management is very essential to drive relevant targeted traffic to the Web site. Through effective Google AdWords Management, the ad can appear higher in the SERPs even though the Web owner has bid less than the other competitors.

The process flow of Google AdWords is shown in figure 8.5.

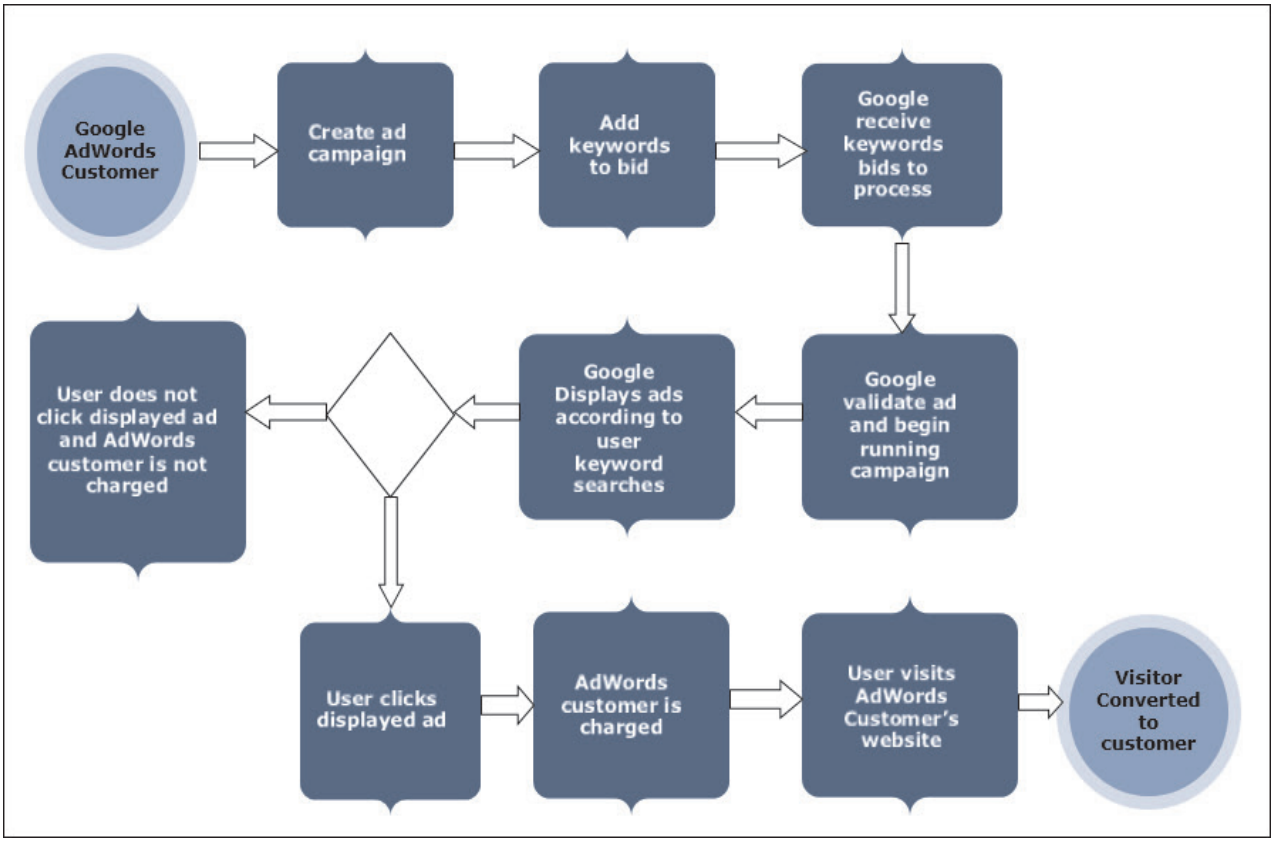


Figure 8.5: Google AdWords - Process Flow

8.4.5 Advantages of Google AdWords

Google AdWords provides the following advantages when used for advertising:

1. **Ads reach out to people as they search and browse**
   - As people search for terms or browse Web sites on a topic, the related products and services are displayed alongside the Google search results or on its partner Web sites.
   - It is possible to specify the Web sites and geographic locations where the ads must appear.
   - Google AdWords is supported by the Google Display Network (GDN), which is the largest network of Web sites and apps where ads can appear.

According to a statistic, the GDN reaches 80% of Internet users in the US.

2. **Controlled budget**
   - Advertising with Google AdWords is pocket friendly as it applies the Cost-Per-Click (CPC) bidding mechanism where a charge applies only when a visitor clicks the ad and not for just displaying the ad.
   - Users can specify the budget for the advertising and decide how much or little they want to spend per month. There is no minimum spending amount.

3. **Monitor ad response**
   - Google AdWords enables users to view statistical reports about their ad performance. They can get information about how many visitors are viewing their ads, from which geographic locations, and in turn how much business the ads are giving them.
   - Users can use AdWords tools to edit and improve their ads.

8.4.6 Upcoming Features of Google AdWords

Google is planning to introduce the following features and modifications to their AdWords tool:

- **New App Ads**

  Most Web sites look to provide their services and products in the form of mobile apps for greater reachability to its customers. Google is working on introducing a new mobile app promotion feature to be built into AdWords and Google Display Network, which helps display ads for mobile apps on search result pages and YouTube. These mobile app ads will allow Internet users to download the apps by clicking the ad instead of going to the app's landing page.

- **Measuring In-store Transactions for Offline Conversions**

  For organizations that provide products and services both in-store and online, Google is working on building a set of tools to track the in-store activities along with online transactions and club them to study user behavior. Further, the tools will be able to convert this data and provide Webmasters suggestions on optimizing their Web site content and developing more targeted advertisement strategies.

- **Tools for Efficiency and Scale**

  Google is developing a set of new tools that will help in bulk management of a large number of advertising campaigns easily in a matter of a few clicks. Webmasters will be able to view customized analysis reports regarding the performance of the advertisements and user behavior in a variety of graphical formats.

8.4.7 AdWords Express for Small Businesses

Google provides a tool called AdWords Express that can be used by small businesses who what to mark their presence on the Internet but do not want to spend a lot on advertising strategies and developing Ads.

Google AdWords Express provides a quick and easily way of advertising for business that do not have a Web site. Google AdWords creates a free Google+ page where they can display all the information about their business and its products and services. Users simply need to sign up for a Google AdWords account, enter their business information along with the category of their business, create an ad, and select a monthly budget that they are willing to spend on ads.

For users who want more than just an advertising medium to gain more business, there is AdWords. AdWords provides larger number reports and tools to control the ad campaigns for a Web site. It is possible to use a single account to create multiple ads for a variety of products and services and manage them together.

8.5 Affiliate Programs

An affiliate program works through a network of Web site owners, known as affiliates. The Web site owner, who pays affiliates for sending traffic to their site, is known as merchant or advertiser.

Affiliate programs work to generate traffic and refer it to the merchant's Web site. For this, the online merchant Web site pays commission to the affiliate Web sites for the service.

This payment is based on:

- The number of traffic the affiliate Web site sends to the merchant's site
- Or, the number of people perform some action
- Or, the number of visitors converted to customers
- Or, according to the number of visitors who are interested in merchant site's banner ads

Generally, the affiliate site, the merchant site, and the customers are the parties associated with an affiliate program transaction.

Figure 8.6 depicts how an affiliate program works.

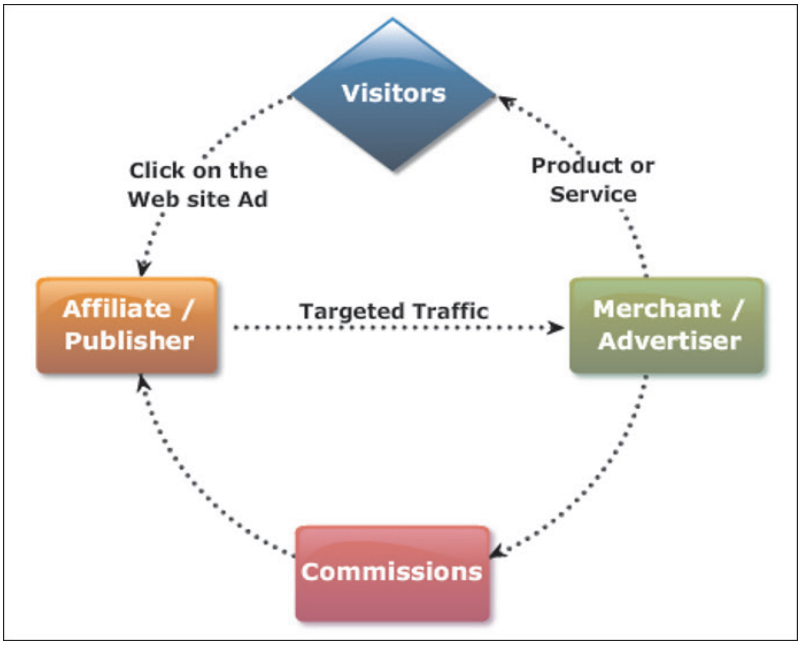


Figure 8.6: Affiliate Programs

Affiliate marketing is a huge marketing channel and for many online businesses, it has become the key growth driver. The benefits of affiliate marketing also includes lower customer acquisition cost, improved exposure and brand awareness, better reach, business focus, reduced marketing costs, and measurable results.

8.5.1 PPC Affiliate Programs

PPC Affiliate Program is an online affiliate tool that enables Web site publishers to earn money by displaying ads on their Web sites. PPC model consists of three core elements - keywords, ads, and bid. These three elements are combined together to generate relevant ads based on the search queries generated by the users.

PPC affiliate programs are designed to increase traffic to advertisers who want their products to be advertised through other Web sites. PPC program is used by two kinds of people on Internet: one is the advertiser and the other is the publisher.

A publisher is a person who is responsible for displaying ads on a Web site by joining PPC Affiliate Programs like Google AdSense. When someone clicks the ads appearing on the Web site, the publisher will get paid some percentage of the amount that the merchant or advertiser has paid to the affiliate for the ads.

The basic requirements that must be fulfilled before a publisher can join PPC Affiliate Programs are as follows:

- A computer that connects to the Internet
- An e-mail address

- One Web site or blog related to the area of interest Niche/Keyword except any pornographic, drugs related, hate-related, violent, or illegal content
- Some spare time - Invest some time and effort to earn online through PPC program

For an effective PPC program, design the Web site/blog in such a way that it will attract Internet users. A user-friendly site is always recommended so that the customers will know which one to click. If it does not look professional, most likely, users will just ignore it. The look and feel of the ad along with the visibility is always important.

8.5.2 Google AdSense

With Google AdSense, the non-search engine sites are able to set up accounts with Google and integrate Google ads into their Web site or blogs. Google AdSense is a free method for Web site owners, bloggers, and ad publishers to insert a small amount of HTML into their Web sites and earn money.

Figure 8.7 depicts the way Google AdSense works.

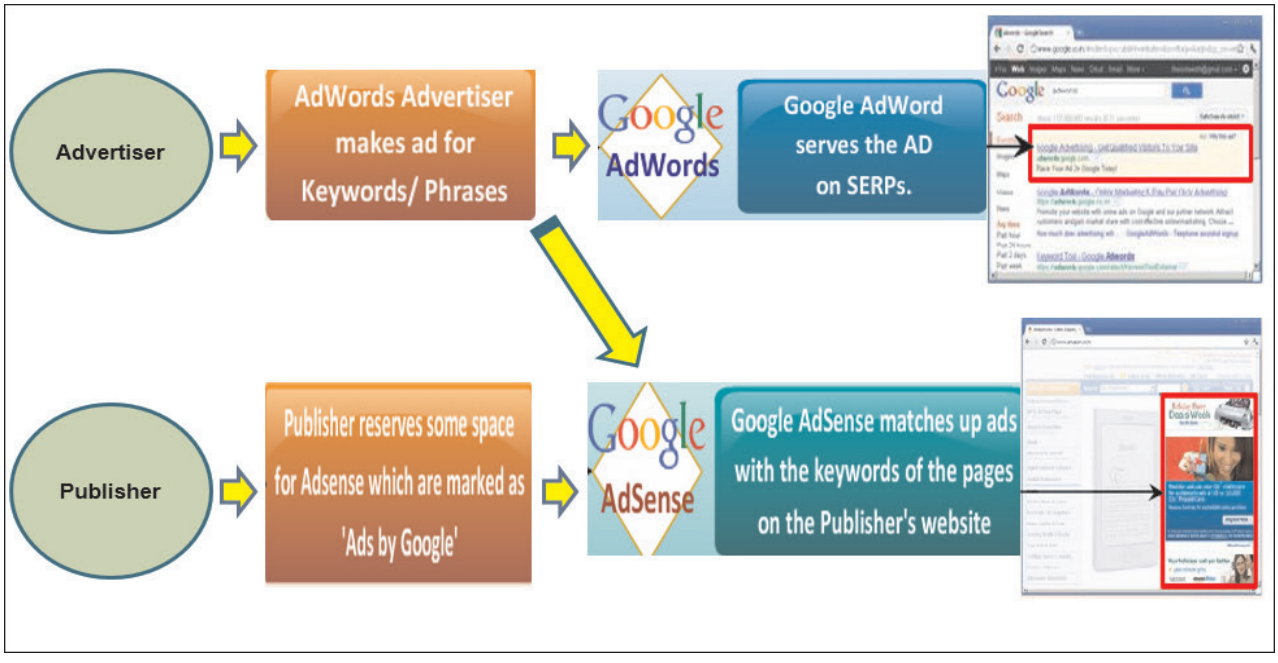


Figure 8.7: Google AdSense - Process Flow

The Web site owner can customize the ads and choose the placement of the ads to match the site's look and feel. Online reports facility is also available to track the success or earnings in real time basis.

The only difference between Google AdWords and Google AdSense is that the Web publishers pay Google to place AdWords on Google's system, while Google pays to the Web publishers to put AdSense on their site.

8.5.3 Advantages of Google AdSense

Advertising using Google AdSense provides the following advantages:

- Allows customizing ads to make them available to users in various sizes and formats such that they are relevant to the users.

- Provides the option to block certain categories of ads or specific advertisers.
- Provides help center support, online learning tools, and recommendations on how to grow the business.
- Does away with the need to maintain advertiser relationships as it provides integrated advertiser support, ad spaces at competitive rates, and more relevant ads for products and services.
- Provides personalized reports that enable users to assess and analyze their ad performance and business from ads. Also, allows users to identify business growth opportunities through ads.
- Integrates seamlessly with **DoubleClick for Publishers** and **Google Analytics**.

8.6 Search Engine Marketing Strategy

An effective SEM strategy with all the key elements is crucial for successful marketing of the Web site.

8.6.1 Key Elements of Successful SEM Plan

Figure 8.8 illustrates the key elements of successful SEM plan.

Figure 8.8: Key Elements of SEM

Following are the key elements of a successful SEM plan:

- **Visual Design**: Design of the Web site is the most important factor to keep visitors on the site and to generate a positive impression. Within few seconds of visiting the Web site, a visitor makes a judgment about the Web site based on the eye-catching design. It is essential to have a visually appealing Web site with good use of color, easy to read text, meaningful graphics, and quality photography.

- **User Interface**: For a good functional Web site, prominent and logical user interface is necessary throughout the entire Web site. The target audience needs to take into consideration when planning the user interface for the Web site. A search engine friendly user interface is vital to find all the important information on the Web site.
- **Primary Navigation**: Keep links to the key areas throughout various pages of the Web site to ensure primary means of navigation. Consistency in easy to find link location with minimal scroll is the key.
- **Content**: Visitors always look for relevant and informative content that will help them to make a decision. Uses of short, organized, and updated original content are required to convince a visitor.
- **Functionality**: Ensure an error-free Web site without any typos, incorrect grammar, punctuation, and spelling mistakes. Every component of the Web site including site search, contact forms, hyperlinks, and sign-up should work properly.
- **Consistent Layout**: For usability, a consistent site layout is extremely important. Visitors are always curious. So repeat certain elements throughout the Web site to provide potential visitors more information about the Web owner's business and expertise.
- **Contact Information**: An e-mail address, a contact form, a business phone line, or a mailing address are the simple information a prospective customer look for to get in touch with the business owner.
- **Sitemap**: A structured Site map with the list of all the pages of a Web site is an invaluable tool for the Web site.
- **Separate Design from Content**: This can be done by linking to an external CSS file that specifies the background images, fonts, colors, and so on, while including the written content in HTML format.
- **Cross Browser Compatibility**: For displaying content, different Web browsers have different rules. Therefore, it is important to make the Web site compatible across platforms.
- **Web Optimized Images**: The images should be saved in a compressed format to ensure proper screen resolution.

8.6.2 Developing an SEM Plan

A comprehensive and logically organized SEM strategy is essential to ensure highest rank in search engines.

Figure 8.9 illustrates the various ways of SEM drives.

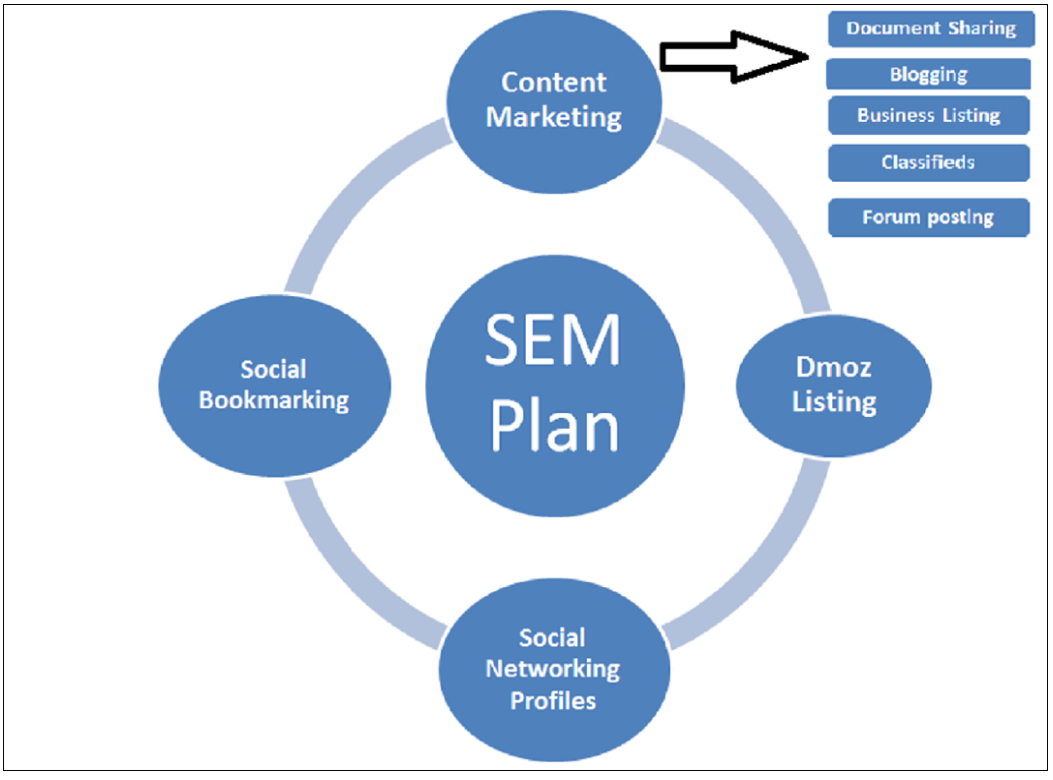


Figure 8.9: Various SEM Drives

Following are the essentials for a successful SEM plan:

1. Define the goals and key performance indicators
2. Check the current search engine ranking
3. Research about the target audience for the Web site
4. Analyze keywords and choose right keywords that represent the Web site's most relevant potential content. Validate the keywords by testing and refining them with the keyword selector tools such as Google Keyword Selector and Google Trends
5. Improve the Web design, Web page content, and architecture
6. Identify related sites for listing, link-building, partnerships, and Internet advertising opportunities
7. Publish and circulate the Web site in picture sites, video, and blogs
8. Develop an online lead generation program through ad networks, promotions, sponsorships, white papers, and newsletter ads
9. Circulate through press releases, blog posts, social tagging, online press room, and a Web-based PR distribution service
10. Use tools like Google Analytics to track and analyze performance for better results

8.6.3 Attributes of an SEM Consultant

Internet or online marketing is getting faster and complex and search engines are constantly changing their algorithms. Therefore, the expertise of an SEM consultant is required to keep up with the technology.

An SEM consultant can support:

- Increasing traffic into the Web site
- Achieving high rankings on the SERPs
- Maximizing ROI
- Using the power of updated technologies to drive effective SEM strategies

Following attributes are required to be an efficient SEM consultant:

- The ability to integrate updated knowledge and information into the current scenarios is a key skill.
- Accordingly, the ability to be a creative and critical thinker will help maximize the benefits for an SEM professional.
- SEM is a continuous process, not the result of a single strategy. So, continuous attention to numerous small details is required by the SEM consultant to make it a successful campaign. SEM consultant needs to be a team player, because Internet or online marketing is much about getting things done by others. In addition, an analytical mind competent of processing numeric information and decisions making based on number analysis is required.
- Educational background and suitable experience in related field are also important factors while looking into the capability of an SEM consultant. The key is finding the right combination of candidate's personality, aptitude, skills, and background can make a great value addition for the company or Web site.

Check Your Progress

1. Which of the following are the important factors of a PPC program?

(A)	Monthly budget
(B)	PPC Agent
(C)	Target keywords or phrases
(D)	All of the above

(A)	A	(C)	C
(B)	B	(D)	D

2. Which of the following are the three parties in an affiliate program transaction?

(A)	Customer
(B)	Affiliate Site
(C)	Merchant Site
(D)	SEO Consultant

(A)	A, C, D	(C)	A, D
(B)	B, D	(D)	B, C

Check Your Progress

3. Using which of the following methods can a publisher earn money?

(A)	Google AdWords
(B)	Google AdSense
(C)	Google Analytics
(D)	Google Ad Planner

(A)	A	(C)	C
(B)	B	(D)	D

4. Match the columns.

	SEM Related Concepts		Description
(a)	Pay-per-click	(C)	Brings traffic for free.
(b)	Search Engine Marketing	(D)	Arrangements in which an online merchant Web site pays Google commission to send them traffic.
(c)	Organic Search		A marketing method that gives the opportunity to buy search engine placement of a Web site by bidding on different keywords or keyword phrases.
(d)	Affiliate Program		To drive prospective visitors and turn them to customers.

(A)	a-2, b-1, c-4, d-3	(C)	a-1, b-4, c-3, d-2
(B)	a-4, b-3, c-2, d-1	(D)	a-3, b-4, c-1, d-3

Check Your Progress

5. Which of these are the key elements of a successful SEM plan?

(A)	Sitemap
(B)	Content
(C)	Layout
(D)	Affiliate Programs

(A)	A, C, D	(C)	A, D
(B)	B, D	(D)	A, B, C

6. Which is more important in Google AdWords while creating a campaign?

(A)	Quality Score
(B)	Keywords
(C)	Click-through-rate
(D)	Both A and C

7. Consider you have a startup online business. What will you choose to start with your business marketing?

(A)	SEO
(B)	SEM
(C)	Email Marketing
(D)	Adsense

Answer

1.	D
2.	A
3.	C
4.	D
5.	D
6.	D
7.	B

Summary

- SEM is the process of improving a Web site's visibility in search engine rankings.
- The goal of SEM is to drive prospective visitors and turn them to customers.
- SEO involves modifying the Web site content and design to improve its visibility in search engine results, while SEM is a medium for marketing products and services offered by a Web site.
- There are two types of SEM to reach and attract more customers, namely, Organic (Search Engine Optimization) and Paid (Pay-Per-Click).
- Pay-Per-Click marketing is a marketing method that gives the opportunity to buy search engine placement of a Web site by bidding on different keywords or keyword phrases.
- Google AdWords will soon introduce certain enhancements to the tool that will help promote mobile apps, manage ads efficiently on a large scale, and measure in-store transactions for offline conversions.
- Affiliate programs work to generate traffic and refer it to the merchant's Web site.
- The difference between Google AdWords and Google AdSense is that the Web publishers pay Google to place AdWords on Google's system, while Google pays to the Web publishers to put AdSense on their site.
- An effective SEM strategy with all the key elements is crucial for successful marketing of the Web site.

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09

Mobile SEO

Welcome to the Session, **Mobile SEO**.

This session begins with an introduction to mobile SEO and continues to explain the technologies and tools used to develop mobile sites. The session also explains the importance of Sitemaps and the best practices to be followed for mobile SEO.

In this Session, you will learn to:

- Explain mobile SEO
- Explain the technologies and tools used for development of mobile sites
- Explain mobile Sitemaps
- List the best practices for mobile SEO

Session
09

Mobile SEO

9.1 Introduction

Mobile SEO is the process of optimizing the natural search results for a mobile Web site for increasing the traffic towards the mobile Web site. The display format as well as the management of mobile sites is different from desktop sites. It requires a different expertise in design, development, and management of mobile sites. A user views these mobile Web sites on mobile devices.

9.2 Mobile Sites

Nowadays, with the introduction of smartphones and the development of operating systems such as Google Android for mobiles, the demand and usage of mobile Web sites is also increasing. Special Web sites are being developed specifically to be viewed on mobile phones. These Web sites are known as mobile Web sites. Mobile Web sites are developed with the aim of faster download and quick access. For example, along with the main Web site **www.google.com**, Google has developed a mobile Web site **www.m.google.com** for mobile users.

A mobile Web site tries to provide all the functionalities that are available on the regular Web site but with limitations. Some of the functionalities obtained using technologies such as JavaScript or Flash, are not available on mobile devices and thus, these technologies are not used by mobile Web site developers. As mobile phones have limited display area, a mobile Web site would try to eliminate content that requires more space such as images or videos. Also, the manner in which a mobile Web site is displayed is different to suit the smaller screens of mobile devices.

9.3 Types of Mobile Devices

The mobile devices currently available in the market are classified in two categories:

1. **Smartphones**: With respect to Internet capability, the browsers in smartphones are able to display regular desktop Web sites with most of its features if not all. These phones are also capable of rendering Web sites coded using HTML5. These include Android and iOS phones among others.
2. **Feature Phones**: Browsers in feature phones cannot generate regular desktop Web pages. These browsers also do not support cHTML (iMode), WML, WAP, and so forth.

Tablets are not considered as mobile devices. It is recommended that Web sites be optimized to cater to both tablets and desktops.

9.4 Responsive Designs for Mobile Sites

A Web site with responsive design has one HTML code for the content but uses different CSS to display the content in different layouts and formats depending on the type of device on which the content is being displayed. Instead of using multiple pages with varied layouts for various types of devices, Google

recommends using a responsive design for content design. This also saves users from unnecessary redirects.

When using responsive design, ensure that the Googlebots and other search engine crawlers are allowed to crawl and index all pages including the CSS and JavaScript as well as the images used in the Web site. Google recommends including the following CSS media query to their Web pages:

```
@media only screen and (max-width: 640px) {...}
```

The `max-width` value will vary depending on the screen size of the target device. Crawlers read this information and search results are able to refer users the right page to smartphone, tablet, and laptop screen resolutions.

9.4.1 Benefits of Using a Responsive Design

Following are the benefits of using responsive design for Web sites:

- Makes it easier for users to access and share Web site content over various devices.
- Single URL can be used to refer to Web site from any device. This also helps Google algorithms index the Web site content better and refer the right pages to users, in search results.
- Responsive designs do not require redirection to multiple pages based on device. This results in faster loading of Web pages.
- Redirecting users tends to result in errors. Responsive design does away with these errors and enhances user experience.
- Using responsive design reduces the number of Web pages, thus, enabling Web crawlers index more of the Web site's contents and keep the indexed data updated.

9.4.2 Dynamic Serving and Vary HTTP Header

When using a responsive design, it is recommended to use the Vary HTTP header to indicate to Googlebots and Web servers that the Web site is using a responsive design and a single HTML page from one URL is being presented in different layouts and formats. The Vary HTTP header is an indication which ensures that the Web servers double check their cache and display the mobile version of the page to the mobile users instead of the desktop version or vice versa. Similarly, Googlebots are able to identify the mobile version of the content faster and crawl directly to index them.

When a Web browser sends a HTTP user request to a Web server to view a Web page with the Vary HTTP header, the server returns a response with the Vary HTTP header.

The Web browser will then check the Vary field value and compare with the header of its cached pages. If they match then it serves the Web page from its cache, else it gets the fresh page from the Web server and updates its cache.

9.4.3 Separate Mobile Sites and Bidirectional Link Annotations

Some Web sites might have separate URLs for their desktop site and mobile sites. To enable Web crawlers and Web servers identify that the two URLs are related and contain the same content albeit for different devices, it is recommended to use bidirectional link annotations. Following two steps need to be performed to include bidirectional link annotations:

1. In the `<head>` section of the desktop page, include the `rel="alternate"` attribute to a link tag and point to the URL of the corresponding mobile Web site.

2. In the `<head>` section of the mobile page, include the `rel="canonical"` attribute to a link tag and point to the URL of the corresponding mobile Web site.

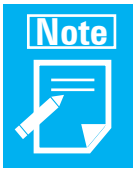
For example, suppose that the desktop URL is **http://aptechwebsite.com/home-1** and the corresponding mobile URL is **http://m.aptechwebsite.com/home-1**. The bidirectional link annotations will be included as follows:

```
Desktop page <head> section

<link rel="alternate" media="only screen and (max-width: 640px)"
href="http://m.aptechwebsite.com/home-1" >


Mobile page <head> section

<link rel="canonical" href="http://www.aptechwebsite.com/home-1" >
```



Ensure that **robots.txt** is not blocking Googlebots and other crawlers from accessing any of the page elements on the Web site. This will make sure that the crawlers are able to fully access and accurately index all pages on the Web site and also note whether the Web site uses responsive design or redirected URLs for its desktop and mobile sites.

Redirection Techniques

In case of Web sites that use different URLs for displaying on different user devices, the users can be redirected to the correct Web site by using one of the following two techniques:

- **HTTP redirection:** HTTP redirection is used to redirect users to the appropriate device-specific URLs depending on the user-agent specified in the HTTP request headers. Ensure that the redirected URL matches the URL specified with the `rel="alternate"` attribute value in the `<link>` tag on the Web site pages.

HTTP 301 or HTTP 302 - either of these status codes can be used when redirecting Web sites. Google however recommends using HTTP 302.

- **JavaScript redirection:** Instead of using HTTP redirection, JavaScript can be used to redirect users to the device-specific URLs. However, the downside of using JavaScript redirection is the time lag in redirection. When Web pages are redirected using JavaScript, the Web page needs to first load onto the client side and run the JavaScript to trigger the redirection.

9.4.4 Possible Errors with Mobile Web Sites

When developing and maintaining Mobile Web sites, ensure that the following errors do not creep in and degrade the user experience:

- **Unsupported Videos:** Avoid including videos and other content in formats such as Flash, as they are not supported by all mobile devices. It is recommended to use HTML5 standard tags to include videos in mobile Web sites.
- **Faulty Redirects:** Often users are directed to mobile Web sites based on the user-agent value in HTTP header requests. Ensure that faulty redirects do not happen and the users are not forcefully redirected to an irrelevant URL on a mobile Web site even when they want to access the desktop version from their mobile device.
- **Smartphone-only 404s:** If a user accesses the desktop version of the Web site from a smartphone, redirect the user to the smartphone-friendly version of the Web site instead of displaying a 404 or soft 404 page. If there is no smartphone-friendly version of the page, let the desktop version load instead of annoying the user with a 404-error page.
- **Irrelevant links:** Remove irrelevant links from the Web page, such as a link to the desktop version from the mobile version page and vice versa.
- **Distracting App Download Promotions:** When promoting the Web site apps to users, make sure that they do not distract the users from the content on the page. For example, there may be app download interstitials, which should be avoided. On the Internet, interstitials mean Web pages displayed before or after an expected content page, such as advertisements or confirmation dialogs to verify the user's age. Keep the app promotion banners simple and small and ensure that they point to correct app store link for download.

9.4.5 Improving Performance and User Experience of Mobile Web Sites

Mobile Web sites are for users who want to access information quickly on the go. When developing mobile version of Web sites, it is important to keep them light and optimize the pages to minimize the loading time. Ideally, mobile pages should display in under one second. Google PageSpeed Insight is a tool that can be used to analyze page-loading issues in a Web site.

Following are some of the ways for improving mobile sites:

- **Remove extra windows:** While handling multiple windows in desktop browsers is very easy, the same is very difficult in case of mobile browsers and makes navigating very cumbersome for users. It is therefore recommended to remove from mobile sites any code or script that generates extra windows which the user needs to close before accessing the site content. These include, JavaScript pop-us, survey requests, ad banners, and app download promotions.
- **Provide device-appropriate functionality:** Mobile devices do not have certain plug-ins or support for certain features. Do not include in mobile sites, content that requires certain plug-ins or software such as Adobe Flash that is not supported on mobile operating systems. Do not display mobile site versions on tablets. Serve either the desktop version or a separate tablet version as the screen resolutions are different for both. Make sure that mobile site provides all the services that are available on the desktop site.
- **Monitor traffic and user-experience of mobile-pages:** Constantly track popular pages on the desktop site that get high traffic and ensure that the features on those pages are also available on the mobile sites. Identify pages that are slow and make improvements to improve their speed. Test site pages on the same device on which the users are expected to view them to be able to identify issues on the site pages better.
- **Optimize for improving page speed:** Use optimized images for mobile sites such as cropped or compressed images to display them better and help the page load faster on mobile devices. Use asynchronous scripts, make use of browser caching, and avoid landing page redirects to improve page speed.
- **Track cross-device workflow and include new value-add features:** Study user behavior across various devices and track their visits, revisits, adding to cart and wish list actions online as well as the in-store visits. Try to include new features that make use of technologies such as GPS, camera, and social networking, to provide better user experience.

With the increase of mobile Web sites, the requirement for mobile SEO has been created. When a search query is entered by a user, mobile SEO, a technique, helps to find the site and rank it higher than other sites. In other words, SEO helps to get traffic for the Web site from search engines. Hence, mobile Web site developers are not only concerned with optimization of mobile sites but also optimization of mobile applications. In fact, there are different tools and technologies that are being used to develop applications to enhance optimization for mobile Web sites. The process of mobile SEO starts with the development of the mobile Web site. For developing a mobile Web site, different mobile markup languages are used. After a mobile Web site is developed, it has to be validated to check that the code is rendered correctly. Different validators are available online that help to validate a mobile Web site. Finally, to view how the pages are displayed on different mobile devices, emulators are used.

9.4.6 Mobile Markup Languages

Web sites for mobiles are created using different markup languages such as eXtensible HyperText Markup Language (XHTML), Compact HyperText Markup Language (cHTML), and Wireless Markup Language (WML). XHTML is a subset of HTML that includes XHTML Basic and XHTML Mobile Profile (XHTML MP).

This language is majorly used in European and North American region. cHTML is another HTML subset that is used mainly in Japan and Europe. WML is the oldest of the mobile markup languages, which is still being used globally. Google supports and is compatible with the following markup languages for mobile SEO:

- **XHTML Basic 1.1**

  This language includes the basic set of modules that are required for developing a mobile Web site. This language is rich, can be used for content authoring, and includes images, tables, form, and other objects, but does not support all the features present in XHTML. This language was built for devices such as Personal Digital Assistants (PDAs), pagers, and set top boxes. XHTML Basic was developed as a common base language that would be supported by most of the devices.

- **XHTML Basic 1.2**

  This hypertext markup language is developed specifically for mobile phones. Mobile phones had certain resource related constraints that were resolved by using XHTML MP. XHTML MP bought in enhanced functionality and support for the Web developers. This included the addition of presentation elements and style sheets.

- **cHTML**

  Small information-based devices have several hardware restrictions imposed on them such as low memory, low power CPU, negligible secondary storage, small display, and restricted input methods. The cHTML language is a subset of HTML that was specifically developed for such devices that include communicators, PDAs, smartphones, and so on. This is a HTML based language and hence, can perform and support most of the functions of HTML.

- **WML**

  This language was developed to support Wireless Application Protocol (WAP) based devices. This was one of the oldest mobile markup languages. WML supports text, images, hyperlinks, forms, and so on.

9.4.7 Validators and Other Tools

After the pages are developed using the markup language, check whether the pages render correctly on mobile devices. Validators are used to check the syntax validity of languages used for developing a Web site. Validators are commonly used for HTML and CSS Web pages. There are validators that are available for mobile Web sites also. Some of these validators are as follows:

- **TAW mobileOK Basic -** The tool is used to analyze a mobile Web site to know if the best practices for Web site designing and development have been followed. These best practices are in line with the best practices that have been provided by the World Wide Web Consortium (W3C), which is an international community that defines the standards for the WWW.

Figure 9.1 displays the TAW mobileOK Basic validator.

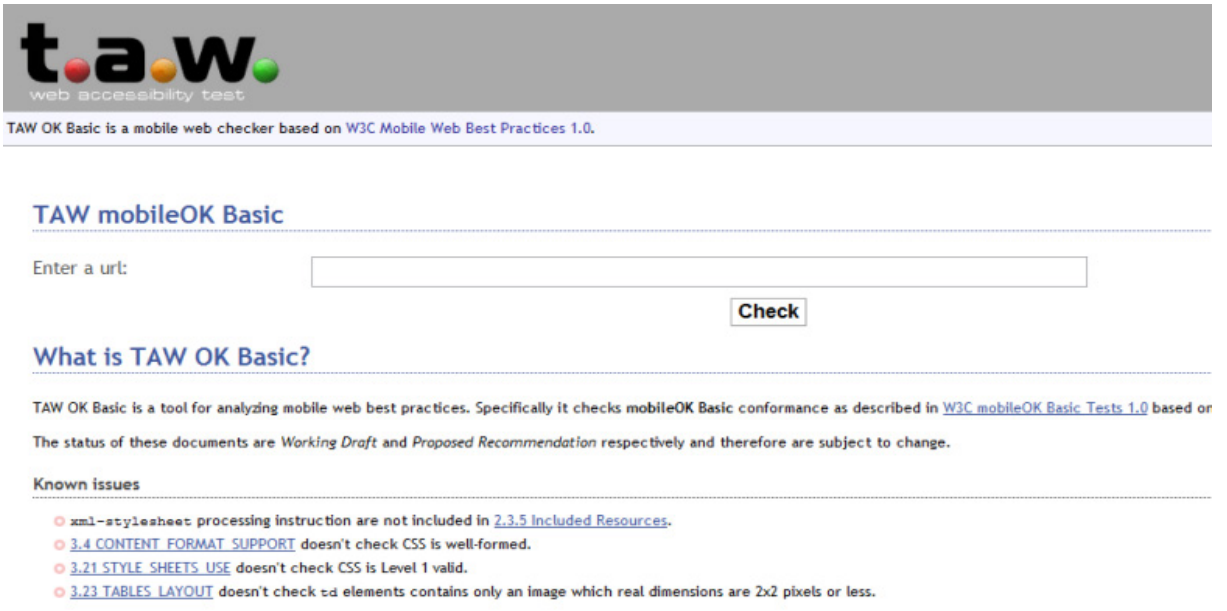


Figure 9.1: TAW mobileOK Basic Validator

- **Mobile-friendly XHTML Validator -** This tool checks whether the Web site is mobile friendly. In other words, the Web pages undergo various tests. If the Web page passes all the tests, it is conferred as a mobileOK Web page. Figure 9.2 displays the W3C mobileOK Checker validator.

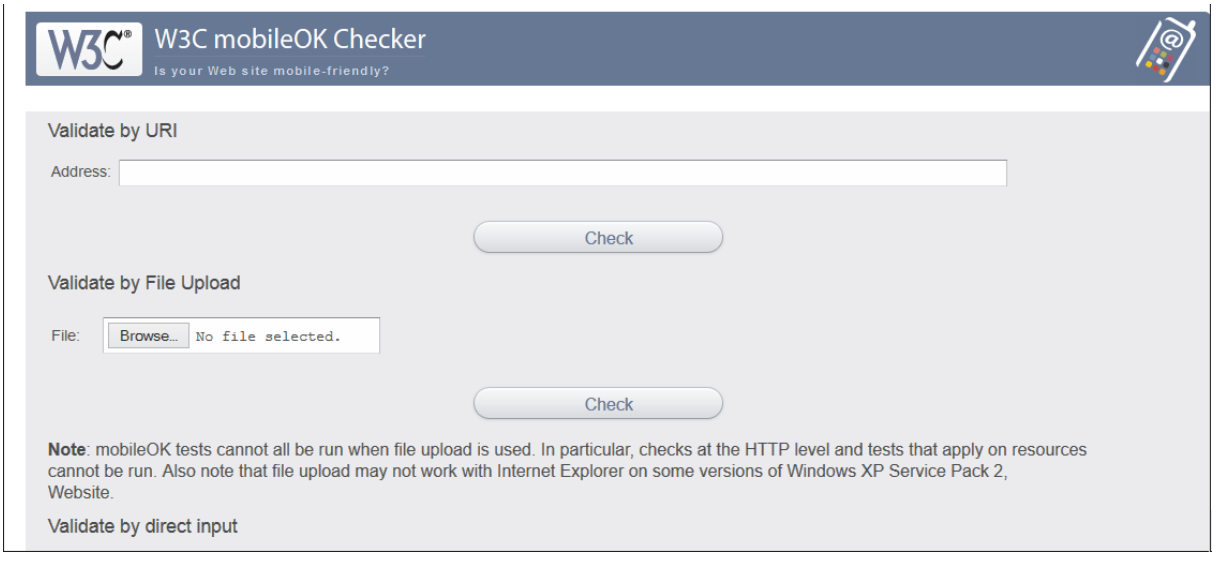


Figure 9.2: W3C mobileOK Checker Validator

- **Mobile-readiness Checker (.mobi) -** This tool is an advanced checking tool that checks the mobile readiness of Web pages according to the industry standards and best practices. This tool not only rates the Web pages but also provides a detailed analysis report of the Web pages.

This enables the developer to understand how well the site will perform on a mobile device. Figure 9.3 displays the mobiReady validator.

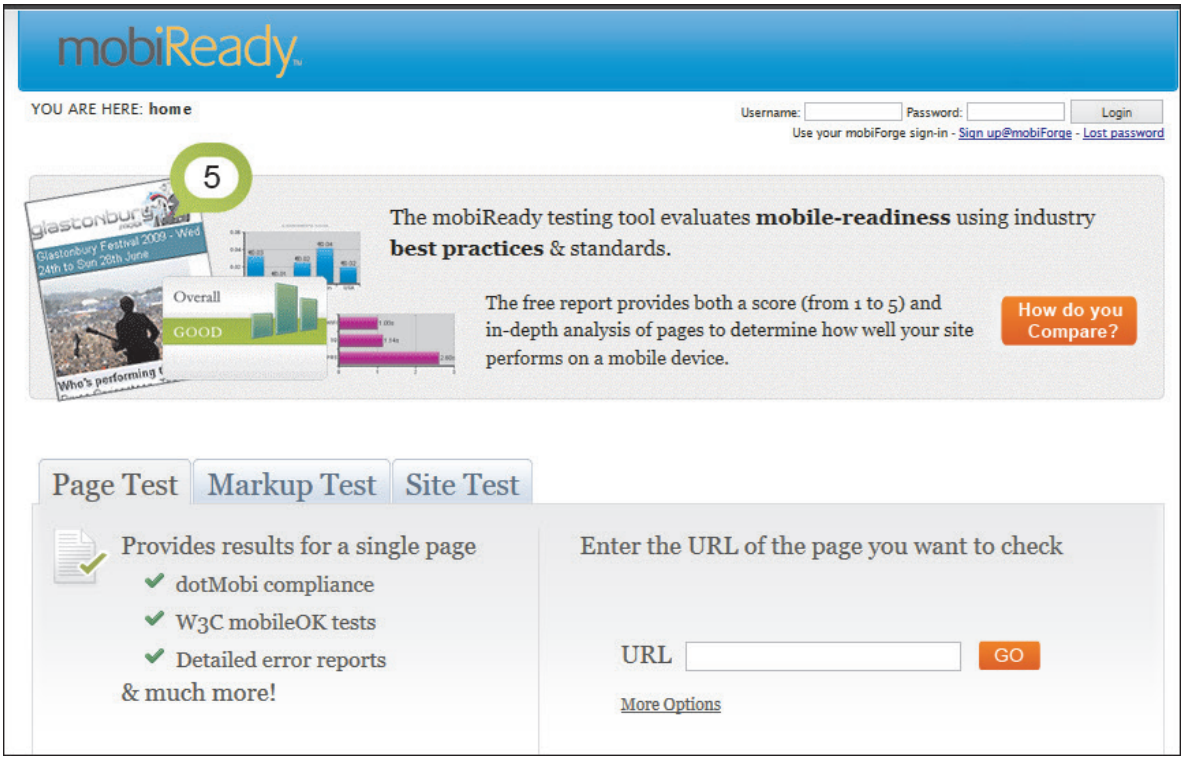


Figure 9.3: mobiReady Validator

- **WurML (shadowplay.net) -** WurML is a command-line tool. It acts as a robot that crawls the Web site to help Web developers in testing the WAP-based applications. It checks for cookies, headers, and other customizable variables on a mobile site. It also navigates through the links present in a valid WML document.

9.4.8 Emulators

After the completeness and validity of the Web pages are checked, the developers need to check how the Web pages will appear on a mobile phone. Emulators are used to test a mobile Web page and see how it would appear on an actual mobile screen. There are various emulators that can be used before uploading the mobile site on the Web. Some of these emulators are as follows:

- **wmlbrowser Add-on -** This is an add-on provided by Mozilla that allows to simulate the WML pages on the computer-based Mozilla Firefox browser.
- **XHTML Mobile Profile Add-on -** The mime-type for XHTML MP based applications is not natively supported by Firefox. As a workaround for this, Mozilla provides the XHTML MP add-on that will support the mime-type.
- **User Agent Switcher Add-on -** This is an add-on provided by Mozilla that adds a tool bar and menu for switching the browser user agent.

- **.mobi Emulator -** This emulator helps to view a mobile Web site in the same way as it would appear on a mobile phone. To view the site, the user types the URL of the Web site in the dialog box provided, click Submit, and the site would be shown in the same way as it would appear on a mobile phone.
- **i-mode Emulator -** This emulator has been developed for developers and devices that use the i-mode technology. i-mode is a mobile Internet technology service provided by Nippon Telegraph and Telephone (NTT) DoCoMo. i-mode includes a variety of services in addition to regular wireless Internet. These services include e-mail, financial services, weather forecasting, sports reports, ticketing, and so on.
- **Openwave Phone Simulator –** Openwave Phone Simulator is a free software development kit for creating innovative mobile applications.
- **Opera Mobile Emulator -** Opera Mobile Emulator is one of the most popular free emulators. It helps make sure that a site looks good and works exactly as it should in mobile and tablet browsers. The Opera Mobile Emulator is available for Windows, Linux and Mac desktop environments. Launching the Opera Mobile Emulator displays a Profile Selector as shown in figure 9.4.

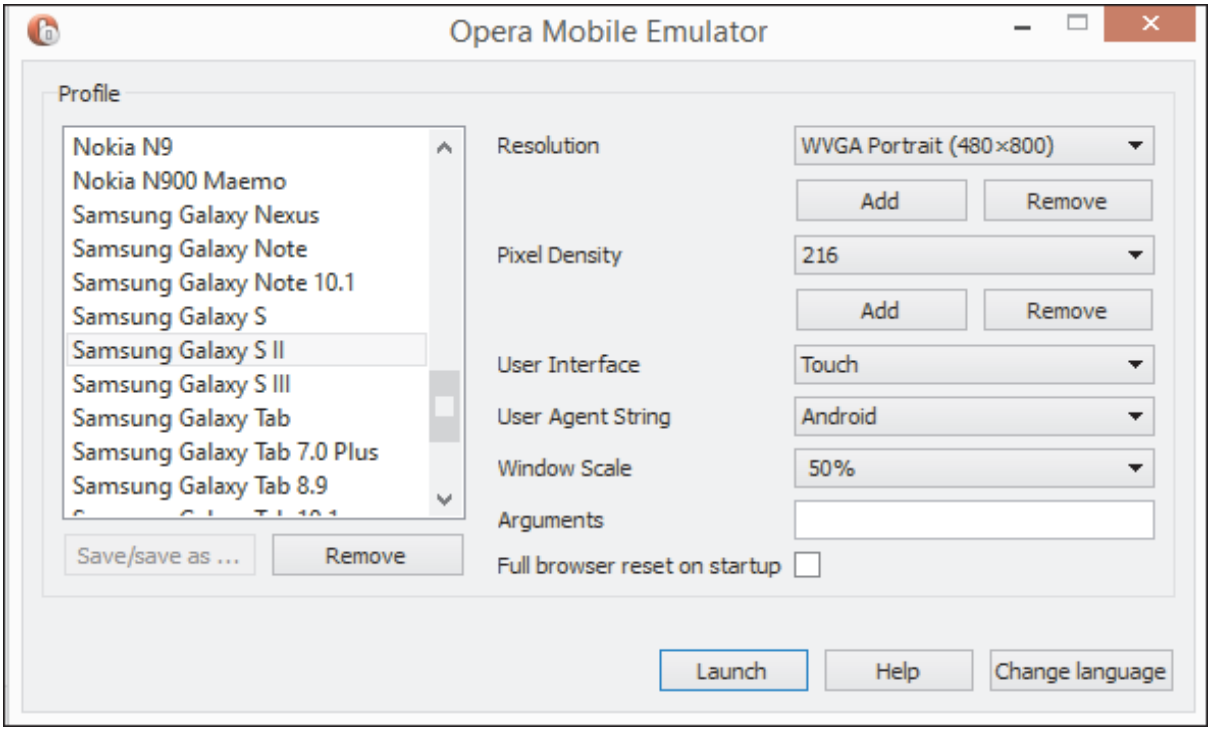


Figure 9.4: Opera Mobile Emulator Profile Selector Screen

Using the Profile Selector options, one can test the site in different phone configurations. Figure 9.5 displays the Google home page on the emulator.

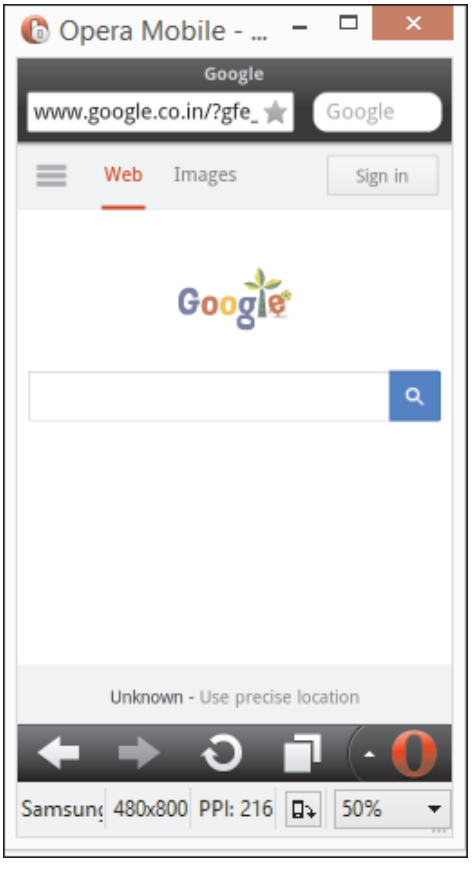


Figure 9.5: Opera Mobile Emulator

Mobile Page Analyzer

Mobile Page Analyzer is a tool that takes Web site and Web page URLs as input, analyzes the content, and returns data as to whether the site is mobile friendly and error free. It can accept up to 50 page URLs as inputs, read and analyze their code, and identify the following parameters:

- If there are any faulty or irrelevant mobile redirects
- If the site generates any mobile 404 errors
- Whether mobile-specific Meta tags are present
- Whether desktop pages are redirecting correctly to mobile-friendly pages
- Whether the page is too heavy to load fast on mobile devices
- The type of mobile redirect being used, 301 or 302 redirect
- Whether the Web site uses a responsive design
- Whether the desktop and mobile pages are tagged with canonical and alternate tags

Mobile Redirect Viewer

Mobile Redirect Viewer is a tool that helps identify pages that render faulty redirects or 404 errors. Entering a Web site URL displays simultaneously, how the site is displayed on the various devices such as iPhone, Android, iPad, and Blackberry. It lists the redirects coded for the various devices such as smartphones, tablets, and desktops. It helps track if mobile redirects have been configured properly according to SEO best practices. It identifies whether:

- The Vary tags contain proper User Agent flag
- A page uses server or browser redirects
- The page takes the benefit of page caching to improve the speed
- The redirect URLs point to the correct links with relevant content

9.5 Mobile Sitemaps

Mobile Web sites also require to be crawled upon by crawler of different search engines so that the sites can be indexed. This enhances the visibility of the sites to the users of the search engine. To help the crawlers to easily access and crawl the mobile Web site, mobile Site maps must be created.

Site maps can be defined as an XML file that contains a list of URLs present in a mobile Web site. In other words, Site maps help to specify all the pages that are present on a Web site to a search engine. A search engine such as Google may not crawl to all the pages of a Web site, but if a Site map is provided, all the pages are revealed to the search engine. A Site map is created using XML and they are often called XML Site maps. A Web developer must not only create a Site map, but also submit it to Google. This helps Google to discover URLs of pages that may not be detected in the crawling process. Hence, a Site map helps to improve the crawl ability of search engines to access all pages of a Web site and will enable re-crawling when there are any changes made on the site. Mobile Site maps also help to discover new Web sites and new content that may have been added to the indexed pages of the existing Web sites.

Site maps also provide metadata to Google. These metadata help Google to identify whether the content is a video, image, news article, source code, and so on. Also, additional information such as details on page updates can be provided to Google on a regular basis by using Site maps.

It is recommended to provide a separate Site map for Submit information related to news.

Following types of Web sites would benefit when Site maps are used:

- Web sites that contain dynamic content
- Web sites that have content such as images that will not be crawled on by Googlebot
- Web sites that are recently released and that contain less number of links

- Web sites that contain pages that are not linked or poorly linked

Google follows the Sitemap Protocol 0.9 that has been provided by **sitemaps.org**. This protocol is compatible with other search engines. Hence, the Web sites using this protocol in their Site maps are enhanced in a better way for mobile SEO.

The entire protocol can be viewed on the link **http://www.sitemaps.org/protocol.html**

The mobile crawlers used by Google uses the Request Header along with the HTTP "Accept" header to verify the developer's mobile site. It suggests to the developer's mobile site that it should return mobile content wherever available, instead of regular HTML. If the mobile site meets the necessary standards, the search engine will directly point to the mobile Web site instead of the regular site.

9.5.1 Adding Mobile URL to Sitemaps

A mobile Site map uses a protocol named, Site map. It requires a URL name and a specific tag. The format for a mobile Site map is different as compared to a normal Site map. Assuming that a Site map is to be created for an HTML file present at the URL, m.aptechlearningservices.com, its code will be as shown in Code Snippet 1.

Code Snippet 1:

```
<?xml version="1.0" encoding="UTF-8" ?>
<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"
xmlns:mobile="http://www.google.com/schemas/sitemap-mobile/1.0">
<url>
<loc>http://m.aptechlearningservices.com/imagefile.html </loc>
<mobile:mobile/>
</url>
</urlset>
```

Following guidelines must be considered when developing a mobile Site map:

- Provide only mobile Web content in the URL links for a Site map. If non-mobile Web content is provided, Google will dismiss the site. Hence, it is recommended to create a separate Site map for non-mobile content.
- Provide a unique name to each mobile Site map.
- Create a separate configuration file for a mobile Site map if the Site map Generator is used.
- Ensure that the `<mobile:mobile/>` tag is present for effective crawling.
- Use only single Site map even if multiple markup languages are used in the mobile Web site development.

- After the Site map is created, it must be submitted to Google for crawling by using the Webmaster Tools.

9.5.2 Troubleshooting Issues Related to Mobile Sitemaps

Mobile Site maps are comparatively different from normal Web-based Site maps. Hence, there are certain issues faced by developers using mobile Site maps. Table 9.1 describes a list of common issues that can occur with mobile Site maps.

Issue	Description
Unsupported features	Some features are not supported by all mobile phones. Some of these features are as follows: - Table size and depth - Image size - Div depth - JavaScript - Meta tags - Iframe tags - Object tags - Cookies - Embedding To resolve this, the mobile page layout and design must be simple and clear.
Submitting non-mobile URLs	Google does not support submission of non-mobile URL for crawling using the mobile crawler. Non-mobile URLs can be submitted for normal Web-based crawling only.
Illegal tags	If the mobile Web site contains illegal tags, these would not be supported on the mobile. To avoid the use of illegal tags, the markup language must be checked for validity using the various validators.
Unsupported markup language	Google only supports XHTML Basic, XHTML MP, WML, and cHTML markup languages. No other languages are supported by Google for mobile Web sites.
Unsupported encoding	Google supports UTF8 as its preferred encoding type. For Japanese sites, Shift-JIS encoding type is preferred. It is preferred if any one of these encoding types are used for a mobile Web site.
Large Document Size	Large documents take time to load on a mobile site. Hence, it is preferable to avoid large documents for a mobile site. It is recommended to keep the Web site size below 25 KB for best user experience.

Table 9.1: Common Issues with Mobile Sitemaps

9.6 Googlebot-Mobile for Smartphones

With the rise in the use of smartphones, Googlebot-Mobile can now crawl content and Web sites related to smartphones too. The user-agent strings for Googlebot-Mobile are as follows:

- **Feature phones Googlebot-Mobile:**
  - SAMSUNG-SGH-E250/1.0 Profile/MIDP-2.0 Configuration/CLDC-1.1 UP. Browser/6.2.3.3.c.1.101 (GUI) MMP/2.0 (compatible; Googlebot-Mobile/2.1; +http://www. google.com/bot.html)
  - DoCoMo/2.0 N905i(c100;TB;W24H16) (compatible; Googlebot-Mobile/2.1; +http://www. google.com/bot.html)
- **Smartphone Googlebot-Mobile:**
  - Mozilla/5.0 (iPhone; CPU iPhone OS 6_0 like Mac OS X) AppleWebKit/536.26 (KHTML, like Gecko) Version/6.0 Mobile/10A5376e Safari/8536.25 (compatible; Googlebot/2.1; +http://www.google.com/bot.html)

Googlebot mobile for smartphone crawlers use **robots.txt,** robots meta tag, and HTTP header directives to crawl and index information. For example, the following **robots.txt** crawler directive can be used to block all crawling by the Googlebot for smartphones user-agent and use the regular Googlebot.

```
User-agent: Googlebot
Disallow: /
```

Similarly, the following **robots.txt** directive can be used to block crawling by Google's feature phone crawlers.

```
User-agent: Googlebot-Mobile
Disallow: /
```

Googlebot-mobile for smartphones will help to improve the user experience of smartphone users. If a search result URL on a smartphone directs to another smartphone specific URL, then the user is directly pointed to the smartphone specific URL. This helps to avoid time delay and latency.

Note that each user-agent is device specific. Hence, user-agents for all the targeted devices must be considered for better optimization.

9.7 Best Practices

There are various best practices that can be implemented on the mobile site for mobile SEO. Some of these best practices are as follows:

- **Mobile Keywords -** Mobile keywords must be comparatively short as compared to regular Web searches. The mobile-specific phrases and categories of keywords must be identified and added to the content of the mobile Web site. The Google keyword tool can be used to view the month-wise data for mobile-related keyword searches. It can also be used to view the competition environment. Based on these search phrases, optimized content can then be built.

- **Mobile Content -** The content of the mobile Web site must be very effective. It should direct the mobile search of a user to the desired mobile Web site. This ensures effective mobile marketing. The content must be developed after researching on the interests and needs of the customer. Also, using social topics and specific keywords would help in getting better search results.
- **Positioning -** Location and/or specific task are the two major search factors on the basis of which Google search results are displayed. In mobile search, the results are not extensive but only a few results are displayed. Hence, organization must try to appear in search result specific to location and/or task. For example, if a user has to buy computer hardware in Chicago, then a mobile with the keywords 'Computer hardware' and 'Chicago' would provide the names of only those shops located in Chicago. This factor must be considered for positioning of a business.
- **Transcoding -** If a mobile Web site displayed on a mobile device is not user friendly, Google alters the site to display a mobile friendly result. This process is called transcoding. A mobile site may not be user-friendly due to Flash or Java content or large size of the file or image. In this case, Google would alter the page to remove the unsupported content and put forward a transcoded page on a temporary URL. This page would be as per the guidelines and best practices of mobile Web sites.

  In practice, this means that Google decides to impose mobile Web site presentation standards and if the site does not conform to the standards, Google takes the content and repurposes its design, layout, and format in a way that is best suited to the user and mobile device.

  Transcoding may change the entire display of the Web site. It may also eliminate certain important information and figures. Hence, transcoding has both its advantages and disadvantages. To avoid transcoding, the Web site developer must include the following tag in the header:

  ```
  <link rel="alternate" media="handheld" href="alternate_page.htm" />
  ```

  This should redirect the Web site viewer to a mobile optimized Web site.
- **Mobile Filters -** The Google Analytics tool provides a mobile filter, which provides details about the number of visitors accessing the site via mobile. It shows the details of the users such as the mobile operating systems, mobile devices, and network service providers. There is also a Mobile Apps Software Development Kit (SDK) that is provided by Google for implementing Google Analytics from within mobile applications.
- **On-site Best Practices -** Best practices for mobile sites, even on-page best practices must be followed. These best practices include content optimization, Image Alt description, and so on.
- **Layout of the Mobile Web Site -** The layout of the mobile Web site that is developed must be checked on various emulators. This helps to check the layout of the site on different devices. A developer can see how the site is actually displayed on a mobile phone by using emulators.
- **Mobile URL -** Earlier, Webmasters used to provide different URLs to identify mobile site as compared to regular Web sites. The `rel=canonical` tag was introduced to overcome this issue of dual URL creation. Hence, now the same content is served to Googlebot and Googlebot-Mobile using a single URL. The advantage of this is that it creates a single destination to build links and facilitates sharing through social media for mobiles.
- **Mobile Device -** Google Analytics can be used to view the information as to which mobile phone device is used most by the users to access the mobile Web site. Normally, smartphones are compatible with most of the Web site.

If the audience is viewing the site on any other phone, it should be checked that the Web site is displayed appropriately on the phone. To see this, emulators can be used. Note that Flash content is not supported by mobile phones and this type of content must be avoided.

- **Accessibility -** Provide accessibility of the mobile Web site not only to the users but also to Web search engines. Allow crawlers to crawl through the mobile Web site. The site must also be readable and understandable by the crawler. For this, the correct **robots.txt** file must be used and the pages that are indexed should exist in public domain and so on.
- **Submit the Site -** A mobile Web site must be submitted to the relevant search engines. It should also appear on business directories and listings to guide visitors towards the mobile site.
- **Validate the Site -** Use only the markup languages that are supported by Google. Also, run the mobile Web site through different validators to avoid facing issues at a later stage.
- **Localized information –** Ensure that the mobile sites are sensitive to the location and time zone from which the site is being accessed and display content appropriately. The location and time is at many times very important to mobile Internet users.
- **Enable sharing on social media –** Provide widgets and functionality to allow visitors to the site to share the content on social media and recommend the site's products and services.
- **Avoid hiding content in apps –** While apps are useful and encourage more mobile users to access the Web site and use its services, do not include any exclusive content in apps. There should be no information that is not available on the mobile site and available only in the apps. This is because search engine crawlers cannot access content in apps. Thus, reducing the visibility of site to appropriate users.

9.8 Mobile Content Strategy

This is the mobile age where a significant population of Internet users are using mobile devices. People what to access information on the go and instantly. To be successful in the competition to capture the market, it is therefore wise to have a mobile content strategy. Following steps will help build a mobile content strategy for a Web site:

1. **Mobile First:** With half of all Internet searches happening on mobile devices, having Web content accessible on mobile devices is critical to maintain visibility on the Internet. Create content that is mobile-friendly with small chunks of text, light content, graphical and attractive content. Innovative content presentation methods that make navigation and accessing content on mobile devices easy and hassle-free should be the primary focus.
2. **Know your Audience:** To develop effective mobile content strategies and design and efficient mobile site, it is important to understand the behavior of the visitors to the Web site. Identify the percentage of users accessing the site on mobile devices and the type of devices they use, the type of content they are accessing on the site, the type of activities they perform on the site, the other sites and social media that they tend to access from the site. The answers to these questions will help devise ways to provide better and more useful user experience on the mobile site.

3. **Formatting and Layout:** Mobile devices typically have small screens. The formatting and layout of content on mobile Web sites should therefore, be legible and easy to read. Instead of adding long paragraphs of text, it would be useful to integrate some visual content, such as images, infographics, videos, and charts supported by crisp, effective, and to-the-point content. The content should be attractive and invite the user to read through.

Check Your Progress

1. Google supports _____ as its preferred encoding type.

(A)	UTF8	(C)	WurML
(B)	XHTML MP	(D)	cHTML

2. Which of the following is one of the oldest mobile markup languages supported by Google?

(A)	WML	(C)	XHTML MP
(B)	XHTML Basic	(D)	cHTML

3. Which of the validator provides a score for the Web pages and a detailed analysis report of the Web pages?

(A)	TAW mobileOK Basic	(C)	W3C mobileOK Checker
(B)	mobiReady	(D)	WurML

4. Which of the following option is a mobile emulator?

(A)	i-mode	(C)	TAW mobileOK Basic
(B)	WurML	(D)	mobiReady

5. Google follows the _____ that has been provided by **sitemaps.org**.

(A)	Protocol Sitemap 9.0	(C)	Sitemap Protocol 0.9
(B)	Protocol Sitemap 0.9	(D)	Sitemap Protocol 9.0

6. Which are the common SEO best practices and improvement suggestions for mobile sites?

(A)	Add Popups	(C)	Use Flash to display content that is attractive and effective
(B)	When redirecting users to a mobile site, makes sure to redirect them to the right page	(D)	No need to maintain total parity between the desktop and mobile sites

(A)	B, C	(C)	A, C, D
(B)	C, D	(D)	A, B, D

Check Your Progress

7. Which of the following are features of Mobile Page Analyzer?

A. Identifies if there are any faulty or irrelevant mobile redirects

B. Identifies whether the redirect URLs point to the correct links with relevant content

C. Takes the benefit of page caching to improve the speed

D. Identifies whether mobile-specific Meta tags are present

(A)	B, C	(C)	A, C, D
(B)	C, D	(D)	A, B, D

Answer

1.	A
2.	A
3.	B
4.	C
5.	C
6.	A
7.	D

Summary

- Web sites for mobiles are created using different markup languages such as XHTML Basic, XHTML MP, cHTML, and WML.
- Ensure that mobile and desktop versions of Web pages redirect appropriately.
- Validators are used to check the validity of syntax of a Web site.
- Emulators are used to test a mobile Web page and see how it would appear on an actual mobile screen.
- Site maps help to elaborate all the pages that are present on a Web site to a search engine.
- A Web user must use the Google Webmaster Tools service to submit a Site map to Google.
- All pages on a mobile site must contain links to its desktop site home page.
- Include user agent settings to enable search engines to direct the users to the appropriate page depending on the device they are using.
- Google Analytics tool provides a mobile filter that provide details about the number of visitors accessing the site via mobile.

10 SEO and Content Marketing

Welcome to the Session, **SEO and Content Marketing**.

This session describes the concept of SEO content marketing, the components of SEO content marketing, and the importance of local SEO. This session also introduces the various social networking media that serve as a platform for SEO content marketing.

In this Session, you will learn to:

- Identify the importance of SEO content development
- Describe the guidelines for developing effective SEO content strategy
- List the components of SEO content
- Describe the importance of Local SEO content
- Describe social media as a medium of business success
- Identify the popular social networking media

Session
10

SEO and Content Marketing

10.1 SEO Content and Content Marketing

While developing SEO content will increase the visibility of the Web site and bring more user traffic, if the content is not engaging enough, it will not compel them to share it with others, or revisit again. Content marketing is the technique of creating engaging content that markets the products and services offered on the Web site and compels visitors to not only absorb that information but also share with their friends and people in their community. Inducing such sharing of content that can potentially bring in new customers and business is called content marketing. Content Marketing and SEO complement each other in the online success of a business.

10.2 Developing SEO Content Marketing Strategy

SEO content marketing requires a significant research, analysis, and careful drafting of content that keeps the users on the Web site and attracts them to read through. Following are the five guiding principles for devising an effective SEO content marketing strategy:

1. **Identify the goals**

   The first step is to identify the goals or the Web site or business. Determine whether the purpose of the Web site is to increase the sales of your products or services or the Web site is purely information based and makes money from ads placed on it. Depending on the goals, the content of the Web site, its placement, and design will vary.

2. **Identify your audience**

   Use analytical tools and surveys to determine the type of visitors that the Web site gets. Identify the kind of users who would likely be interested in the services, products, or information on your Web site. Design content to attract these kind of Internet users to your Web site.

3. **Schedule tasks**

   Once the goals and the type of audience for the Web site have been identified, the next step is to build an editorial calendar. An editorial calendar defines when the site content will be published. It also helps identify the interval at which the content will be updated or new content will be added to the Web site. Fresh, updated content is a key aspect of SEO.

4. **Analyze the Web site**

   Constantly monitor and analyze the Web site using SEO tools. Track information such as page views, back links, comments, social shares, and so on. Based on these data and analysis, refine and improve your Web site content.

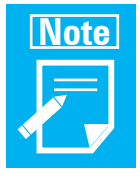
5. **Perform regular updates**

   Study the analysis reports and identify the features on the Web site that draw the most visitors. Apply those features in as many pages as possible. Also, regularly update the content of the Web site and optimize them for maximum user traffic.

10.3 Components of SEO Content Marketing

A search engine optimized content includes the following elements:

- **Articles**: Articles can be included independent of blogs as part of the news about the Web site or as part of newsletters.
- **Product information pages**: Most optimized site content contain pages that provide information about the products and services that they offer. The products and their description serve as keywords for search engine results. They also serve as marketing tools when used as Pay Per Click (PPC) landing pages.
- **Graphs and Charts**: Information can also be depicted graphically using a special set of images, graphs, and charts along with some text. These are called infographics. Infographics help pack a lot of information using a few graphics and are very attractive. Thus, they are capable of driving lot of visitor traffic. However, since they are in the form of images, search engines cannot read them and display in search engine search results effectively.
- **Blogs**: Blogs help keep the audience connected with the Web site. Information in blogs is updated frequently and can be used to keep the visitors informed of the new products or services being launched, or about new achievements and events. Blogs can include articles, lists, guides, videos, slideshows, infographics, and other such attractive media of presenting content to the visitors. They serve as an SEO content as well as a marketing medium.
- **Guided Content (also called Guides)**: Often Web sites like to include detailed specifications about the products and services they offer. Such information runs into several pages and is typically highly text-based. These are called guides. Guides are often displayed as excerpts on the main page and need to be either downloaded as PDF files or require registration to view the full text.



Having a registration process might deter users to access the guide and adversely affect the visitor traffic, thus proving to be a negative SEO technique.

- **Videos**: Videos are an attractive way of presenting information on Web sites. Sometimes, it is more effective to explain a concept through a video than through text. However, search engine crawlers are unable to capture the information shown in the videos. Hence, pages tend to include more text than videos. To overcome this issue, it might be a good idea to include a text transcript of the video in the content.
- **Glossary**: Often, Internet users like to search for brief information or definition of terms. It is therefore useful to include a list of keyword terms as a glossary in the Web site. It will help capture the visits of several such users to the Web site and thus increase traffic.
- **Lists**: Lists are typically pieces of information that contain a bullet list or numbered list of instructions, or summarized information that readers can scan through quickly to grab information.

10.4 Local SEO Content

Businesses with physical presence or virtual presence both need to understand the importance of creating a local presence. That is, visibility for Internet users in specific geographical areas. Different geographic areas have different types of demands, preferences, and social norms. It immensely helps a business to customize their Web content according to its various target audiences. Local SEO is especially important for businesses that serve the local customers such as doctors, dentists, lawyers, restaurants, handymen, grocery stores, and so on.

Including the following types of information in the Web site help make it optimized for local SEO:

1. **Business category**: State the category of business or the business categories that the Web site deals with. Figure 10.1 displays the business details form for creating a Google+ page that requires specifying the category of the business.

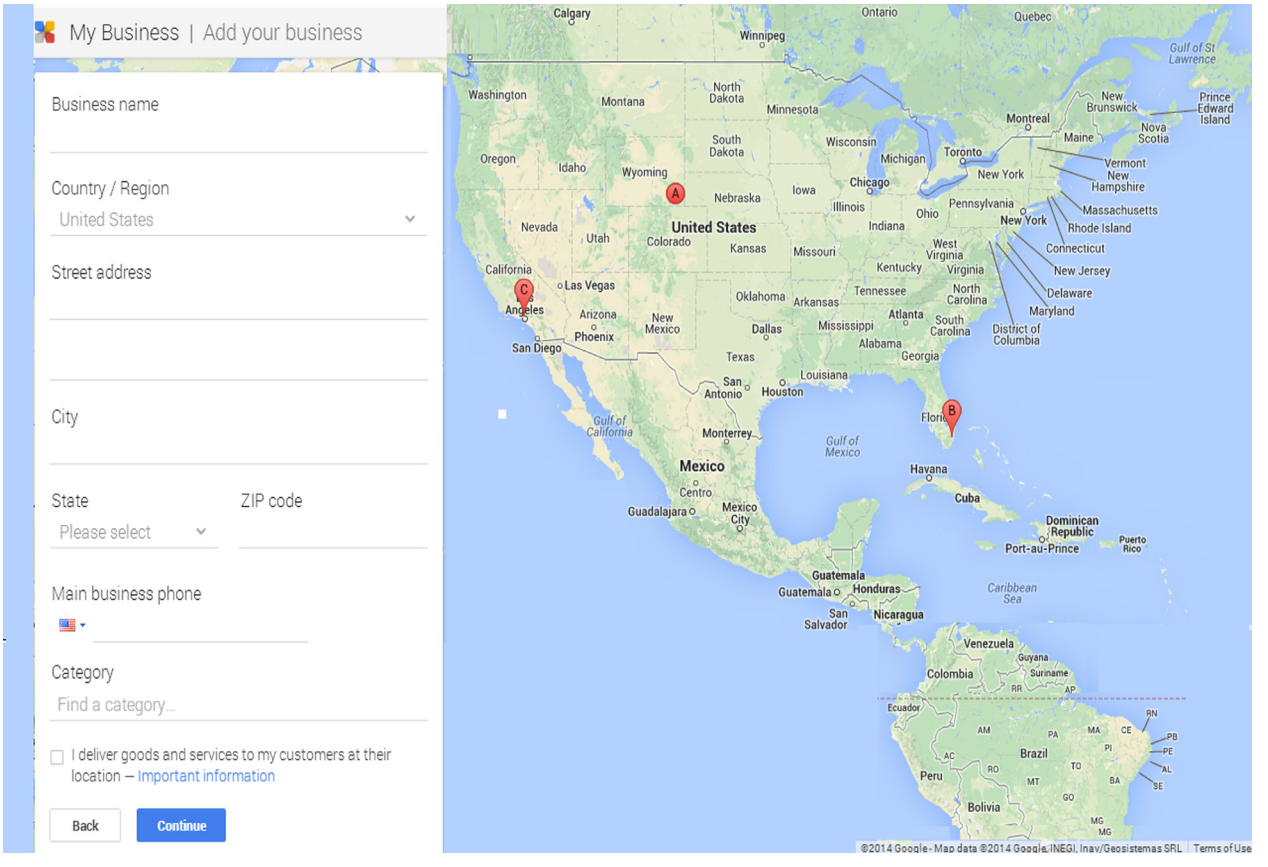


Figure 10.1: Specify Business Category

2. **Physical address of business**: Specify the address of the business including the name of the city in which it is located. Visitors searching for businesses in their locality offering the services that the Web site offers, can then easily be directed here. It is also important to include the city and state of your business in the Title tags, Meta tag descriptions, and site content.

For example, figure 10.2 displays a list of flower vendors in California with their name, address and phone number.

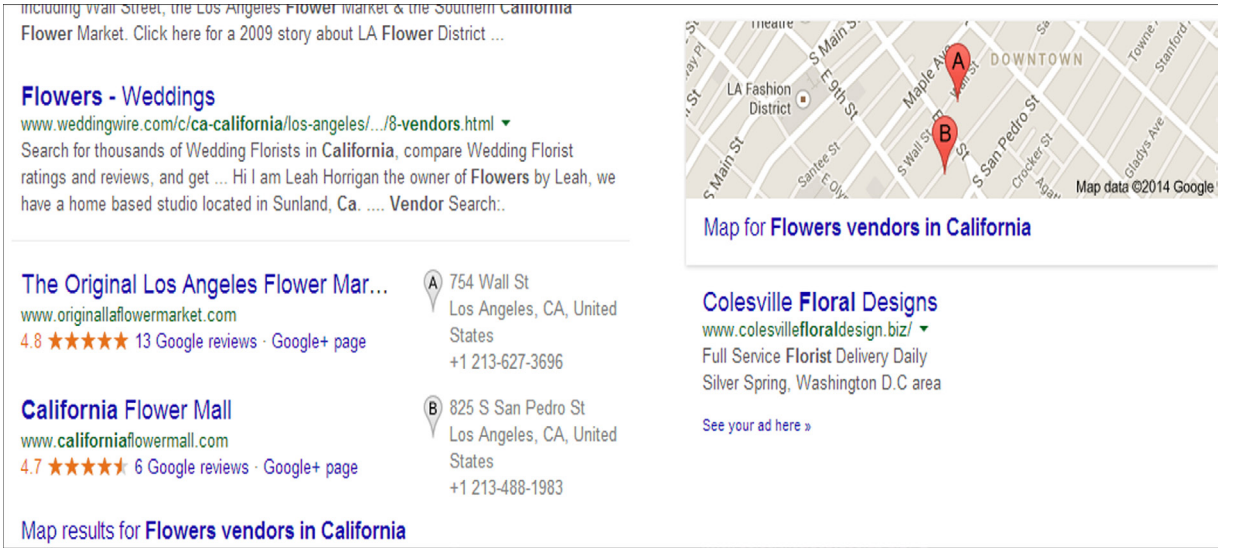


Figure 10.2: Specifying Physical Address of Business

Consistent Name, Address, and Phone number (NAP): Ensure that the business NAP is consistent across all listings on the Internet including local business directories. If there is a change in any of these particulars, make sure that they are updated at all places to maintain the flow of visitors and customers to your business. Figure 10.3 displays different reviews for the same business **Joetheartofcoffee**.

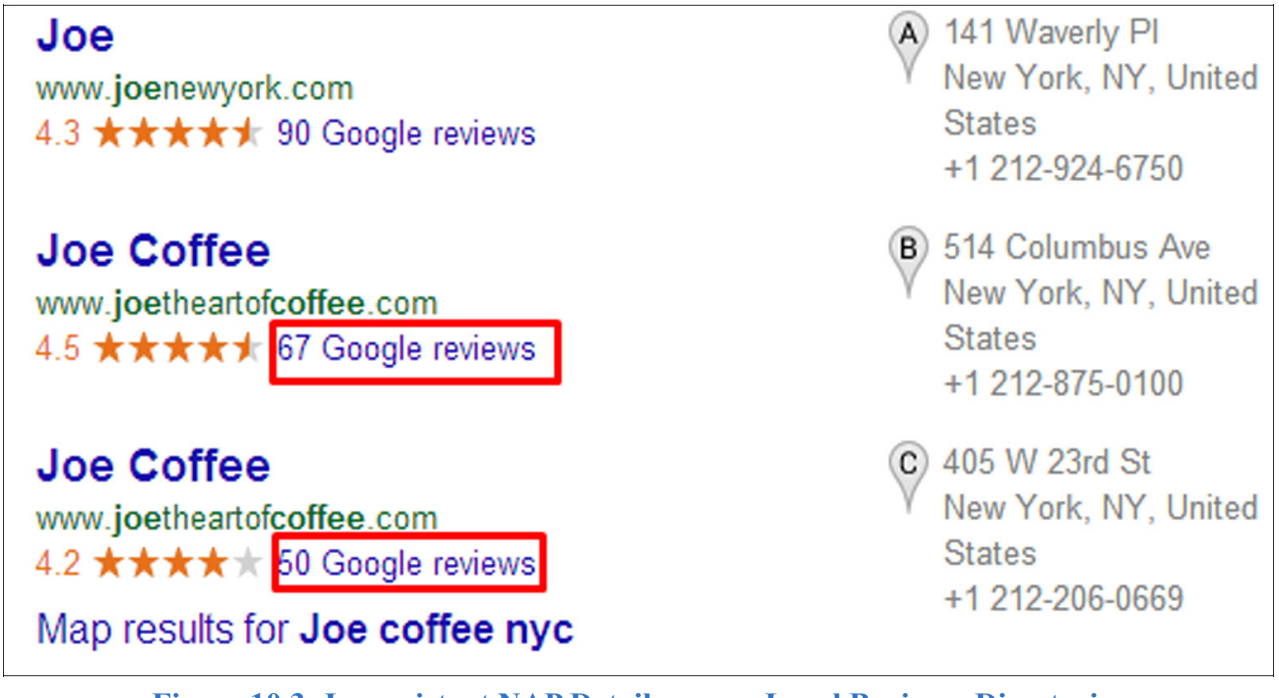


Figure 10.3: Inconsistent NAP Details across Local Business Directories

3. **Domain authority of Web site**: Domain authority of a Web site refers to how strongly a domain name influences the Web site's search results ranking. This depends on factors such as how old a domain name is, the number of inbound links that the domain name has from quality Web sites, and the size of the Web site on the domain. As a Webmaster or SEO consultant, you should include appropriate domain authority information for the business or the site. For example, figure 10.4 displays the domain authority for Walmart Store.

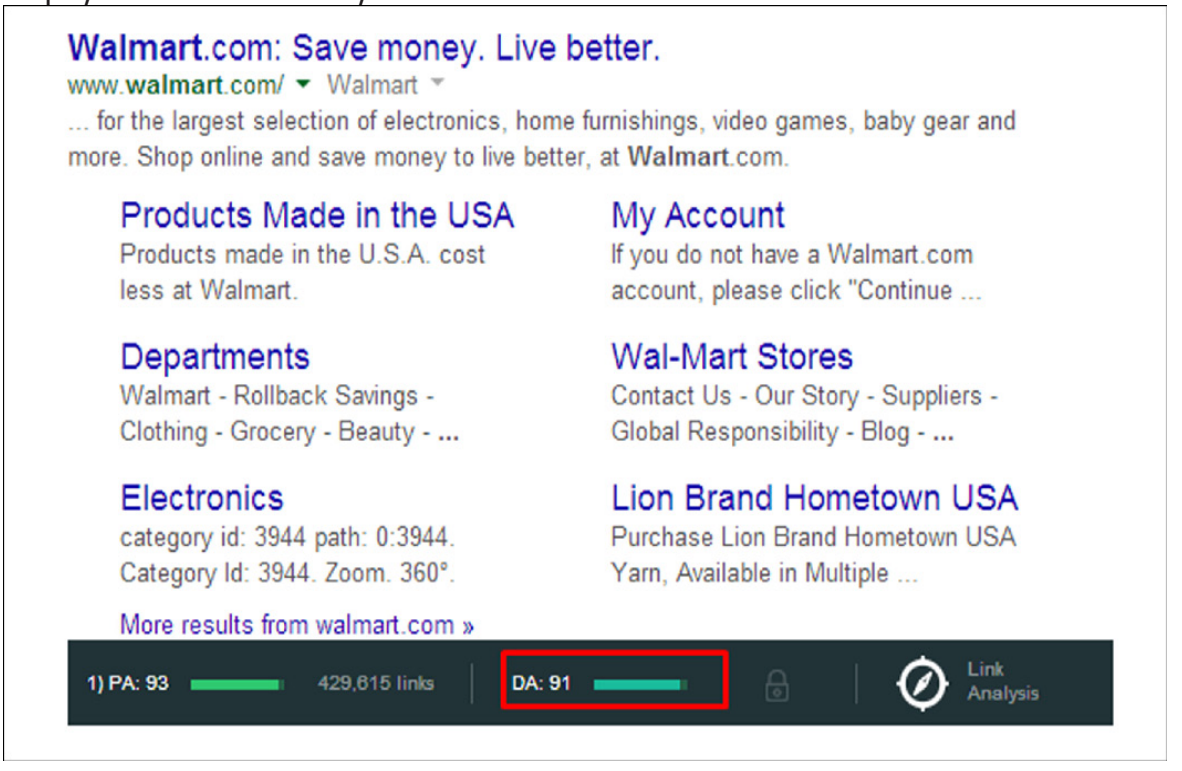


Figure 10.4: Domain Authority of Web Site

4. **Local Google Plus page**: A good local SEO technique is to have a Google+ page for the business with ownership verified. Google+ pages have a high visibility in Google search results. Figure 10.5 shows an example of a local page.

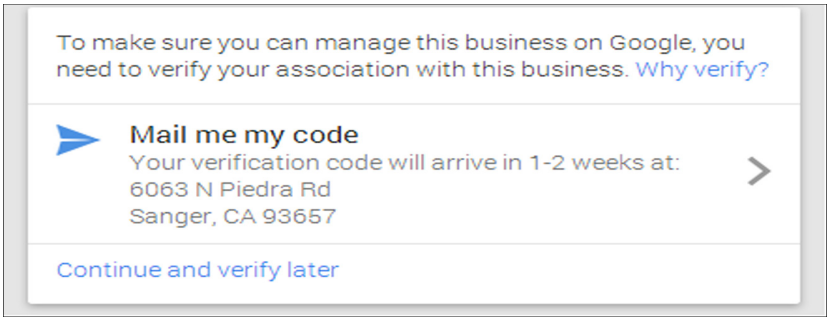


Figure 10.5: Creating User-verified Local Google+ Page

5. **Product/service keyword in business title**: A good SEO practice is to include the product or service offered in the name of the business.

For example, a Cleaners and Laundry business will have a greater search count, better visibility in search results, and higher visitor traffic if it has the words 'Cleaners' included in its name. Figure 10.6 depicts an example for this.

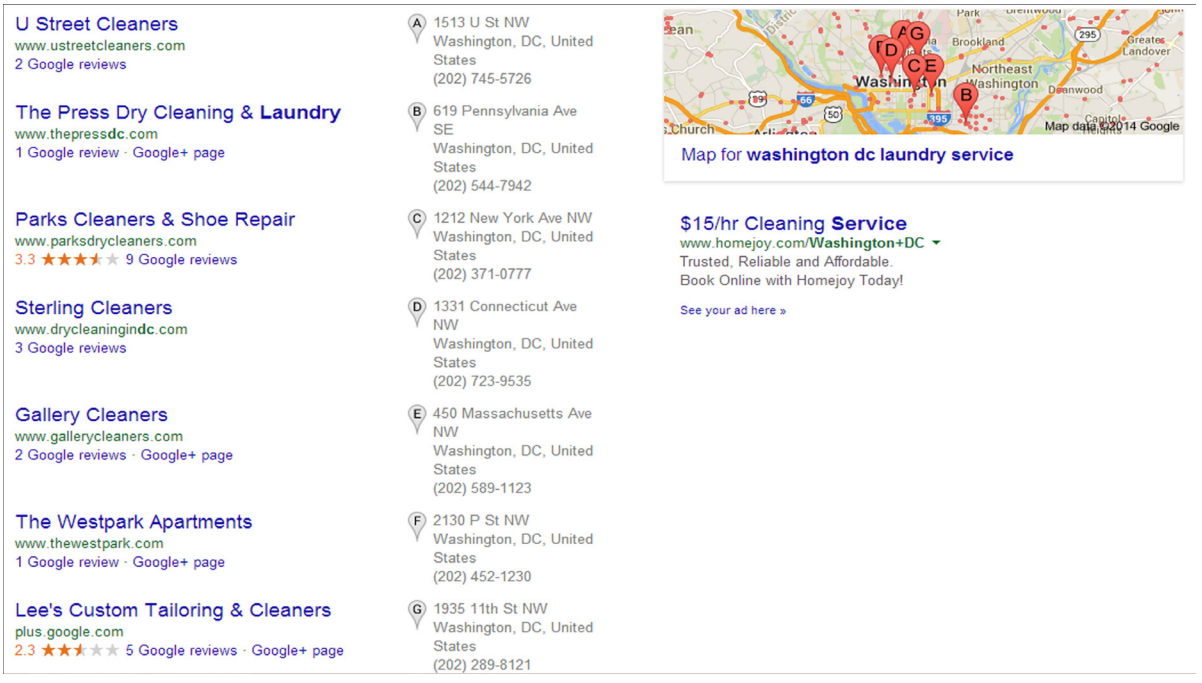


Figure 10.6: Specifying Product/Service Keyword in Business Title

6. **Local area code**: Include area code in phone numbers on Google+ pages as well as, the Web site and local directory listings. Figure 10.7 displays a Google+ page of a business in Washington DC that has its area code included with its phone number.

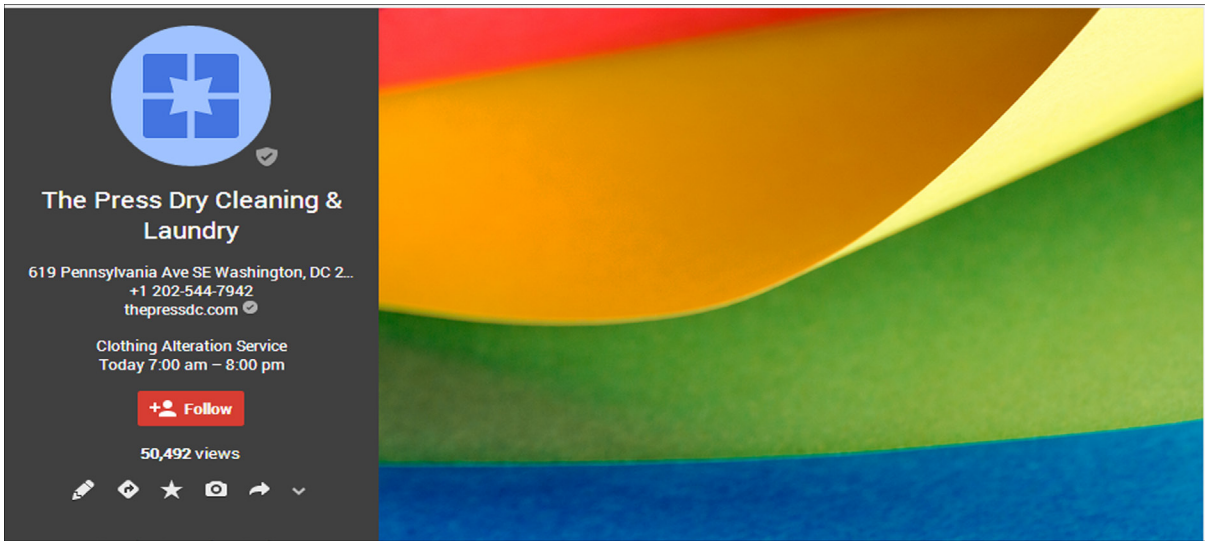


Figure 10.7: Specifying Area Code in Business Phone Number

7. **Schema local markup and Keyhole Markup Language (KML)**: Including schema local markup to design the site content and a KML file in the Web site helps make it easier for search engine crawlers to identify and index the Web site for displaying in search results.

8. **Reviews**: Try to get as many reviews for the Web site products and services. The reviews can be on the Web site pages or on other known Web sites that contain reviews for products and services offered by businesses. The more the number of reviews, especially good reviews, the greater will be the incoming user traffic and visibility on the Internet.

10.5 Understanding Social Media

Social media are the set of Web sites or virtual platforms on which users create and share information with each other on the Internet. It is a virtual platform for social interaction among friends, colleagues, and peers. On the social media, people share their views, opinions, and links to information on a wide variety of topics.

Social media therefore acts as a very important medium for businesses of all types and sizes to reach out to Internet users and market their products or services. It is a very good medium to win customer advocacy.

Following are the various aspects that businesses should take care of for success through the social media:

- **Building relationships and engagements**: Use social media to interact and build relationships with customers. This will help build customer advocacy and loyalty for the business.
- **Gather feedback**: When interacting with customers on social media, make sure the business is open to receive customer opinions, feedback, and comments. These act as feedback for the business. Paying attention to customer interaction on social media might help catch any issues or demands of the customers and help fix them at an early stage. They provide valuable development feedback and help reduce customer service cost.
- **Integration**: Integrate social media to the marketing and Public Relations (PR) strategies of the business. They can also serve as a medium for providing customer service.

10.6 Building Social Media Content

Following are the various types of content that users can place on social media:

- **Related content**: Not every content has to be directly related to the business products or services. Businesses can post information on topics that are related to their line of business. For example, a designer who creates products from recycled material can share tips for recycling at home or office.
- **Responses and promotions**: Give responses to customer comments and opinions. Share information on related topics posted by other users. Promote any events at the organization such as an outstation trip for the employees or a blood camp organized by the business.
- **Job openings**: Social media is a good medium to advertise job openings and find talented resources for working with the organization.

10.7 Types of Social Media

Following are the various social media platforms available on the Internet:

- **Facebook**: Facebook is one of the most popular and widely used social networking tool today. Not just individuals, several businesses, eminent personalities, and even political parties are now a part of Facebook. Facebook allows interaction in groups, both private and public. It allows posting events and building fan pages.
- **Twitter**: Twitter is a revolutionary social networking tool that allows its users to share brief 140 character information bits on the Internet. Twitter can be very useful for businesses in brand building. Since the number of characters that can be entered in a message is restricted, it takes a lot of creativity and skill to use the right kind of words to reach the customers. With short information bits, Twitter tends to overflow with information sharing and interactions. To keep track of all user opinions, comments, and feedback, respond swiftly and immediately to your followers.
- **Google+**: Google+ provides a platform to share a large amount of information in a variety of formats, such as articles, videos, slideshows, and images. Google+ pages act as mini Web sites. In fact, for small businesses, it substitutes for a Web site. Google+ pages help build localized SEO content and with the right kind of information about the business, increase presence in Google search results.
- **LinkedIn**: LinkedIn is one of the largest professional networking tools. It helps businesses reach out to potential employees as well as potential business collaborators all over the world. It also enables sharing content with the community. It is a great avenue to find new clients and grow. Businesses can share information about their company's policies, principles, and social values. It can be used to market the products and services that the businesses offer and gather recommendations from the community.
- **YouTube**: YouTube is a highly used social network tool to share information in video format. Several businesses have started using it as a complementary tool along with other social networking tools such as Facebook and Google+ as well as for their Web sites. Many businesses use YouTube to share their product or service advertisements and even video guides on how to use their products.
- **Pinterest**: Pinterest provides an attractive means of sharing information through a series of images. Businesses are now using this social networking tool to advertise their products and their uses. For instance, a paper trimmer manufacturer can display the various models of paper trimmers that they offer. They can also post information about the various Do It Yourself (DIY) projects that can be done using that product. Thus, attracting customers into purchasing their product.
- **Blogs**: There are a wide variety of blogging sites that are available on the Internet, Blogger and WordPress being the most popular ones. Blogs are like journals that contain articles organized by date of publication. Businesses can use blogs to inform their customers about new products or developments/improvements in their products and services. Blogs can also be used to inform customers about events and promotions. Information can be shared in the form of text, images, videos, or infographics.

10.8 Guidelines for Using Social Media

Following are the key features of social media that need to be taken care off when using it:

- **Appropriate content**: Content posted on social networking sites has a very wide reach. It does not necessarily stay restricted to a group of followers. People can share it not just with their own extended network on the same site, but also on other networking sites. It is therefore very important to pay attention to the type of content that is posted on a social networking site. Do not post inappropriate content that might hurt the sentiments or values of any community.
- **Post timings**: Pay attention to the types of information that get the most attention from the network community. Also, gather information about the time and geographic locations that has your followers most active. This can be used to build better marketing strategies and direct interaction points with the customers.
- **Credible information**: Check facts before posting any piece of information on the site. Make sure that no malicious or spam content is posted on the site. Also, ensure that the content uses correct grammar and spellings in the content.
- **Enable social sharing**: Set up blogs and Web sites to be linked to the other social networking profiles of the business. Also, enable users to share the content across various social networking sites.

Check Your Progress

1. Content marketing is the technique of:

(A)	creating ads for products and services offered on a Web site	(C)	creating engaging content that compels visitors to share with their friends and people in their community
(B)	creating content that will increase the visibility of the Web site on the Internet and generate more traffic	(D)	adding content to social networking sites

2. Including schema local markup and KML to design Web site content helps:

(A)	gather information regarding incoming user traffic	(C)	improve the domain authority of the Web site
(B)	make it easier for search engine crawlers to identify and index the Web site for displaying in search results	(D)	display Web site in local directory listings

3. Which of the following indicate the benefits of customer interaction, opinions, feedback, and opinions on social media networks?

(A)	increases the visitor traffic on Web sites	(C)	helps gather feedback for the business
(B)	helps improve the page ranking of Web sites	(D)	helps build customer advocacy and loyalty for business

(A)	A, B	(C)	A, C, D
(B)	C, D	(D)	B

Answer

1.	C
2.	B
3.	B

Summary

- Content marketing is the technique of presenting Web site content such that it induces extensive sharing of content by visitors, thus potentially bring in new customers and business.
- Search engine optimized content can be created by including product pages, blog posts, articles, guides, videos, and infographics.
- Local SEO is very important for businesses that serve the local customers such as doctors, dentists, lawyers, restaurants, handymen, grocery stores, and help create local presence in the market.
- Social media are the set of Web sites or virtual platforms for social interaction and sharing information among friends, colleagues, and peers.
- Social media acts as a very useful medium for businesses of all types and sizes to reach out to Internet users and market their products or services.
- Facebook, Twitter, Google+, LinkedIn, YouTube, Pinterest, and blogs are some of the popular social networking media.

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